



US012414460B2

(12) **United States Patent**
Song et al.

(10) **Patent No.:** **US 12,414,460 B2**

(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **ORGANIC LIGHT EMITTING DIODE AND ORGANIC LIGHT EMITTING DEVICE INCLUDING THE SAME**

(71) Applicant: **LG DISPLAY CO., LTD.**, Seoul (KR)

(72) Inventors: **In Bum Song**, Paju-si (KR); **Seung Hee Yoon**, Paju-si (KR); **Jin Ho Park**, Paju-si (KR); **Dong Hyeok Lim**, Paju-si (KR)

(73) Assignee: **LG Display Co., Ltd.**, Seoul (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 949 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **17/260,178**

(22) PCT Filed: **Dec. 21, 2019**

(86) PCT No.: **PCT/KR2019/018262**

§ 371 (c)(1),

(2) Date: **Jan. 13, 2021**

(87) PCT Pub. No.: **WO2020/138871**

PCT Pub. Date: **Jul. 2, 2020**

(65) **Prior Publication Data**

US 2021/0320265 A1 Oct. 14, 2021

(30) **Foreign Application Priority Data**

Dec. 28, 2018 (KR) 10-2018-0172061

(51) **Int. Cl.**

H10K 85/60 (2023.01)

H10K 50/13 (2023.01)

H10K 50/18 (2023.01)

(52) **U.S. Cl.**

CPC **H10K 85/615** (2023.02); **H10K 50/13** (2023.02); **H10K 85/622** (2023.02);

(Continued)

(58) **Field of Classification Search**

CPC .. H10K 85/615; H10K 85/626; H10K 85/658; H10K 50/13

See application file for complete search history.

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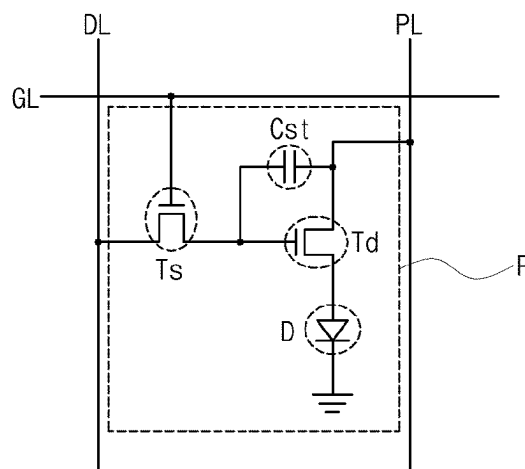
Primary Examiner — Kregg T Brooks

(74) *Attorney, Agent, or Firm* — Morgan, Lewis & Bockius LLP

(57) **ABSTRACT**

The present disclosure relates to an OLED that includes a first electrode; a second electrode facing the first electrode; a first emitting material layer including a first host, a second host and a blue dopant and positioned between the first and second electrodes; a first electron blocking layer including an electron blocking material of an amine derivative and positioned between the first electrode and the first emitting material layer; and a first hole blocking layer including at

(Continued)



least one of a first hole blocking material and a second hole blocking material and positioned between the second electrode and the first emitting material layer, wherein the first host is an anthracene derivative, and the second host is a deuterated anthracene derivative, and wherein the first hole blocking material is an azine derivative, and the second hole blocking material is a benzimidazole derivative.

37 Claims, 3 Drawing Sheets

(52) U.S. Cl.

CPC **H10K 85/624** (2023.02); **H10K 85/633** (2023.02); **H10K 85/654** (2023.02); **H10K 85/6572** (2023.02); **H10K 85/6574** (2023.02); **H10K 50/18** (2023.02); **H10K 50/181** (2023.02); **H10K 85/658** (2023.02)

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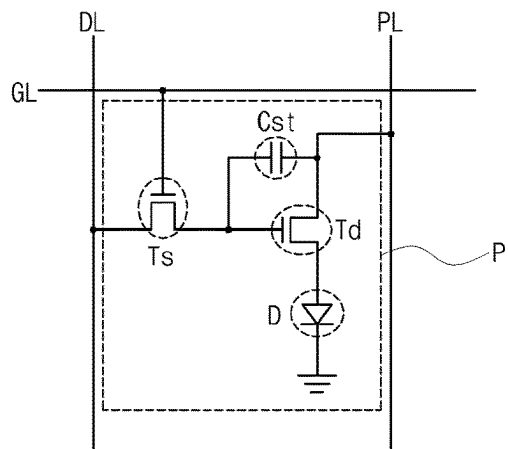
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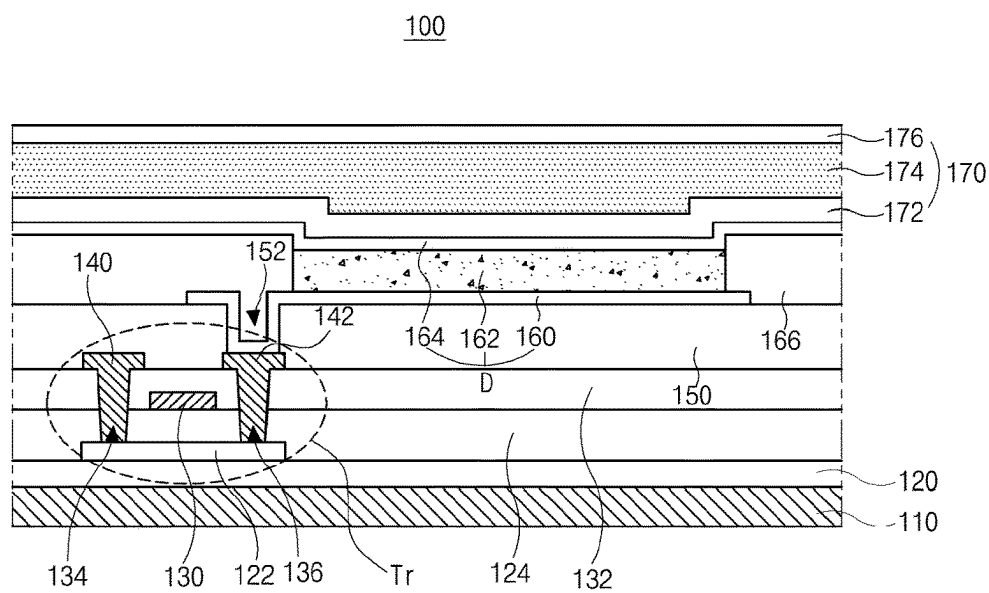
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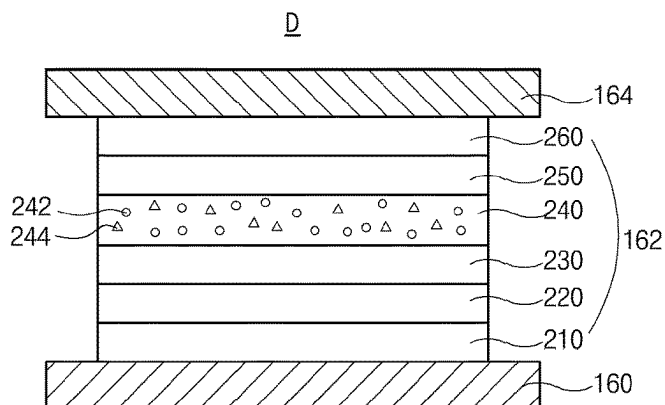
[Fig. 1]



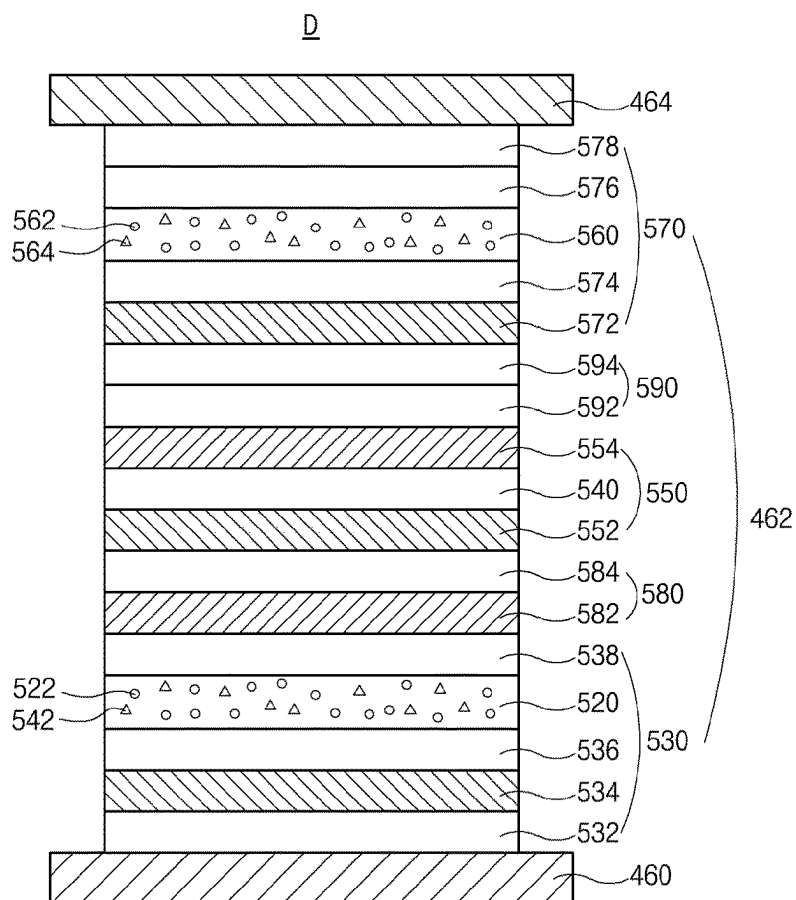
[Fig. 2]



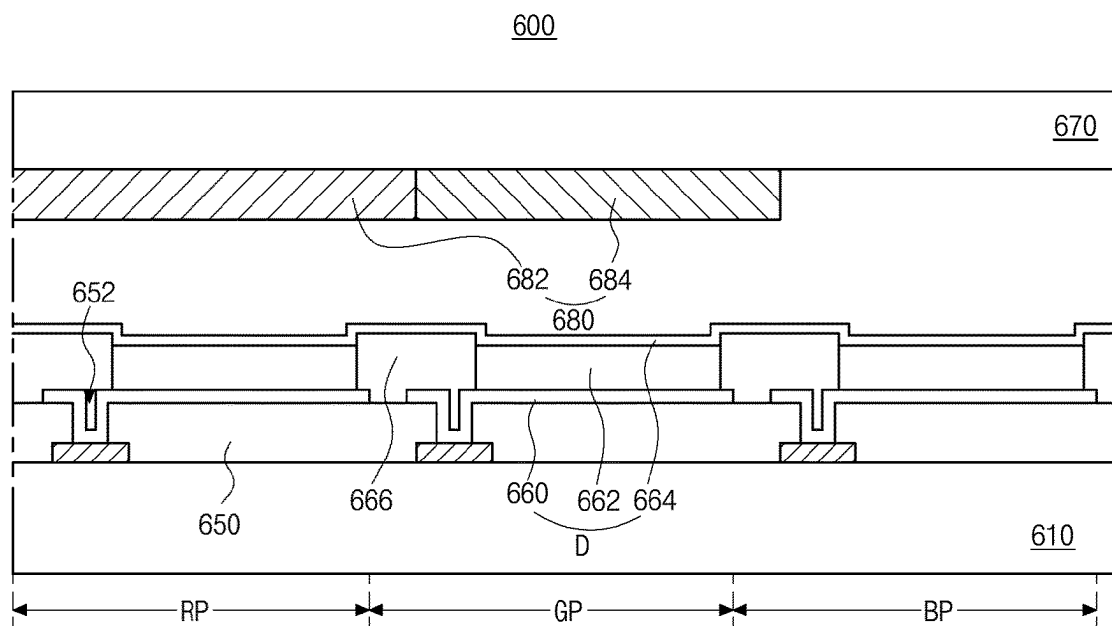
[Fig. 3]



[Fig. 6]



[Fig. 7]



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ORGANIC LIGHT EMITTING DIODE AND ORGANIC LIGHT EMITTING DEVICE INCLUDING THE SAME

TECHNICAL FIELD

The present disclosure relates to an organic light emitting diode (OLED), and more specifically, to an OLED having enhanced emitting efficiency and lifespan and an organic light emitting device including the same.

BACKGROUND ART

As requests for a flat panel display device having a small occupied area have been increased, an organic light emitting display device including an OLED has been the subject of recent research and development.

The OLED emits light by injecting electrons from a cathode as an electron injection electrode and holes from an anode as a hole injection electrode into an emitting material layer (EML), combining the electrons with the holes, generating an exciton, and transforming the exciton from an excited state to a ground state. A flexible substrate, for example, a plastic substrate, can be used as a base substrate where elements are formed. In addition, the organic light emitting display device can be operated at a voltage (e.g., 10V or below) lower than a voltage required to operate other display devices. Moreover, the organic light emitting display device has advantages in the power consumption and the color sense.

The OLED includes a first electrode as an anode over a substrate, a second electrode, which is spaced apart from and faces the first electrode, and an organic emitting layer therebetween.

For example, the organic light emitting display device may include a red pixel region, a green pixel region and a blue pixel region, and the OLED may be formed in each of the red, green and blue pixel regions.

However, the OLED in the blue pixel does not provide sufficient emitting efficiency and lifespan such that the organic light emitting display device has a limitation in the emitting efficiency and the lifespan.

DISCLOSURE OF INVENTION

Technical Problem

Accordingly, the present disclosure is directed to an OLED and an organic light emitting device including the OLED that substantially obviate one or more of the problems due to the limitations and disadvantages of the related art.

An object of the present disclosure is to provide an OLED having enhanced emitting efficiency and lifespan and an organic light emitting device including the same.

Additional features and advantages of the disclosure will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by practice of the disclosure. The objectives and other advantages of the disclosure will be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.

Solution to Problem

According to an aspect, the present disclosure provides an OLED that includes a first electrode; a second electrode

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facing the first electrode; a first emitting material layer including a first host, a second host and a blue dopant and positioned between the first and second electrodes; a first electron blocking layer including an electron blocking material of an amine derivative and positioned between the first electrode and the first emitting material layer; and a first hole blocking layer including at least one of a first hole blocking material and a second hole blocking material and positioned between the second electrode and the first emitting material layer, wherein the first host is an anthracene derivative, and the second host is a deuterated anthracene derivative, and wherein the first hole blocking material is an azine derivative, and the second hole blocking material is a benzimidazole derivative.

As an example, in the first emitting material layer, a weight % ratio of the first host to the second host is 1:9 to 9:1.

As an example, in the first emitting material layer, the weight % ratio of the first host to the second host is 1:9 to 7:3.

As an example, in the first emitting material layer, the weight % ratio of the first host to the second host is 3:7.

As an example, in the first emitting material layer, the weight % ratio of the first host to the second host is 7:3.

The OLED may include a single emitting part or a tandem structure of a multiple emitting parts.

The tandem-structured OLED may emit blue color or white color.

According to another aspect, the present disclosure provides an organic light emitting device comprising the OLED, as described above.

For example, the organic light emitting device may be an organic light emitting display device or a lightening device.

It is to be understood that both the foregoing general description and the following detailed description are examples and are explanatory and are intended to provide further explanation of the disclosure as claimed.

Advantageous Effects of Invention

An emitting material layer of an OLED of the present disclosure includes a first host of an anthracene derivative and a second host of a deuterated anthracene derivative such that an emitting efficiency and a lifespan of the OLED and an organic light emitting device including the OLED are improved.

In addition, an electron blocking layer of an OLED of the present disclosure includes an amine derivative as an electron blocking material, and a hole blocking layer of the OLED includes at least one of an azine derivative and a benzimidazole derivative as a hole blocking material. Accordingly, the lifespan of the OLED and an organic light emitting device is further improved.

BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the disclosure, are incorporated in and constitute a part of this specification, illustrate implementations of the disclosure and together with the description serve to explain the principles of embodiments of the disclosure.

FIG. 1 is a schematic circuit diagram illustrating an organic light emitting display device of the present disclosure.

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FIG. 2 is a schematic cross-sectional view illustrating an organic light emitting display device according to a first embodiment of the present disclosure.

FIG. 3 is a schematic cross-sectional view illustrating an OLED having a single emitting part for the organic light emitting display device according to the first embodiment of the present disclosure.

FIG. 4 is a schematic cross-sectional view illustrating an OLED having a tandem structure of two emitting parts according to the first embodiment of the present disclosure.

FIG. 5 is a schematic cross-sectional view illustrating an organic light emitting display device according to a second embodiment of the present disclosure.

FIG. 6 is a schematic cross-sectional view illustrating an OLED for the organic light emitting display device according to the second embodiment of the present disclosure.

FIG. 7 is a schematic cross-sectional view illustrating an organic light emitting display device according to a third embodiment of the present disclosure.

MODE FOR THE INVENTION

Reference will now be made in detail to aspects of the disclosure, examples of which are illustrated in the accompanying drawings.

FIG. 1 is a schematic circuit diagram illustrating an organic light emitting display device of the present disclosure.

As illustrated in FIG. 1, a gate line GL and a data line DL, which cross each other to define a pixel (pixel region) P, and a power line PL are formed in an organic light display device. A switching thin film transistor (TFT) Ts, a driving TFT Td, a storage capacitor Cst and an OLED D are formed in the pixel region P. The pixel region P may include a red pixel, a green pixel and a blue pixel.

The switching thin film transistor Ts is connected to the gate line GL and the data line DL, and the driving thin film transistor Td and the storage capacitor Cst are connected between the switching thin film transistor Ts and the power line PL. The OLED D is connected to the driving thin film transistor Td. When the switching thin film transistor Ts is turned on by the gate signal applied through the gate line GL, the data signal applied through the data line DL is applied a gate electrode of the driving thin film transistor Td and one electrode of the storage capacitor Cst through the switching thin film transistor Ts.

The driving thin film transistor Td is turned on by the data signal applied into the gate electrode so that a current proportional to the data signal is supplied from the power line PL to the OLED D through the driving thin film transistor Tr. The OLED D emits light having a luminance proportional to the current flowing through the driving thin film transistor Td. In this case, the storage capacitor Cst is charge with a voltage proportional to the data signal so that the voltage of the gate electrode in the driving thin film transistor Td is kept constant during one frame. Therefore, the organic light emitting display device can display a desired image.

FIG. 2 is a schematic cross-sectional view illustrating an organic light emitting display device according to a first embodiment of the present disclosure.

As illustrated in FIG. 2, the organic light emitting display device 100 includes a substrate 110, a TFT Tr and an OLED D connected to the TFT Tr. For example, the organic light emitting display device 100 may include a red pixel, a green pixel and a blue pixel, and the OLED D may be formed in each of the red, green and blue pixels. Namely, the OLEDs

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D emitting red light, green light and blue light may be provided in the red, green and blue pixels, respectively.

The substrate 110 may be a glass substrate or a plastic substrate. For example, the substrate 110 may be a polyimide substrate.

A buffer layer 120 is formed on the substrate, and the TFT Tr is formed on the buffer layer 120. The buffer layer 120 may be omitted.

A semiconductor layer 122 is formed on the buffer layer 120. The semiconductor layer 122 may include an oxide semiconductor material or polycrystalline silicon.

When the semiconductor layer 122 includes the oxide semiconductor material, a light-shielding pattern (not shown) may be formed under the semiconductor layer 122. The light to the semiconductor layer 122 is shielded or blocked by the light-shielding pattern such that thermal degradation of the semiconductor layer 122 can be prevented. On the other hand, when the semiconductor layer 122 includes polycrystalline silicon, impurities may be doped into both sides of the semiconductor layer 122.

A gate insulating layer 124 is formed on the semiconductor layer 122. The gate insulating layer 124 may be formed of an inorganic insulating material such as silicon oxide or silicon nitride.

A gate electrode 130, which is formed of a conductive material, e.g., metal, is formed on the gate insulating layer 124 to correspond to a center of the semiconductor layer 122.

In FIG. 2, the gate insulating layer 124 is formed on an entire surface of the substrate 110. Alternatively, the gate insulating layer 124 may be patterned to have the same shape as the gate electrode 130.

An interlayer insulating layer 132, which is formed of an insulating material, is formed on the gate electrode 130. The interlayer insulating layer 132 may be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride, or an organic insulating material, e.g., benzocyclobutene or photo-acryl.

The interlayer insulating layer 132 includes first and second contact holes 134 and 136 exposing both sides of the semiconductor layer 122. The first and second contact holes 134 and 136 are positioned at both sides of the gate electrode 130 to be spaced apart from the gate electrode 130.

The first and second contact holes 134 and 136 are formed through the gate insulating layer 124. Alternatively, when the gate insulating layer 124 is patterned to have the same shape as the gate electrode 130, the first and second contact holes 134 and 136 is formed only through the interlayer insulating layer 132.

A source electrode 140 and a drain electrode 142, which are formed of a conductive material, e.g., metal, are formed on the interlayer insulating layer 132.

The source electrode 140 and the drain electrode 142 are spaced apart from each other with respect to the gate electrode 130 and respectively contact both sides of the semiconductor layer 122 through the first and second contact holes 134 and 136.

The semiconductor layer 122, the gate electrode 130, the source electrode 140 and the drain electrode 142 constitute the TFT Tr. The TFT Tr serves as a driving element. Namely, the TFT Tr may correspond to the driving TFT Td (of FIG. 1).

In the TFT Tr, the gate electrode 130, the source electrode 140, and the drain electrode 142 are positioned over the semiconductor layer 122. Namely, the TFT Tr has a coplanar structure.

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Alternatively, in the TFT Tr, the gate electrode may be positioned under the semiconductor layer, and the source and drain electrodes may be positioned over the semiconductor layer such that the TFT Tr may have an inverted staggered structure. In this instance, the semiconductor layer may include amorphous silicon.

Although not shown, the gate line and the data line cross each other to define the pixel, and the switching TFT is formed to be connected to the gate and data lines. The switching TFT is connected to the TFT Tr as the driving element.

In addition, the power line, which may be formed to be parallel to and spaced apart from one of the gate and data lines, and the storage capacitor for maintaining the voltage of the gate electrode of the TFT Tr in one frame may be further formed.

A passivation layer 150, which includes a drain contact hole 152 exposing the drain electrode 142 of the TFT Tr, is formed to cover the TFT Tr.

A first electrode 160, which is connected to the drain electrode 142 of the TFT Tr through the drain contact hole 152, is separately formed in each pixel. The first electrode 160 may be an anode and may be formed of a conductive material having a relatively high work function. For example, the first electrode 160 may be formed of a transparent conductive material such as indium-tin-oxide (ITO) or indium-zinc-oxide (IZO).

When the OLED device 100 is operated in a top-emission type, a reflection electrode or a reflection layer may be formed under the first electrode 160. For example, the reflection electrode or the reflection layer may be formed of aluminum-palladium-copper (APC) alloy.

A bank layer 166 is formed on the passivation layer 150 to cover an edge of the first electrode 160. Namely, the bank layer 166 is positioned at a boundary of the pixel and exposes a center of the first electrode 160 in the pixel.

An organic emitting layer 162 is formed on the first electrode 160. The organic emitting layer 162 may have a single-layered structure of an emitting material layer including an emitting material. To increase an emitting efficiency of the OLED D and/or the organic light emitting display device 100, the organic emitting layer 162 may have a multi-layered structure. For example, the organic emitting layer 162 may include the EML, an electron blocking layer (EBL) between the first electrode 160 and the EML, and a hole blocking layer (HBL) between the EML and the second electrode 164.

The organic emitting layer 162 is separated in each of the red, green and blue pixels. As illustrated below, the organic emitting layer 162 in the blue pixel includes a first host of an anthracene derivative (compound) and a second host of a deuterated anthracene derivative such that the emitting efficiency and the lifespan of the OLED D in the blue pixel are improved.

In addition, the EBL includes an amine derivative as an electron blocking material, and the HBL includes at least one of an azine derivative and a benzimidazole derivative as a hole blocking material. Accordingly, the lifespan of the OLED D and an organic light emitting device 100 is further improved.

A second electrode 164 is formed over the substrate 110 where the organic emitting layer 162 is formed. The second electrode 164 covers an entire surface of the display area and may be formed of a conductive material having a relatively low work function to serve as a cathode. For example, the second electrode 164 may be formed of aluminum (Al), magnesium (Mg) or Al—Mg alloy.

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The first electrode 160, the organic emitting layer 162 and the second electrode 164 constitute the OLED D.

An encapsulation film 170 is formed on the second electrode 164 to prevent penetration of moisture into the OLED D. The encapsulation film 170 includes a first inorganic insulating layer 172, an organic insulating layer 174 and a second inorganic insulating layer 176 sequentially stacked, but it is not limited thereto. The encapsulation film 170 may be omitted.

A polarization plate (not shown) for reducing an ambient light reflection may be disposed over the top-emission type OLED D. For example, the polarization plate may be a circular polarization plate.

In addition, a cover window (not shown) may be attached to the encapsulation film 170 or the polarization plate. In this instance, the substrate 110 and the cover window have a flexible property such that a flexible display device may be provided.

FIG. 3 is a schematic cross-sectional view illustrating an OLED having a single emitting unit for the organic light emitting display device according to the first embodiment of the present disclosure.

As illustrated in FIG. 3, the OLED D includes the first and second electrodes 160 and 164, which face each other, and the organic emitting layer 162 therebetween. The organic emitting layer 162 includes an EML 240 between the first and second electrodes 160 and 164, an EBL 230 between the first electrode 160 and the EML 240, and an HBL 250 between the EML 240 and the second electrode 164.

The first electrode 160 may be formed of a conductive material having a relatively high work function to serve as an anode. The second electrode 164 may be formed of a conductive material having a relatively low work function to serve as a cathode.

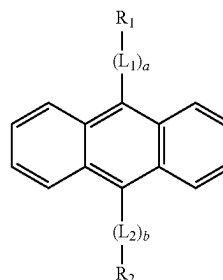
The organic emitting layer 162 may further include a hole transporting layer (HTL) 220 between the first electrode 160 and the EBL 230.

Moreover, the organic emitting layer 162 may further include a hole injection layer (HIL) 210 between the first electrode 160 and the HTL 220 and an electron injection layer (EIL) 260 between the second electrode 164 and the HBL 250.

The EML 240 includes a first host 242 of an anthracene derivative, a second host 244 of a deuterated anthracene derivative and a blue dopant (not shown) and provides blue emission.

The compound of the first host 242 may be represented by Formula 1:

Formula 1



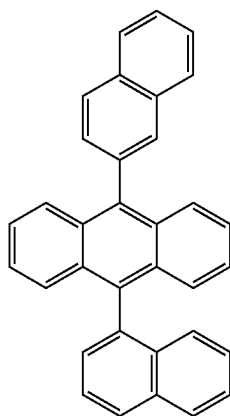
In Formula 1, each of R_1 and R_2 is independently C_6 – C_{30} aryl group or C_5 – C_{30} heteroaryl group, and each of L_1 and L_2 is independently C_6 – C_{30} arylene group. Each of a and b is an integer of 0 or 1, and at least one of a and b is 0.

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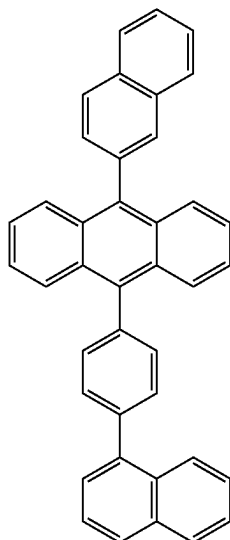
For example, R_1 may be phenyl or naphthyl, R_2 may be naphthyl, dibenzofuranyl or fused dibenzofuranyl. Each of L_1 and L_2 may independently be phenylene.

In an exemplary embodiment, the first host **242** may be a compound being one of the followings in Formula 2:

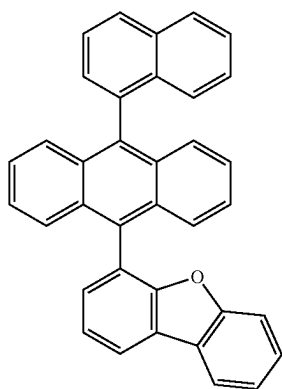
Formula 2



Host 1



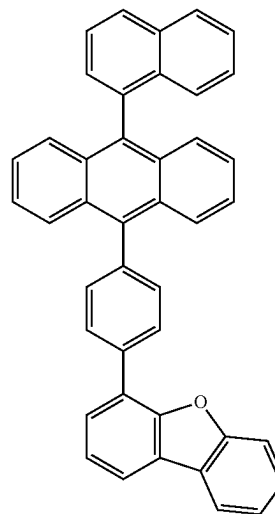
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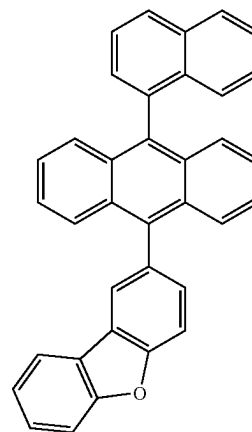
Host 3

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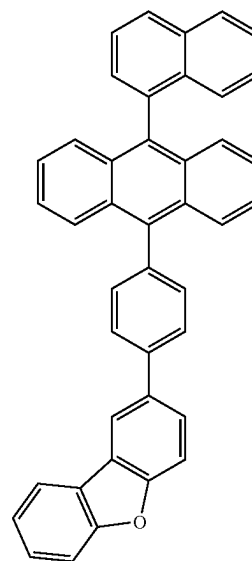
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Host 4



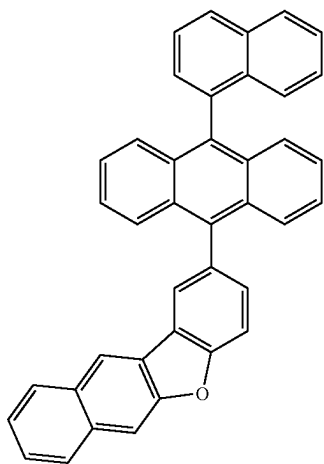
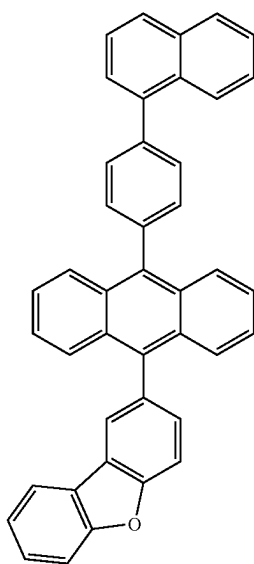
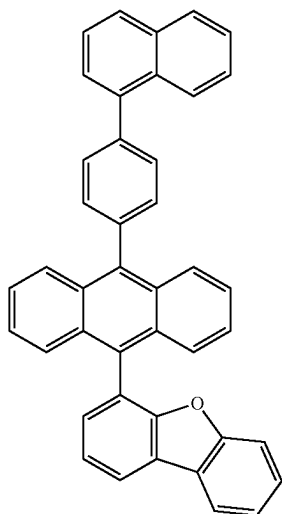
Host 5



Host 6

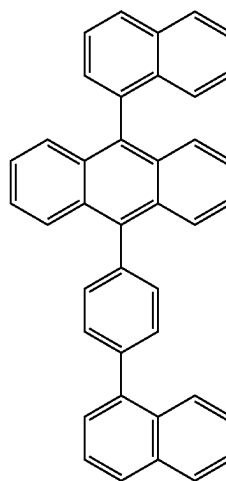
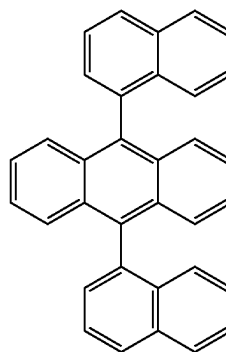
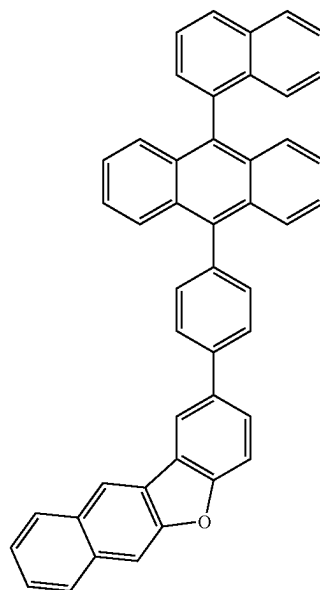
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Host 7

Host 10

Host 8

Host 11

Host 9

Host 12

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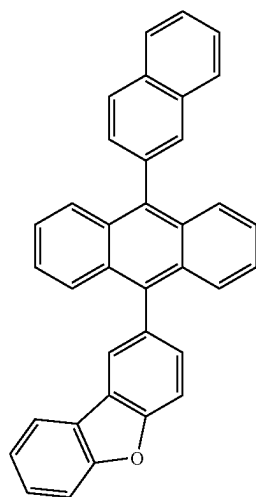
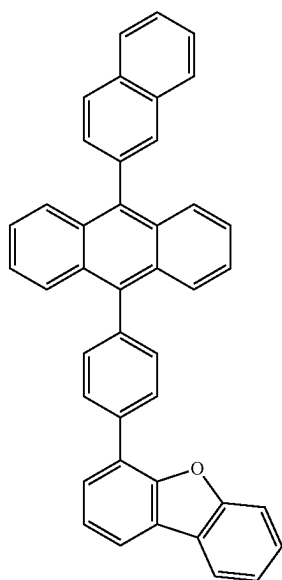
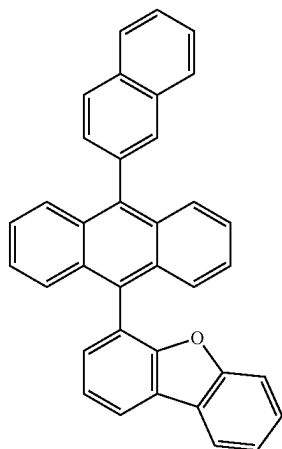
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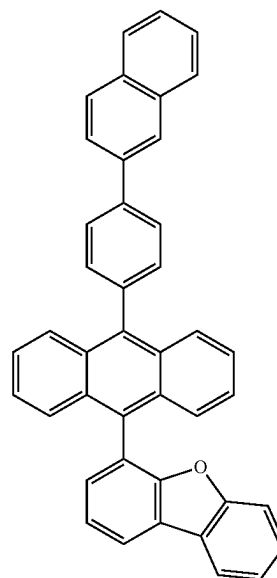
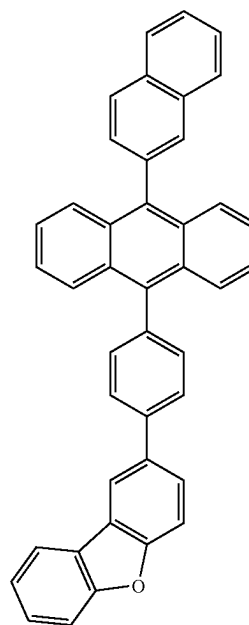
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12

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Host 13

Host 16

Host 14

Host 17

Host 15

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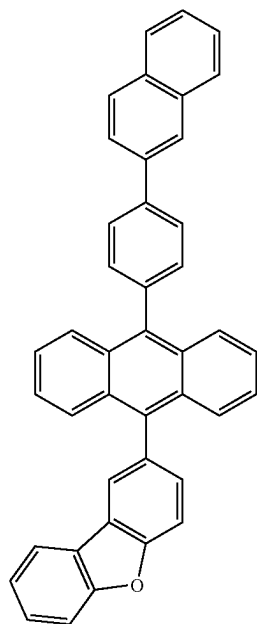
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-continued



14

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Host 18

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Host 19

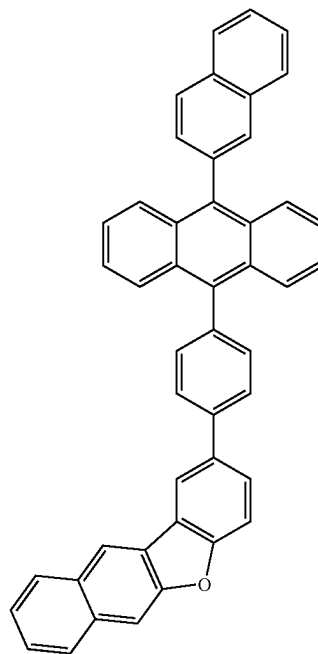
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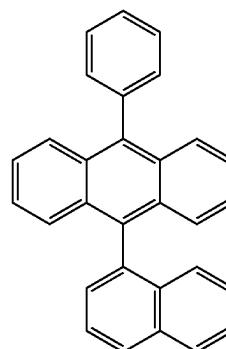
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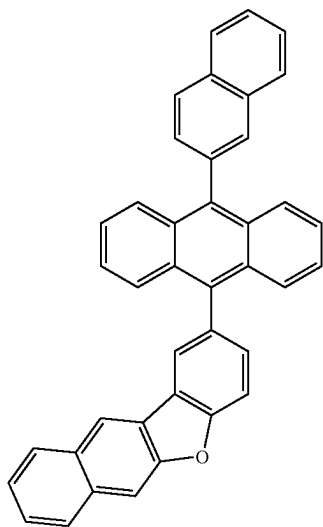
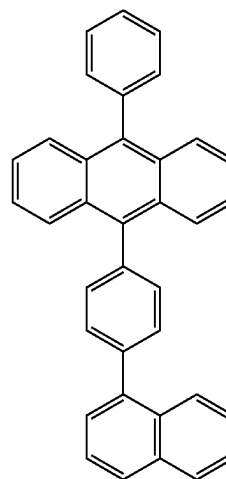
Host 20



Host 21

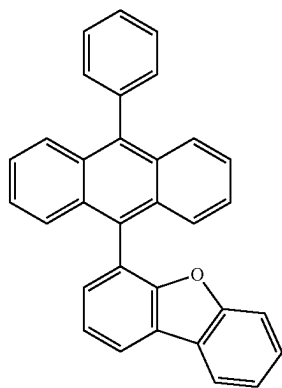


Host 22



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-continued



Host 23

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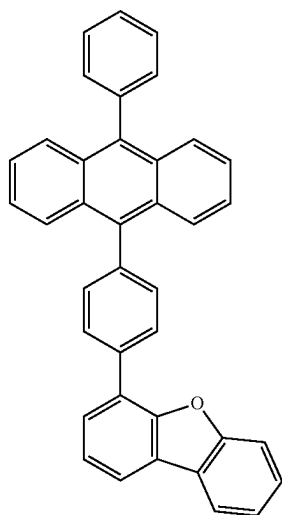
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Host 24

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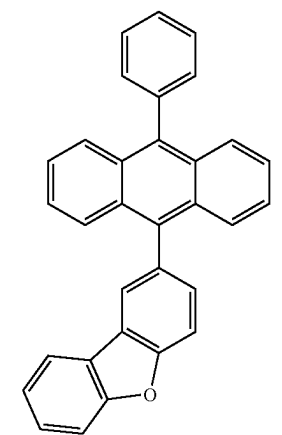


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Host 25

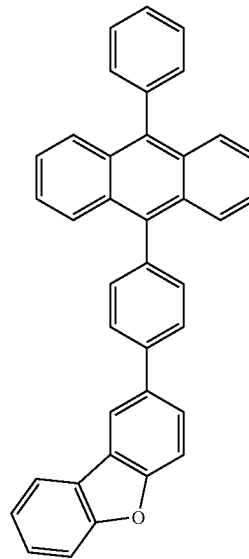
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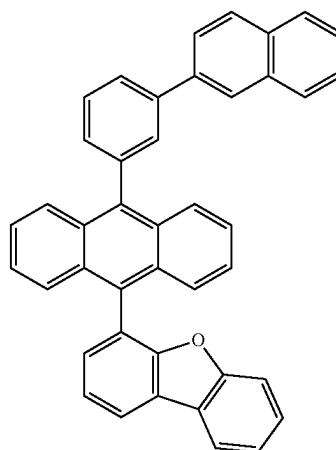
16

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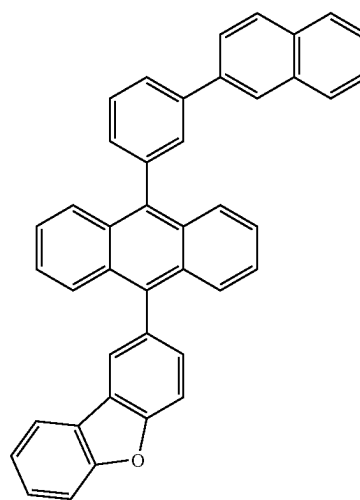


Host 26

Host 27

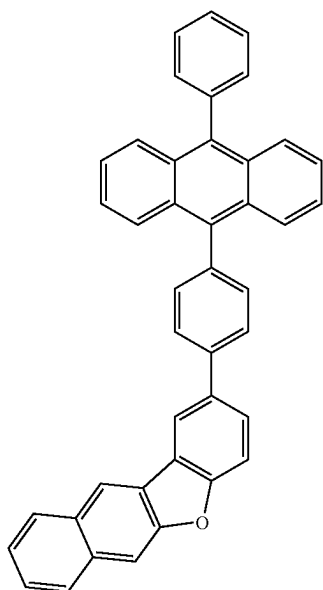
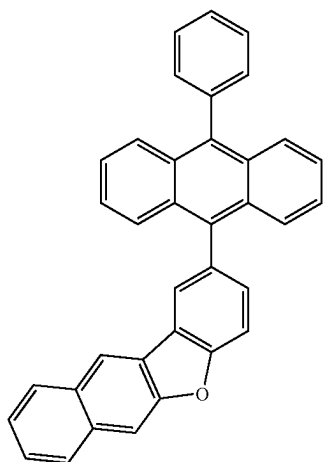


Host 28



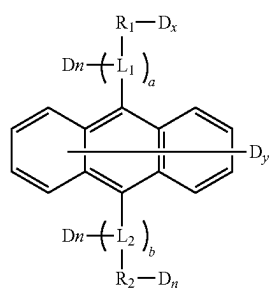
17

-continued



The second host **244** may be a deuterated compound of the first host **242**. Namely, a hydrogen atom of the first host **242** may be substituted by a deuterium atom to form the second host **244**. A part or all of the hydrogen atoms of the compound of the first host **242** may be substituted by the deuterium atom.

For example, the compound of the second host **244** may be represented by Formula 3:

**18**

Host 29

In Formula 3, the definition of R_1 , R_2 , L_1 , L_2 , a and b is same as Formula 1. In Formula 3, D_x , D_y , D_m and D_n denote a number of the deuterium atom, and each of x , y , m and n is independently a positive integer. For example, a summation of x , y , m and n may be 15 to 29.

In an exemplary embodiment, the second host **244** of Formula 3 may be a compound being one of the followings in Formula 4:

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15 Formula 4

Host 30

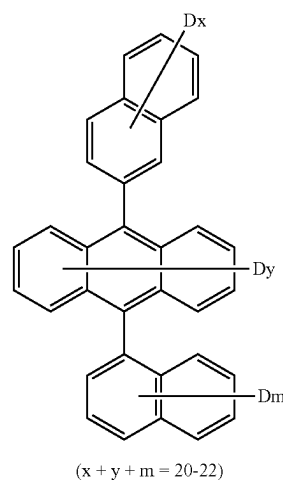
Host 31

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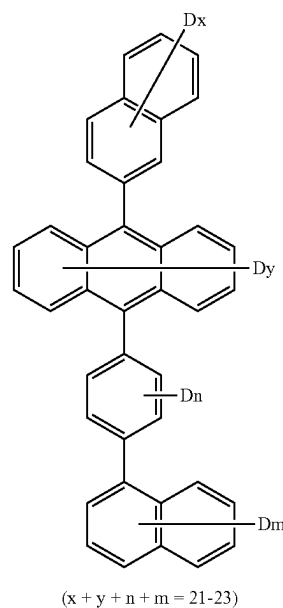


Host 32

Formula 3

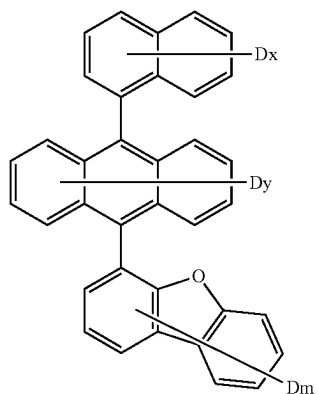
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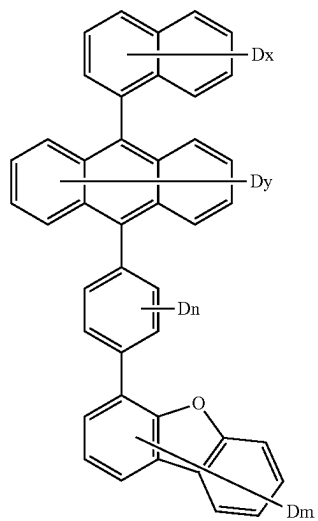


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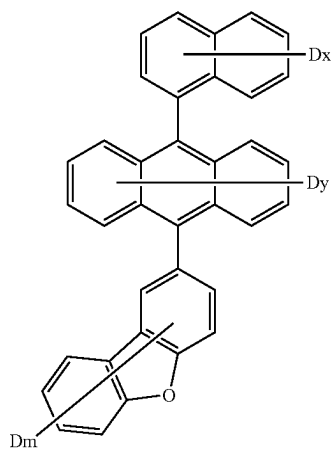
-continued



$$(x + y + m = 20-22)$$



$$(x + y + n + m = 21-23)$$



$$(x + y + m = 20-22)$$

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-continued

Host 33

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Host 34

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Host 35

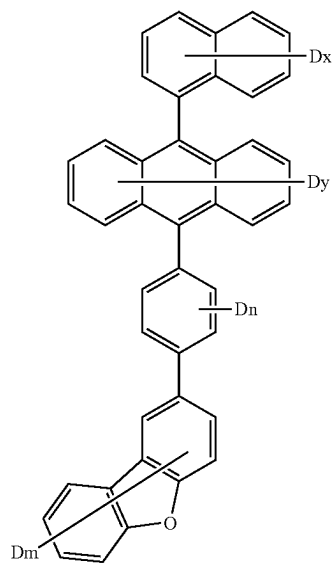
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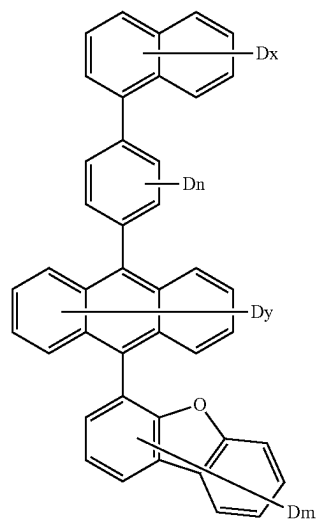
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Host 36



$$(x + y + n + m = 21-23)$$

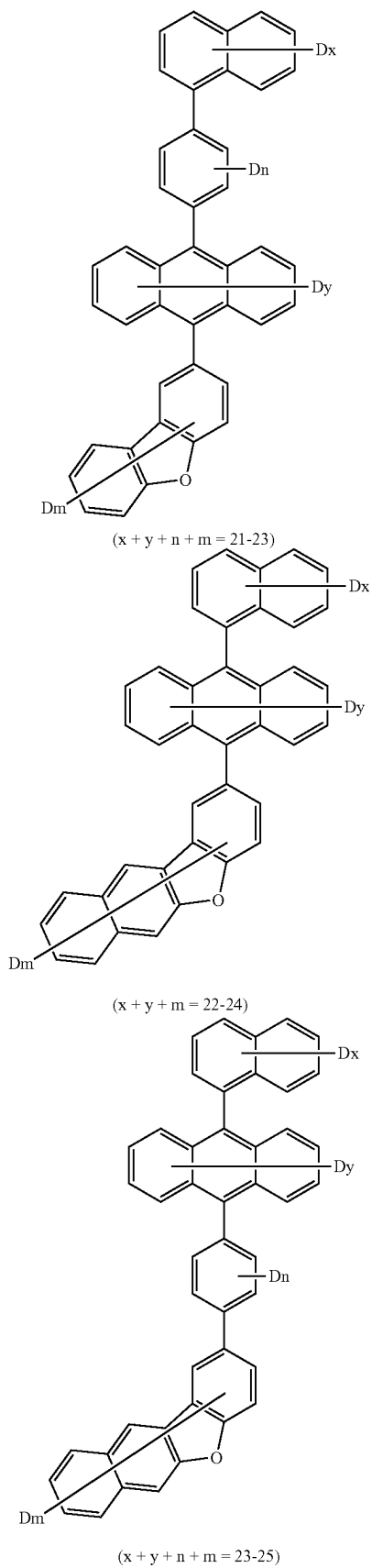
Host 37



$$(x + y + n + m = 21-23)$$

21

-continued



22

-continued

Host 38

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Host 39

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Host 40

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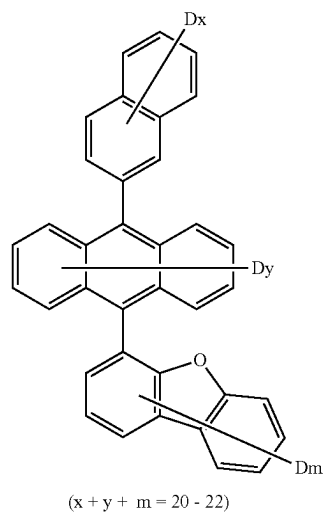
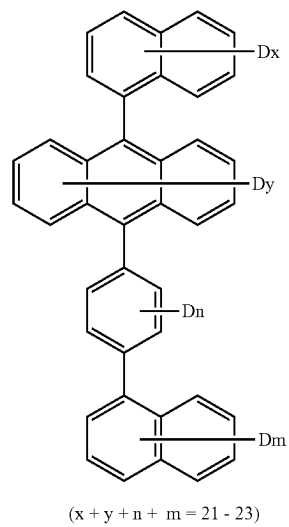
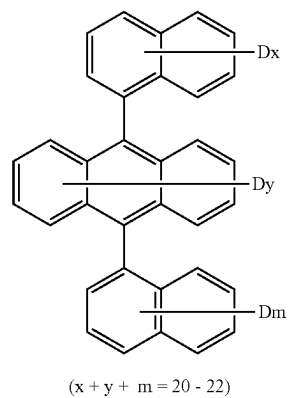
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Host 41

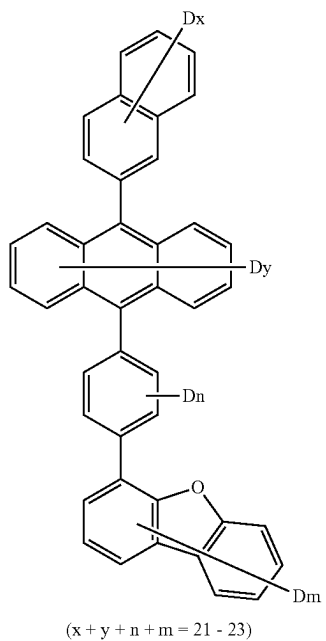
Host 42

Host 43



23

-continued



24

-continued

Host 44

Host 46

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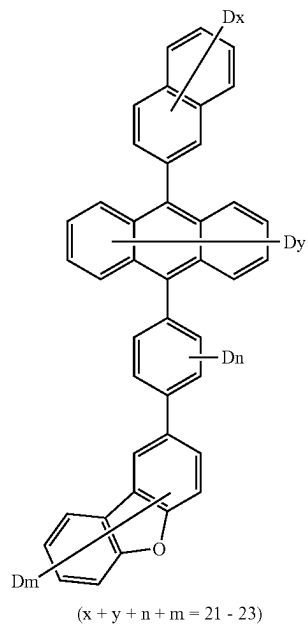
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Host 45

Host 47

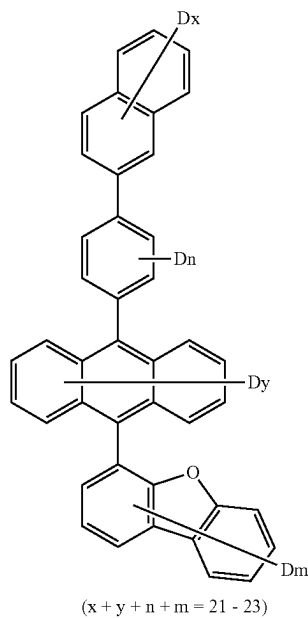
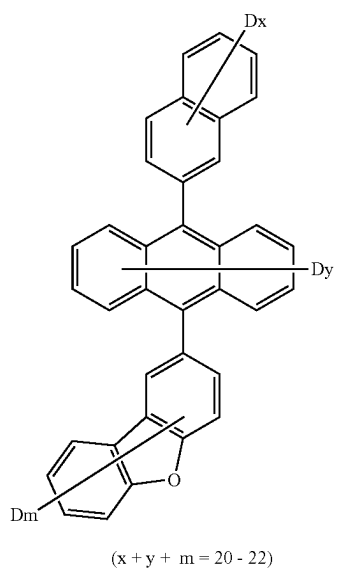
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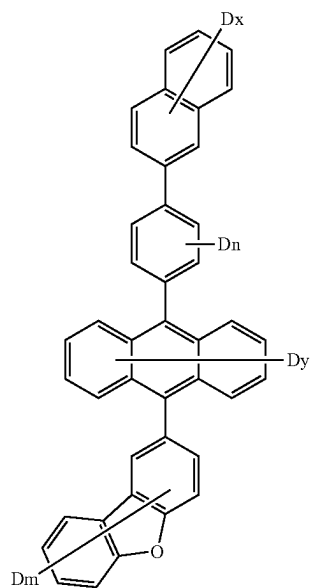
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25

-continued



$$(x + y + n + m = 21 - 23)$$

Host 48

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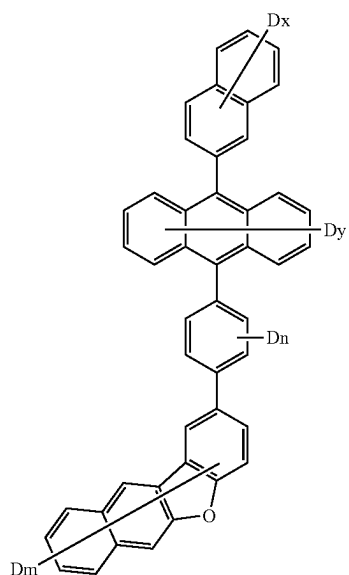
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26

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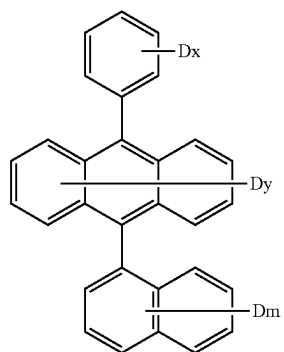
$$(x + y + n + m = 23 - 25)$$

Host 50

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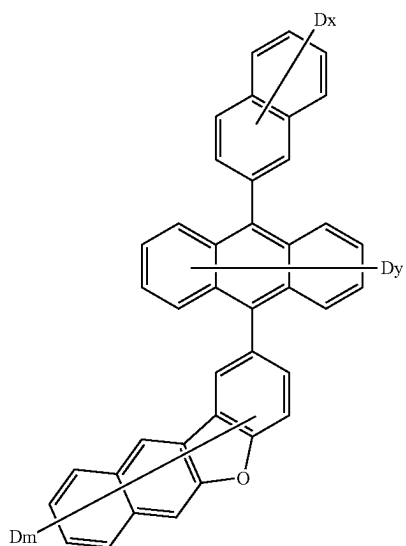


$$(x + y + m = 18 - 20)$$

Host 51

Host 49

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$$(x + y + m = 22 - 24)$$

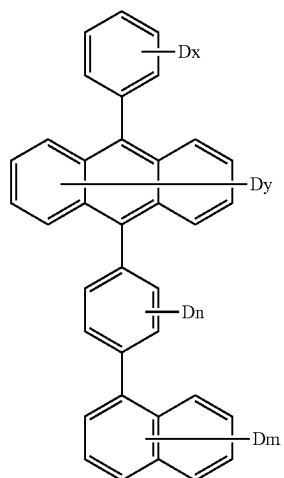
Host 51

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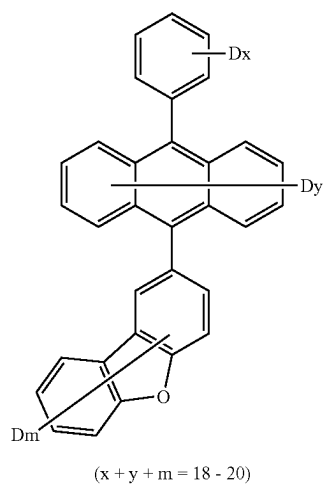
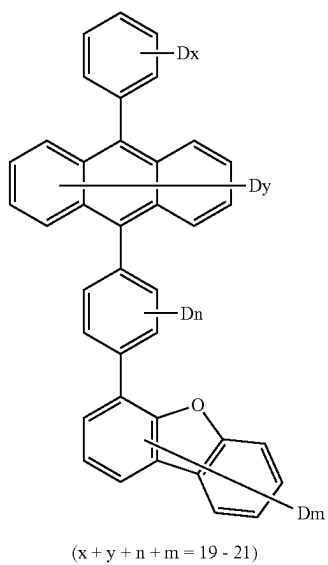
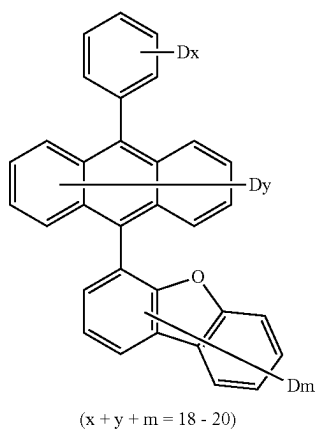


$$(x + y + n + m = 17 - 21)$$

Host 52

27

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**28**

-continued

Host 53

Host 56

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Host 54

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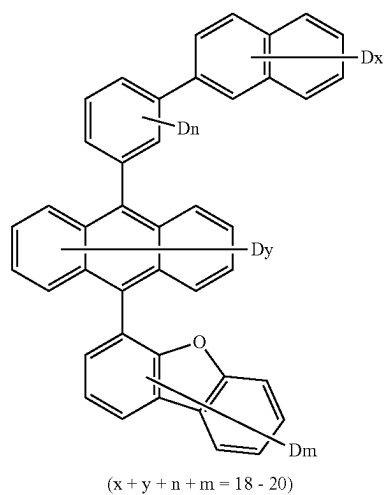
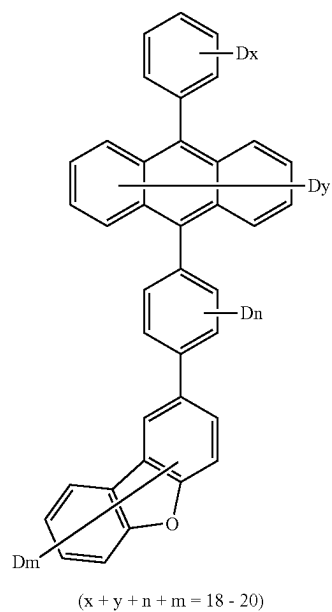
Host 55

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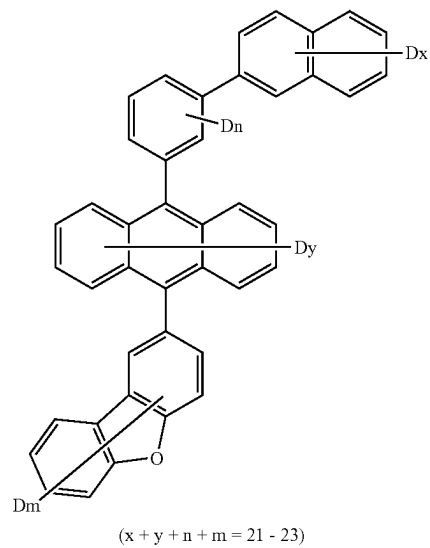
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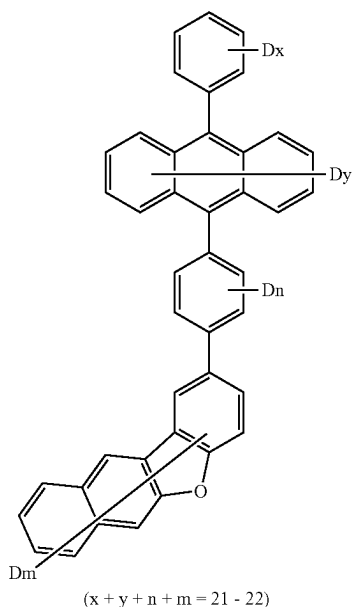
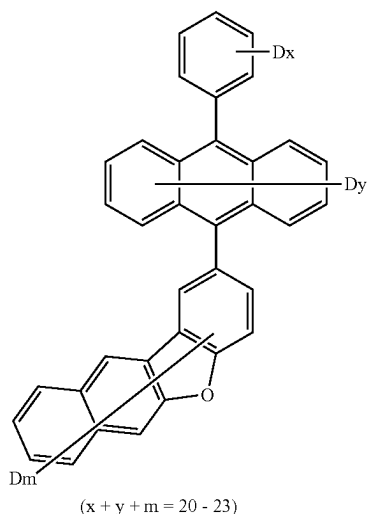
Host 57

Host 58



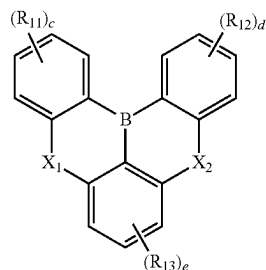
29

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A compound of the blue dopant may be represented by Formula 5-1 or Formula 5-2, but it is not limited thereto.

Formula 5-1



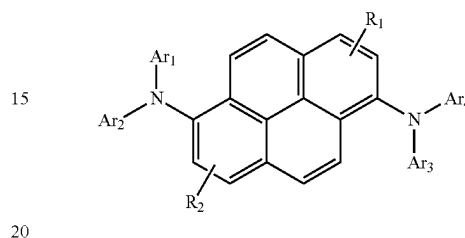
In Formula 5-1, each of c and d is independently an integer of 0 to 4, and e is an integer of 0 to 3. Each of R₁₁ and R₁₂ is independently selected from the group consisting of C₁~C₂₀ alkyl group, C₆~C₃₀ aryl group, C₅~C₃₀ hetero

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aryl group and C₆~C₃₀ aryl amino group, or adjacent two among R₁₁ or adjacent two among R₁₂ form a fused aromatic ring or a hetero-aromatic ring. R₁₃ is selected from the group consisting of C₁~C₁₀ alkyl group, C₆~C₃₀ aryl group, C₅~C₃₀ hetero aryl group and C₅~C₃₀ aromatic amino group. Each of X₁ and X₂ is independently oxygen (O) or NR₁₄, and R₁₄ is C₆~C₃₀ aryl group.

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Formula 5-2



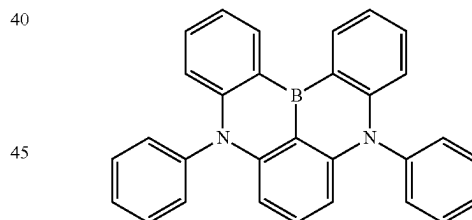
Host 60

In Formula 5-2, each of Ar₁, Ar₂, Ar₃ and Ar₄ is independently selected from the group consisting of C₆~C₃₀ aryl group and C₅~C₃₀ hetero aryl group, and each of R₁ and R₂ is independently selected from the group consisting of hydrogen, C₁~C₂₀ alkyl group and C₆~C₃₀ aryl group. For example, each of Ar₁, Ar₂, Ar₃ and Ar₄ may be independently selected from the group consisting of phenyl, dibenzofuranyl, naphthyl and biphenyl and may be substituted by trifluoromethyl, cyano or fluorine (F). Each of R₁ and R₂ may be independently selected from the group consisting of hydrogen, isopropyl and phenyl.

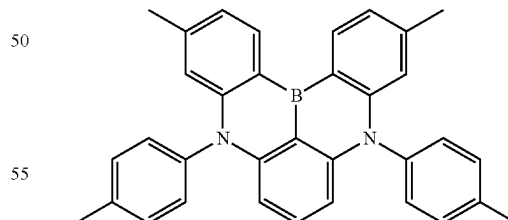
For example, the blue dopant in Formula 5-1 may be a compound being one of the followings in Formula 6-1:

Formula 6-1

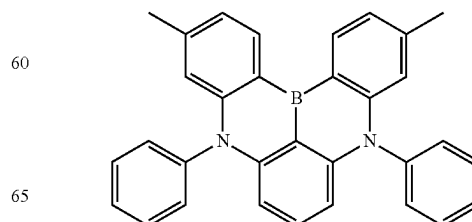
Dopant 1



Dopant 2

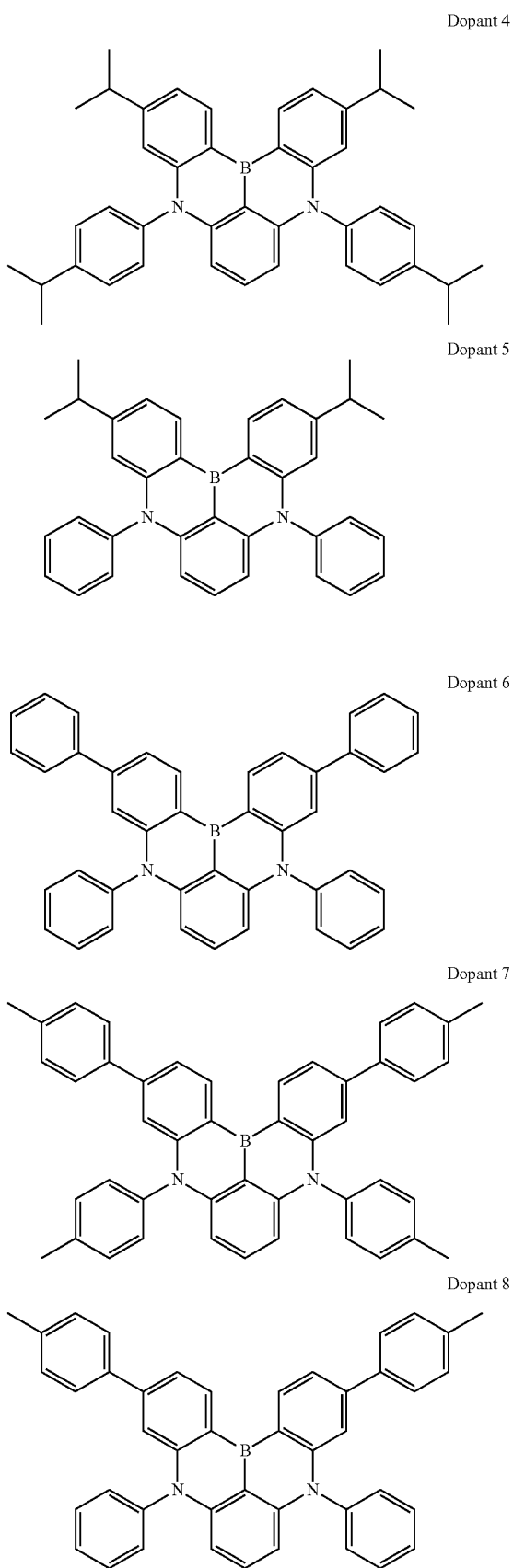


Dopant 3

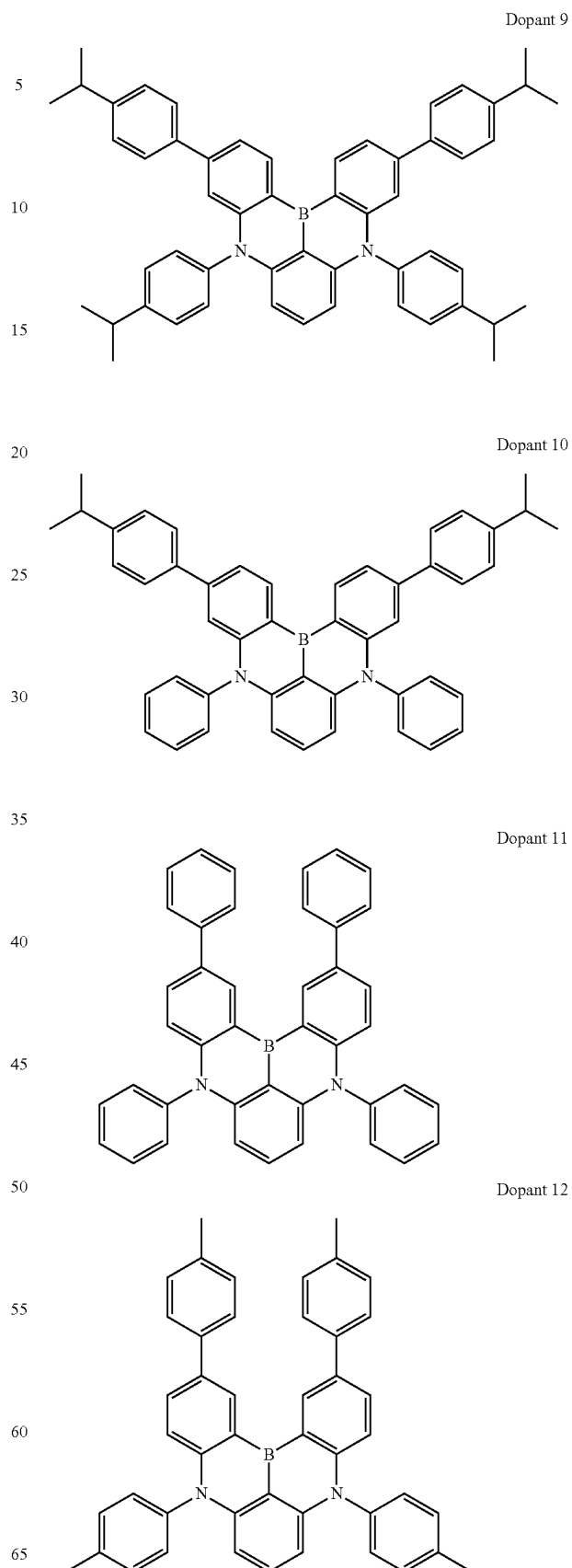


31

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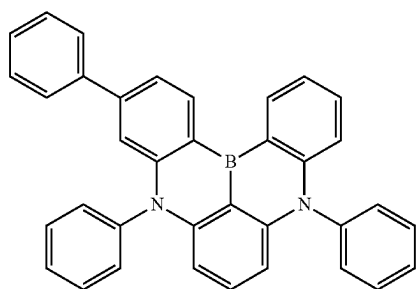
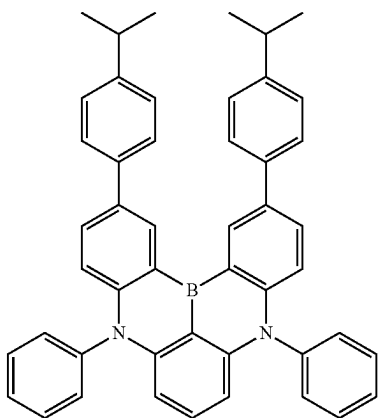
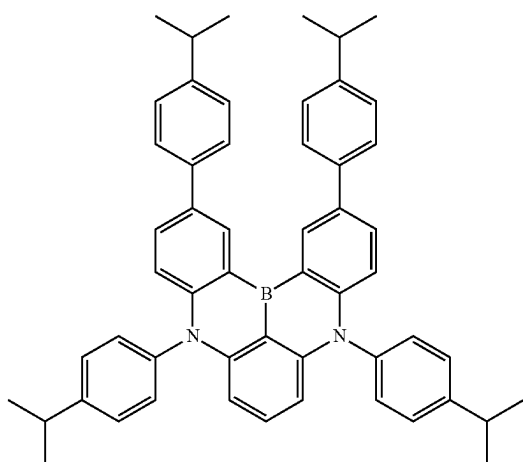
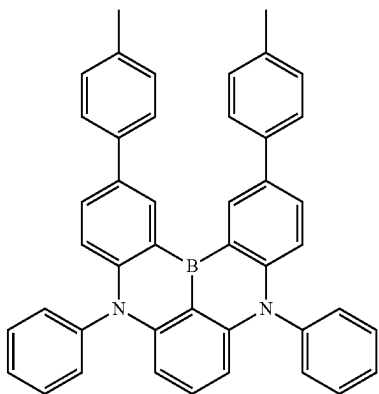
**32**

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33

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**34**

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Dopant 13

Dopant 17

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Dopant 14

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Dopant 15

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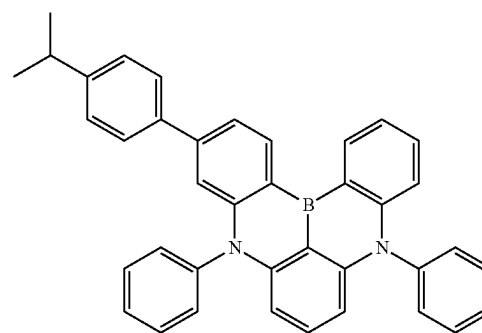
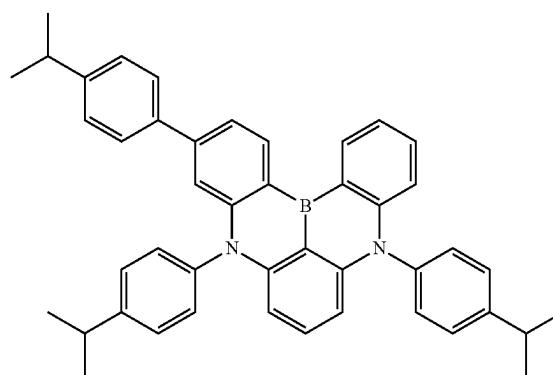
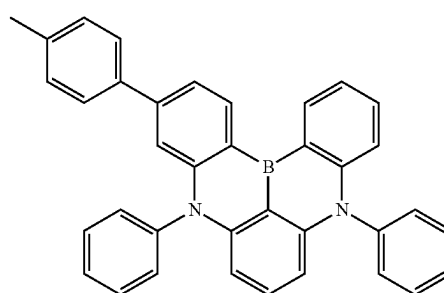
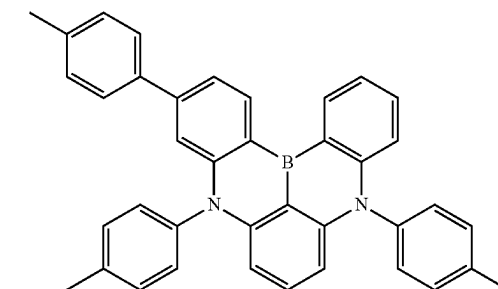
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Dopant 16

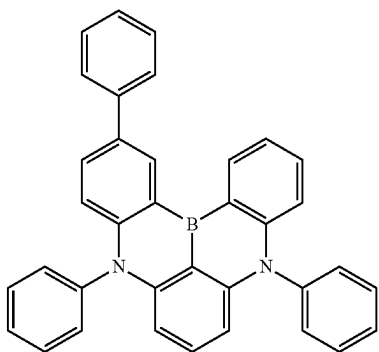
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Dopant 21

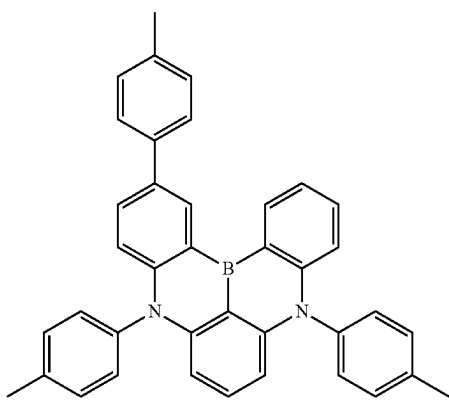
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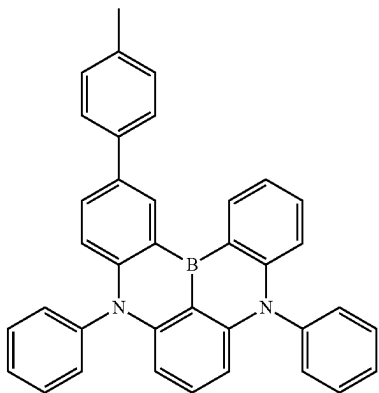
Dopant 22

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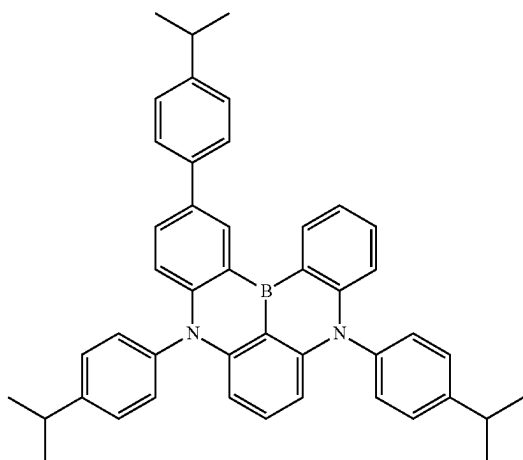
Dopant 23

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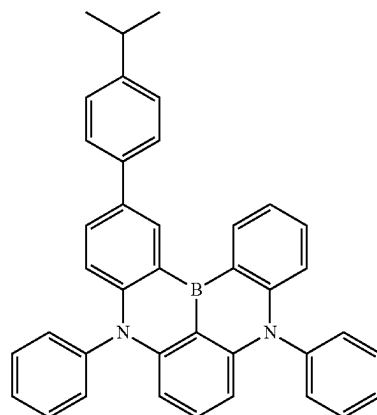
Dopant 24

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36

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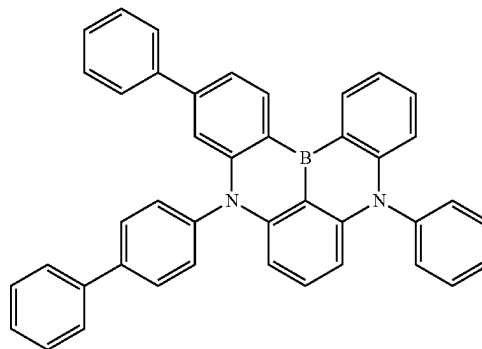


Dopant 25

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Dopant 26

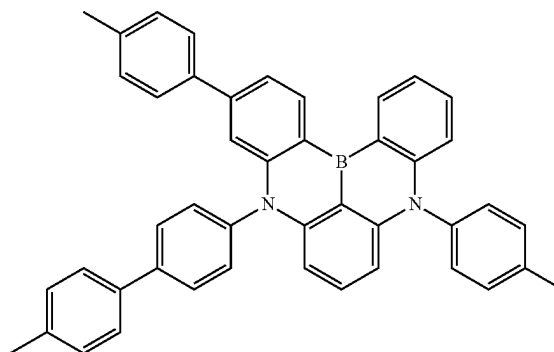
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Dopant 27

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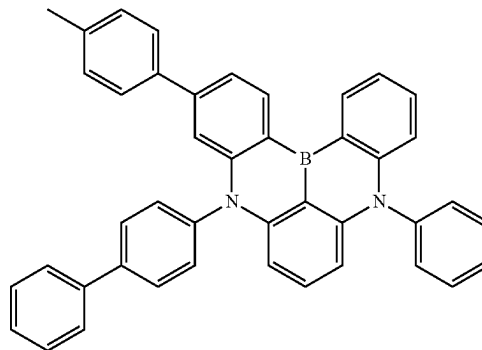
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Dopant 28

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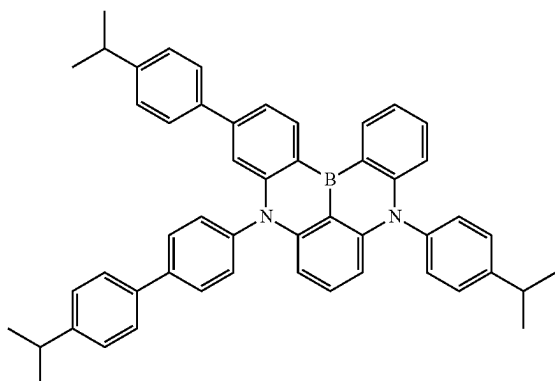
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37

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Dopant 29



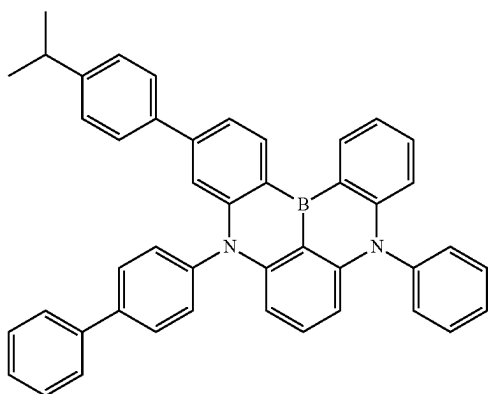
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Dopant 30



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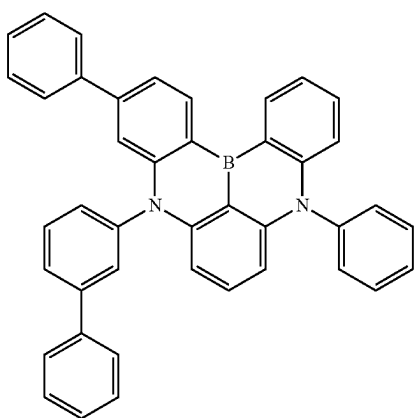
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Dopant 31



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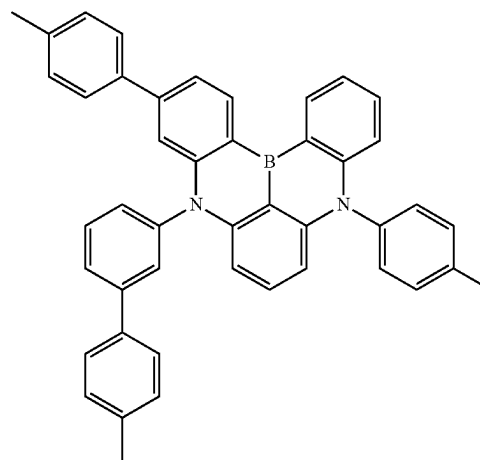
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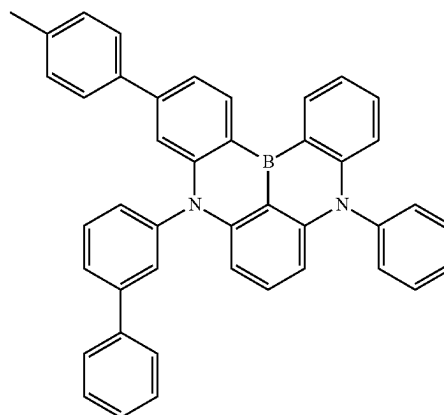
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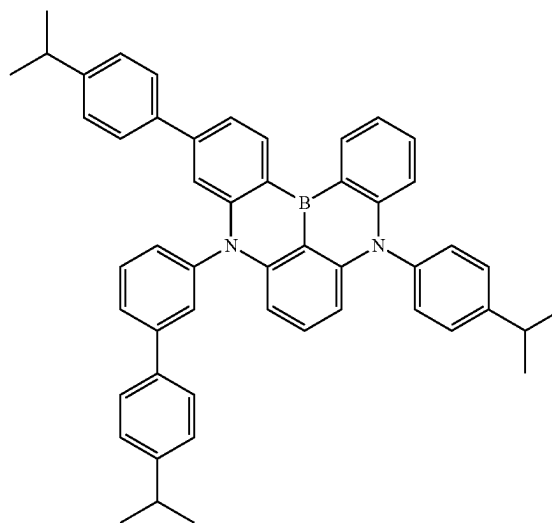
Dopant 32



Dopant 33



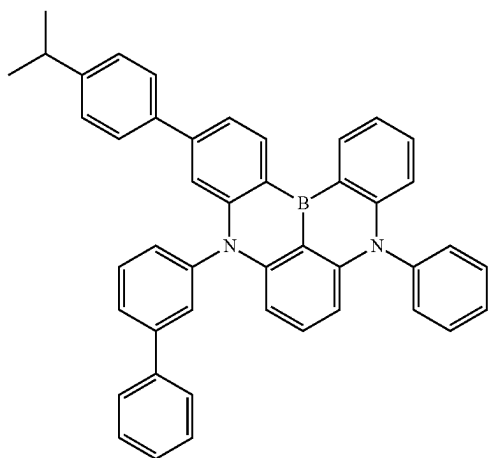
Dopant 34



39

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Dopant 35



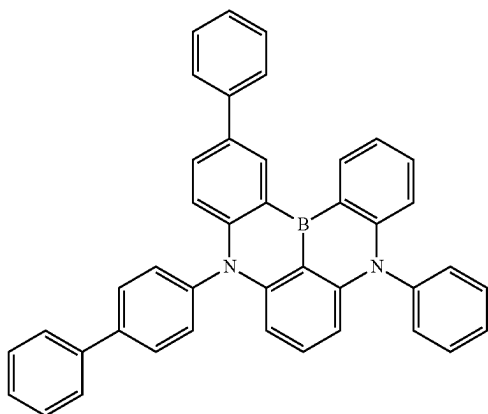
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Dopant 36



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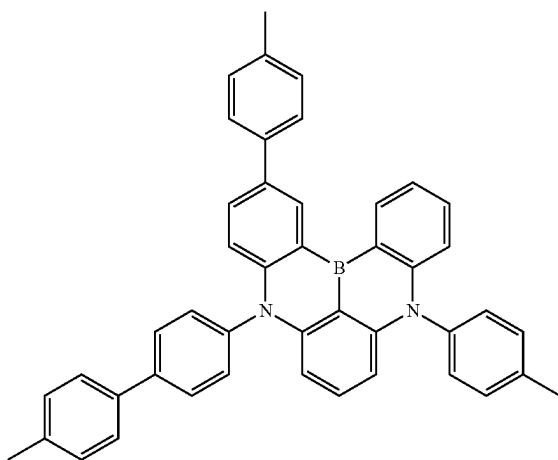
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Dopant 37



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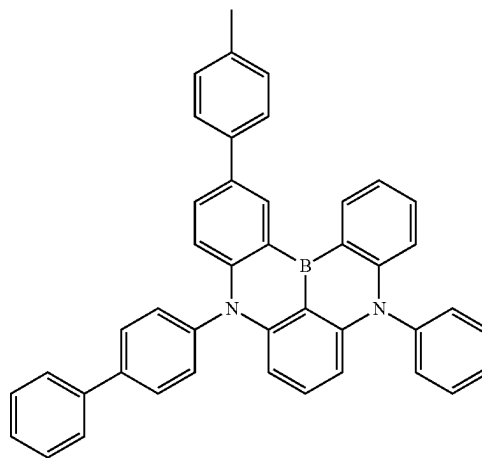
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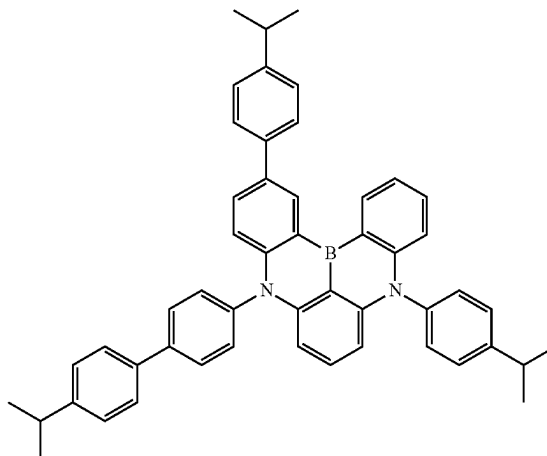
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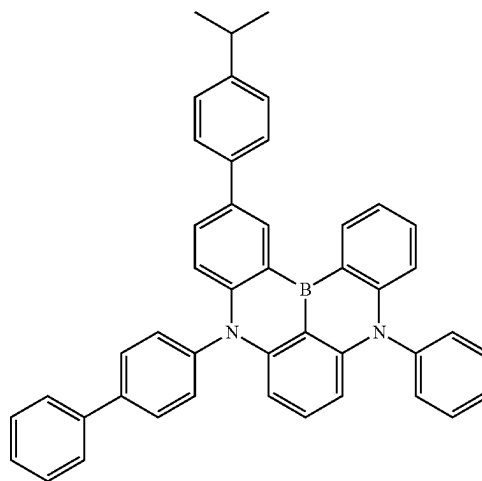
Dopant 38



Dopant 39

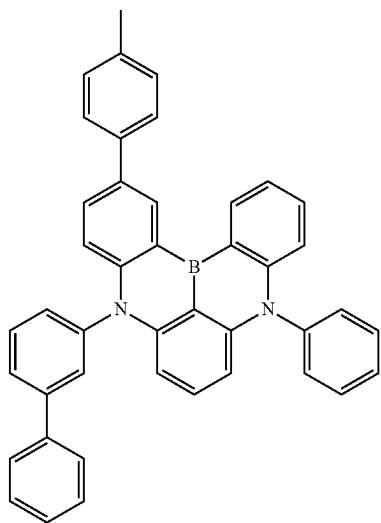
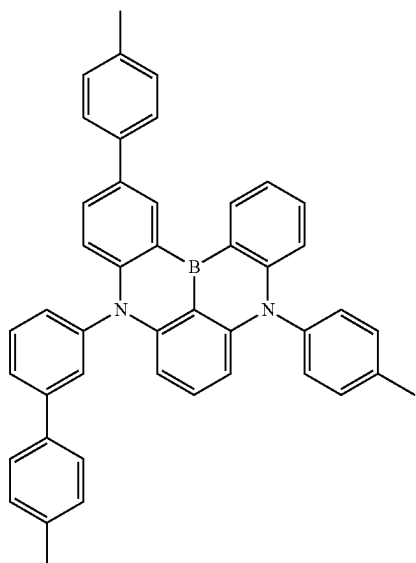
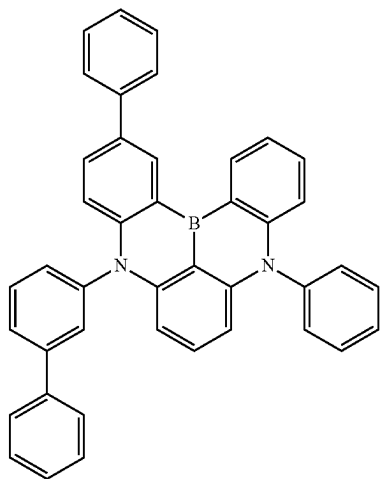


Dopant 40



41

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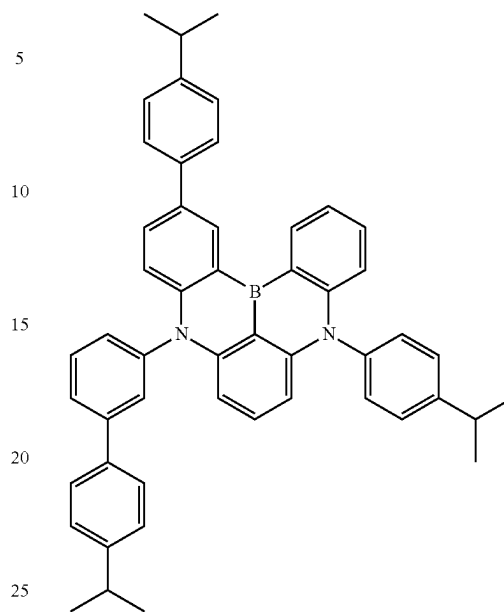


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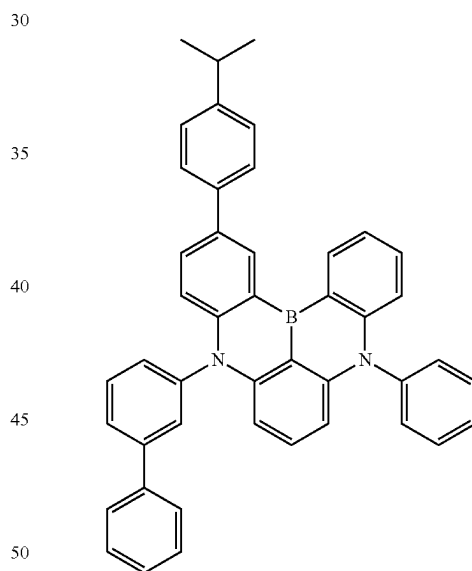
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Dopant 41

Dopant 44

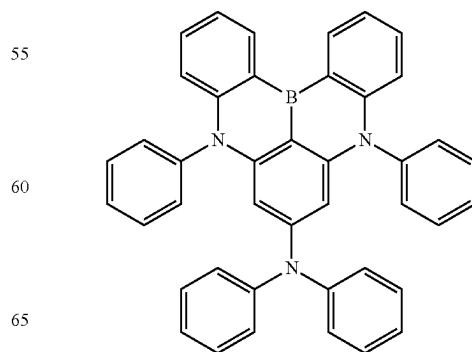


Dopant 42



Dopant 45

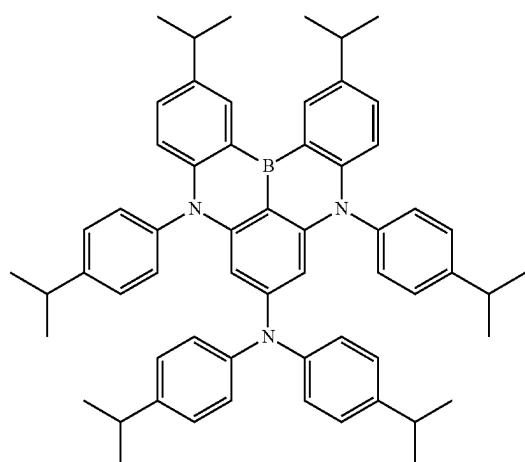
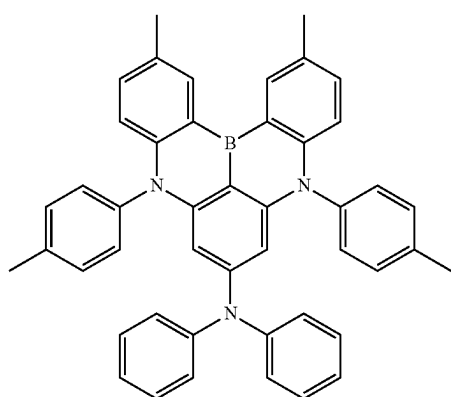
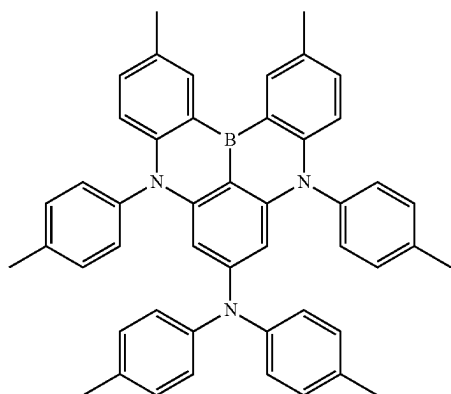
Dopant 43



Dopant 46

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**44**

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Dopant 47

Dopant 50

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Dopant 48

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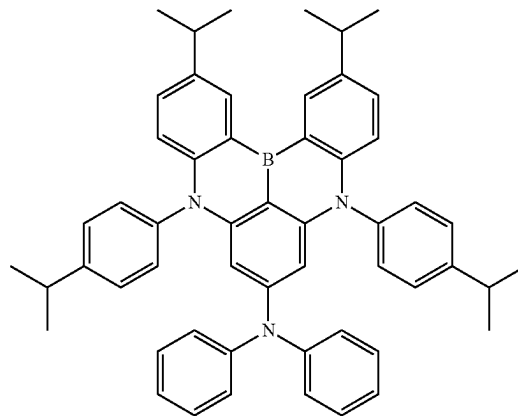
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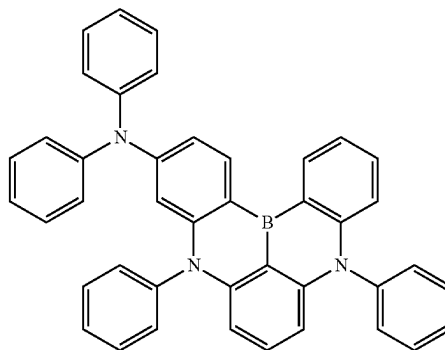
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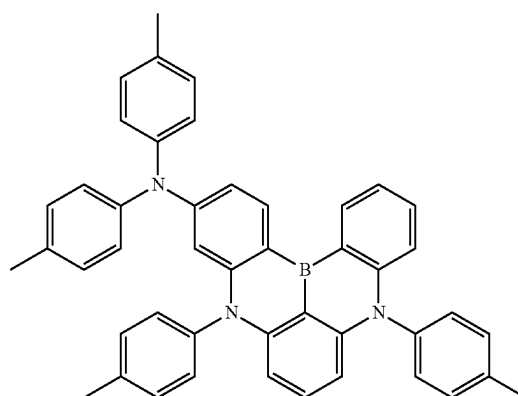
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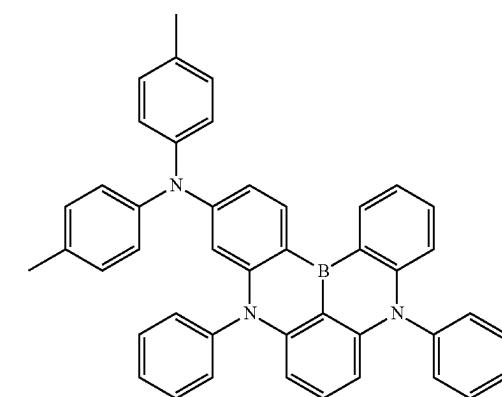
Dopant 51



Dopant 52



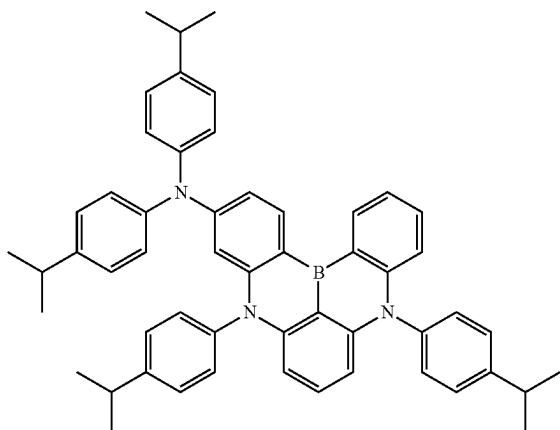
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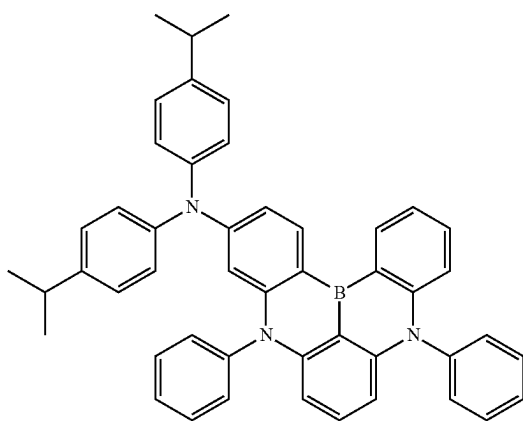
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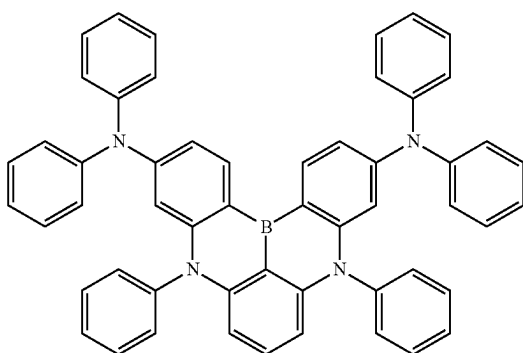
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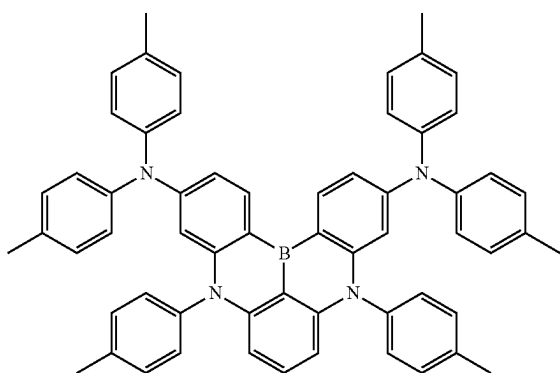
Dopant 55



Dopant 56



Dopant 57

**46**

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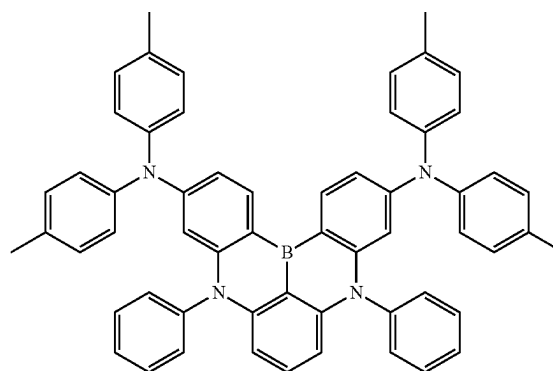
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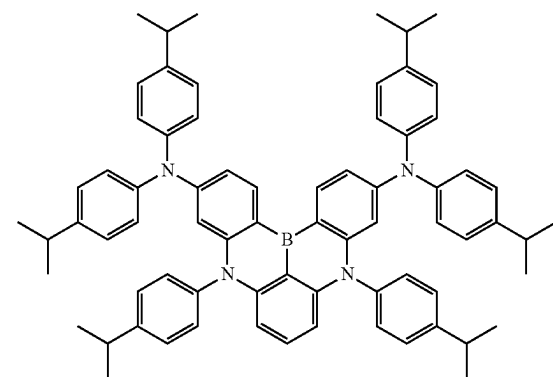


Dopant 59

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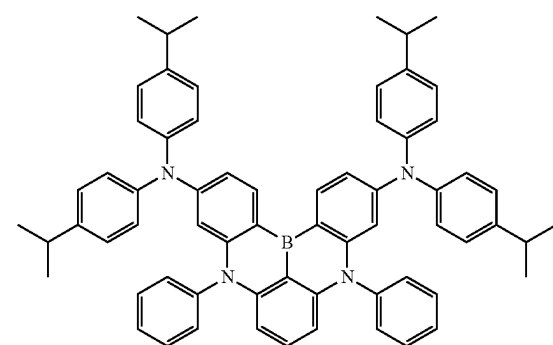


Dopant 60

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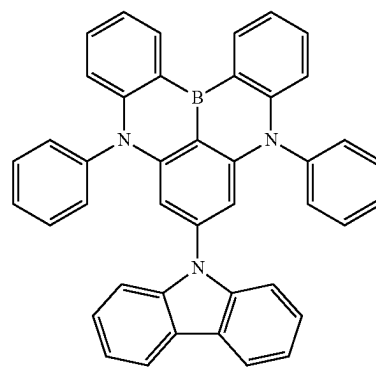


Dopant 61

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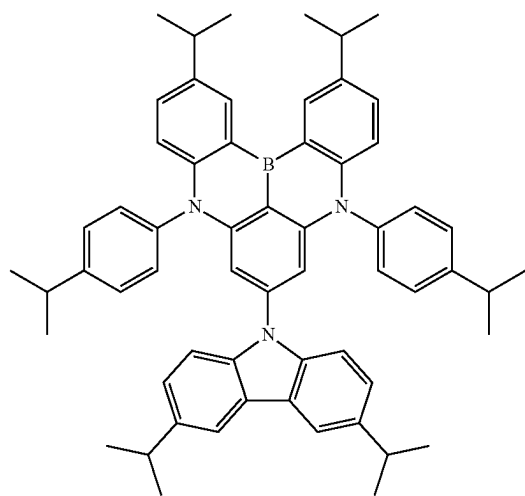
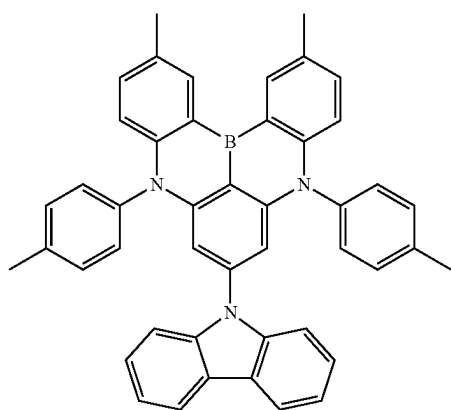
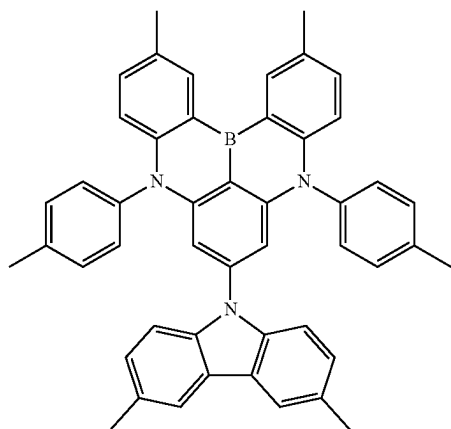
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**48**

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Dopant 62

Dopant 65

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Dopant 63

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Dopant 64

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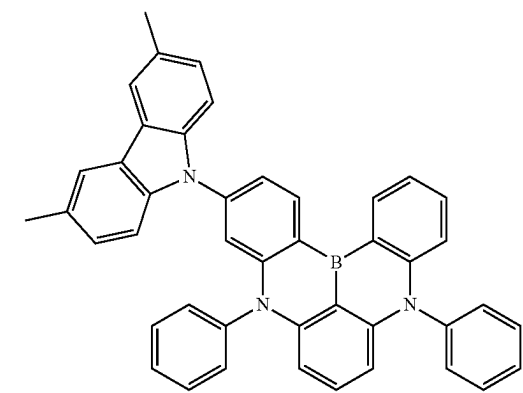
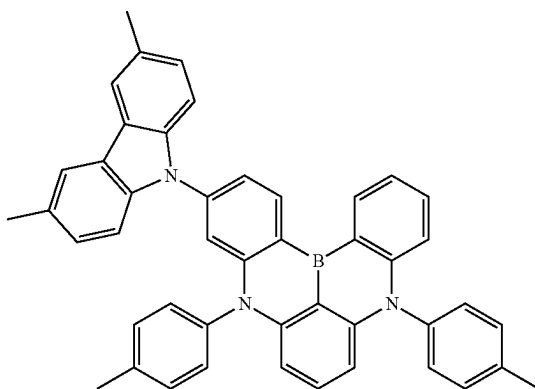
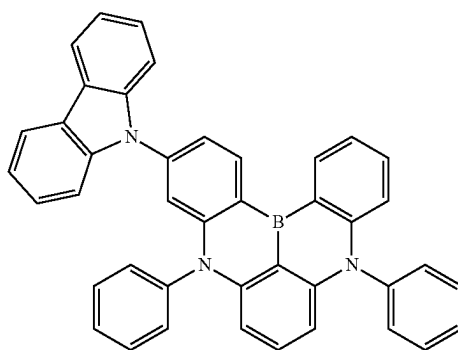
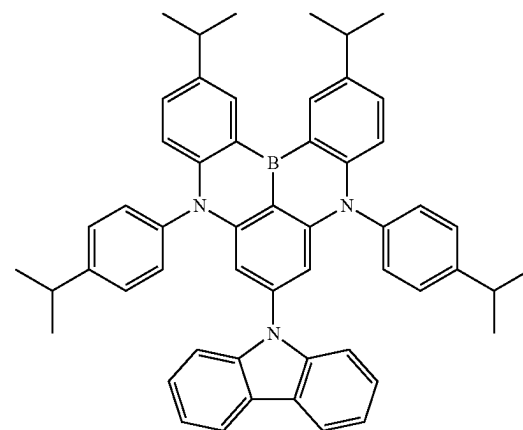
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Dopant 66

Dopant 67

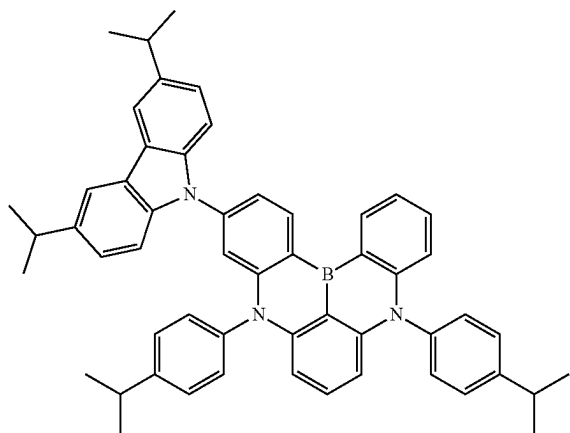
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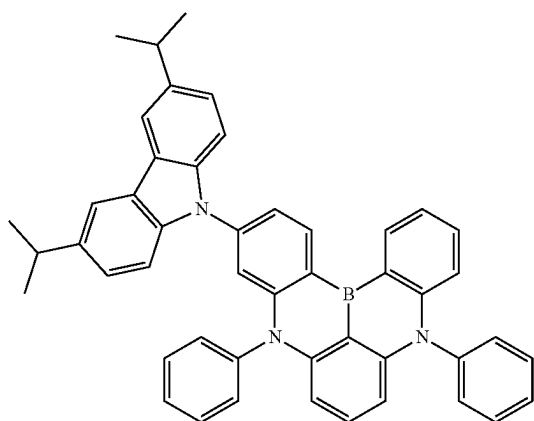
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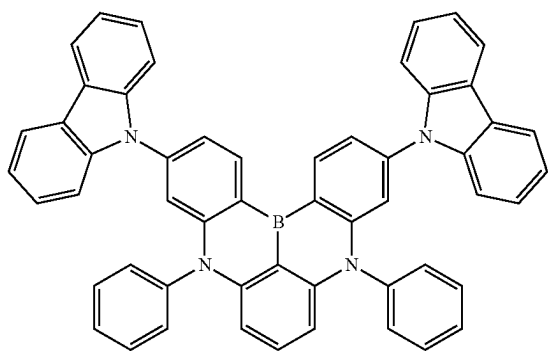
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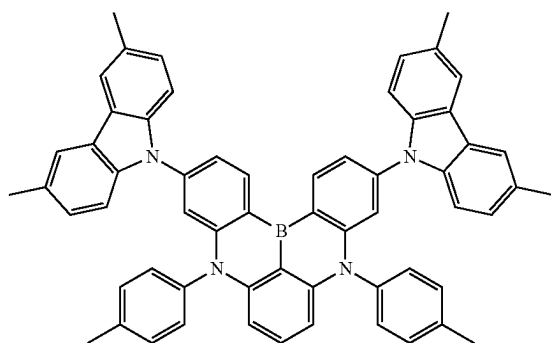
Dopant 70



Dopant 71

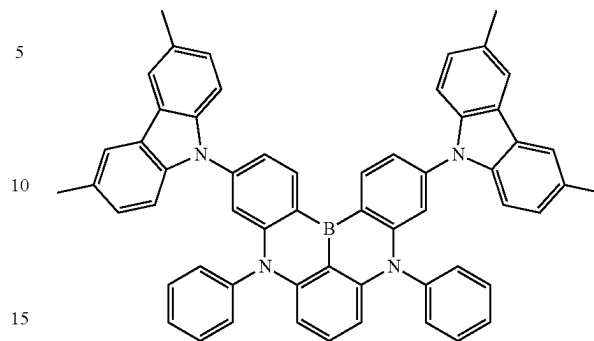


Dopant 72

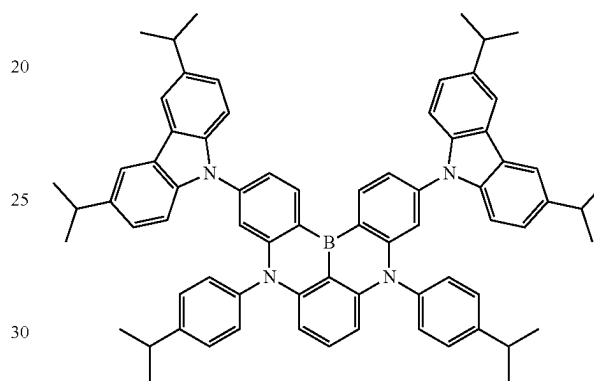
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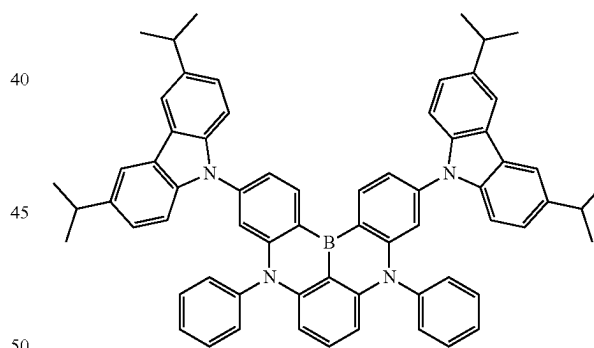
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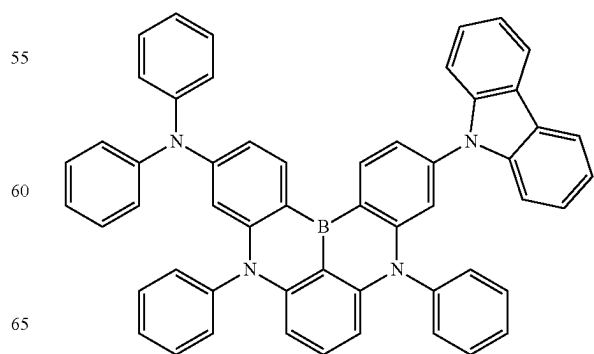
Dopant 74



Dopant 75



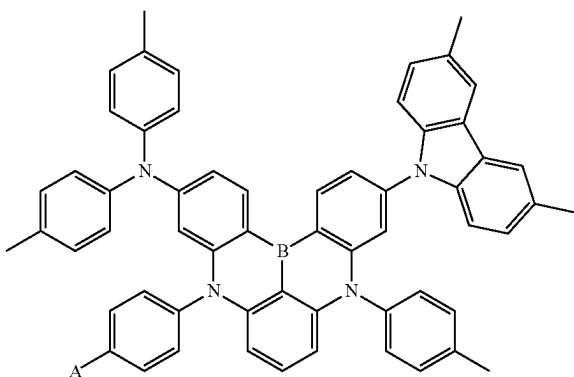
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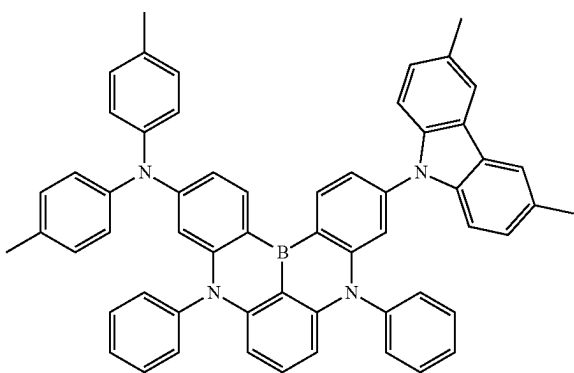
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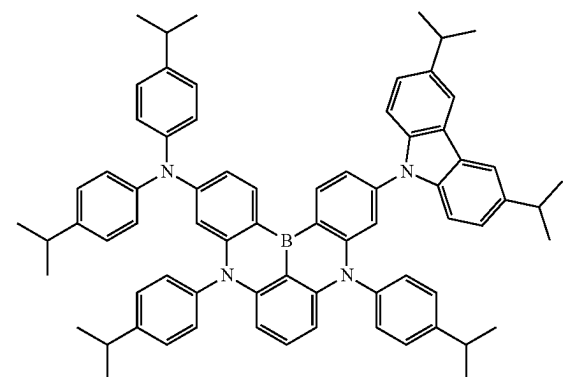
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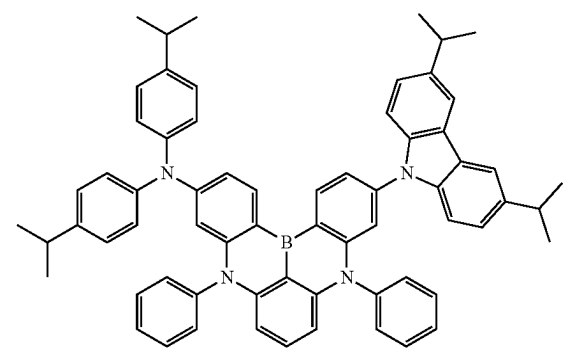
Dopant 78



Dopant 79

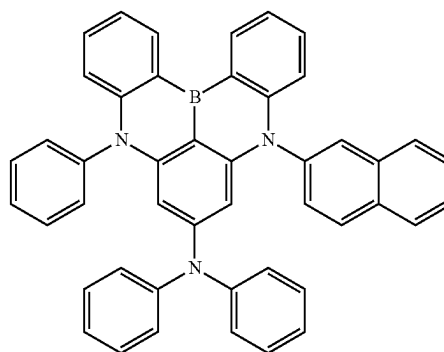


Dopant 80

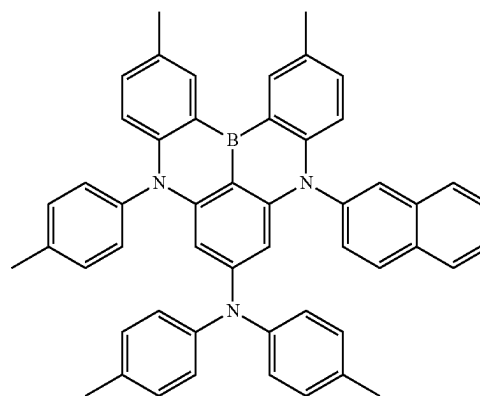
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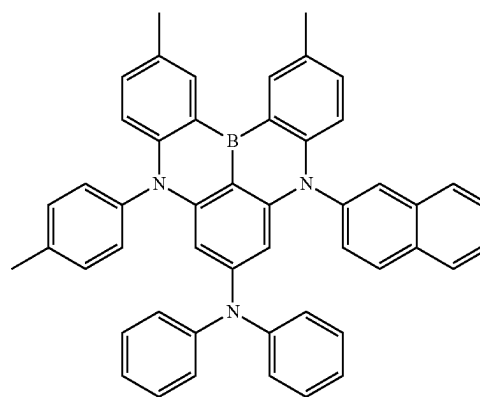
Dopant 81



Dopant 82



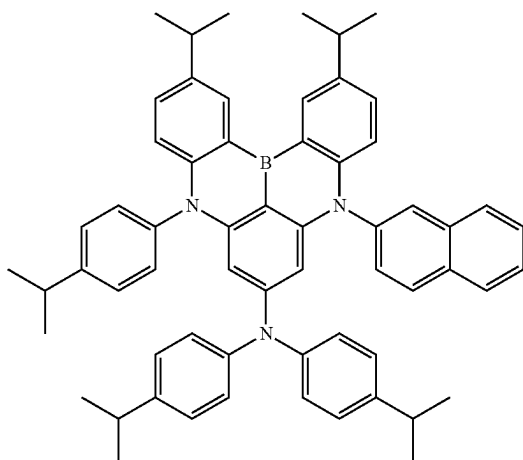
Dopant 83



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Dopant 84



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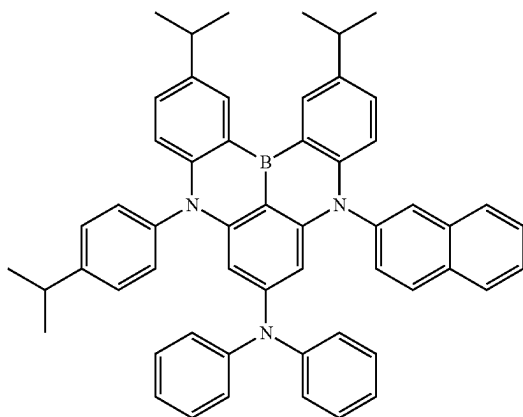
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Dopant 85

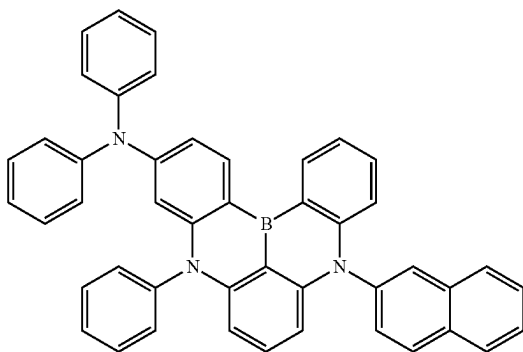


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Dopant 86



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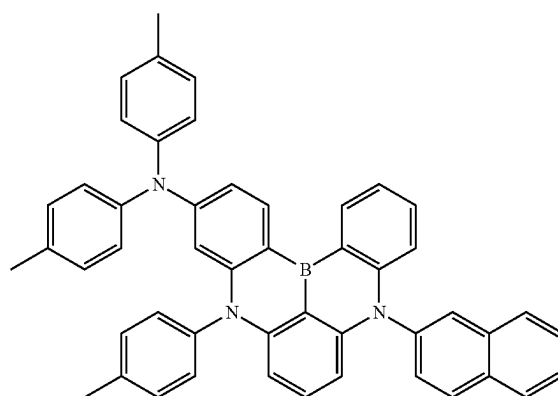
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Dopant 87



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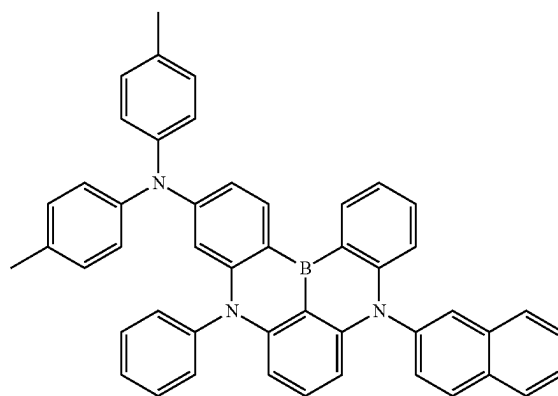
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Dopant 88



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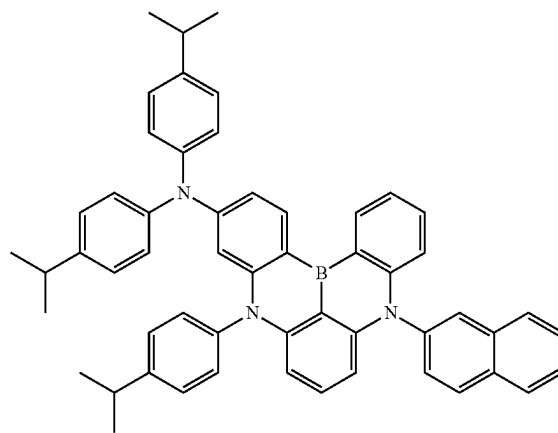
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Dopant 89



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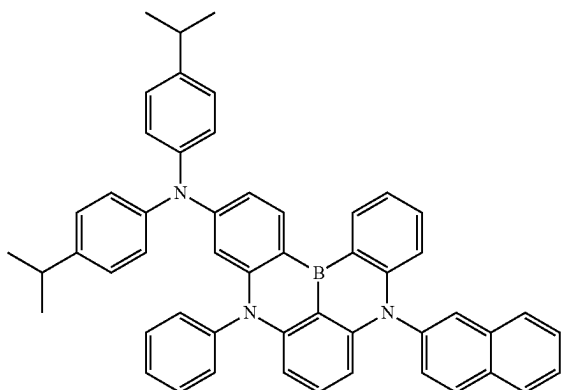
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Dopant 90



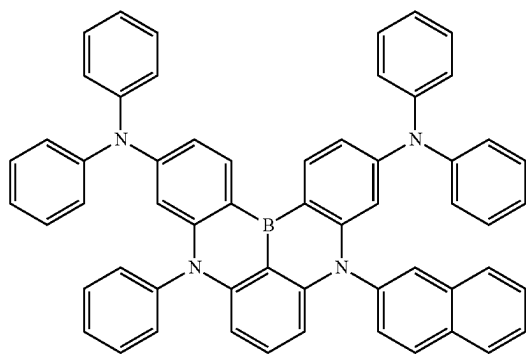
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Dopant 91

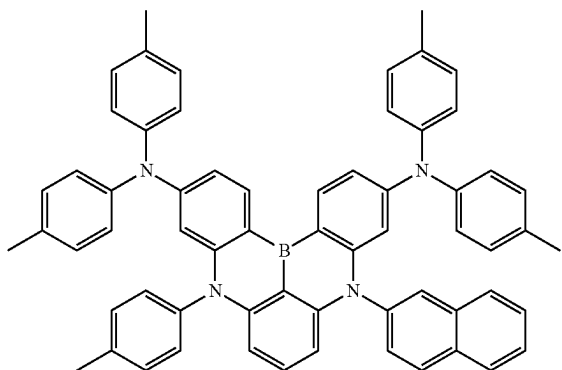


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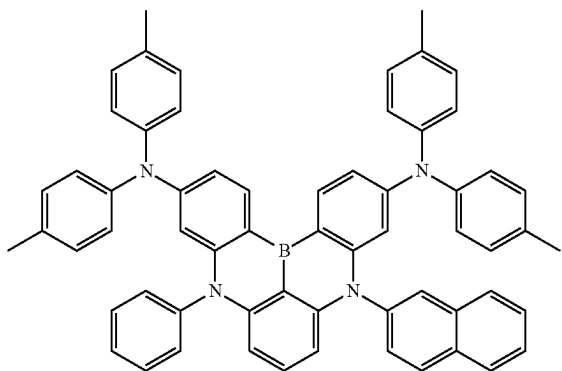
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Dopant 92



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Dopant 93



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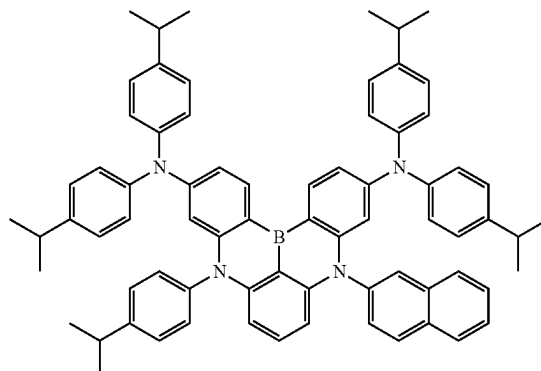
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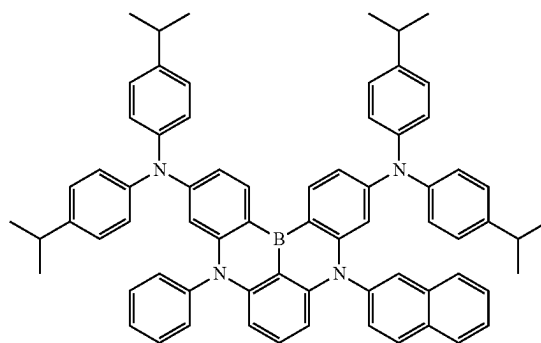
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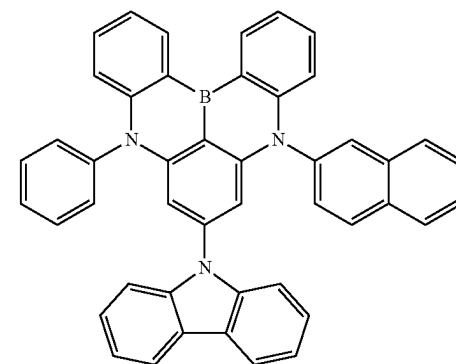
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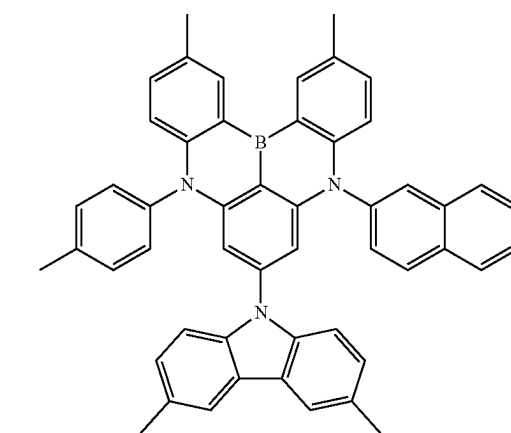
Dopant 95



Dopant 96



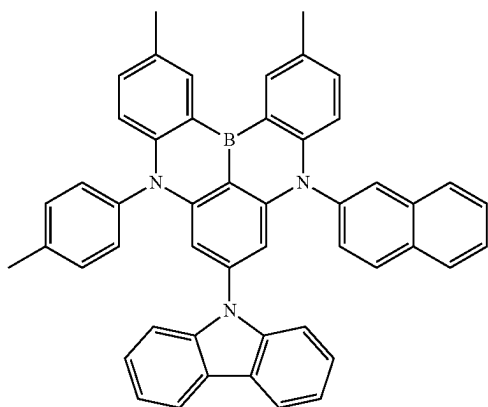
Dopant 97



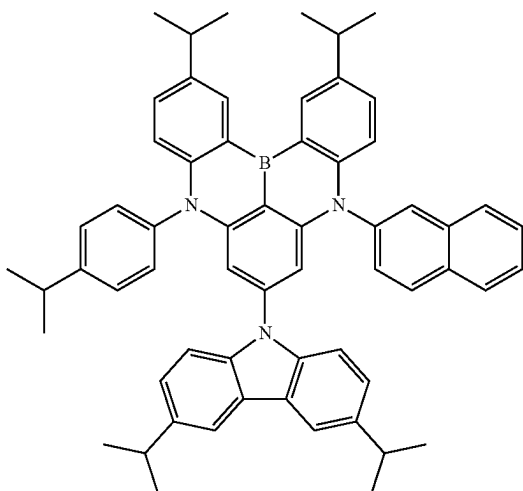
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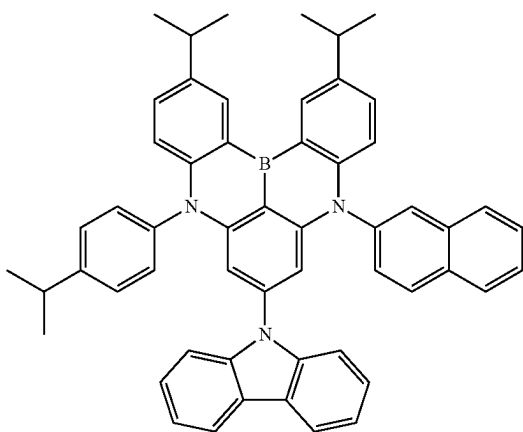
Dopant 98



Dopant 99

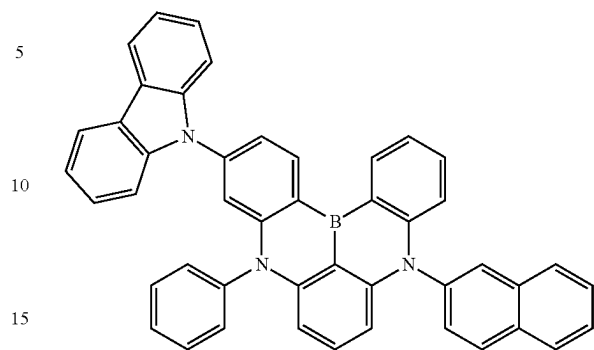


Dopant 100

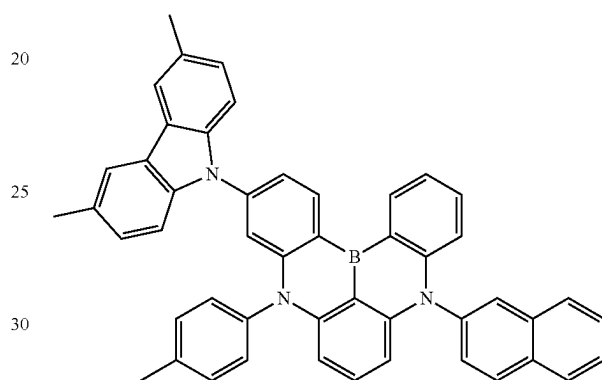
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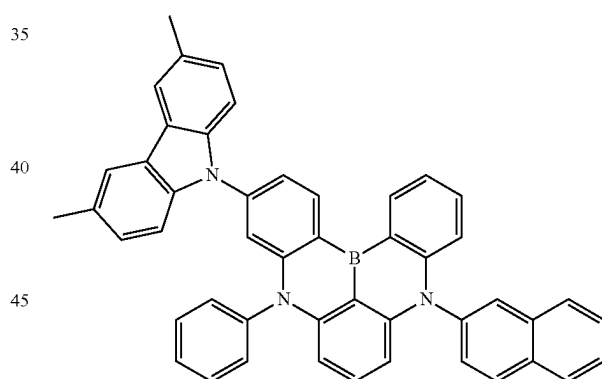
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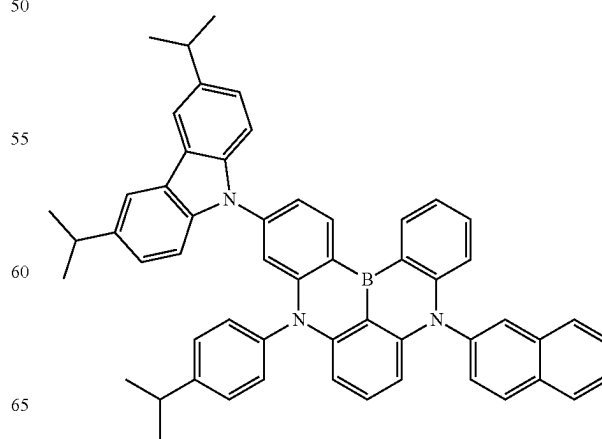
Dopant 102



Dopant 103



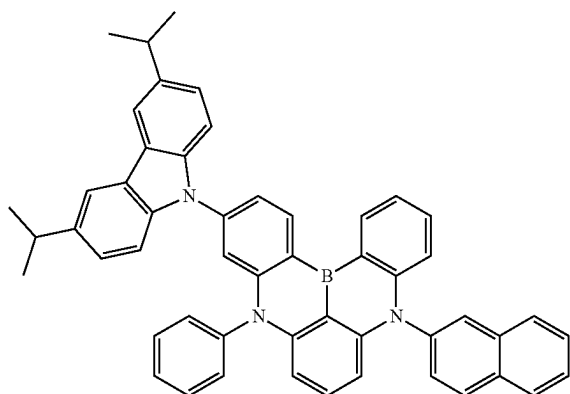
Dopant 104



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Dopant 105



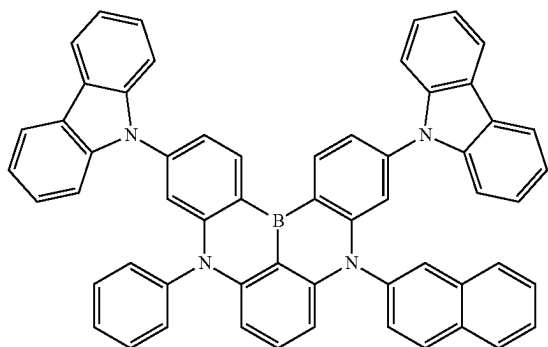
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Dopant 106

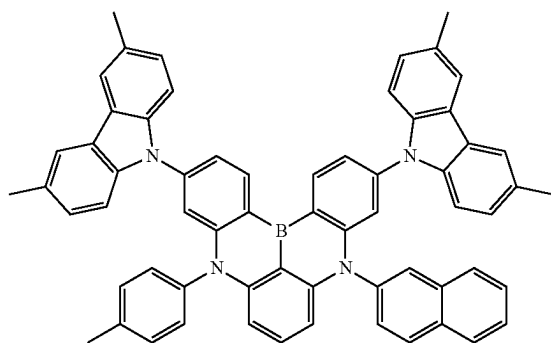


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Dopant 107

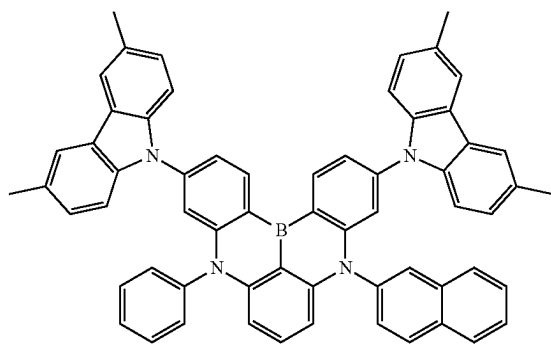


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Dopant 108



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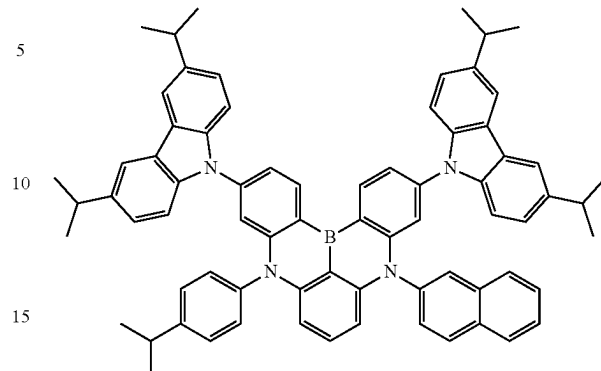
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Dopant 109



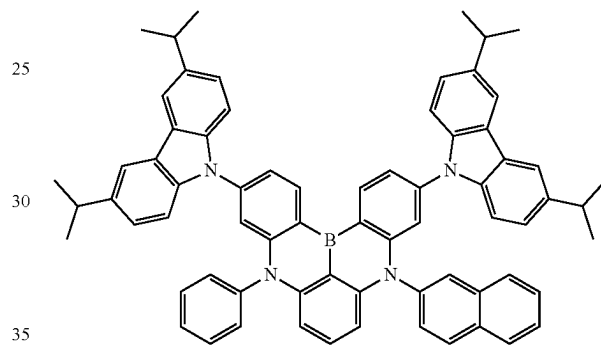
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Dopant 110

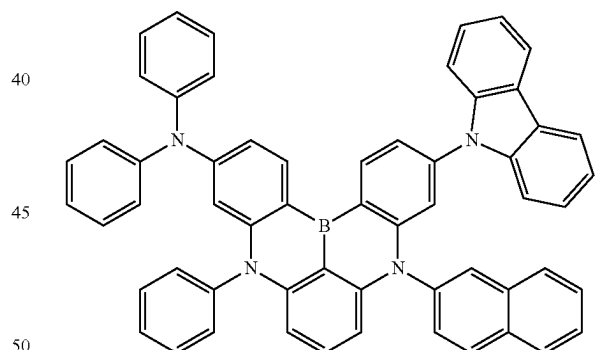


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Dopant 111

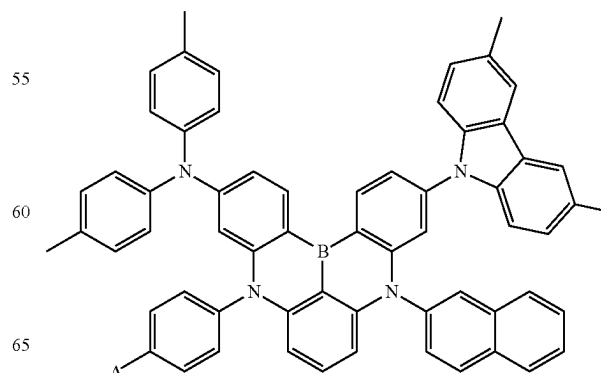


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Dopant 112



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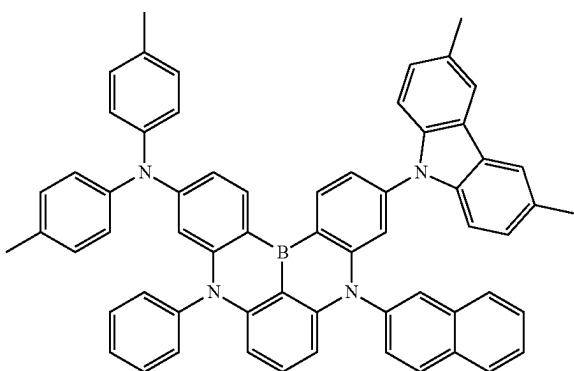
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Dopant 113



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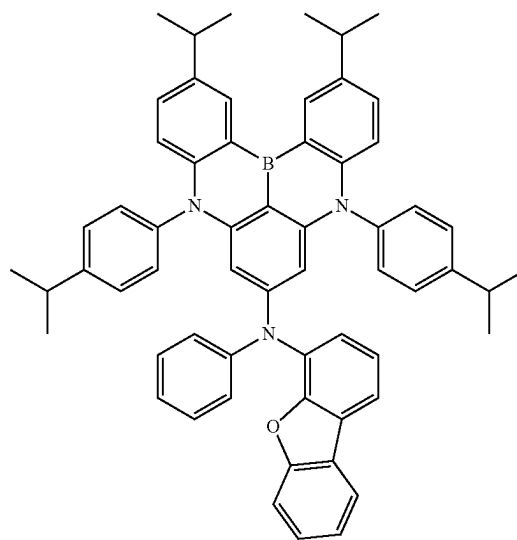
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Dopant 116



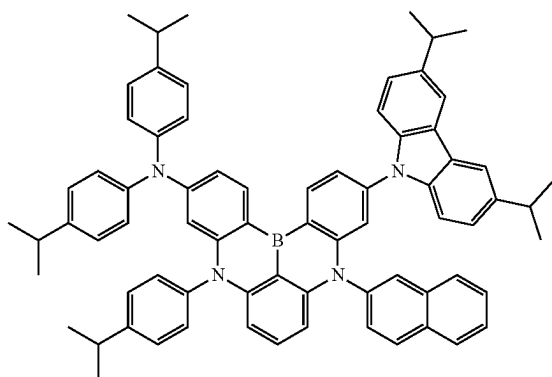
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Dopant 114



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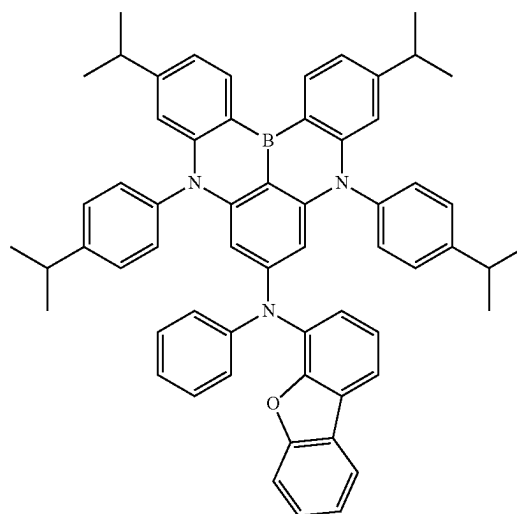
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Dopant 117



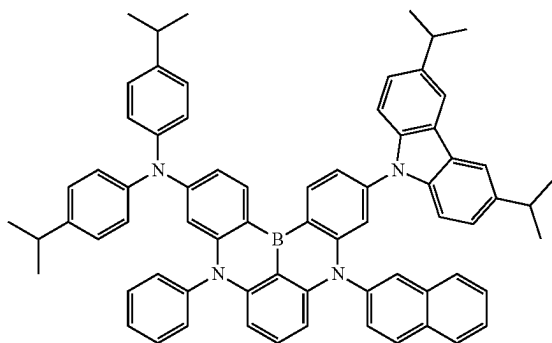
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Dopant 115



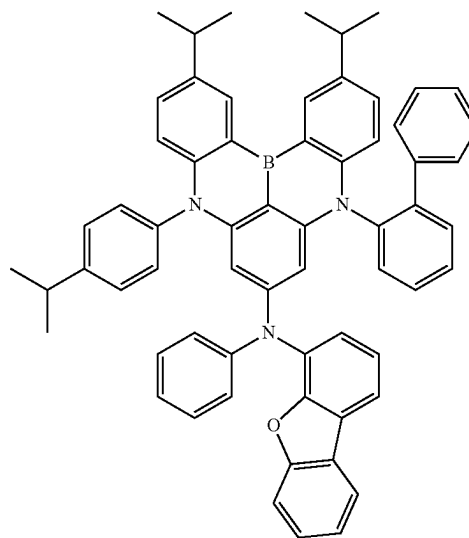
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Dopant 118



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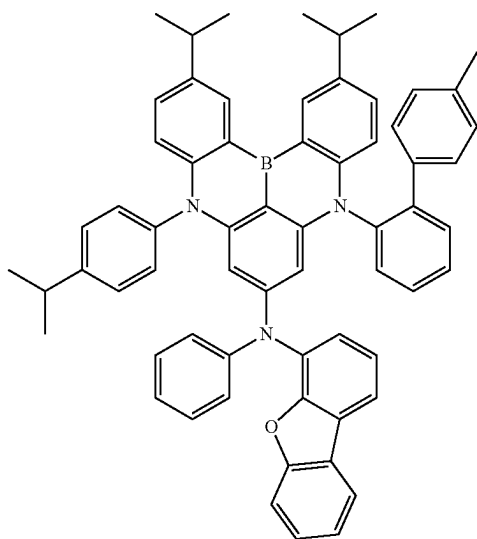
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Dopant 119



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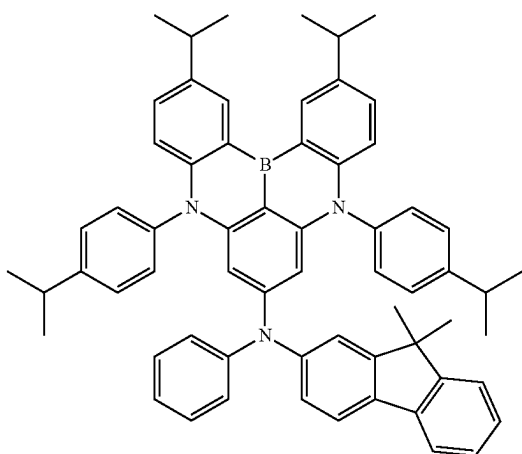
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Dopant 120

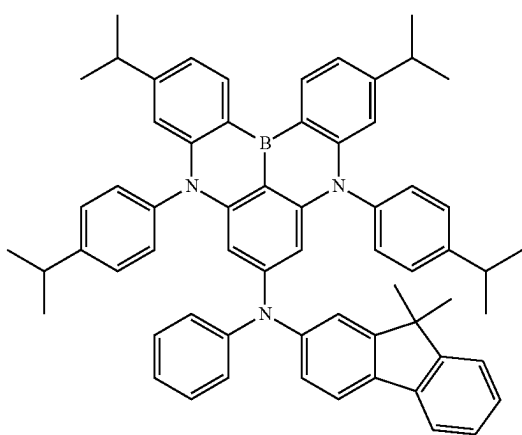


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Dopant 121



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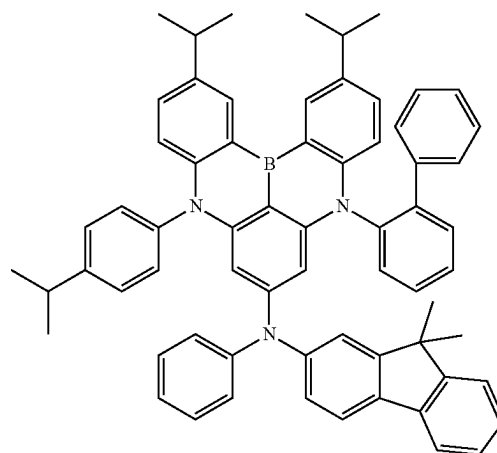
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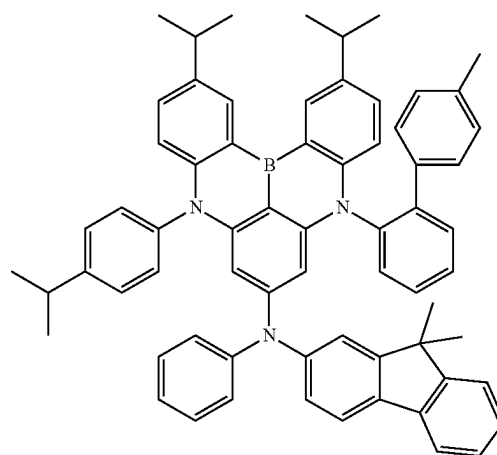
Dopant 122



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Dopant 123



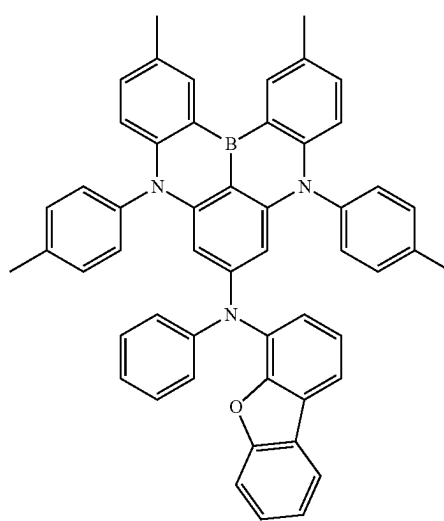
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Dopant 124



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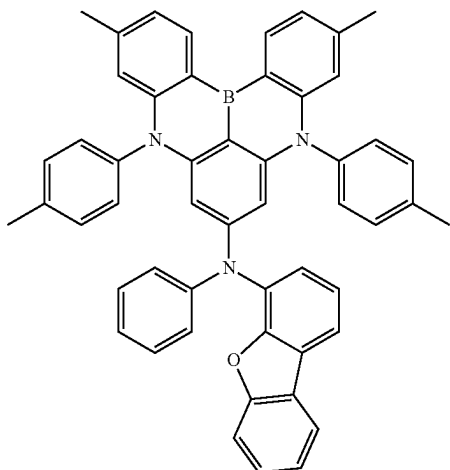
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Dopant 125



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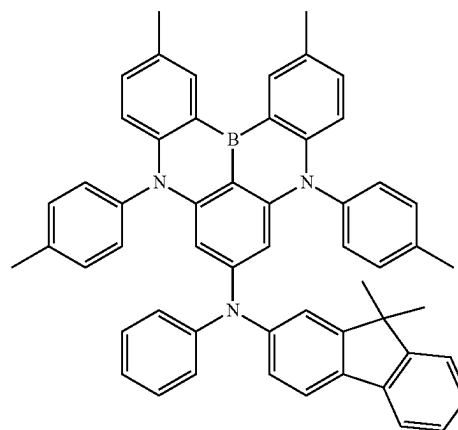
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Dopant 128



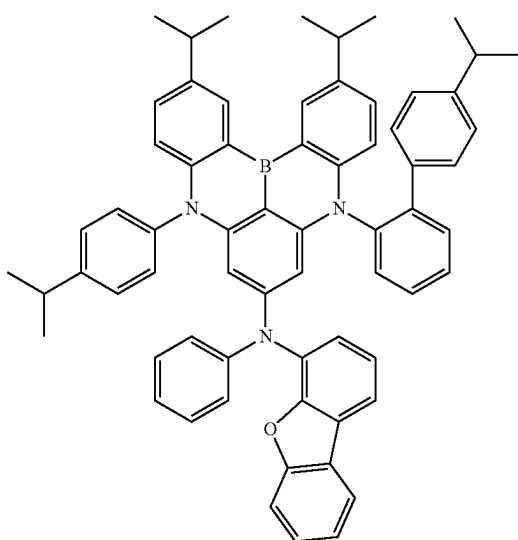
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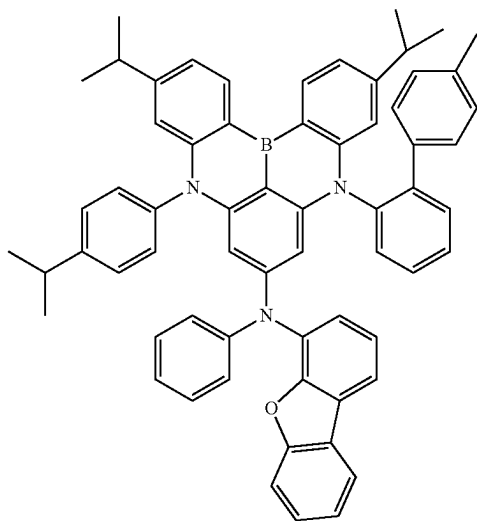
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Dopant 126



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Dopant 127



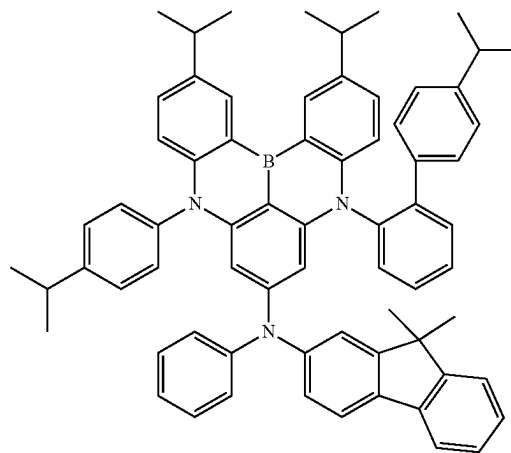
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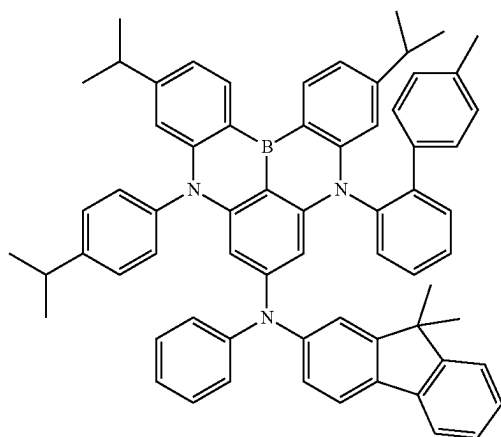
Dopant 130



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Dopant 131

**68**

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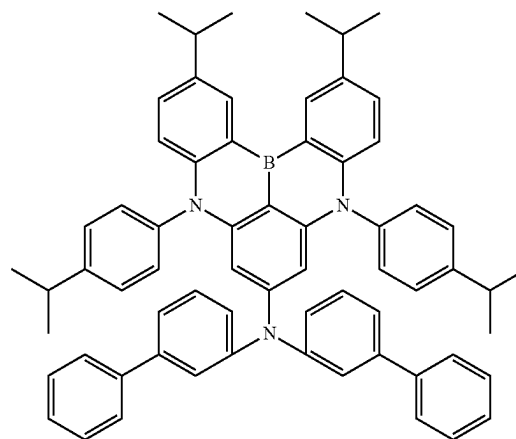
Dopant 134

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Dopant 132

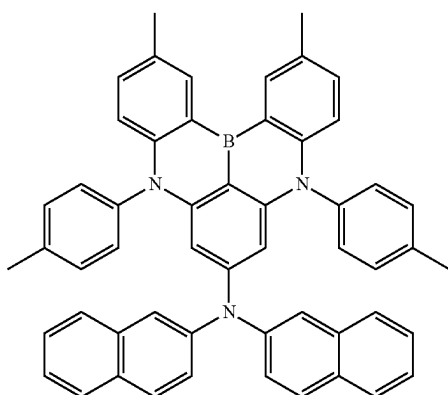
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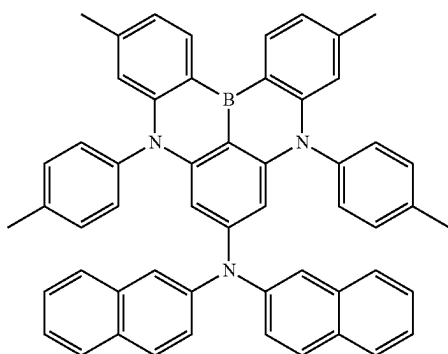
Dopant 135

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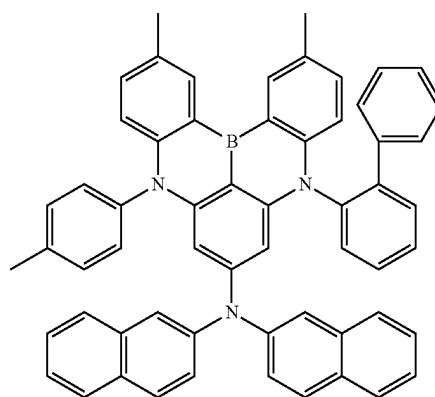
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Dopant 133

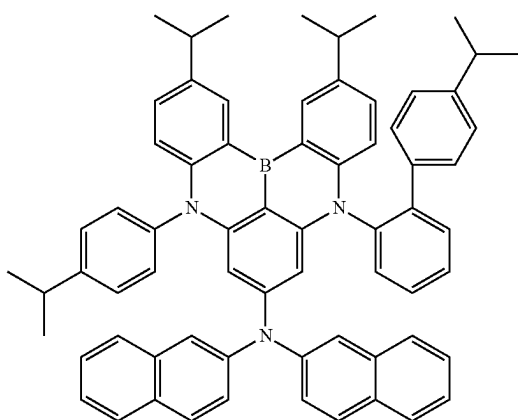
Dopant 136



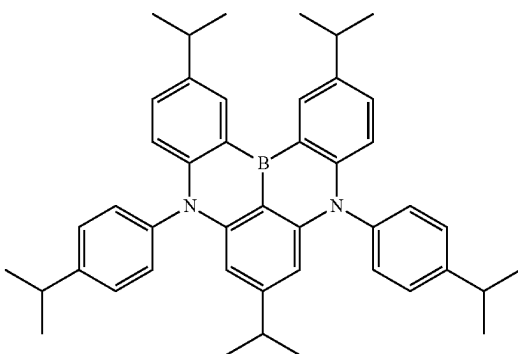
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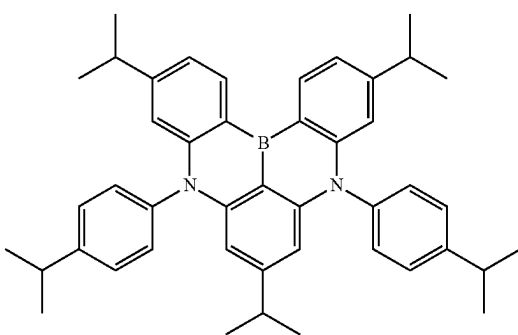
Dopant 137



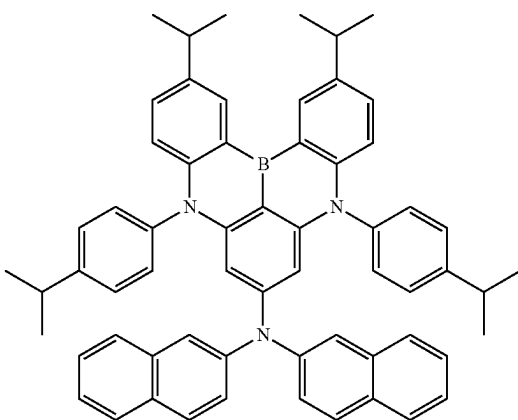
Dopant 138



Dopant 139

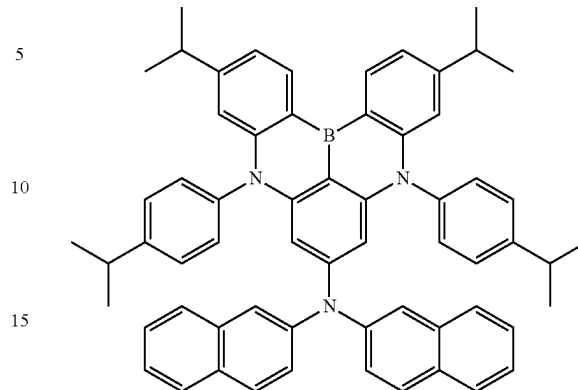


Dopant 140

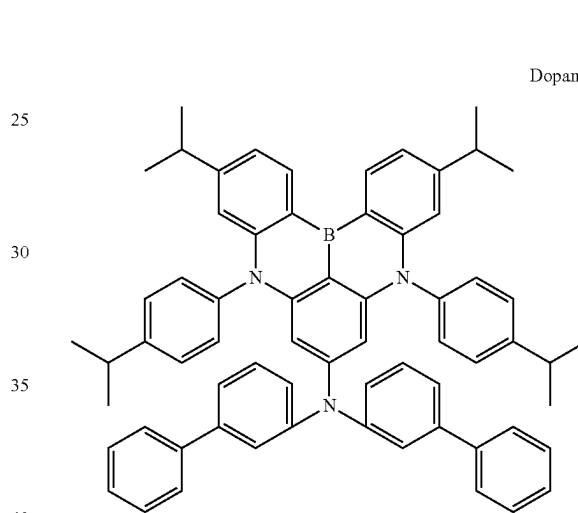
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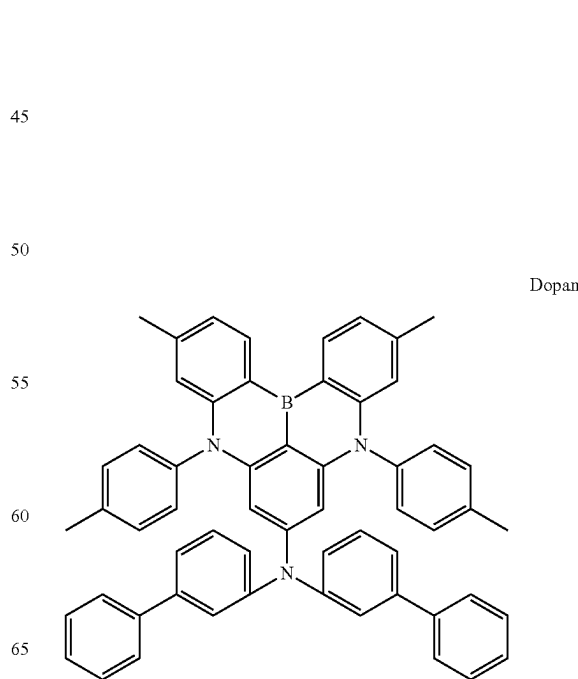
Dopant 141



Dopant 142

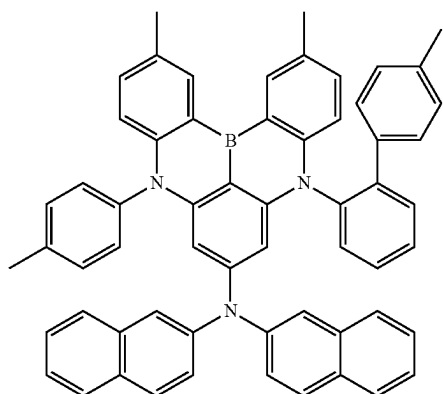


Dopant 143



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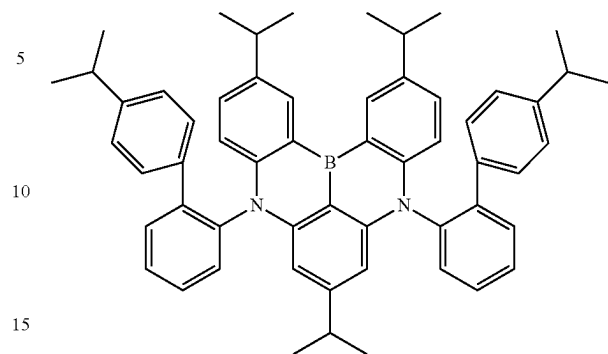
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Dopant 144

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Dopant 148

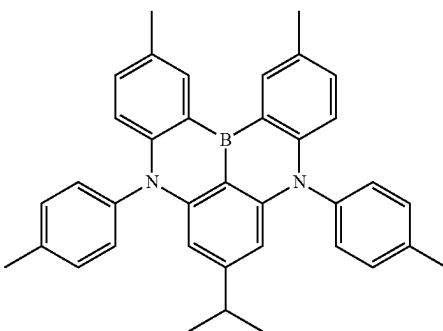
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Dopant 145



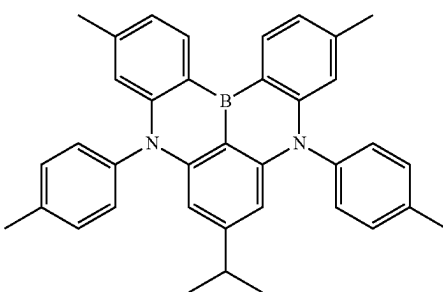
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Dopant 146

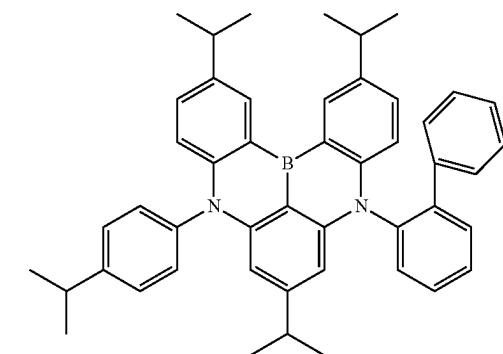


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Dopant 147

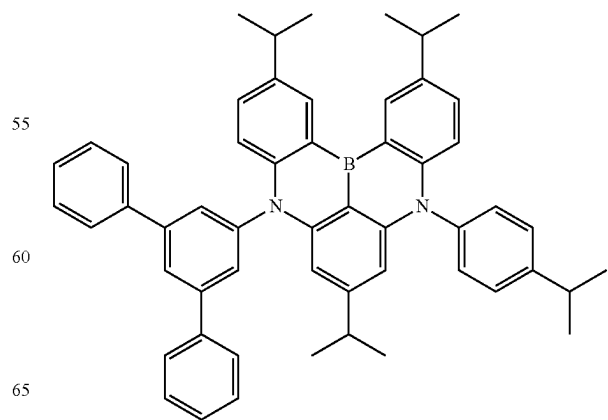


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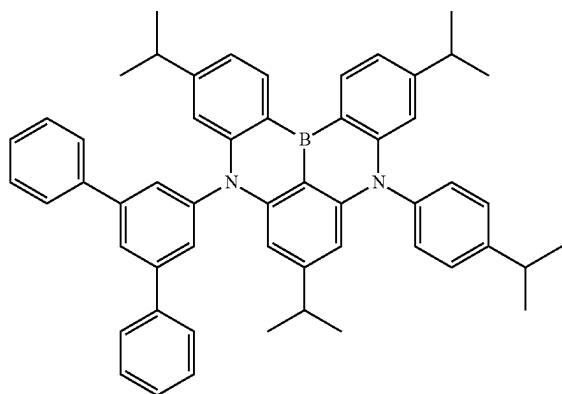
Dopant 150



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Dopant 151

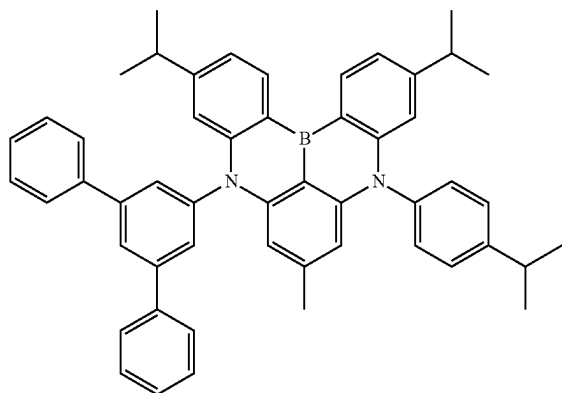


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Dopant 152

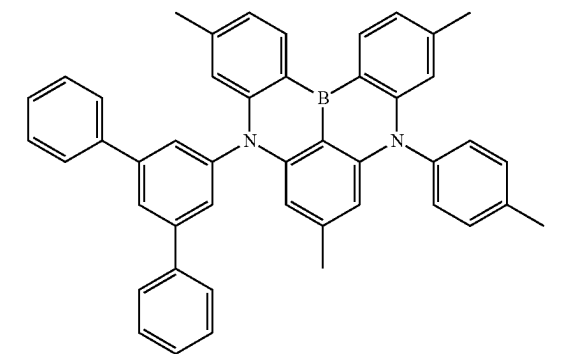


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Dopant 153

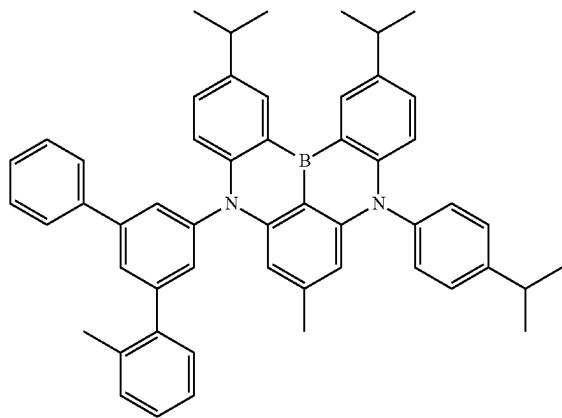


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Dopant 154



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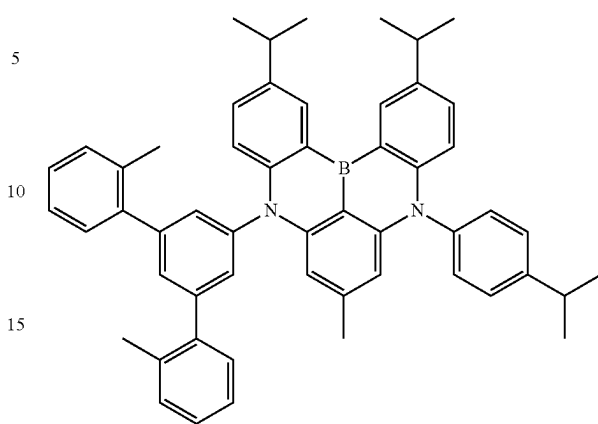
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Dopant 155

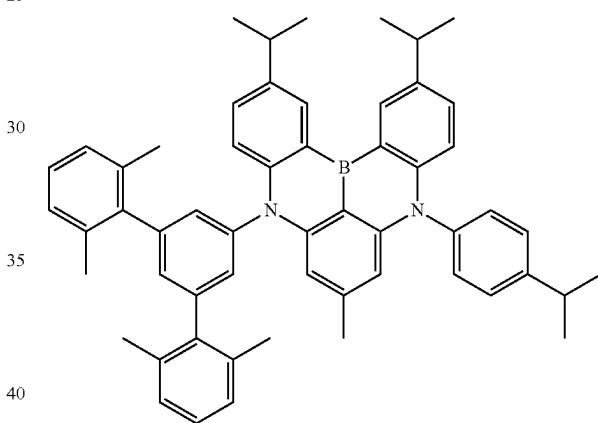


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Dopant 156



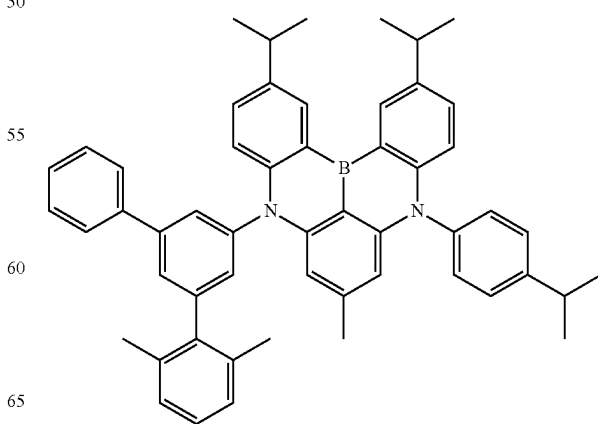
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Dopant 157



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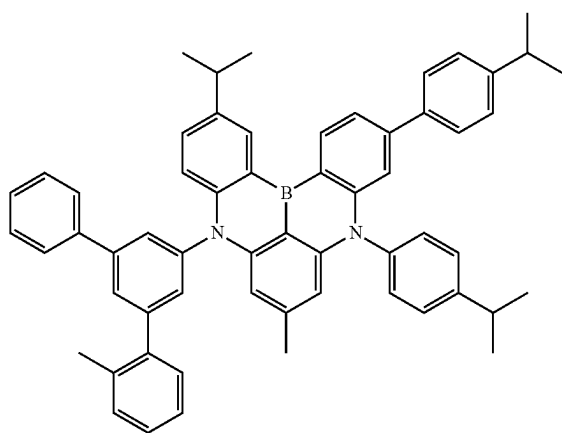
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Dopant 158



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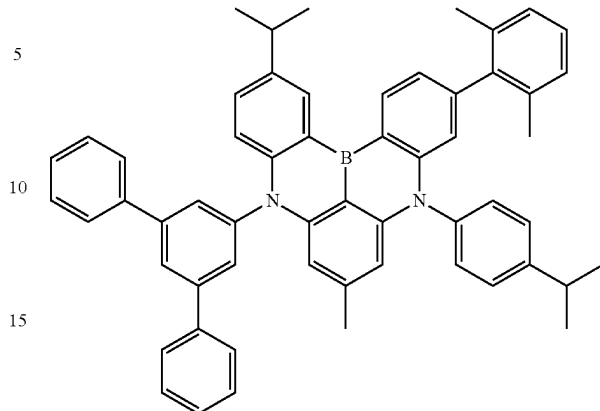
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Dopant 161



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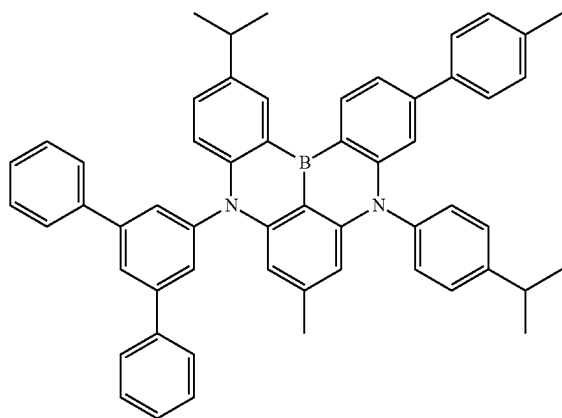
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Dopant 160

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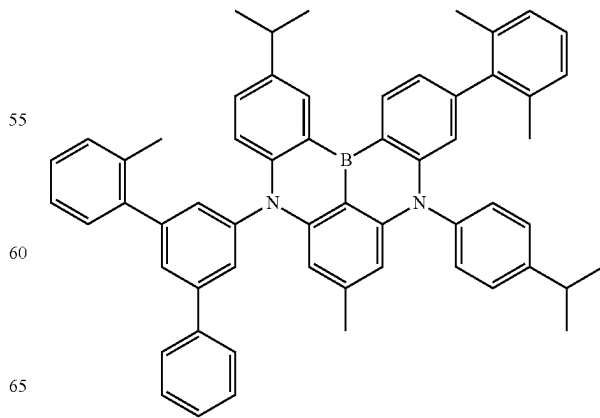


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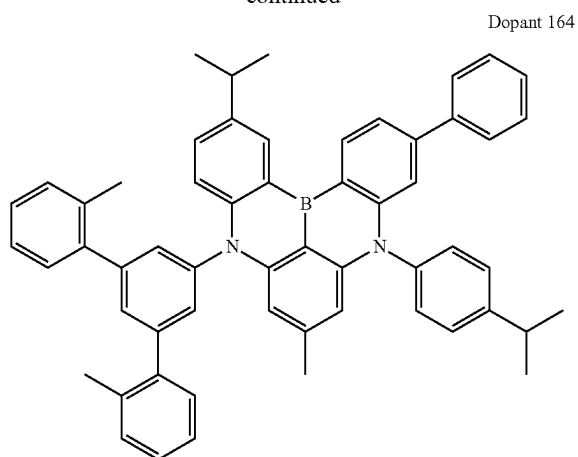
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Dopant 163

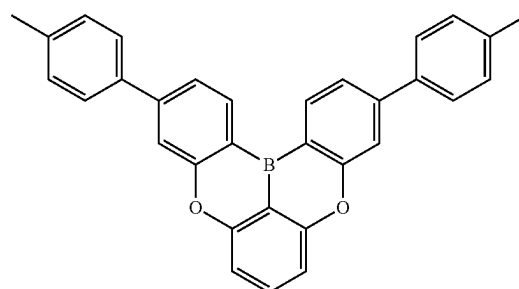
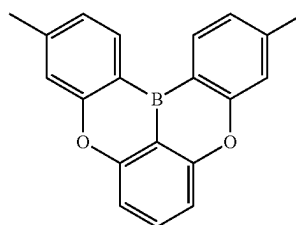
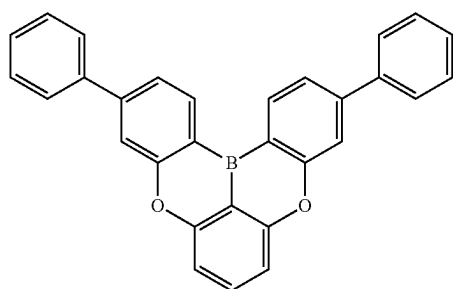
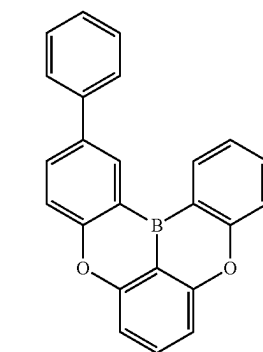
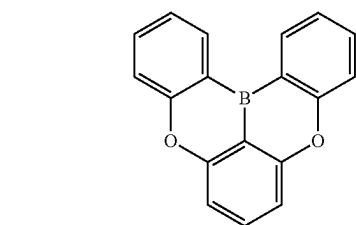
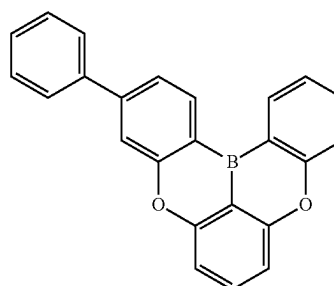
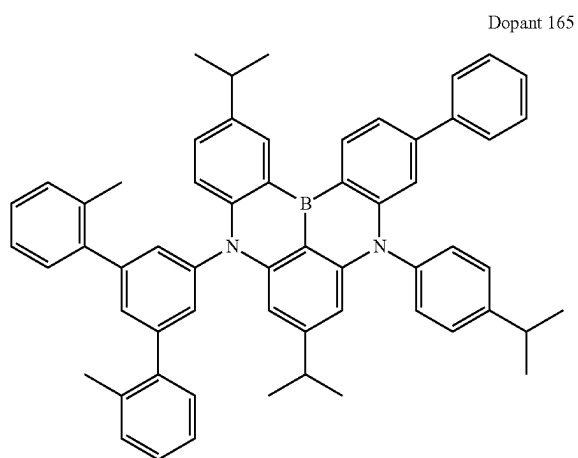
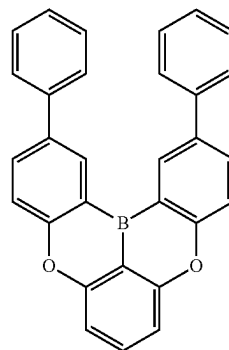


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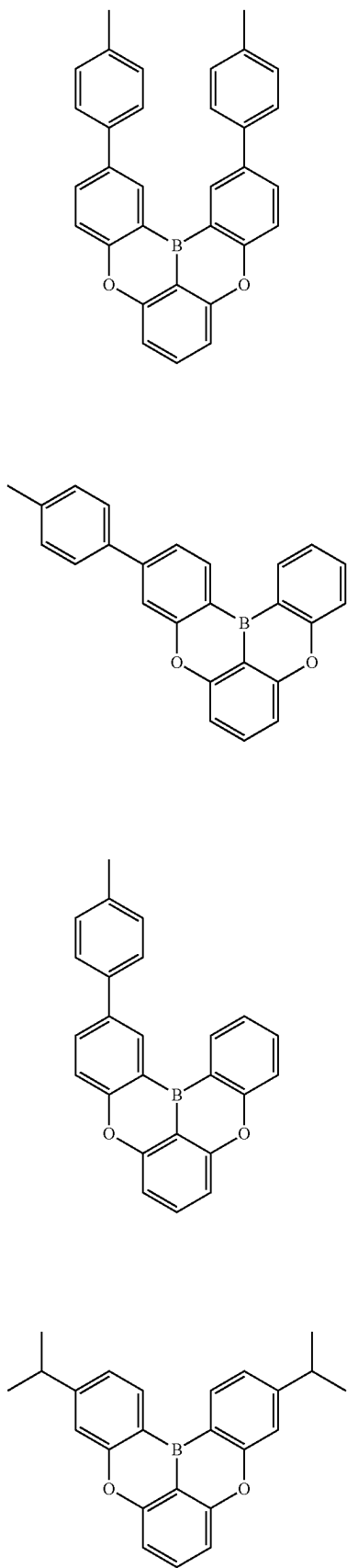
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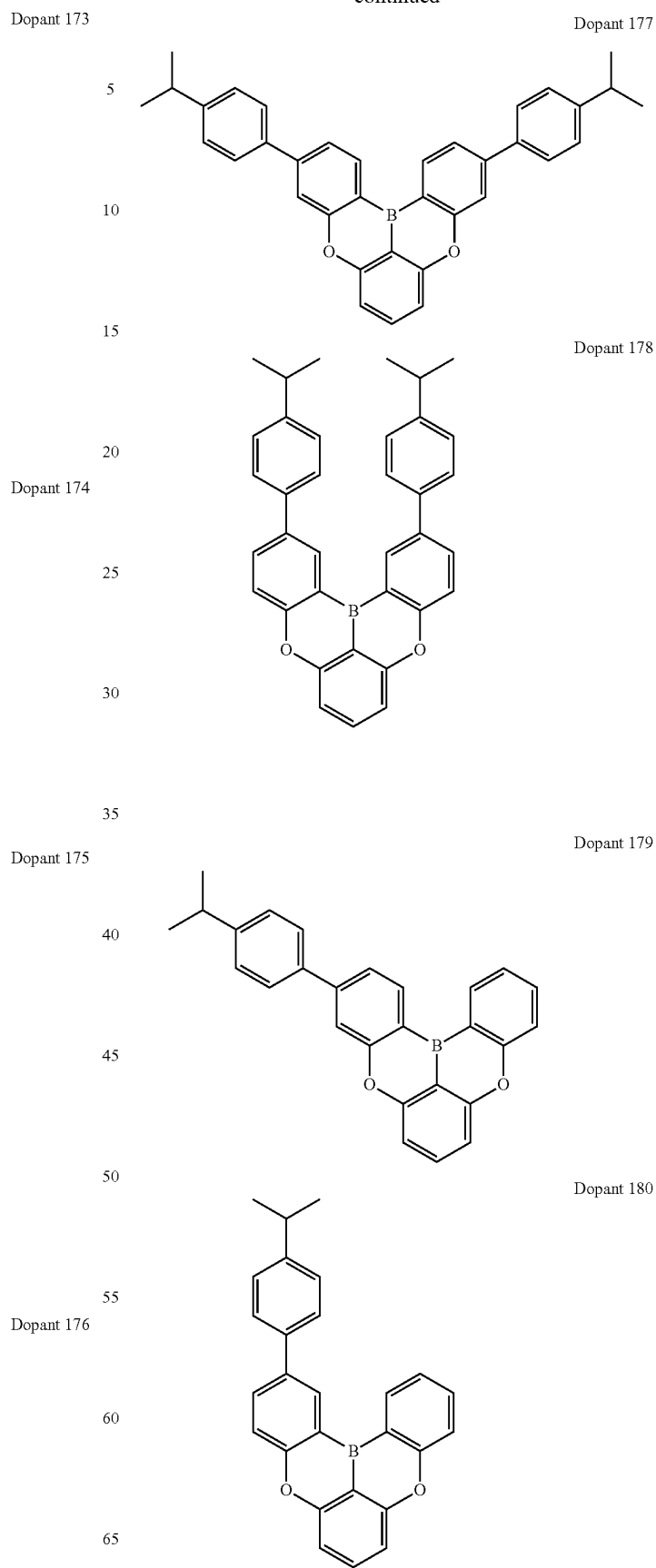
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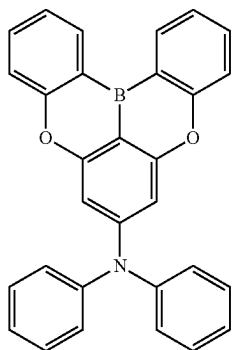
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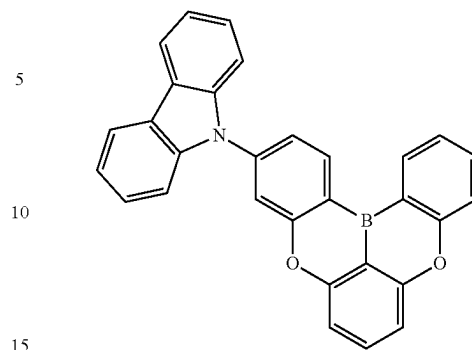
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Dopant 181

82

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Dopant 185

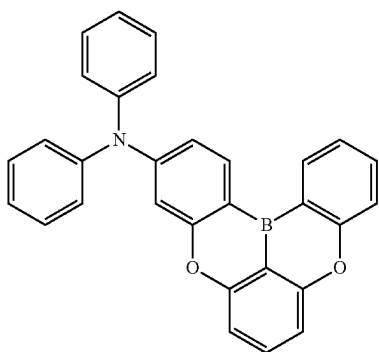
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Dopant 182

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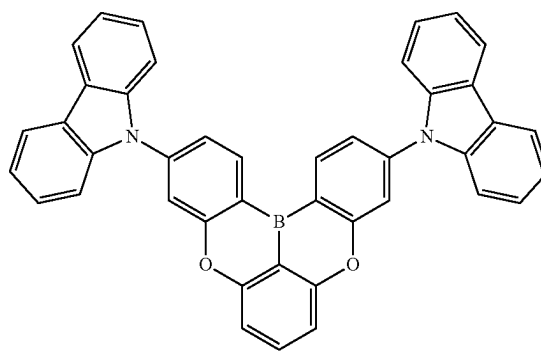


Dopant 186

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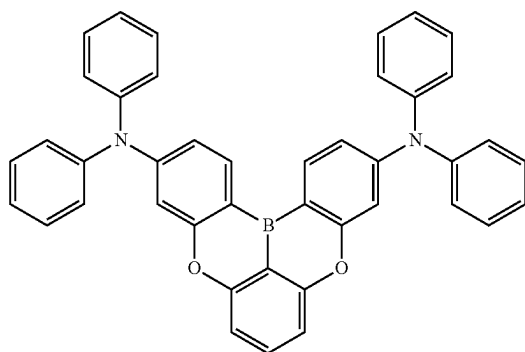


Dopant 183

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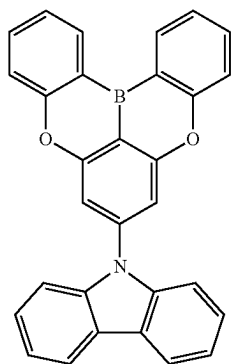
Dopant 187

Dopant 184

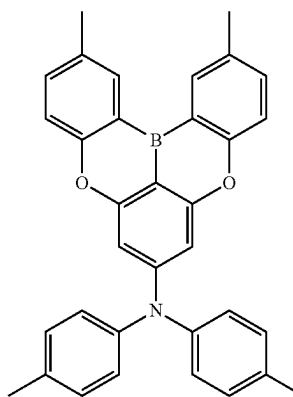
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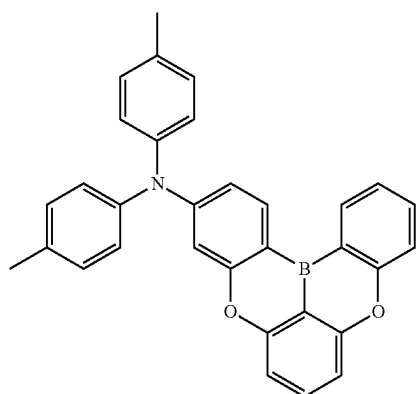


Dopant 188



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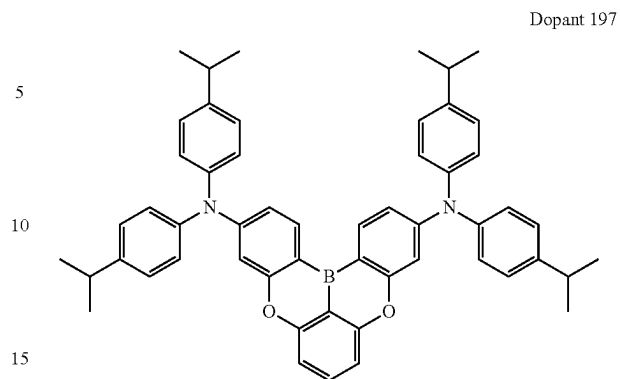
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Dopant 189

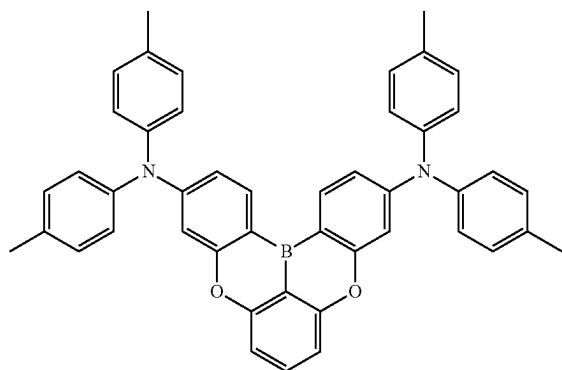
84

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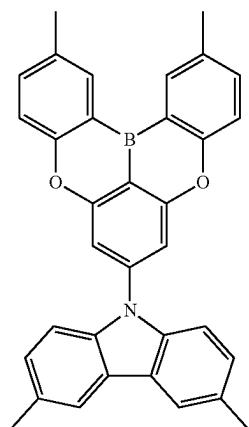


Dopant 197

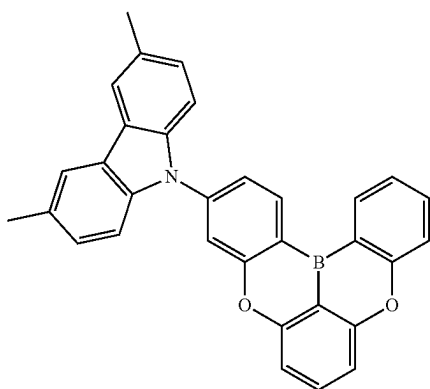
Dopant 190



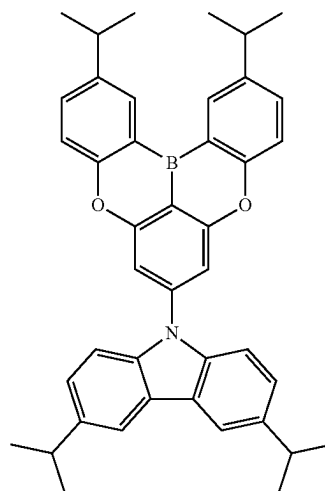
Dopant 191



Dopant 192

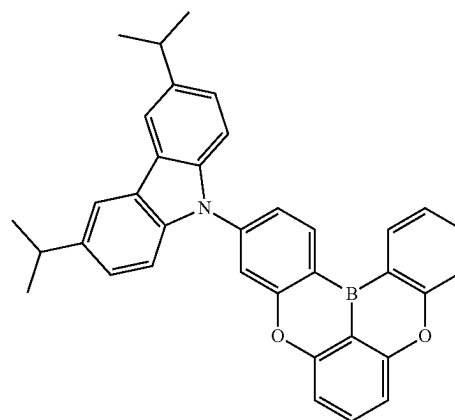


Dopant 198



Dopant 198

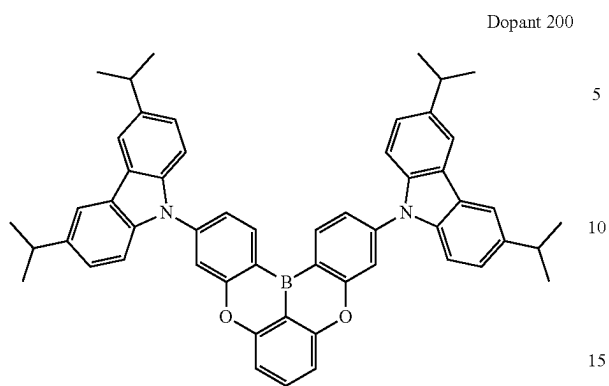
Dopant 199



Dopant 199

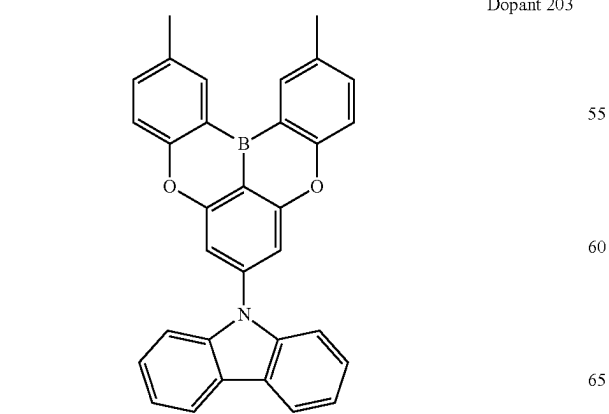
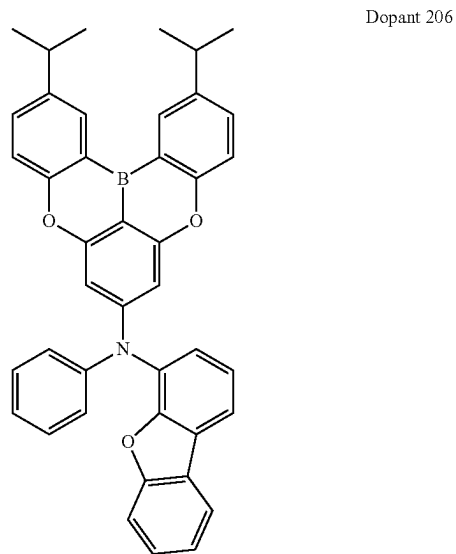
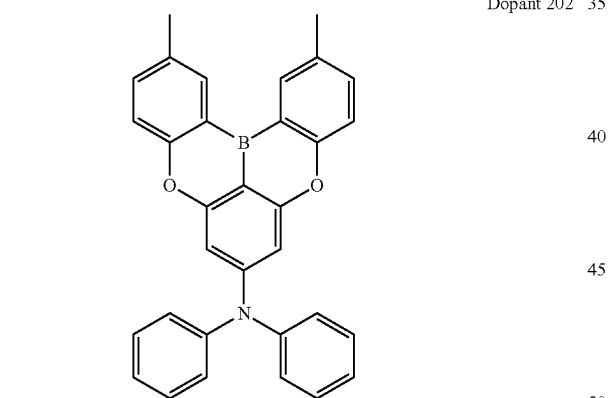
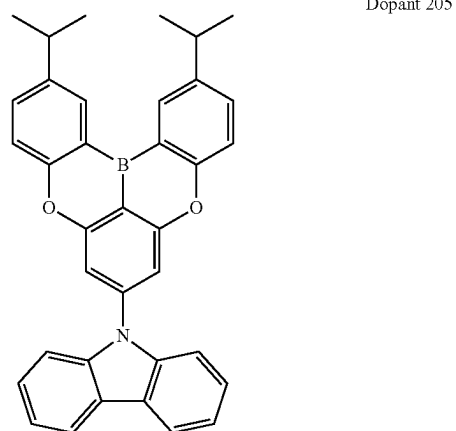
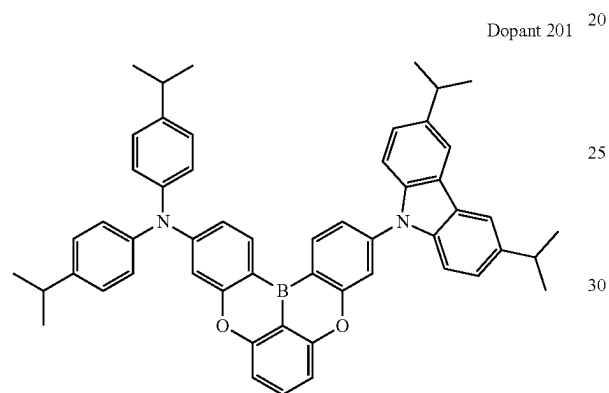
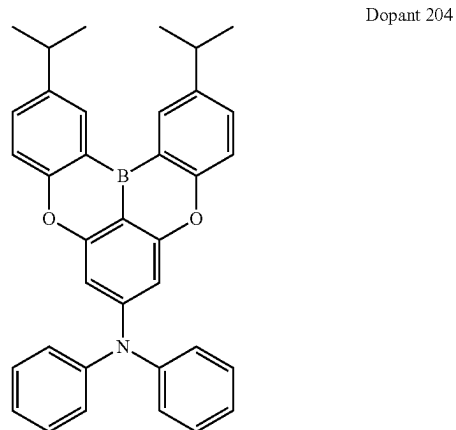
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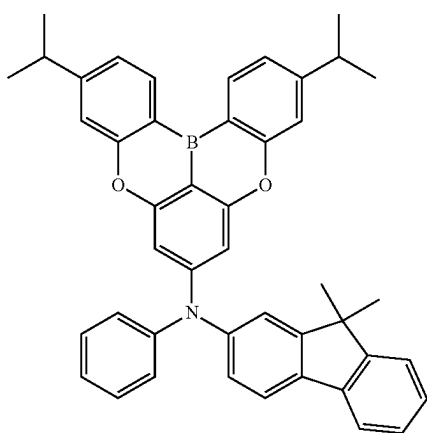
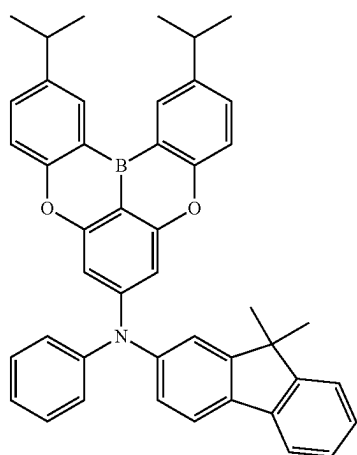
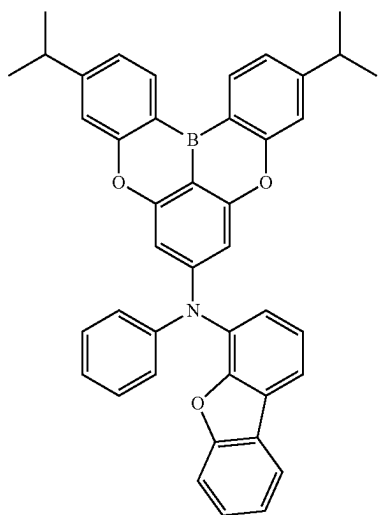
86

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Dopant 207

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Dopant 208

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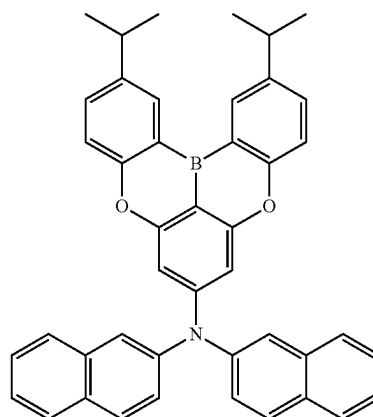
Dopant 209

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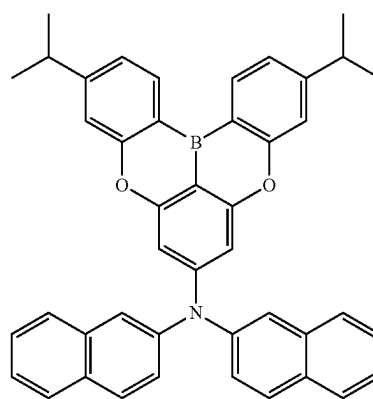
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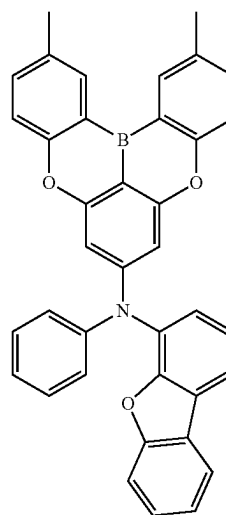
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Dopant 210



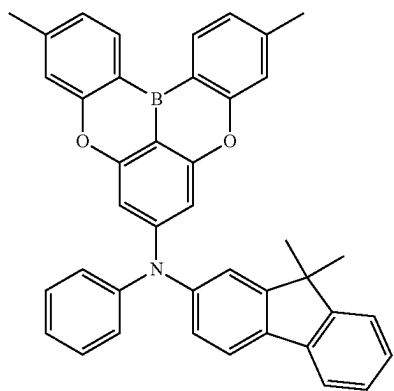
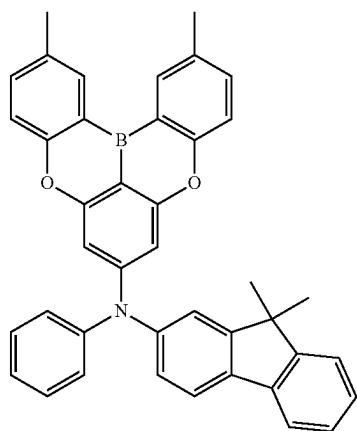
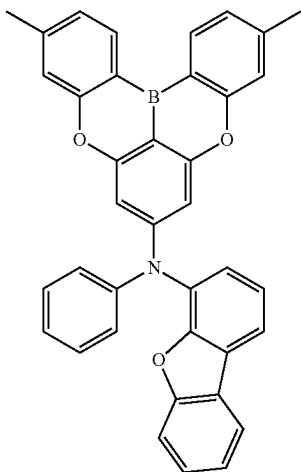
Dopant 211



Dopant 212

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Dopant 213

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Dopant 214

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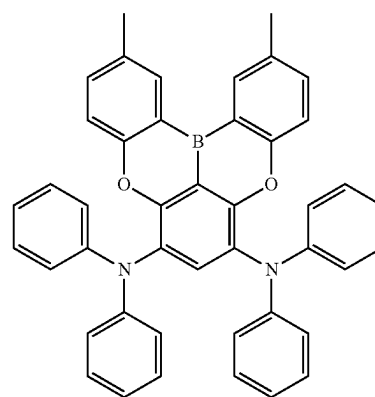
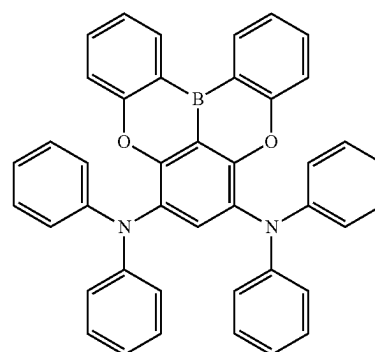
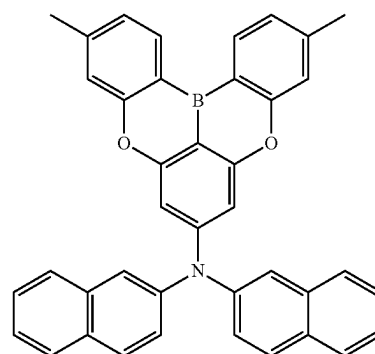
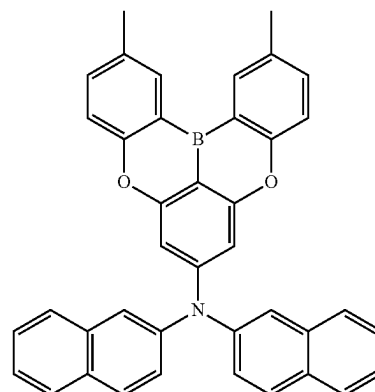
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Dopant 215

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Dopant 216

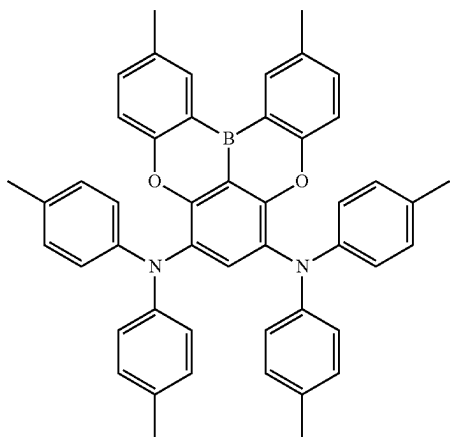
Dopant 217

Dopant 218

Dopant 219

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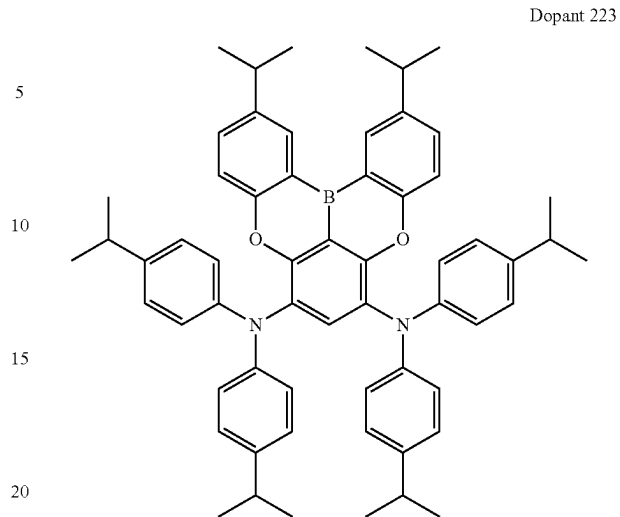
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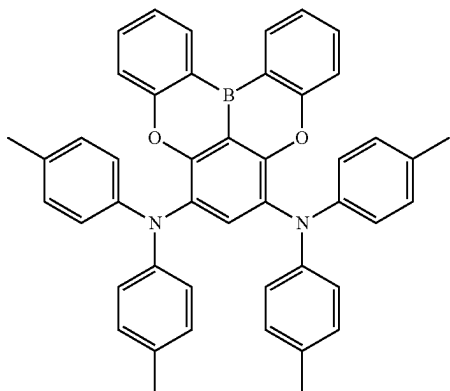
Dopant 220

92

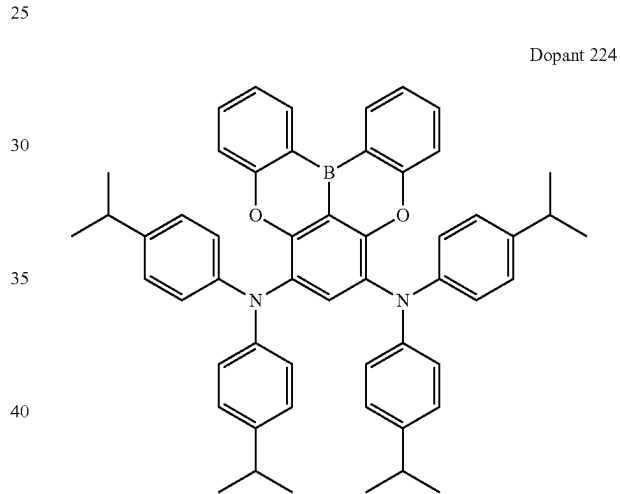
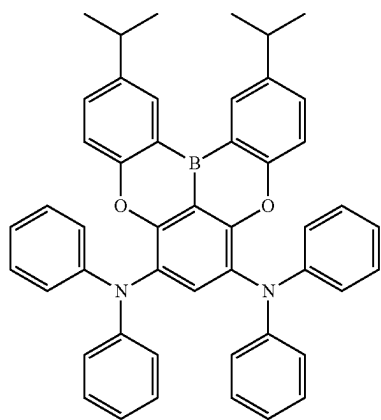
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Dopant 221



Dopant 222



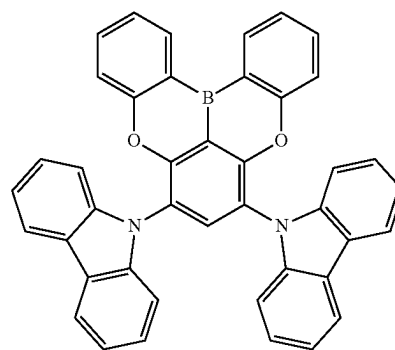
Dopant 225

Dopant 226

Dopant 227

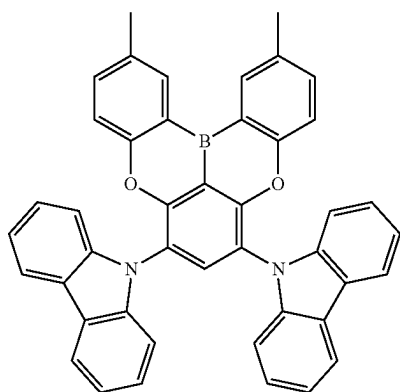
Dopant 228

Dopant 229



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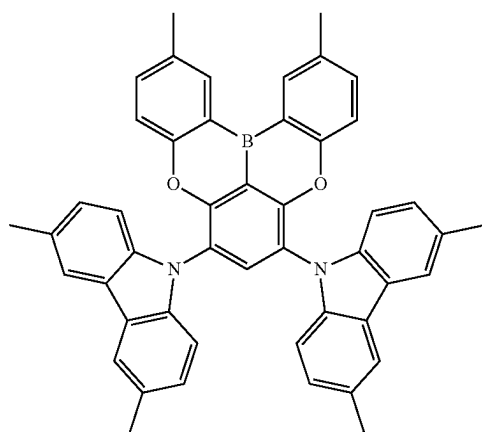


Dopant 226

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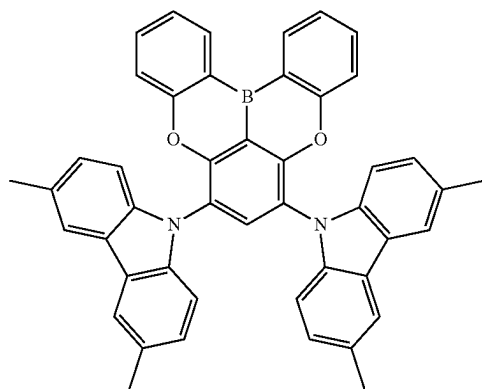


Dopant 227

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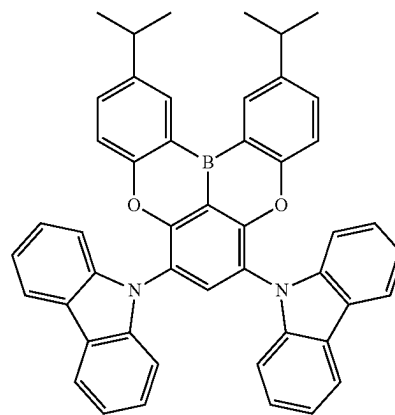
Dopant 228

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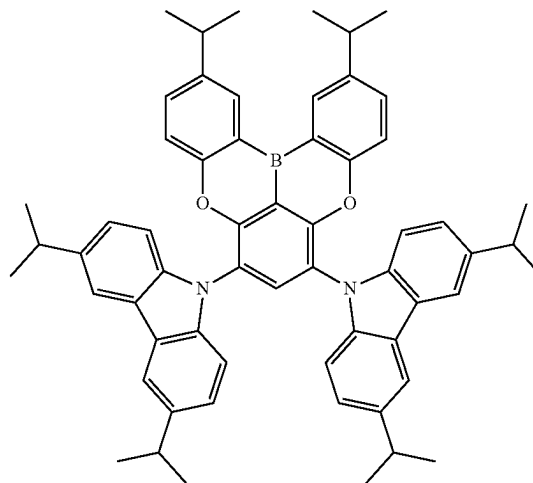
94

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Dopant 229

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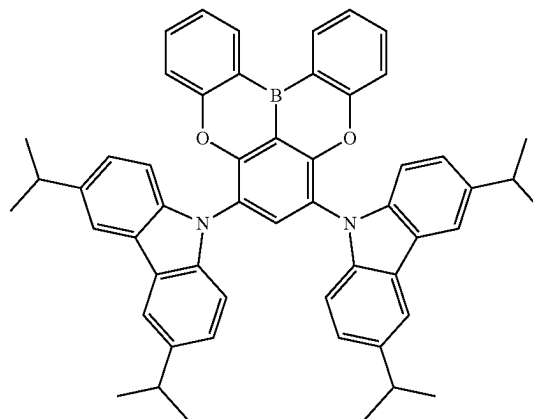
Dopant 230

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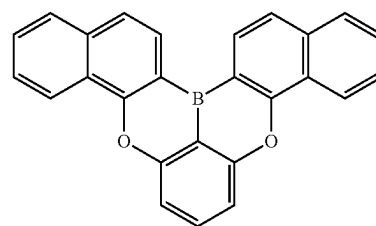
Dopant 231

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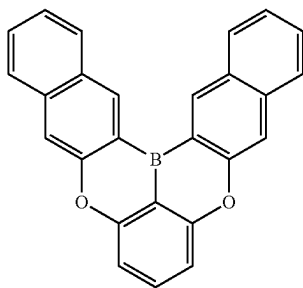
Dopant 232

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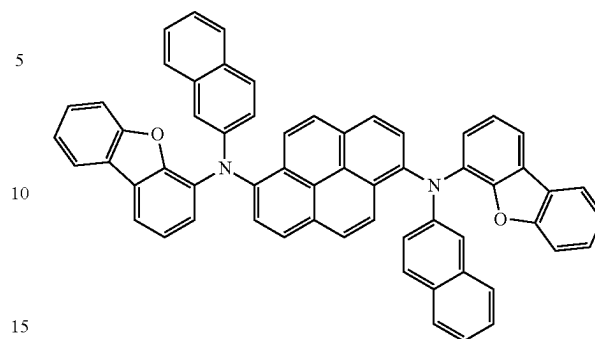


Dopant 233

96

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Dopant 3

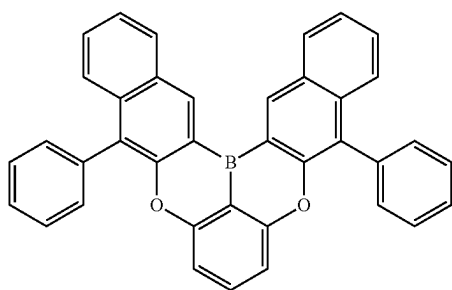


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Dopant 234



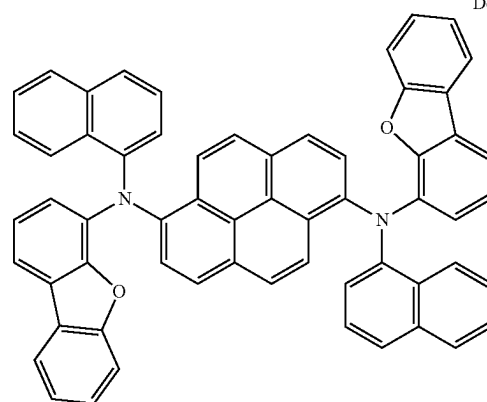
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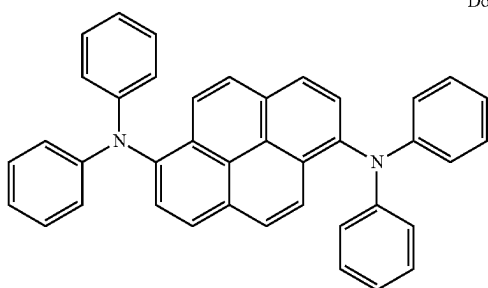
Dopant 4



For example, the blue dopant of Formula 5-2 may be a compound being one of the followings in Formula 6-2:

Formula 6-2

Dopant 1

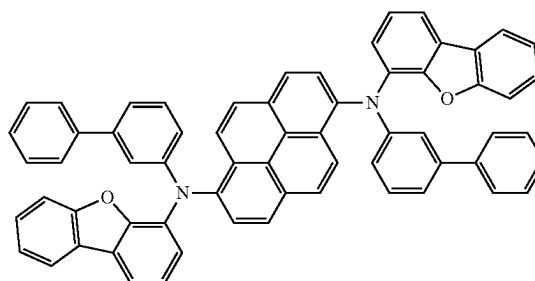


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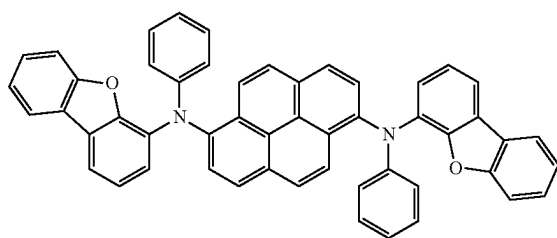
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Dopant 5



Dopant 6

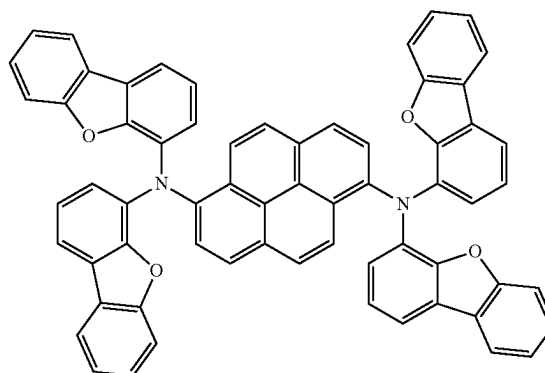
Dopant 2



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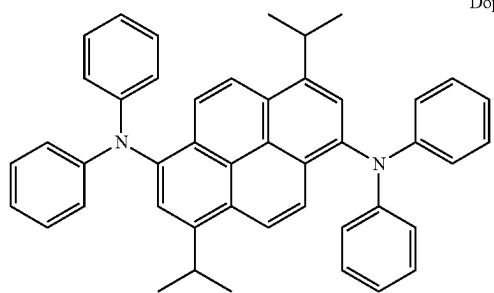
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97

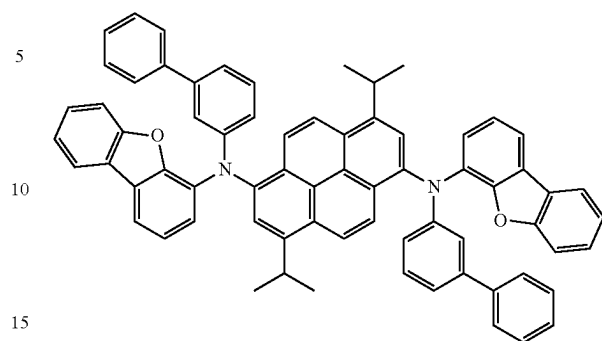
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Dopant 7

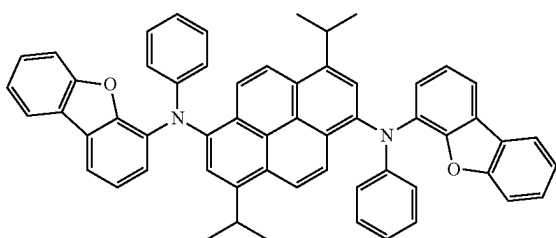
98

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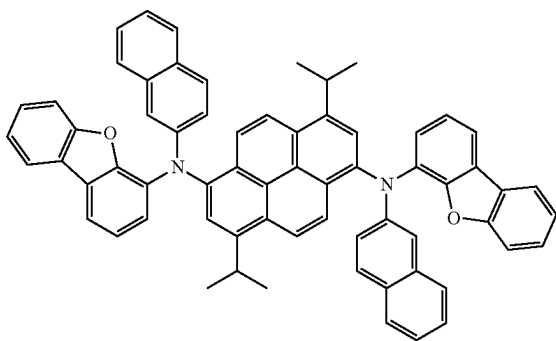
Dopant 11

Dopant 8



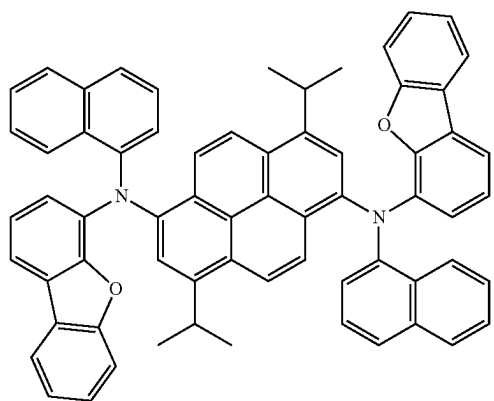
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Dopant 9



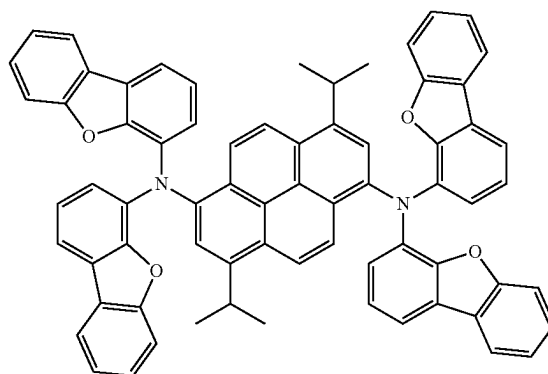
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Dopant 10



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Dopant 12

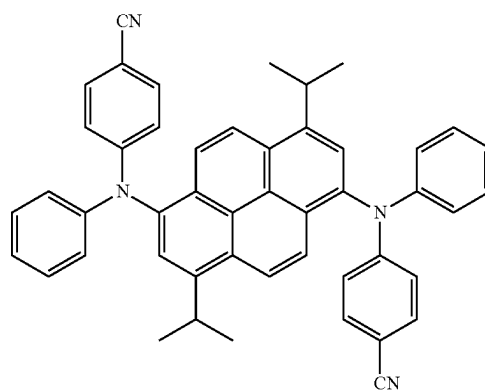


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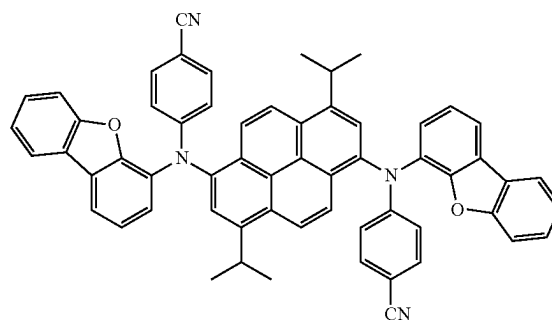
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Dopant 13



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Dopant 14



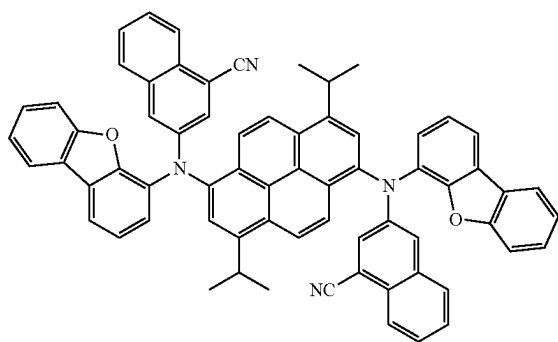
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99

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Dopant 15

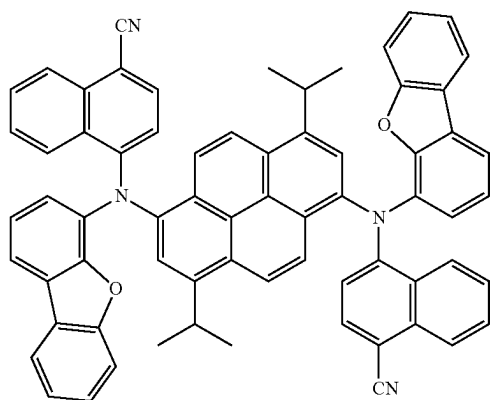


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Dopant 16



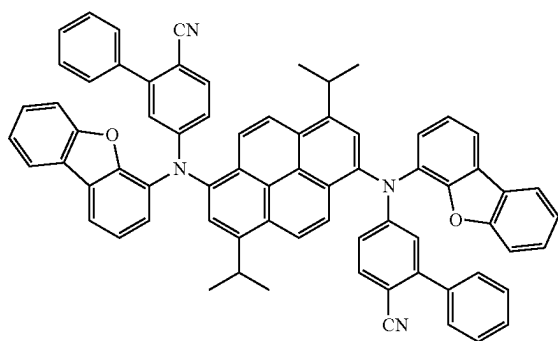
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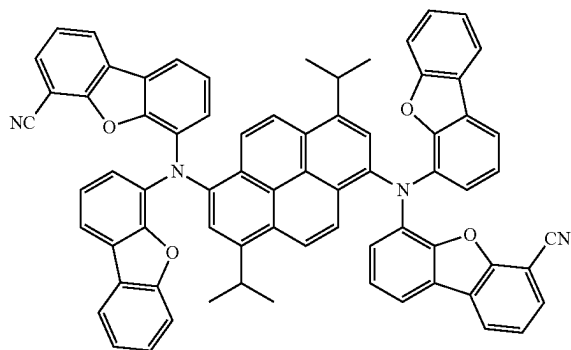
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Dopant 17



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Dopant 18



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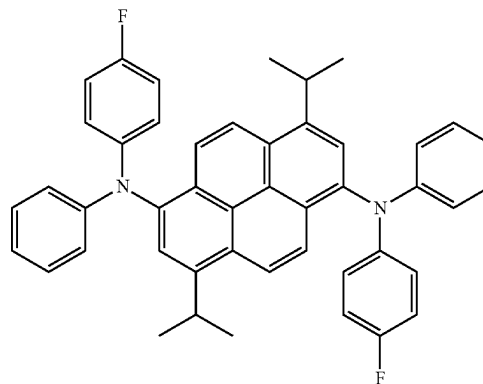
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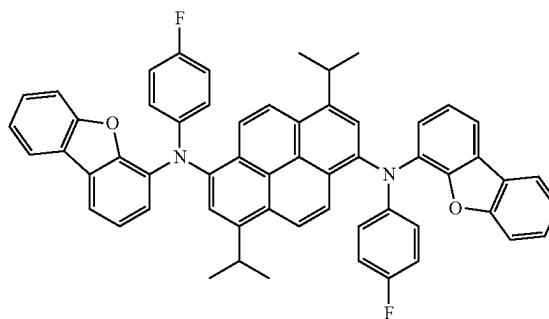
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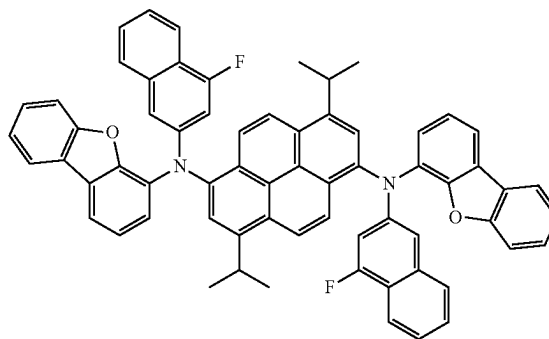
Dopant 19



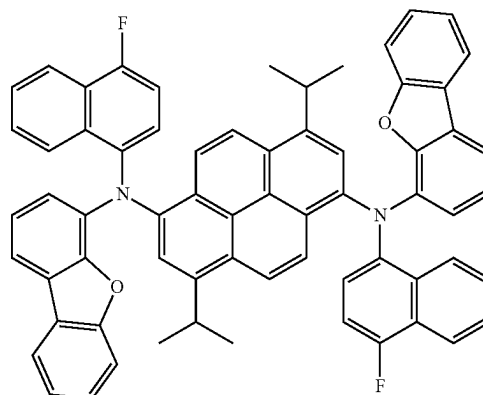
Dopant 20



Dopant 21



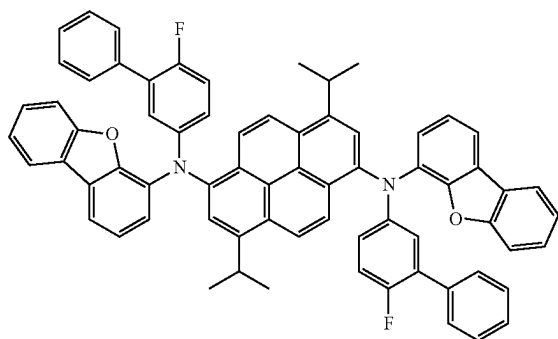
Dopant 22



101

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Dopant 23

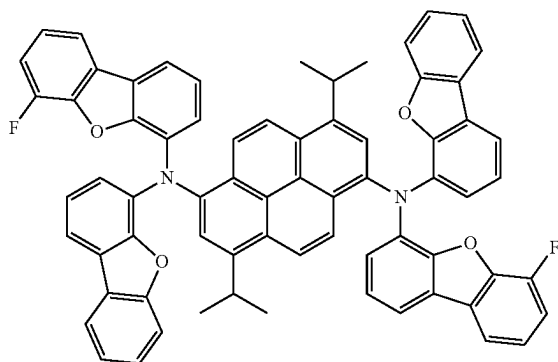


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Dopant 24



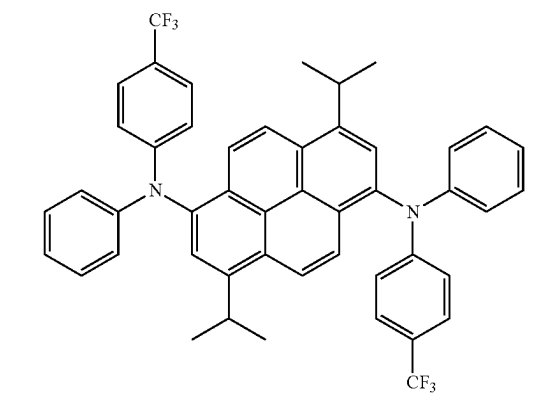
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Dopant 25

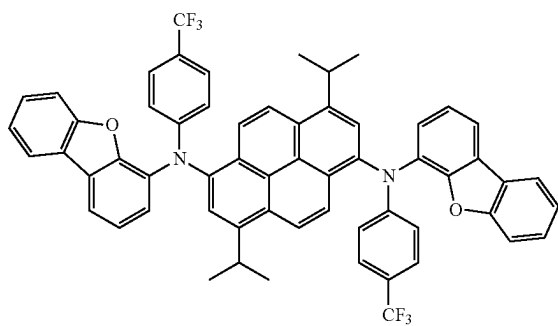


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Dopant 26



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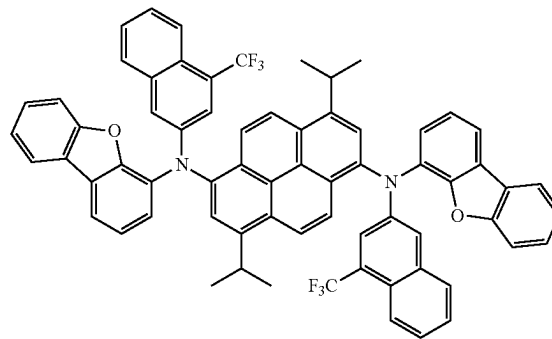
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102

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Dopant 27

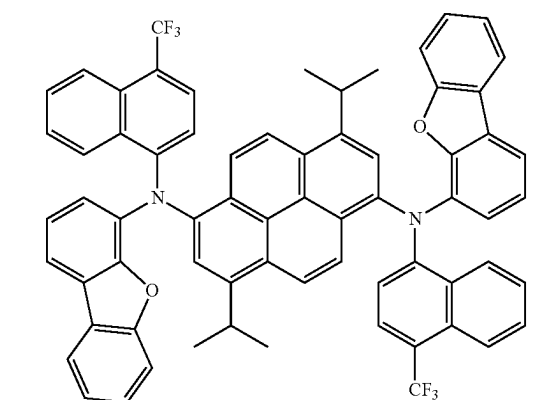


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Dopant 28



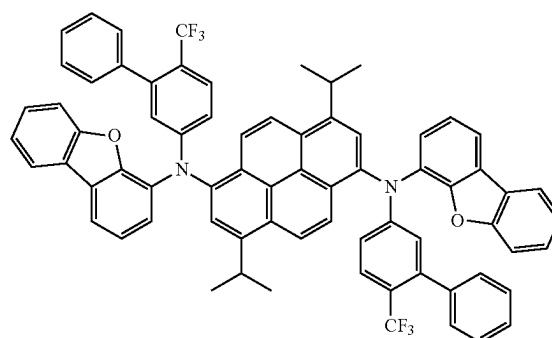
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Dopant 29

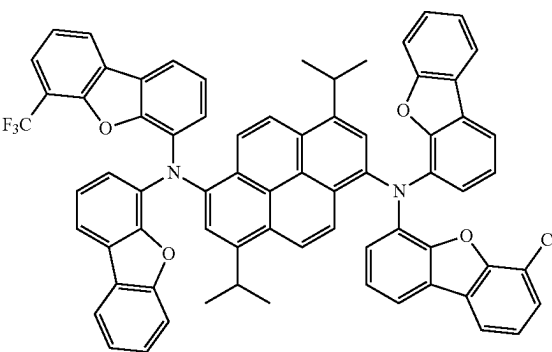


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Dopant 30



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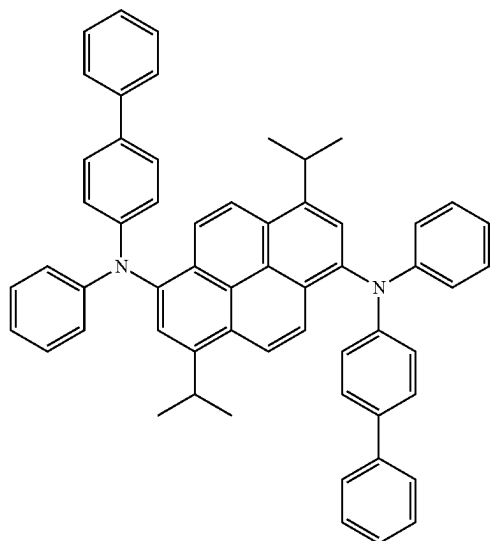
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103

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Dopant 31



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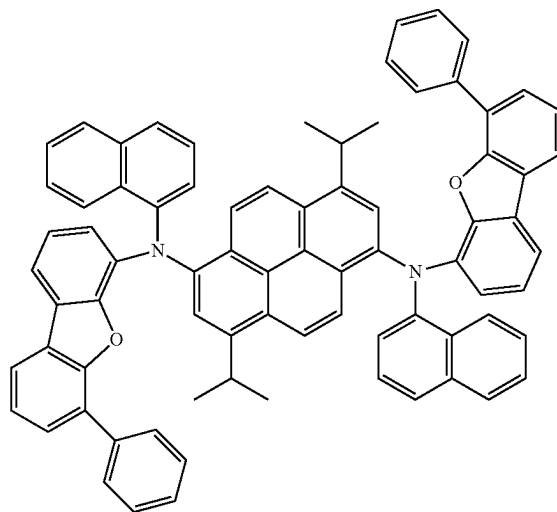
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104

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Dopant 34



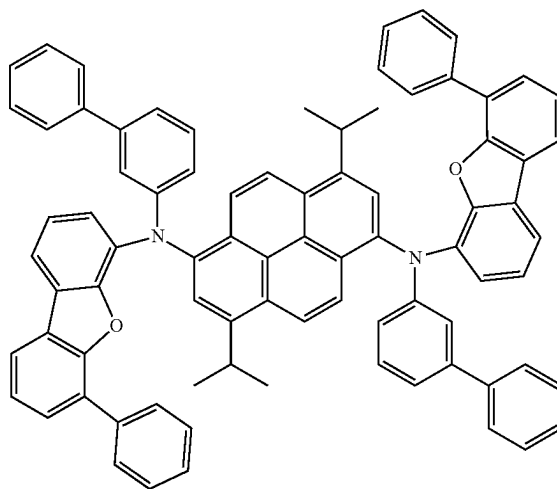
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Dopant 35



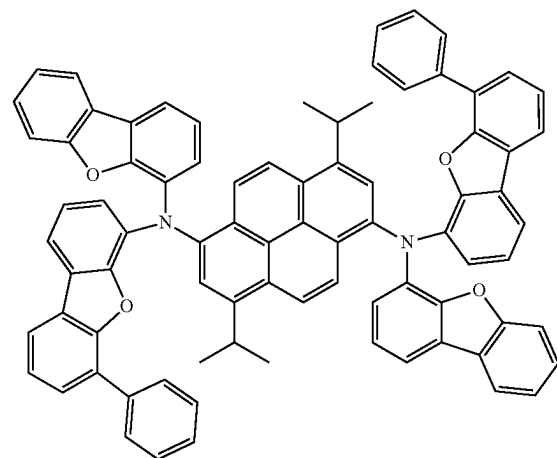
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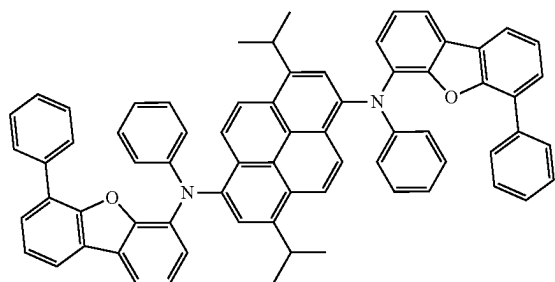
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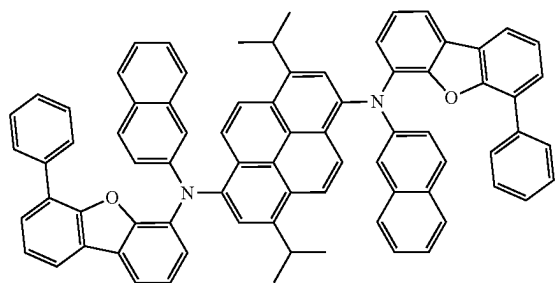
Dopant 36



Dopant 32



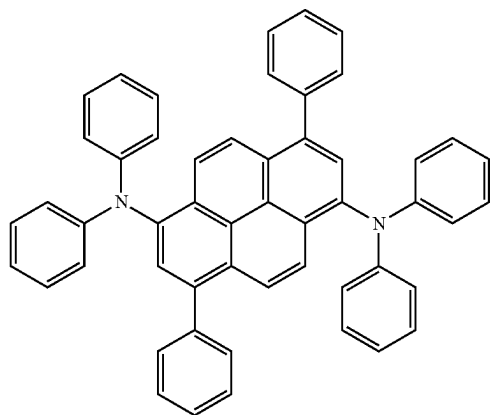
Dopant 33



105

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Dopant 37



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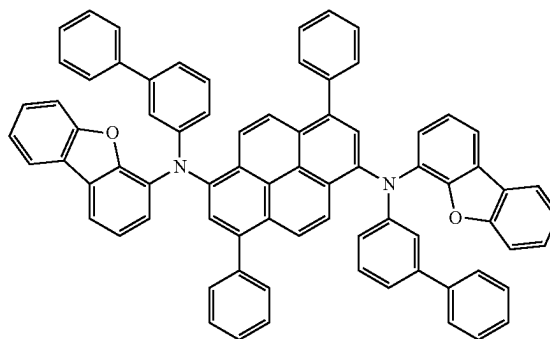
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106

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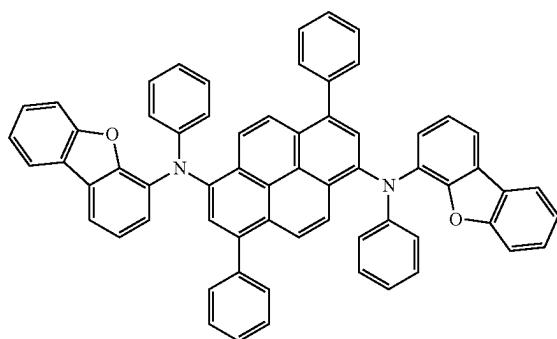
Dopant 41



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Dopant 42

Dopant 38

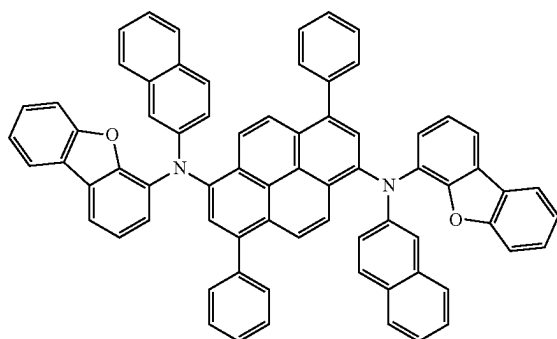


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Dopant 39

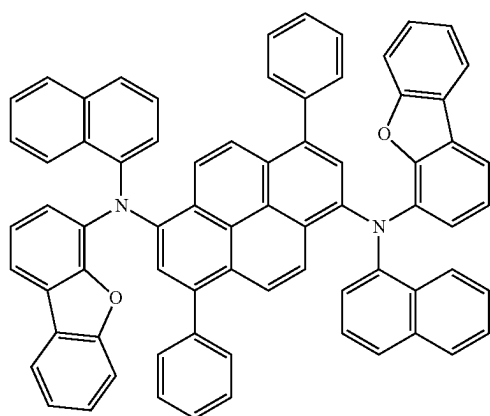


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Dopant 40

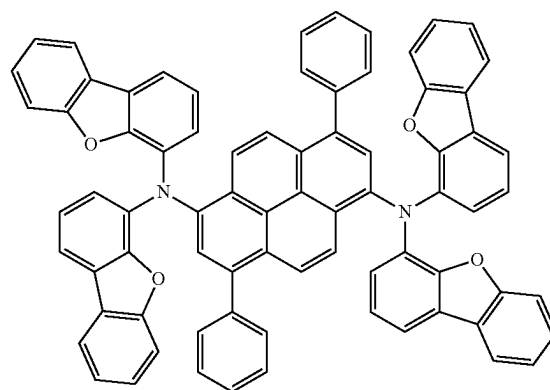


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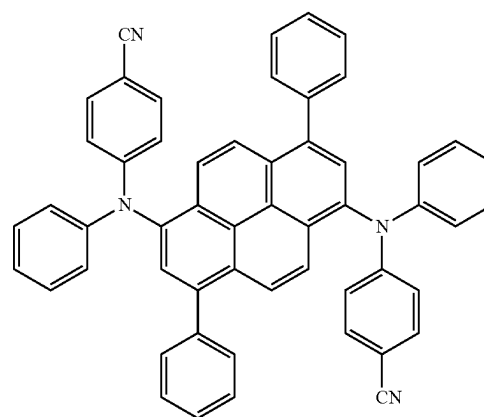
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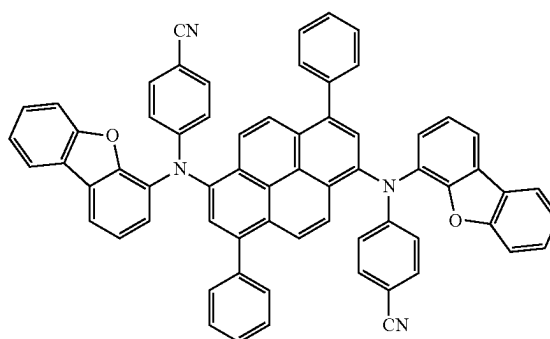
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Dopant 43



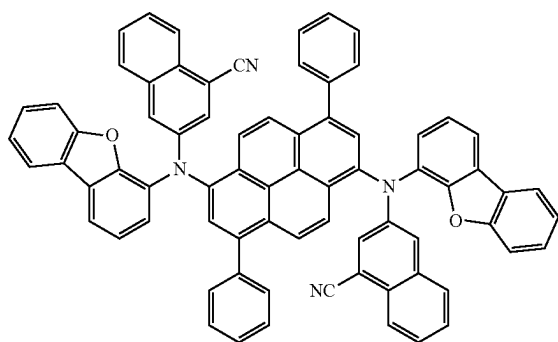
Dopant 44



107

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Dopant 45



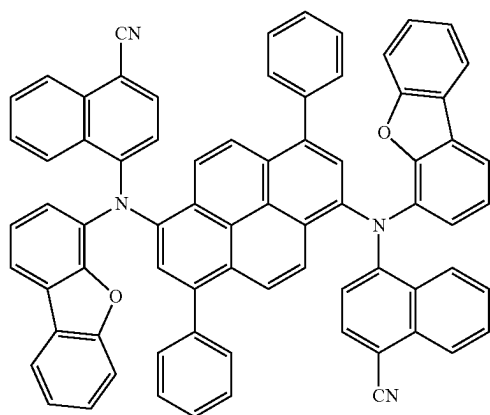
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Dopant 46

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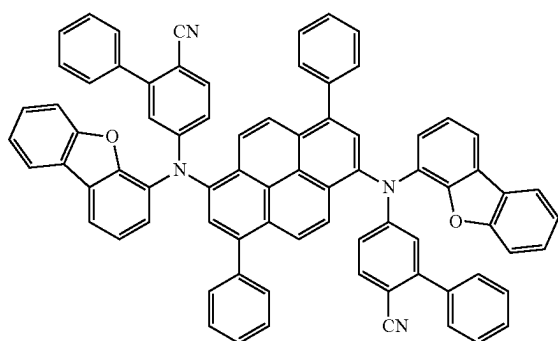
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Dopant 47

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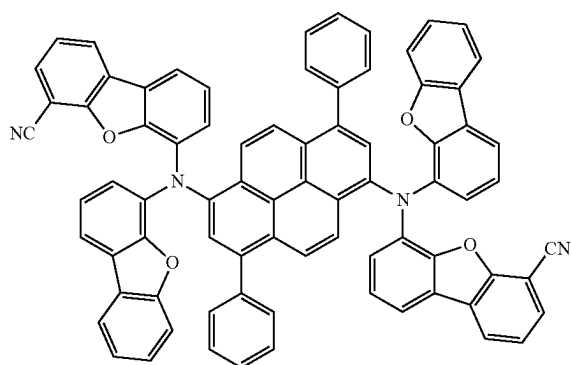


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Dopant 48

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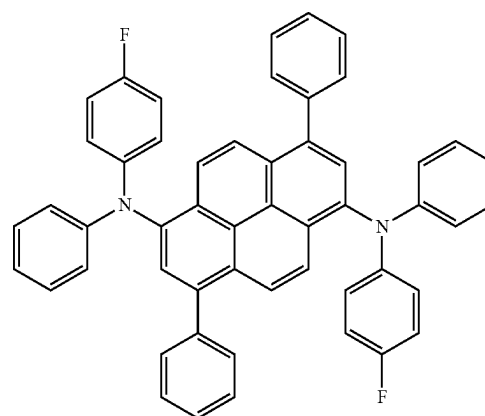
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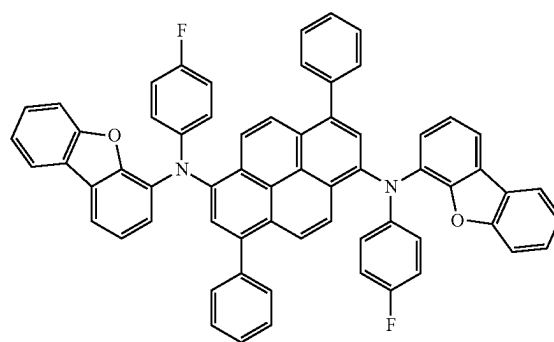
108

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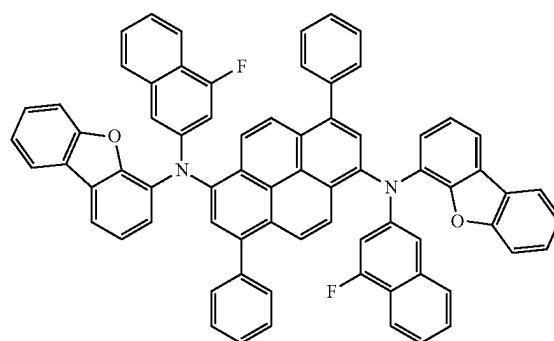
Dopant 49



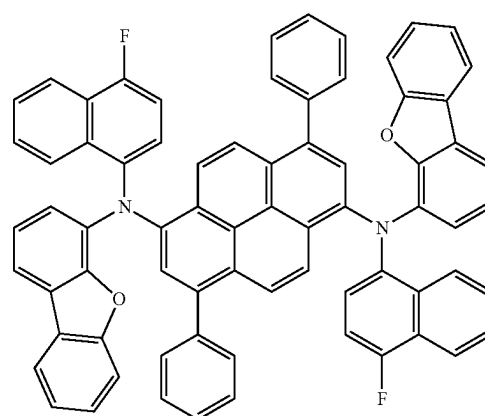
Dopant 50



Dopant 51



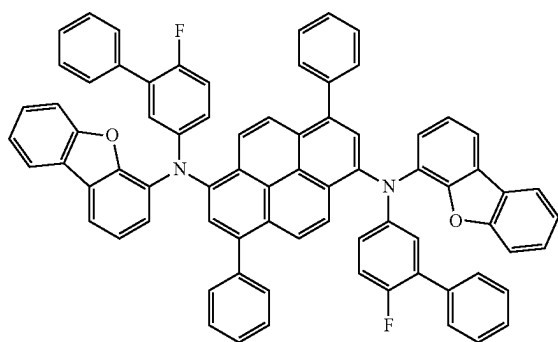
Dopant 52



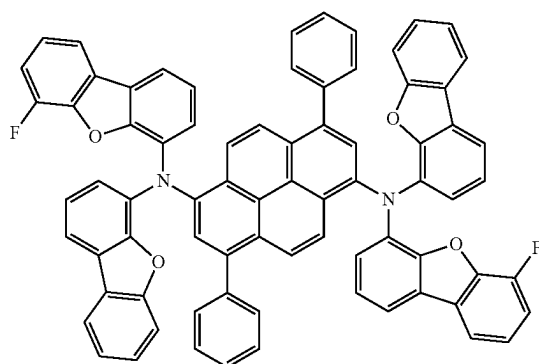
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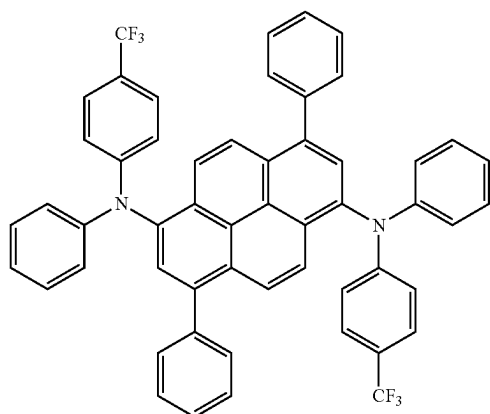
Dopant 53



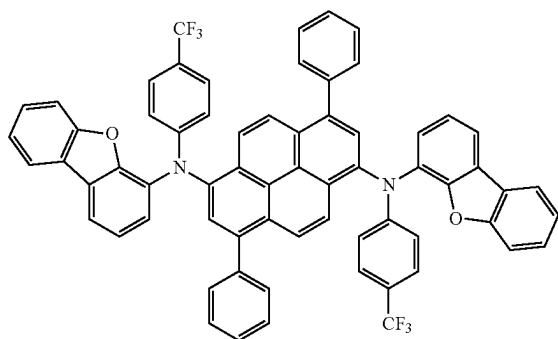
Dopant 54



Dopant 55

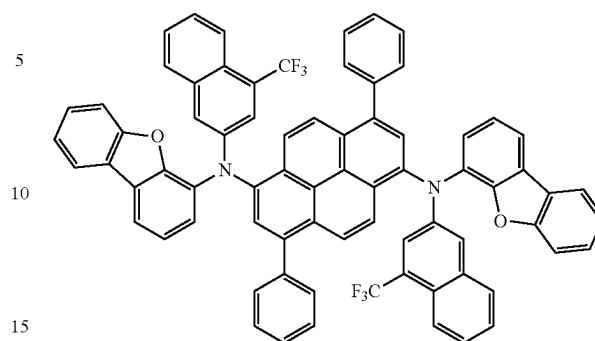


Dopant 56

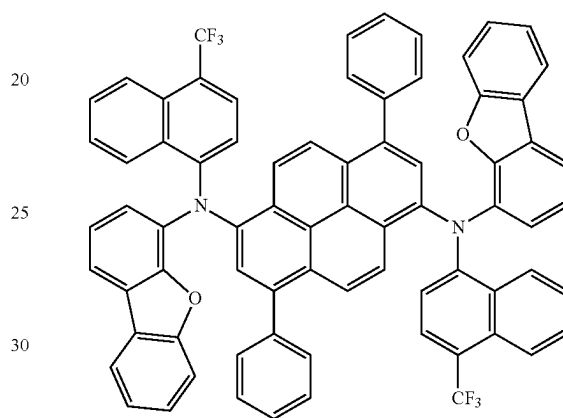
**110**

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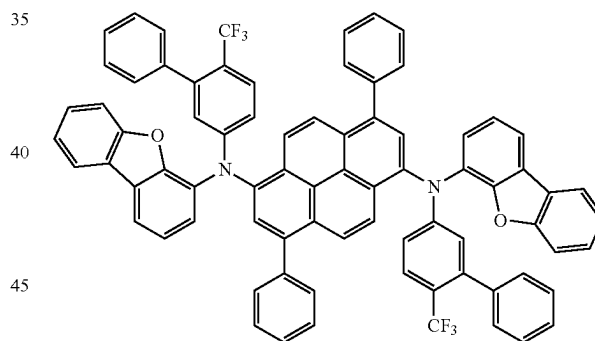
Dopant 57



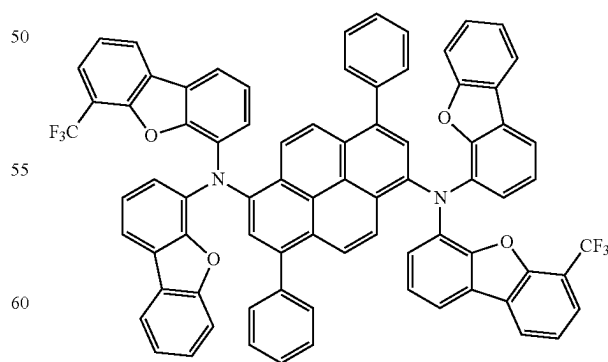
Dopant 58



Dopant 59

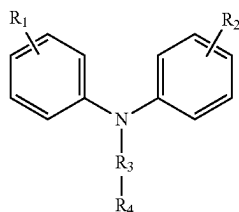


Dopant 60



65 The EBL **230** includes an amine derivative as an electron blocking material. The material of the EBL **230** may be represented by Formula 7:

111

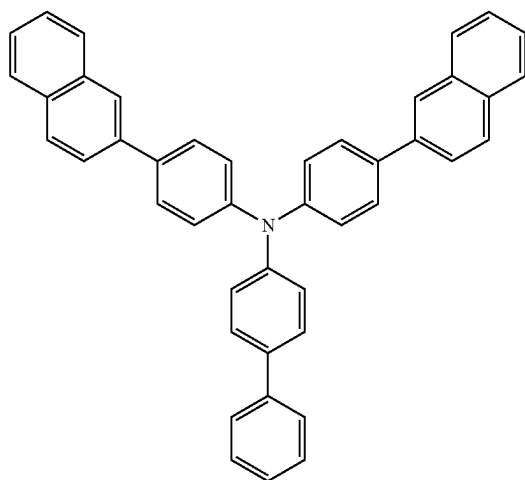


Formula 7

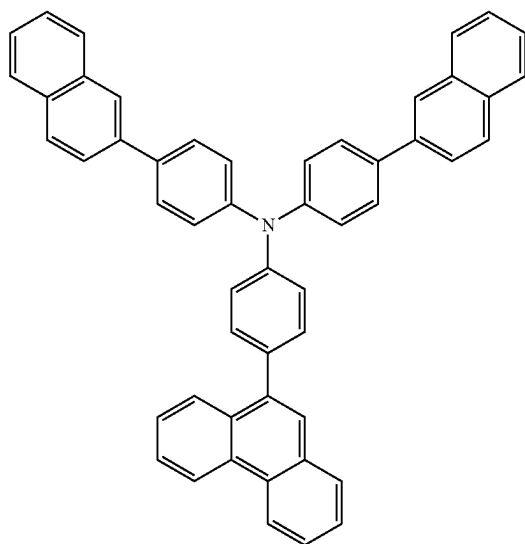
In Formula 7, each of R_1 , R_2 , R_3 and R_4 is independently selected from the group consisting of monocyclic aryl group or polycyclic aryl group, and at least one of R_1 , R_2 , R_3 and R_4 is polycyclic aryl group. For example, two of R_1 , R_2 , R_3 and R_4 may be polycyclic aryl group. The monocyclic aryl group may be phenyl, and the polycyclic aryl group may be a fused-aryl group. The polycyclic aryl group may be an aryl group in which at least two phenyl groups are fused.

The electron blocking material of Formula 7 may be one of the followings of Formula 8:

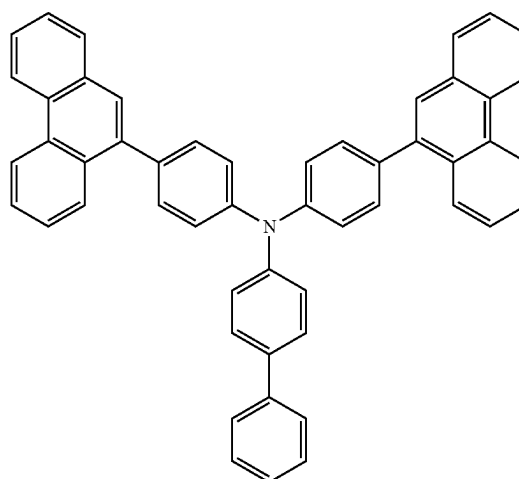
Formula 8



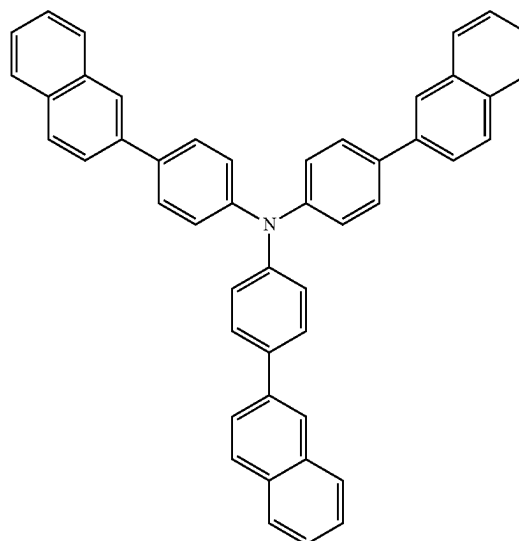
H1



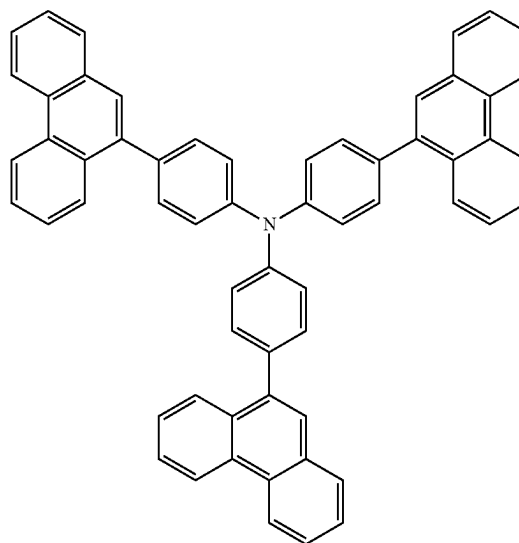
H2



H3



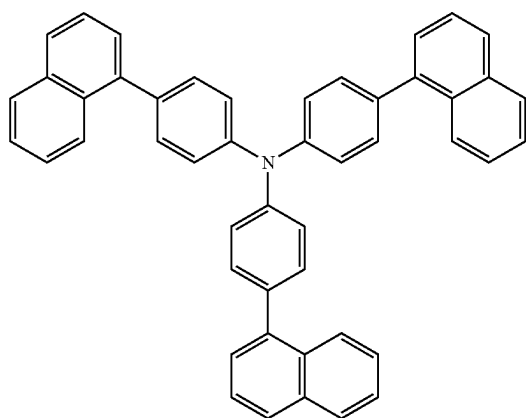
H4



H5

113

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H6

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H7

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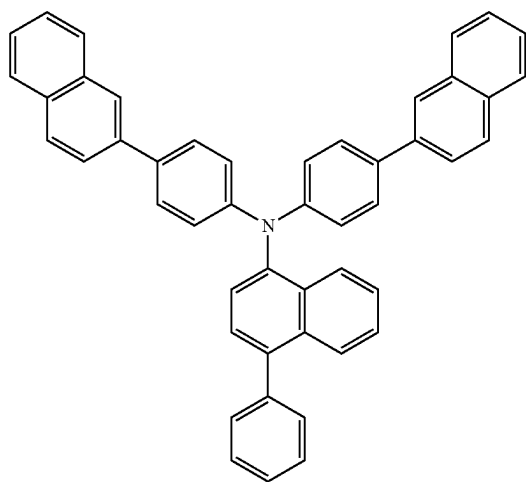
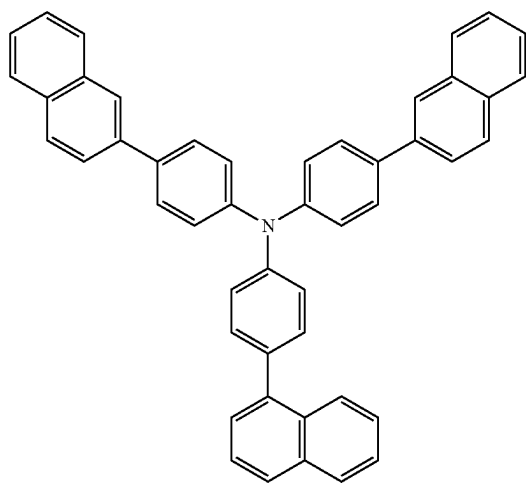
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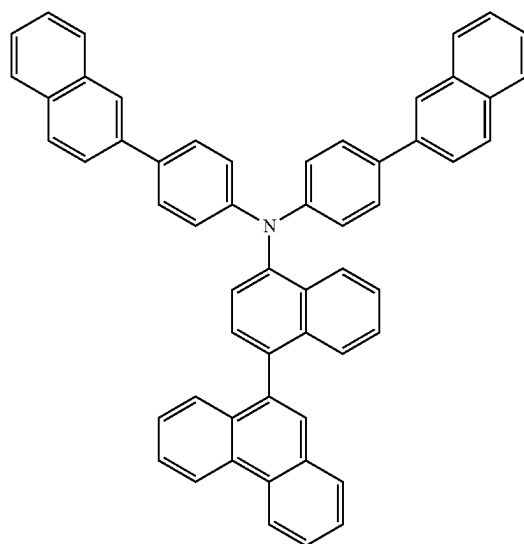
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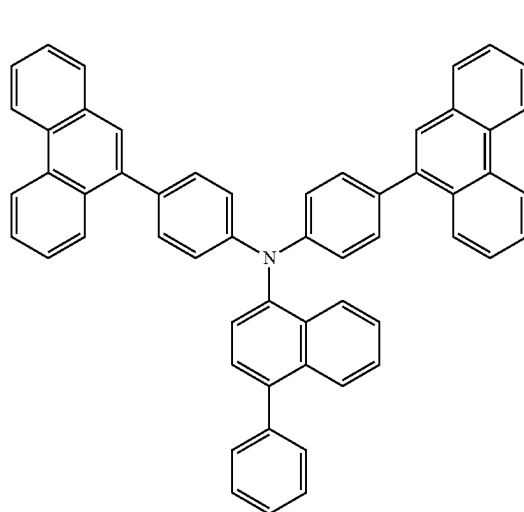
H8

114

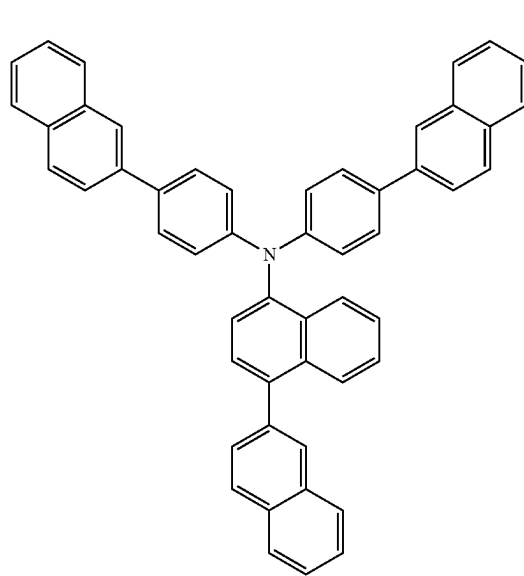
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H9



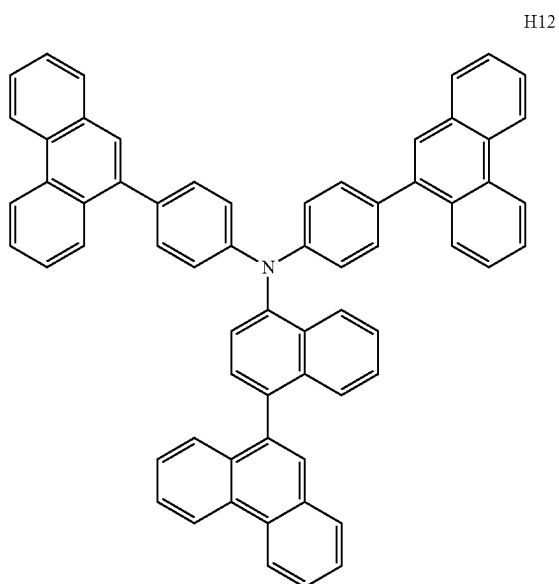
H10



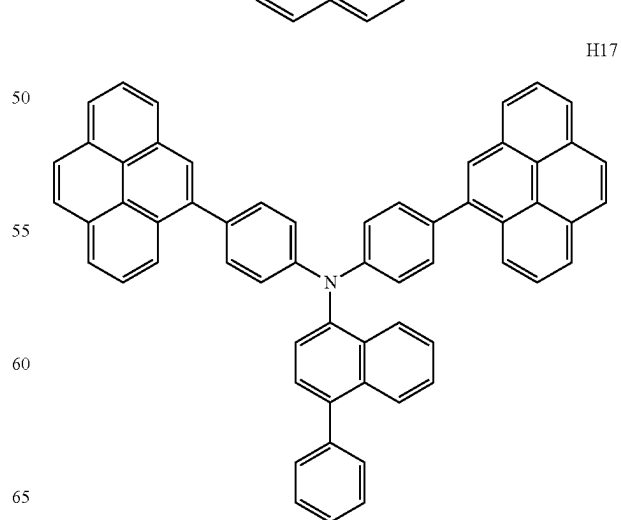
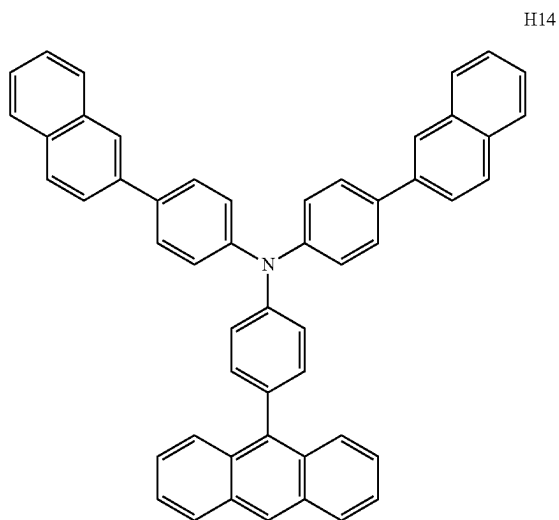
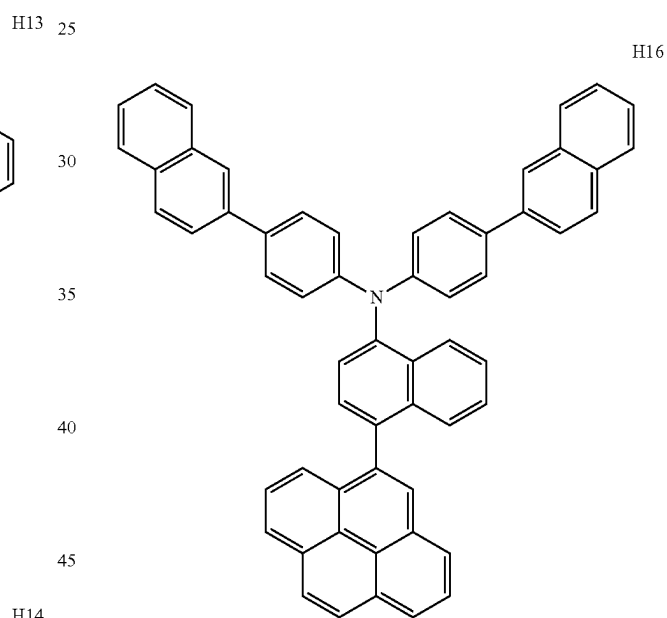
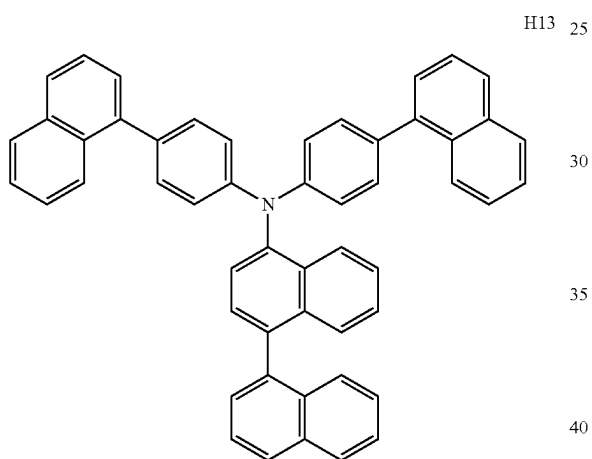
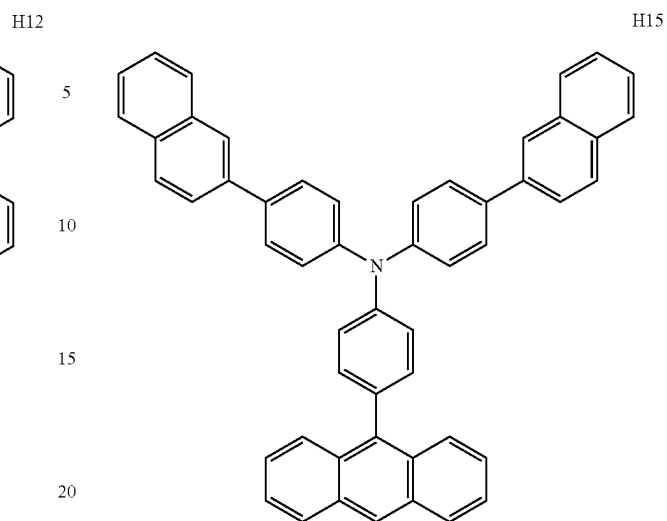
H11

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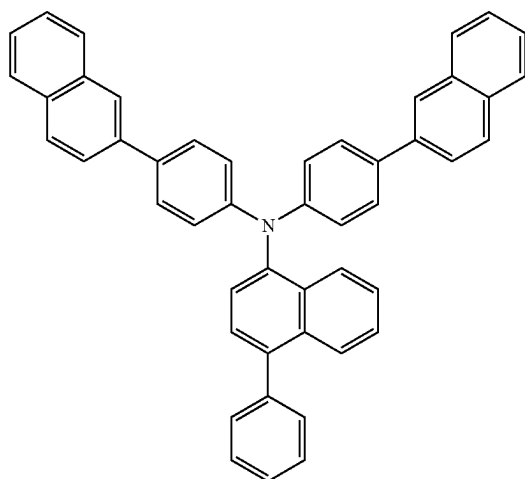
**116**

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117

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H18

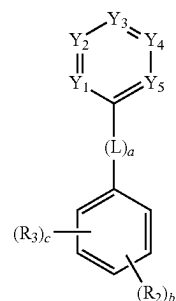
Formula 9

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118



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In Formula 9, each of Y_1 to Y_5 are independently CR_1 or N, and one to three of Y_1 to Y_5 is N. R_1 is independently hydrogen or $C_6\text{--}C_{30}$ aryl group. L is $C_6\text{--}C_{30}$ arylene group, and R_2 is $C_6\text{--}C_{30}$ aryl group or $C_5\text{--}C_{30}$ hetero aryl group. R_3 is hydrogen, or adjacent two of R_3 form a fused ring. "a" is 0 or 1, "b" is 1 or 2, and "c" is an integer of 0 to 4.

The hole blocking material of Formula 9 may be one of the followings of Formula 10:

H19

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Formula 10

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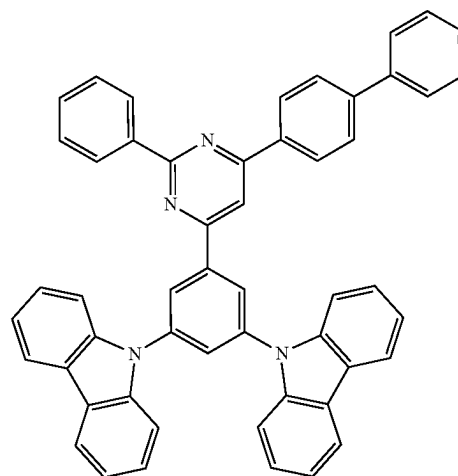
H20

50

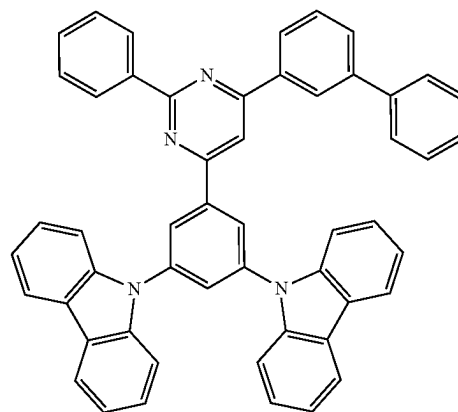
55

60

E1



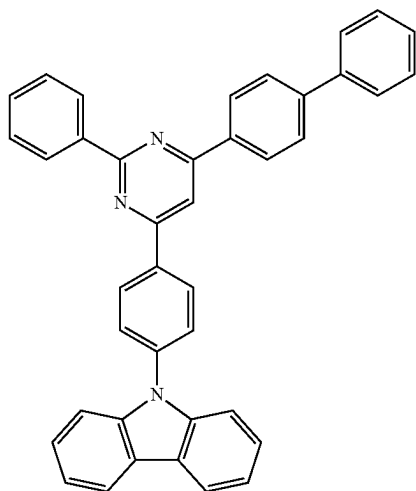
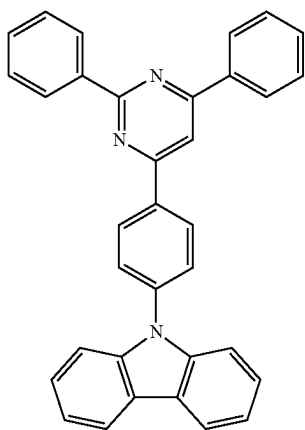
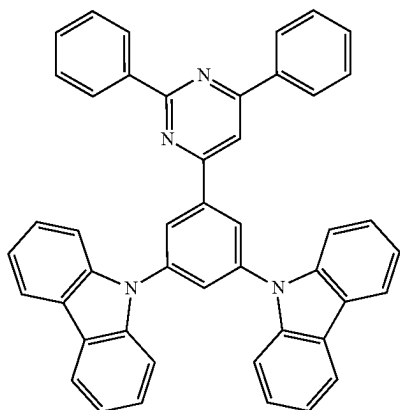
E2



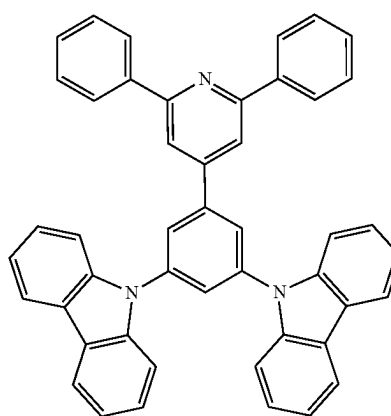
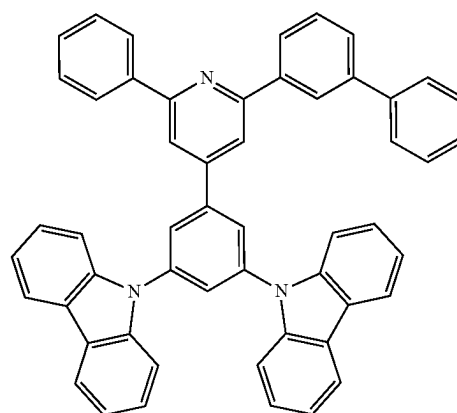
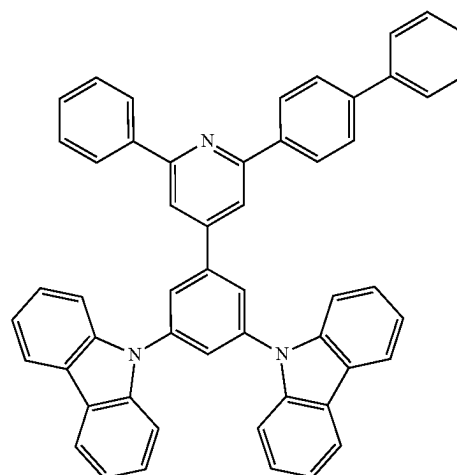
The HBL **250** may include an azine derivative as a hole blocking material. For example, the material of the HBL **250** may be represented by Formula 9:

119

-continued

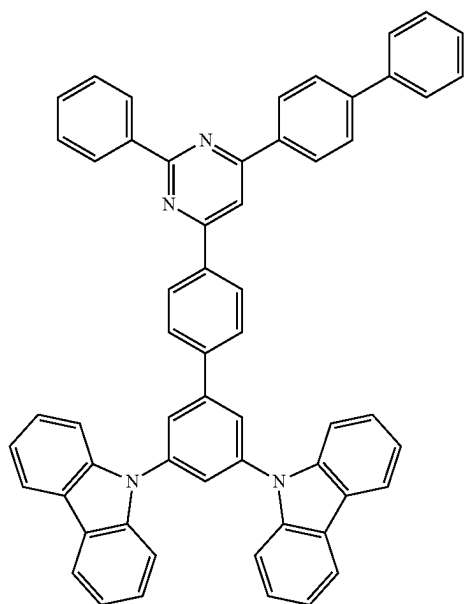
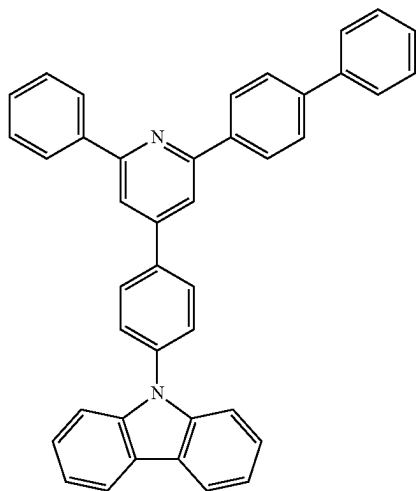
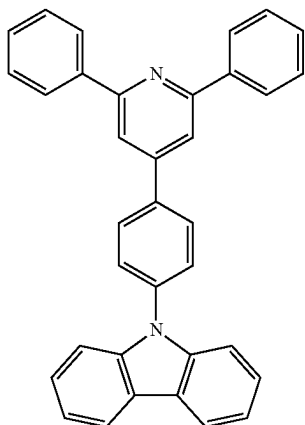
**120**

-continued

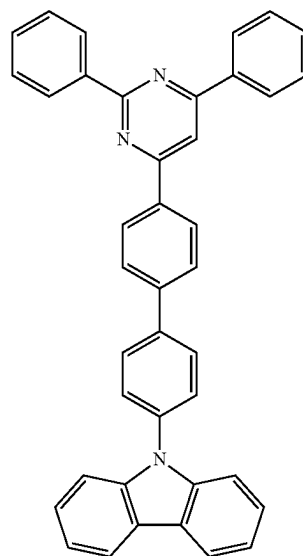
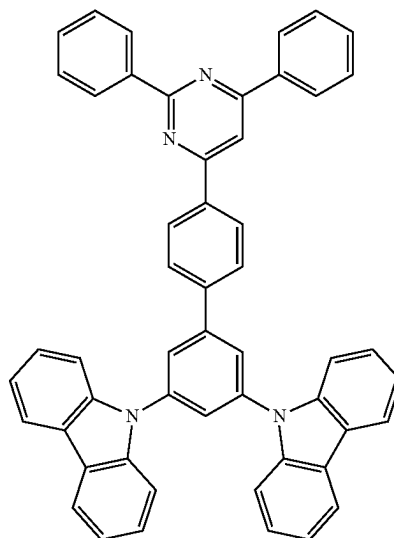


121

-continued

**122**

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E9

E12

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E10

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E11

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E13

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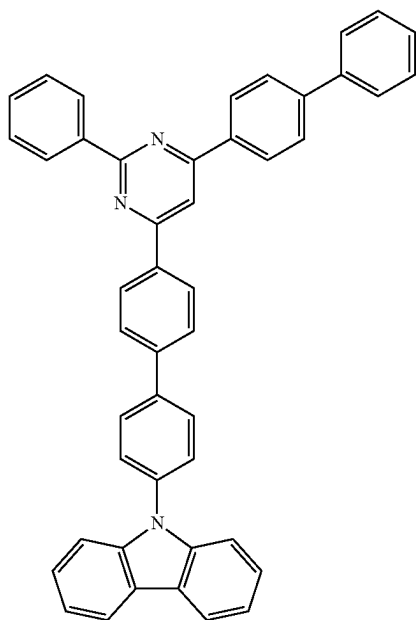
55

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65

123

-continued



E14

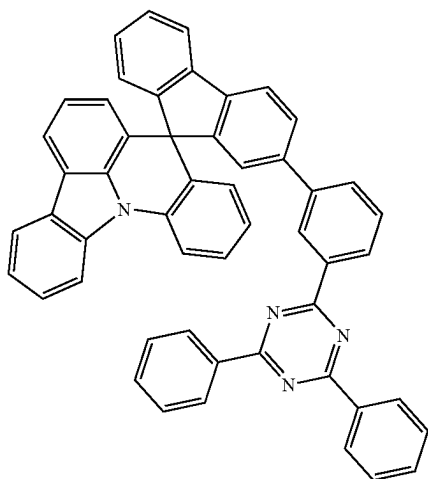
5

10

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20

25



E15

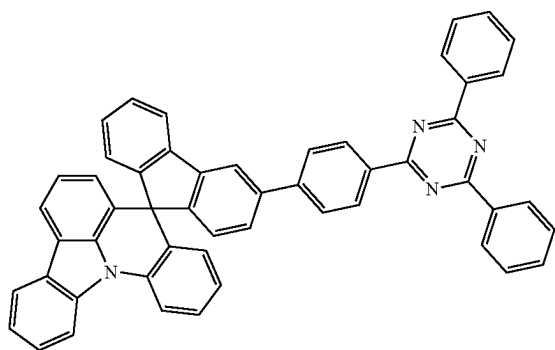
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E16

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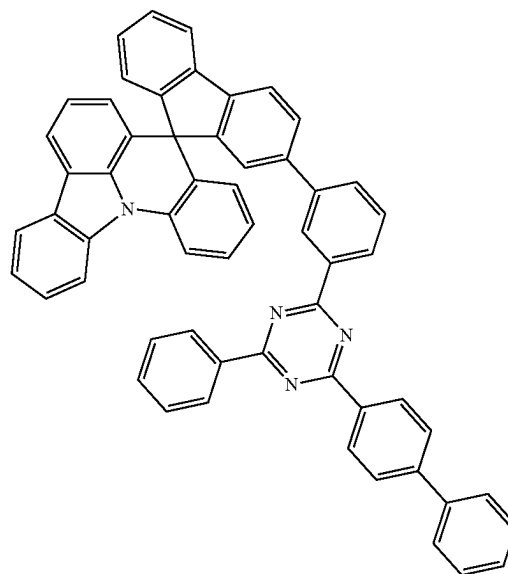
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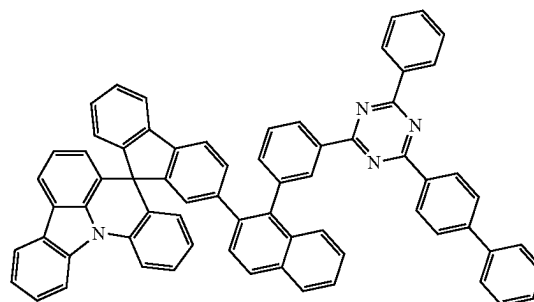
124

-continued

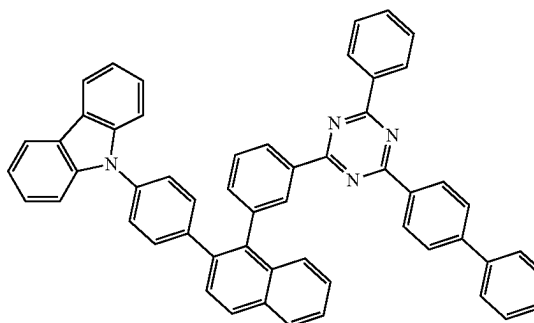
E17



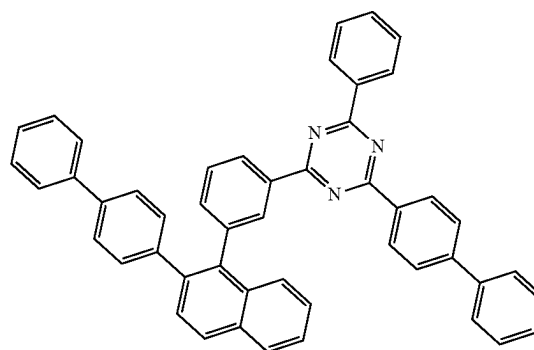
E18



E19



E20

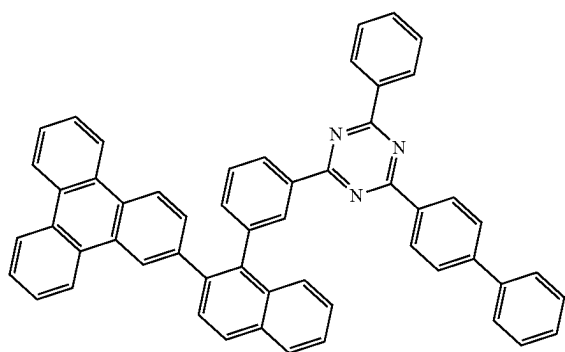


125

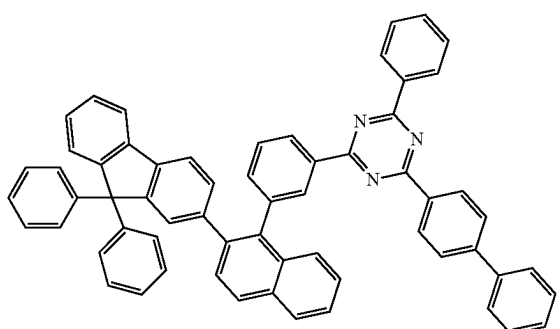
-continued

E21

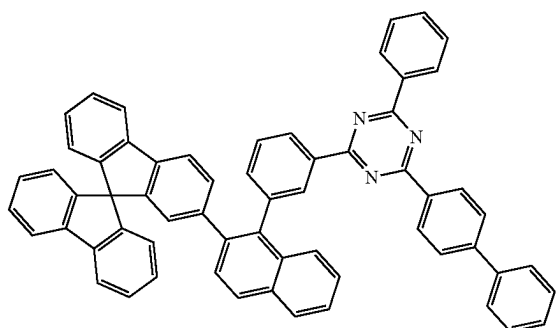
Formula 12



E22

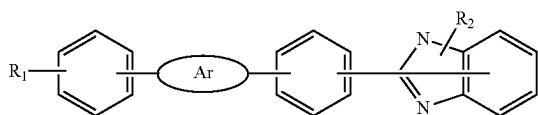


E23



Alternatively, the HBL **250** may include a benzimidazole derivative as a hole blocking material. For example, the material of the HBL **250** may be represented by Formula 11:

Formula 11



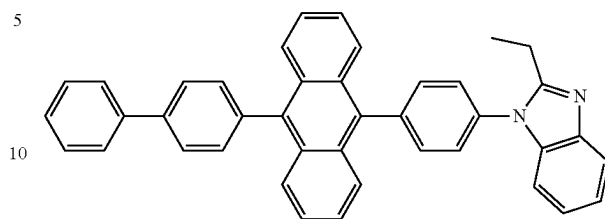
In Formula 11, Ar is C₁₀~C₃₀ arylene group, R₁ is C₆~C₃₀ aryl group or C₅~C₃₀ hetero aryl group, and R₂ is C₁~C₁₀ alkyl group or C₆~C₃₀ aryl group.

For example, Ar may be naphthylene or anthracenylene, R₁ may be benzimidazole or phenyl, and R₂ may be methyl, ethyl or phenyl.

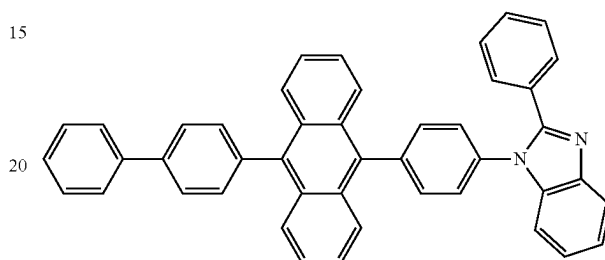
The hole blocking material of Formula 11 may be one of the followings of Formula 12:

126

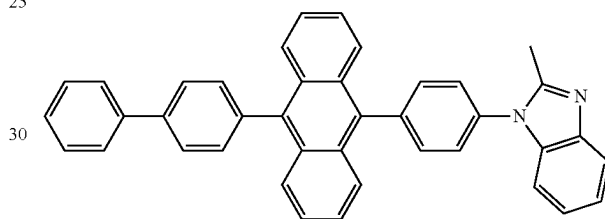
F1



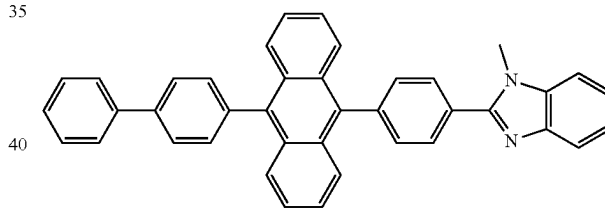
F2



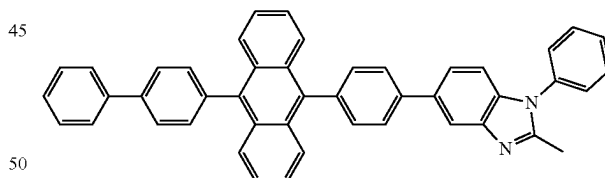
F3



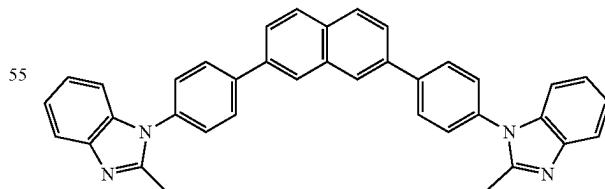
F1



F2



F6



The HBL **250** may include one of the hole blocking material of Formula 9 and the hole blocking material of Formula 11.

In this instance, a thickness of the EML **240** may be greater than each of a thickness of the EBL **230** and a thickness of the HBL **250** and may be smaller than a

127

thickness of the HTL **220**. For example, the EML may have a thickness of about 150 to 250 Å, and each of the EBL **230** and the HBL **250** may have a thickness of about 50 to 150 Å. The HTL **220** may have a thickness of about 900 to 1100 Å. The EBL **230** and the HBL **250** may have the same thickness.

The HBL **250** may include both the hole blocking material of Formula 9 and the hole blocking material of Formula 11. For example, in the HBL **250**, hole blocking material of Formula 9 and the hole blocking material of Formula 11 may have the same weight %.

In this instance, a thickness of the EML **240** may be greater than a thickness of the EBL **230** and may be smaller than a thickness of the HBL **250**. In addition, the thickness of HBL **250** may be smaller than a thickness of the HTL **220**. For example, the EML may have a thickness of about 200 to 300 Å, and the EBL **230** may have a thickness of about 50 to 150 Å. The HBL **250** may have a thickness of about 250 to 350 Å, and the HTL **220** may have a thickness of about 800 to 1000 Å.

The hole blocking material of Formula 9 and/or the hole blocking material of Formula 11 have an electron transporting property such that an electron transporting layer may be omitted. As a result, the HBL **250** directly contacts the EIL **260** or the second electrode **164** without the EIL **260**.

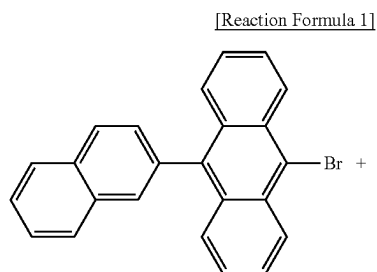
In the OLED D of the present disclosure, a weight % ratio of the first host **242** to the second host **244** may be about 1:9 to about 9:1, preferably about 1:9 to about 7:3. To provide sufficient emitting efficiency and lifespan of the OLED D and the organic light emitting display device, the weight % ratio of the first host **242** to the second host **244** may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host **242** to the second host **244** may be about 7:3. The OLED D and the organic light emitting display device of the present disclosure have advantages in the emitting efficiency and the lifespan.

In addition, when the EML **240** includes the blue dopant of Formula 5-1, an image with narrow full width at half maximum (FWHM) and high color purity is provided.

Moreover, the EBL **230** includes an amine derivative as an electron blocking material, and the HBL **250** includes at least one of an azine derivative and a benzimidazole derivative as a hole blocking material. Accordingly, the lifespan of the OLED D and an organic light emitting device **100** is further improved.

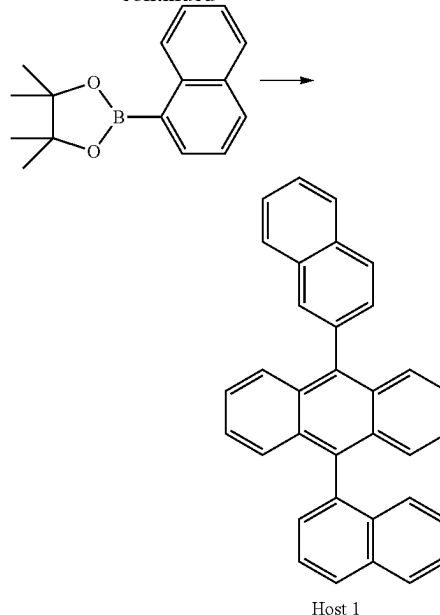
Synthesis of the First Host

1. Synthesis of the Compound Host1



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-continued

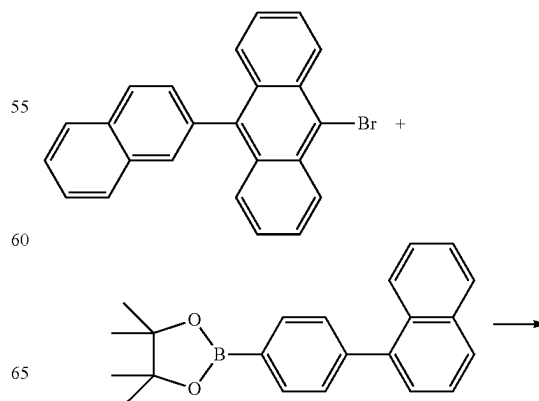


10-bromo-9-(naphthalen-3-yl)anthracene (2.00 g, 5.23 mmol), 4,4,5,5-tetramethyl-2-(naphthalen-1-yl)-1,3,2-dioxaborolane (1.45 g, 5.74 mmol), tris (dibenzylideneacetone) dipalladium(0) ($\text{Pd}_2(\text{dba})_3$) (0.24 g, 0.26 mmol) and toluene (50 mL) were added into the flask (250 mL) in the dry box. The reaction flask was removed from the dry box, and sodium carbonate anhydride (20 mL, 2M) was added into the flask. The reactants were stirred and heated at 90° C. overnight with monitoring the reaction by HPLC (high-performance liquid chromatography). The reaction flask was cooled down to the room temperature, and then an organic layer was separated from an aqueous layer. The aqueous layer was washed with dichloromethane (DCM), and the organic layer was concentrated with the rotary vaporizer to obtain a gray powder. The gray power was purified with alumina, precipitated with hexane and performed column chromatography using silica gel to obtain the compound Host1 of white powder (2.00 g, yield: 89%).

2. Synthesis of the Compound Host2

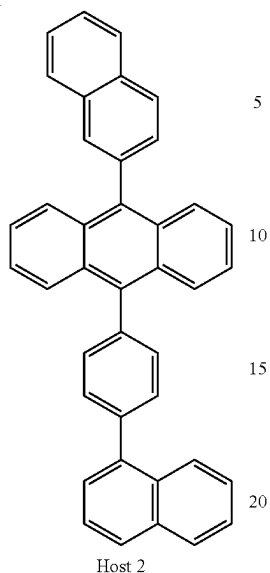
50

[Reaction Formula 2]



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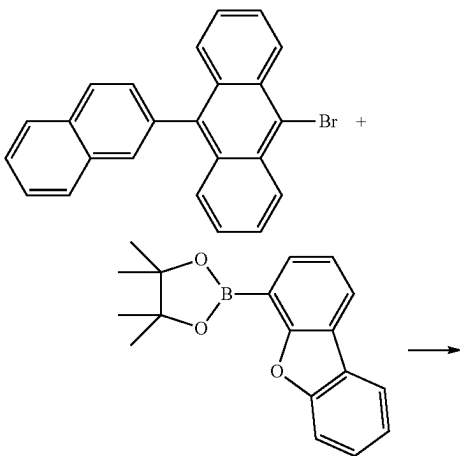
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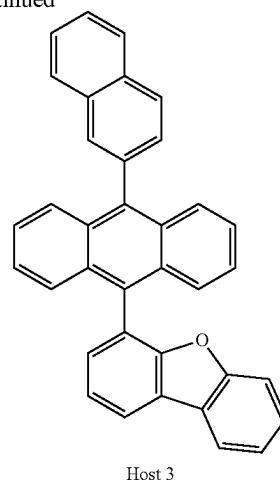
10-bromo-9-(naphthalene-3-yl)-anthracene (2.00 g, 5.23 mmol), 4,4,5,5-tetramethyl-2-(4-(naphthalen-4-yl)phenyl)-1,3,2-dioxaborolane (1.90 g, 5.74 mmol), $\text{Pd}_2(\text{dba})_3$ (0.24 g, 0.26 mmol) and toluene (50 mL) were added into the flask (250 mL) in the dry box. The reaction flask was removed from the dry box, and sodium carbonate anhydride (20 mL, 2M) was added into the flask. The reactants were stirred and heated at 90° C. overnight with monitoring the reaction by HPLC. The reaction flask was cooled down to the room temperature and then an organic layer was separated from an aqueous layer. The aqueous layer was washed with dichloromethane (DCM) twice and the organic layer was concentrated with the rotary vaporizer to obtain a gray powder. The gray power was purified with alumina, precipitated with hexane and performed column chromatography using silica gel to obtain the compound Host2 of white powder (2.28 g, yield: 86%).

3. Synthesis of the Compound Host3

[Reaction Formula 3]

**130**

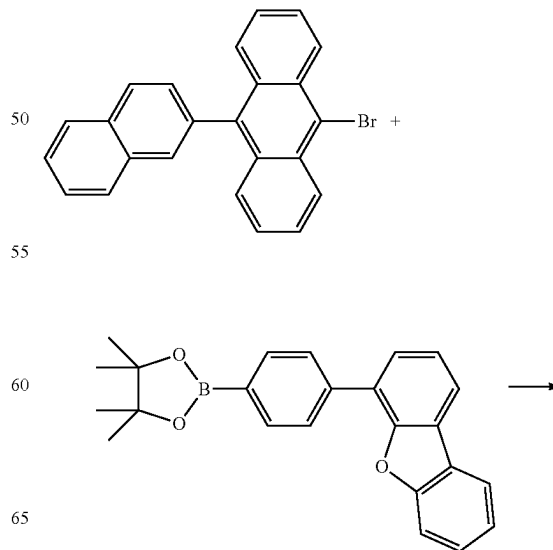
-continued



10-bromo-9-(naphthalene-3-yl)-anthracene (2.00 g, 5.23 mmol), 4,4,5,5-tetramethyl-2-(dibenzofuran-1-yl)-1,3,2-dioxaborolane (1.69 g, 5.74 mmol), $\text{Pd}_2(\text{dba})_3$ (0.24 g, 0.26 mmol) and toluene (50 mL) were added into the flask (250 mL) in the dry box. The reaction flask was removed from the dry box, and sodium carbonate anhydride (20 mL, 2M) was added into the flask. The reactants were stirred and heated at 90° C. overnight with monitoring the reaction by HPLC. The reaction flask was cooled down to the room temperature and then an organic layer was separated from an aqueous layer. The aqueous layer was washed with dichloromethane (DCM) twice and the organic layer was concentrated with the rotary vaporizer to obtain a gray powder. The gray power was purified with alumina, precipitated with hexane and performed column chromatography using silica gel to obtain the compound Host3 of white powder (1.91 g, yield: 78%).

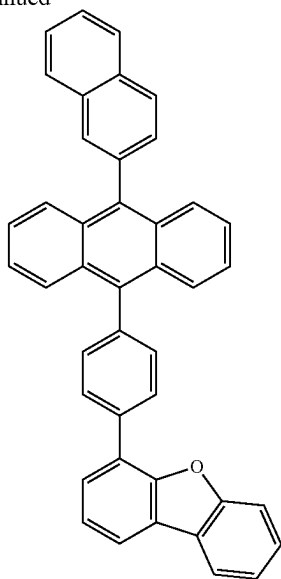
4. Synthesis of the Compound Host4

[Reaction Formula 4]



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-continued



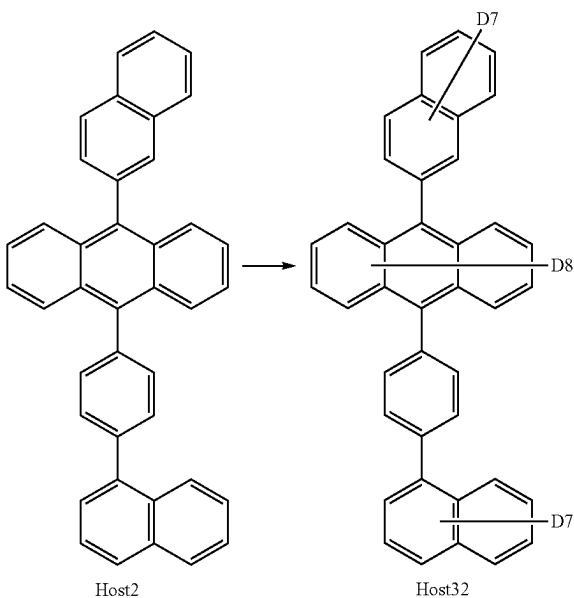
Host 4

10-bromo-9-(naphthalene-3-yl)-anthracene (2.00 g, 5.23 mmol), 4,4,5,5-tetramethyl-2-(dibenzofuran-1-yl)phenyl-1,3,2-dioxaborolane (2.12 g, 5.74 mmol), $\text{Pd}_2(\text{dba})_3$ (0.24 g, 0.26 mmol) and toluene (50 mL) were added into the flask (250 mL) in the dry box. The reaction flask was removed from the dry box, and sodium carbonate anhydride (20 mL, 2M) was added into the flask. The reactants were stirred and heated at 90° C. overnight with monitoring the reaction by HPLC. The reaction flask was cooled down to the room temperature and then an organic layer was separated from an aqueous layer. The aqueous layer was washed with dichloromethane (DCM) twice and the organic layer was concentrated with the rotary vaporizer to obtain a gray powder. The gray power was purified with alumina, precipitated with hexane and performed column chromatography using silica gel to obtain the compound Host4 of white powder (2.31 g, yield: 82%).

Synthesis of the Second Host

1. Synthesis of the Compound Host32

[Reaction Formula 5]



Host2

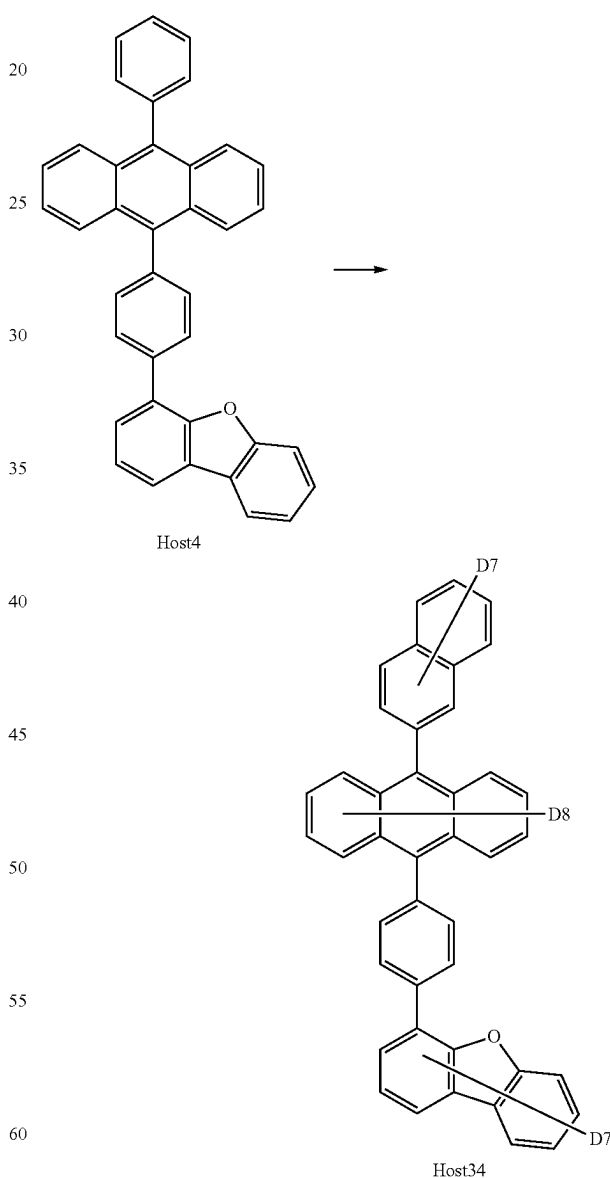
Host32

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Under N_2 condition, AlCl_3 (0.48 g, 3.6 mmol) was added to perdeuterobenzene solution (100 mL) in which the compound Host2 (5 g, 9.87 mmol) was dissolved. The mixture was stirred under the room temperature for 6 hrs, and D_2O (50 mL) was added. After separating the aqueous layer and the organic layer, the aqueous layer was washed with CH_2Cl_2 (30 mL). The obtained organic layer was dried using magnesium sulfate, and the volatiles were removed by rotary evaporation. The crude product was purified by column chromatography to obtain the compound Host32 (4.5 g) of white powder.

2. Synthesis of the Compound Host34

[Reaction Formula 6]



Host34

Under N_2 condition, AlCl_3 (0.48 g, 3.6 mmol) was added to perdeuterobenzene solution (100 mL) in which the compound Host4 (5 g, 9.15 mmol) was dissolved. The mixture was stirred under the room temperature for 6 hrs, and D_2O

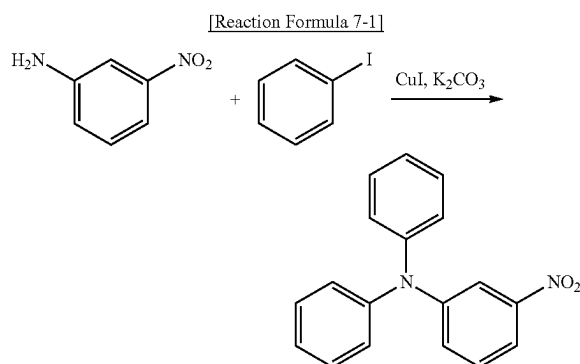
133

(50 mL) was added. After separating the aqueous layer and the organic layer, the aqueous layer was washed with CH_2Cl_2 (30 mL). The obtained organic layer was dried using magnesium sulfate, and the volatiles were removed by rotary evaporation. The crude product was purified by column chromatography to obtain the compound Host34 (4.8 g) of white powder.

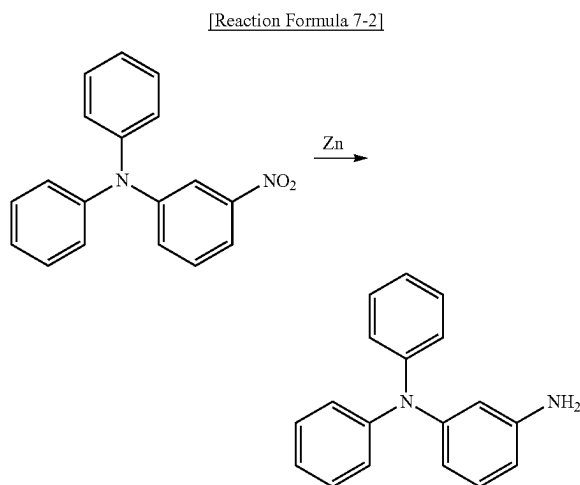
Synthesis of the Blue Dopant

1. Synthesis of the Compound Dopant56

(1) 3-nitro-N, N-diphenylaniline



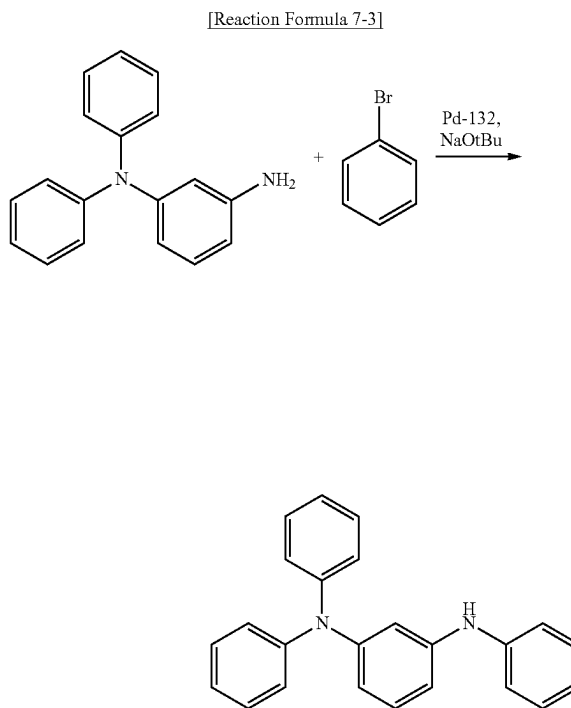
(2) N1,N1-diphenylbenzene-1,3-diamine



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Under N_2 condition, acetic acid cooled in the ice-bath was added and stirred. 3-nitro-N,N-diphenylaniline (44.0 g) was dropped into the solution to avoid significant increase the reaction temperature. After the addition was completed, the mixture was stirred at the room temperature for 30 minutes, and the disappearance of the starting materials was checked. After the reaction was completed, a supernatant liquid was collected by decantation, neutralized with sodium carbonate, and extracted with ethyl acetate. The resultant was purified by silica gel column chromatography (developing solution: toluene/heptane=9/1 (volume ratio)). After removing the solvent by distillation under reduced pressure, heptane was added and reprecipitated to obtain N1,N1-diphenylbenzene-1,3-diamine (36.0 g).

(3) N1,N1,N3-triphenylbenzene-1,3-diamine

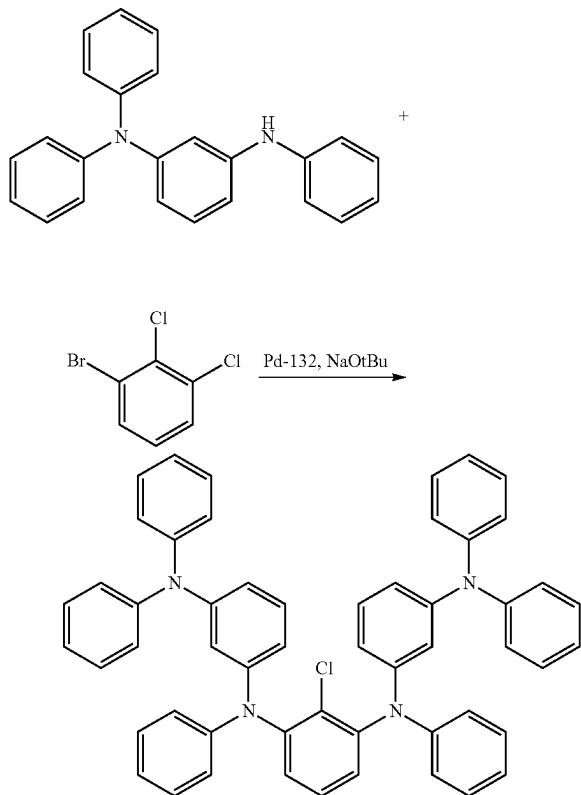


Under N_2 condition, the flask containing N1,N1-diphenylbenzene-1,3-diamine (60.0 g), Pd-132 (1.3 g), NaOtBu (33.5 g) and xylene (300 mL) was stirred by heating at 120°C . The solution of xylene (50 mL), in which bromobenzene (36.2 g) was dissolved, was slowly dropped to the solution, and the mixture was heated and stirred for 1 hour after completion of dropping. After cooling the mixture to the room temperature, water and ethyl acetate were added for liquid separation. The resultant was purified by silica gel column chromatography (developing solution: toluene/heptane=5/5 (volume ratio)) to obtain N1,N1,N3-triphenylbenzene-1,3-diamine (73.0 g).

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(4) N1,N1'-(2-chloro-1,3-phenylene)bis(N1,N3,N3-triphenylbenzene-1,3-diamine)

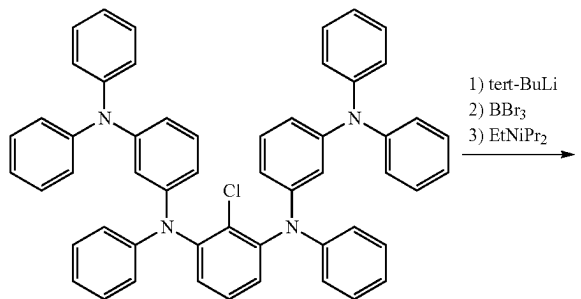
[Reaction Formula 7-4]



Under N₂ condition, the flask containing N1,N1,N3-triphenylbenzene-1,3-diamine (20.0 g), 1-bromo-2,3-dichlorobenzene (6.4 g), Pd-132 (0.2 g), NaOtBu (sodium tert-butoxide, 6.8 g) and xylene (70 mL) was heated and stirred at 120° C. for 2 hrs. After cooling the mixture to the room temperature, water and ethyl acetate were added for liquid separation. The resultant was purified by silica gel column chromatography (developing solution: toluene/heptane=4/6 (volume ratio)) to obtain N1,N1'-(2-chloro-1,3-phenylene)bis(N1,N3,N3-triphenylbenzene-1,3-diamine) (15.0 g).

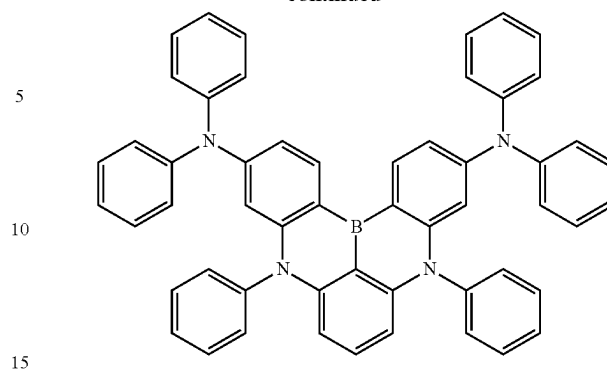
(5) Dopant56

[Reaction Formula 7-5]



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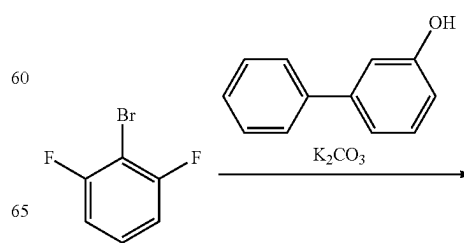


1.7 M tert-butyllithium pentane solution (18.1 ml) was added into the flask containing N1,N1'-(2-chloro-1,3-phenylene)bis(N1,N3-triphenylbenzene-1,3-diamine) (12.0 g) and tert-butylbenzene (100 mL) with cooling in the ice bath under N₂ condition. After heating up to 60° C. and stirring for 2 hrs, the component having boiling point lower than tert-butylbenzene was distilled off under reduced pressure. The mixture was cooled to -50° C., and boron tribromide (2.9 mL) was added. The mixture was heated up to the room temperature and stirred for 0.5 hour. The mixture was cooled again in the ice bath, and N,N-diisopropylethylamine (5.4 mL) was added. The mixture was stirred at the room temperature until the exotherm was finished. The mixture was heated to the temperature of 120° C. and stirred for 3 hrs. The solution was cooled to the room temperature, and an aqueous sodium acetate solution, which was cooled in the ice bath, and ethyl acetate was sequentially added to the solution. The insoluble solid was filtered and phase-separated. Then, the residue was purified by silica gel column chromatography (developing solution: toluene/heptane=5/5 (volume ratio)). The mixture was washed with heated heptane and ethyl acetate, and then reprecipitated with a mixed solvent of toluene and ethyl acetate to obtain Dopant56 (2.0 g).

2. Synthesis of the Compound Dopant167

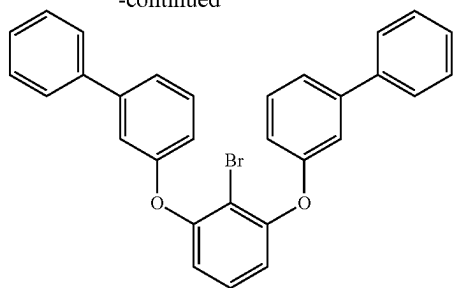
(1) 3,3''-((2-bromo-1,3-phenylene)bis(oxy)) di-1,1'-biphenyl

[Reaction Formula 8-1]



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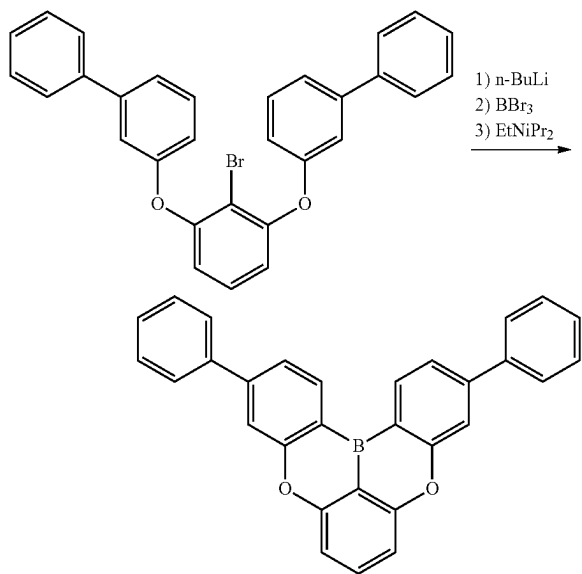
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Under N₂ condition, the flask containing 2-bromo-1,3-difluorobenzene (12.0 g), [1,1'-biphenyl]-3-ol (23.0 g), potassium carbonate (34.0 g) and NMP (130 mL) was heated and stirred at 170° C. for 10 hrs. After the reaction was stopped, the reaction solution was cooled to the room temperature, and water and toluene were added for liquid separation. The solvent was distilled off under reduced pressure, and the resultant was purified by silica gel column chromatography (developing solution: heptane/toluene=7/3 (volume ratio)) to obtain 3,3'-((2-bromo-1,3-phenylene)bis(oxy)) di-1,1'-biphenyl (26.8 g) was obtained.

(2) Dopant167

[Reaction Formula 8-2]



Under N₂ condition, the flask containing 3,3'-((2-bromo-1,3-phenylene)bis(oxy))di-1,1'-biphenyl (14.0 g) and xylene (100 mL) was cooled to -40° C., and n-butyllithium hexane solution (2.6 M, 11.5 mL) was dropped. After heating the mixture to the room temperature, the mixture was cooled again to -40° C., and boron tribromide (3.3 mL) was added thereto. The mixture was heated to the room temperature and stirred for 13 hrs. The mixture was cooled to 0° C., and N,N-diisopropylethylamine (9.7 mL) was added. The mixture was heated and stirred at 130° C. for 5 hrs. The reaction solution was cooled to the room temperature, and an aqueous sodium acetate solution cooled in the ice bath was added and stirred. The precipitated solid was collected by suction

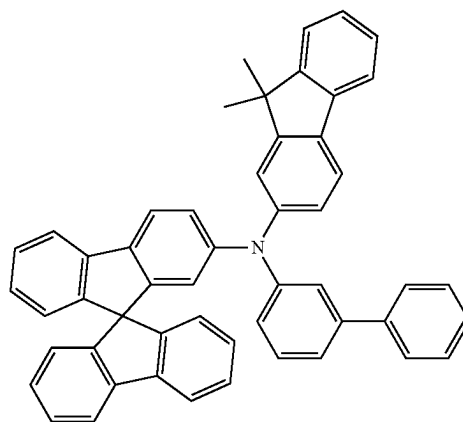
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filtration. The obtained solid was washed with water, methanol and heptane in that order and recrystallized from chlorobenzene to obtain the compound Dopant167 (8.9 g).

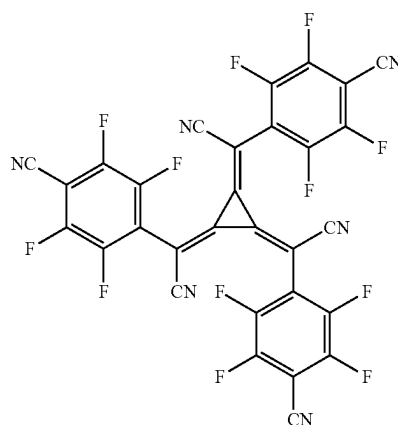
[Organic Light Emitting Diode]

The anode (ITO, 50 Å), the HIL (Formula 13 (97 wt %) and Formula 14 (3 wt %), 100 Å), the HTL (Formula 13, 1000 Å), the EBL (Formula 15, 100 Å), the EML (host (98 wt %) and dopant (2 wt %), 200 Å), the HBL (Formula 16, 100 Å), the EIL (Formula 17 (98 wt %) and Li (2 wt %), 200 Å) and the cathode (Al, 500 Å) was sequentially deposited to form the OLED.

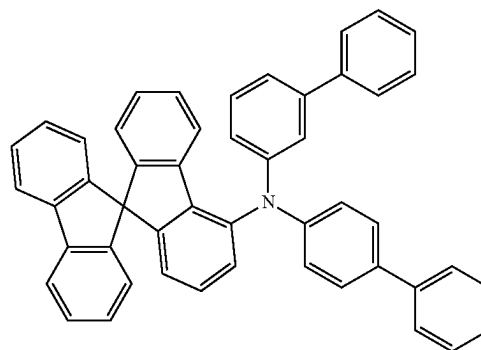
[Formula 13]



[Formula 14]



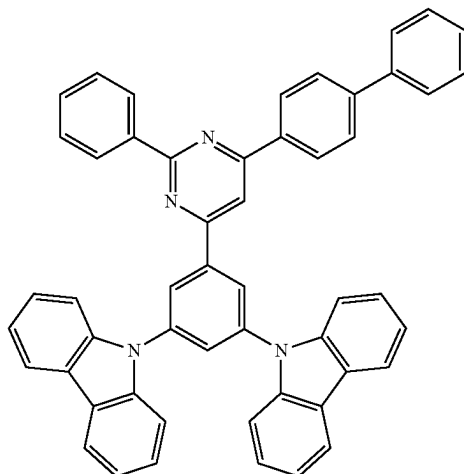
[Formula 15]



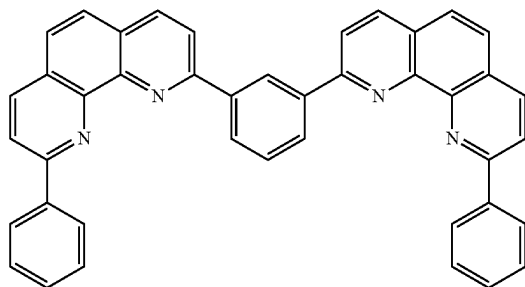
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-continued

[Formula 16]



[Formula 17]

**1. COMPARATIVE EXAMPLES****(1) Comparative Example 1 (Ref1)**

The compound "Dopant56" in Formula 6-1 is used as the dopant, and the compound "Host2" is used as the host.

(2) Comparative Example 2 (Ref2)

The compound "Dopant1" in Formula 6-2 is used instead of the compound "Dopant56" in Formula 6-1 of Comparative Example 1.

(3) Comparative Example 3 (Ref3)

The compound "Host4" is used instead of the compound "Host2" of Comparative Example 1.

(4) Comparative Example 4 (Ref4)

The compound "Host4" is used instead of the compound "Host2" of Comparative Example 2.

(5) Comparative Example 5 (Ref5)

The compound "Host32" is used instead of the compound "Host2" of Comparative Example 1.

(6) Comparative Example 6 (Ref6)

The compound "Host32" is used instead of the compound "Host2" of Comparative Example 2.

140**(7) Comparative Example 7 (Ref7)**

The compound "Host34" is used instead of the compound "Host2" of Comparative Example 1.

(8) Comparative Example 8 (Ref8)

The compound "Host34" is used instead of the compound "Host2" of Comparative Example 2.

2. EXAMPLES**(1) Example 1 (Ex1)**

The compound "Dopant56" in Formula 6-1 is used as the dopant, and the compound "Host2" and the compound "Host32" are used as the host. ("Host2": "Host32"=9:1 (wt % ratio))

(2) Example 2 (Ex2)

The weight % ratio of "Host2" to "Host32" is changed into 7:3 from Example 1. ("Host2": "Host32"=7:3 (wt % ratio))

(3) Example 3 (Ex3)

The weight % ratio of "Host2" to "Host32" is changed into 5:5 from Example 1. ("Host2": "Host32"=5:5 (wt % ratio))

(4) Example 4 (Ex4)

The weight % ratio of "Host2" to "Host32" is changed into 3:7 from Example 1. ("Host2": "Host32"=3:7 (wt % ratio))

(5) Example 5 (Ex5)

The weight % ratio of "Host2" to "Host32" is changed into 1:9 from Example 1. ("Host2": "Host32"=1:9 (wt % ratio))

(6) Examples 6 to 10 (Ex6 to Ex10)

The compound "Dopant1" in Formula 6-2 is used instead of the compound "Dopant56" in Examples 1 to 5.

(7) Example 11 (Ex11)

The compound "Dopant56" in Formula 6-1 is used as the dopant, and the compound "Host4" and the compound "Host34" are used as the host. ("Host4": "Host34"=9:1 (wt % ratio))

(8) Example 12 (Ex12)

The weight % ratio of "Host4" to "Host34" is changed into 7:3 from Example 11. ("Host4": "Host34"=7:3 (wt % ratio))

(9) Example 13 (Ex13)

The weight % ratio of "Host4" to "Host34" is changed into 5:5 from Example 11. ("Host4": "Host34"=5:5 (wt % ratio))

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(10) Example 14 (Ex14)

The weight % ratio of “Host4” to “Host34” is changed into 3:7 from Example 11. (“Host4”：“Host34”=3:7 (wt % ratio))

(11) Example 15 (Ex15)

The weight % ratio of “Host4” to “Host34” is changed into 1:9 from Example 11. (“Host4”：“Host34”=1:9 (wt % ratio))

(16) Examples 16 to 20 (Ex16 to Ex20)

The compound “Dopant1” in Formula 6-2 is used instead of the compound “Dopant56” in Examples 11 to 15.

The properties, i.e., voltage (V), efficiency (Cd/A), color coordinate (CIE), FWHM and lifespan (T95), of the OLEDs manufactured in Comparative Examples 1 to 8 and Examples 1 to 20 are measured and listed in Tables 1 to 4.

TABLE 1

	<u>Host ratio</u>		V	Cd/A	<u>CIE</u>		FWHM	T95
	1st	2nd			x	y		
	host	host					[nm]	[hr]
Ref1	10	0	3.77	4.78	0.1390	0.0612	26	153
Ex1	9	1	3.77	4.78	0.1390	0.0612	26	168
Ex2	7	3	3.77	4.78	0.1390	0.0611	26	184
Ex3	5	5	3.78	4.73	0.1391	0.0610	26	214
Ex4	3	7	3.78	4.68	0.1392	0.0610	26	275
Ex5	1	9	3.78	4.59	0.1393	0.0605	26	298
Ref5	0	10	3.78	4.55	0.1393	0.0605	26	302

TABLE 2

	Host ratio		V	Cd/A	CIE		FWHM	T95
	1st	2nd			x	y		
	host	host					[nm]	[hr]
Ref2	10	0	3.77	6.91	0.1379	0.0974	40	221
Ex6	9	1	3.77	6.91	0.1379	0.0974	40	243
Ex7	7	3	3.77	6.91	0.1379	0.0973	40	265
Ex8	5	5	3.78	6.84	0.1380	0.0972	40	309
Ex9	3	7	3.78	6.77	0.1381	0.0972	40	398
Ex10	1	9	3.78	6.63	0.1382	0.0965	40	431
Ref6	0	10	3.78	6.56	0.1382	0.0965	40	435

TABLE 3

	<u>Host ratio</u>			V	Cd/A	<u>CIE</u>		FWHM	T95
	1st	2nd	x			y			
	host	host					[nm]	[hr]	
Ref3	10	0	3.92	4.98	0.1378	0.0632	26	141	
Ex11	9	1	3.92	4.98	0.1378	0.0632	26	155	
Ex12	7	3	3.92	4.98	0.1378	0.0631	26	169	
Ex13	5	5	3.93	4.93	0.1379	0.0630	26	197	
Ex14	3	7	3.93	4.88	0.1380	0.0630	26	253	
Ex15	1	9	3.93	4.79	0.1381	0.0625	26	274	
Ref7	0	10	3.93	4.75	0.1381	0.0625	26	278	

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TABLE 4

	<u>Host ratio</u>		V	Cd/A	<u>CIE</u>		FWHM	T95
	1st	2nd			x	y		
	host	host			[nm]	[hr]		
Ref4	10	0	3.92	7.11	0.1367	0.0994	40	203
Ex16	9	1	3.92	7.11	0.1367	0.0994	40	224
Ex17	7	3	3.92	7.11	0.1367	0.0993	40	244
Ex18	5	5	3.93	7.04	0.1368	0.0992	40	284
Ex19	3	7	3.93	6.97	0.1369	0.0992	40	366
Ex20	1	9	3.93	6.83	0.1370	0.0985	40	397
Ref8	0	10	3.93	6.76	0.1370	0.0985	40	400

As shown in Tables 1 to 4, the lifespan of the OLED in Examples 1 to 20 using the first host, which is an anthracene derivative non-substituted with deuterium, and the second host, which is an anthracene derivative substituted with deuterium, is significantly increased in comparison to the OLED in Comparative Examples 1 to 4 using the first host without the second host.

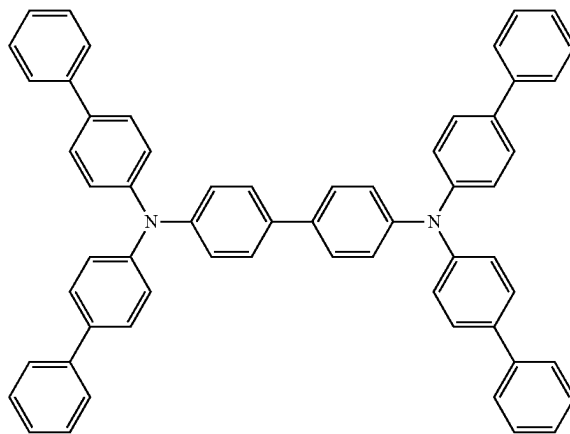
In addition, in comparison to the OLED in Comparative Examples 5 to 8 using the second host without the first host, the lifespan of the OLED in Examples 5, 10, 15 and 20 is slightly decreased, but the emitting efficiency of the OLED in Examples 5, 10, 15 and 20 is improved.

Accordingly, in the OLED of the present disclosure, a weight % ratio of the first host to the second host may be about 1:9 to 7:3. To provide sufficient emitting efficiency and lifespan of the OLED and the organic light emitting display device, the weight % ratio of the first host to the second host may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host to the second host may be about 7:3.

In the OLED D of the present disclosure, the EML 240 includes anthracene derivative as the first host 242 and deuterated anthracene derivative as the second host 244 such that the OLED D and the organic light emitting display device 100 has advantages in the emitting efficiency and the lifespan.

The anode (ITO, 50 Å), the HIL (Formula 18 (97 wt %) and Formula 14 (3 wt %), 100 Å), the HTL (Formula 18, 1000 Å), the EBL (100 Å), the EML (host (98 wt %) and dopant (2 wt %), 200 Å), the HBL (100 Å), the EIL (Formula 17 (98 wt %) and Li (2 wt %), 200 Å) and the cathode (Al, 500 Å) was sequentially deposited to form the OLED.

[Formula 18]



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1. COMPARATIVE EXAMPLES

(1) Comparative Example 9 (Ref9)

The compound of Formula 19 is used to form the EBL, the compound “Dopant56” in Formula 6-1 is used as the dopant, the compound “Host2” in Formula 2 is used as the host, and the compound “E1” in Formula 10 is used to form the HBL.

(2) Comparative Example 10 (Ref10)

The compound “H3” in Formula 8 is used instead of the compound of Formula 19 in Comparative Example 9.

(3) Comparative Example 11 (Ref11)

The compound “F1” in Formula 12 is used instead of the compound “E1” in Formula 10 in Comparative Example 9.

(4) Comparative Example 12 (Ref12)

The compound “F1” in Formula 12 is used instead of the compound “E1” in Formula 10 in Comparative Example 10.

(5) Comparative Examples 13 to 16 (Ref13 to Ref16)

The compound “Dopant167” in Formula 6-1 is used instead of the compound “Dopant56” in Comparative Examples 9 to 12.

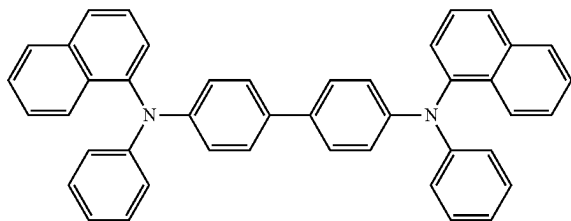
(6) Comparative Examples 17 to 20 (Ref17 to Ref20)

The compound “Host4” in Formula 2 is used instead of the compound “Host2” in Comparative Examples 9 to 12.

(7) Comparative Examples 21 to 24 (Ref21 to Ref24)

The compound “Host4” in Formula 2 is used instead of the compound “Host2” in Comparative Examples 13 to 16.

[Formula 19]



2. EXAMPLES

(1) Example 21 (Ex21)

The compound “H3” of Formula 8 is used to form the EBL, the compound “Dopant56” in Formula 6-1 is used as the dopant, the compound “Host2” and the compound “Host32” are used as the host, and the compound “E1” in Formula 10 is used to form the HBL. (“Host2”: “Host32”=3:7 (wt % ratio))

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(2) Example 22 (Ex22)

The compound “F1” in Formula 12 is used instead of the compound “E1” in Formula 10 in Comparative Example 21.

(3) Example 23 (Ex23)

The compound “Dopant167” in Formula 6-1 is used instead of the compound “Dopant56” in Comparative Example 21.

(4) Example 24 (Ex24)

The compound “Dopant167” in Formula 6-1 is used instead of the compound “Dopant56” in Comparative Example 22.

(5) Example 25 (Ex25)

The compound “H3” of Formula 8 is used to form the EBL, the compound “Dopant56” in Formula 6-1 is used as the dopant, the compound “Host4” and the compound “Host34” are used as the host, and the compound “E1” in Formula 10 is used to form the HBL. (“Host4”: “Host34”=3:7 (wt % ratio))

(6) Example 26 (Ex26)

The compound “F1” in Formula 12 is used instead of the compound “E1” in Formula 10 in Comparative Example 25.

(7) Example 27 (Ex27)

The compound “Dopant167” in Formula 6-1 is used instead of the compound “Dopant56” in Comparative Example 25.

(8) Example 28 (Ex28)

The compound “Dopant167” in Formula 6-1 is used instead of the compound “Dopant56” in Comparative Example 26.

The properties, i.e., voltage (V), efficiency (Cd/A), color coordinate (CIE), FWHM and lifespan (T95), of the OLEDs manufactured in Comparative Examples 9 to 24 and Examples 21 to 28 are measured and listed in Tables 5 and 6.

TABLE 5

	Host ratio		V	Cd/A	CIE		T95
	1st	2nd			x	y	
	host	host					[hr]
Ref9	10	0	3.77	4.70	0.1390	0.0612	45
Ref10	10	0	3.68	4.75	0.139	0.0614	232
Ex21	3	7	3.68	4.65	0.1410	0.0564	418
Ref11	10	0	3.47	5.00	0.1420	0.0562	20
Ref12	10	0	3.38	5.05	0.1420	0.0564	104
Ex22	3	7	3.38	4.95	0.1440	0.0514	188
Ref13	10	0	3.92	4.90	0.1378	0.1212	41
Ref14	10	0	3.83	4.95	0.1378	0.1214	213
Ex23	3	7	3.83	4.85	0.1398	0.1164	383
Ref15	10	0	3.62	5.20	0.1408	0.1162	18
Ref16	10	0	3.53	5.25	0.1408	0.1164	96
Ex24	3	7	3.53	5.15	0.1428	0.1114	172

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TABLE 6

	Host ratio						T95
	1st	2nd			CIE		
	host	host	V	Cd/A	x	y	
Ref17	10	0	3.94	4.62	0.1389	0.0609	41
Ref18	10	0	3.85	4.67	0.1389	0.0612	211
Ex25	3	7	3.85	4.57	0.1409	0.0562	380
Ref19	10	0	3.64	4.92	0.1419	0.0559	18
Ref20	10	0	3.55	4.97	0.1419	0.0562	95
Ex26	3	7	3.55	4.87	0.1439	0.0512	171
Ref21	10	0	4.09	4.82	0.1377	0.1209	38
Ref22	10	0	4.00	4.87	0.1377	0.1212	194
Ex27	3	7	4.00	4.77	0.1397	0.1162	349
Ref23	10	0	3.79	5.12	0.1407	0.1159	17
Ref24	10	0	3.70	5.17	0.1407	0.1162	87
Ex28	3	7	3.70	5.07	0.1427	0.1112	157

As shown in Tables 5 and 6, in comparison to the OLED in Comparative Examples 9, 11, 13, 15, 17, 19, 21 and 23, the properties of the driving voltage, the emitting efficiency and the lifespan of the OLED in Comparative Examples 10, 12, 14, 16, 18, 20, 22 and 24, where the EBL includes the amine derivative of Formula 7, and the HBL includes the azine derivative of Formula 9 or the benzimidazole derivative of Formula 11, are improved.

In addition, when the EML of the OLED includes the first host of Formula 1, which is an anthracene derivative, and the second host of Formula 3, which is an anthracene derivative substituted with deuterium, as Examples 25 to 28, the emitting efficiency and the lifespan are further improved.

[Organic Light Emitting Diode]

The anode (ITO, 50 Å), the HIL (Formula 18 (97 wt %) and Formula 14 (3 wt %), 100 Å), the HTL (Formula 18, 900 Å), the EBL (100 Å), the EML (host (98 wt %) and dopant (2 wt %), 250 Å), the HBL (300 Å), the EIL (LiF, 10 Å) and the cathode (Al, 500 Å) was sequentially deposited to form the OLED.

1. COMPARATIVE EXAMPLES

(1) Comparative Example 25 (Ref25)

The compound of Formula 19 is used to form the EBL, the compound "Dopant56" in Formula 6-1 is used as the dopant, the compound "Host2" in Formula 2 is used as the host, and the compound "E16" in Formula 10 and the compound "F1" in Formula 12 are used to form the HBL. ("E16": "F1"=1:1 (wt % ratio))

(2) Comparative Example 26 (Ref26)

The compound of Formula 20 is used instead of the compound of Formula 19 in Comparative Example 25.

(3) Comparative Example 27 (Ref27)

The compound "H3" in Formula 8 is used instead of the compound of Formula 19 in Comparative Example 25.

(4) Comparative Example 28 (Ref28)

The compound of Formula 19 is used to form the EBL, the compound "Dopant56" in Formula 6-1 is used as the dopant, the compound "Host4" in Formula 2 is used as the host, and

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the compound "E16" in Formula 10 and the compound "F1" in Formula 12 are used to form the HBL. ("E16": "F1"=1:1 (wt % ratio))

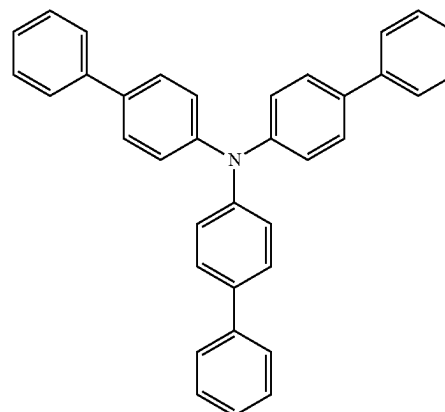
(5) Comparative Example 29 (Ref29)

The compound of Formula 20 is used instead of the compound of Formula 19 in Comparative Example 28.

(6) Comparative Example 30 (Ref30)

The compound "H3" in Formula 8 is used instead of the compound of Formula 19 in Comparative Example 28.

[Formula 20]



2. EXAMPLES

(1) Example 29

The compound "H3" of Formula 8 is used to form the EBL, the compound "Dopant56" in Formula 6-1 is used as the dopant, the compound "Host2" and the compound "Host32" are used as the host, and the compound "E16" in Formula 10 and the compound "F1" in Formula 12 are used to form the HBL. ("Host2": "Host32"=3:7 (wt % ratio), "E16": "F1"=1:1 (wt % ratio))

(2) Example 30

The compound "H3" of Formula 8 is used to form the EBL, the compound "Dopant56" in Formula 6-1 is used as the dopant, the compound "Host4" and the compound "Host34" are used as the host, and the compound "E16" in Formula 10 and the compound "F1" in Formula 12 are used to form the HBL. ("Host4": "Host34"=3:7 (wt % ratio), "E16": "F1"=1:1 (wt % ratio))

The properties, i.e., voltage (V), efficiency (Cd/A), color coordinate (CIE), FWHM and lifespan (T95), of the OLEDs manufactured in Comparative Examples 25 to 30 and Examples 29 and 30 are measured and listed in Table 7.

TABLE 7

	Host ratio				CIE		T95
	1st	2nd	V	Cd/A			
	host	host			x	y	
Ref25	10	0	3.67	4.94	0.1420	0.0600	44
Ref26	10	0	3.55	4.76	0.1362	0.0606	31
Ref27	10	0	3.58	4.99	0.1360	0.0602	227
Ex29	3	7	3.58	4.89	0.1380	0.0552	409
Ref28	10	0	3.84	4.86	0.1430	0.0630	40
Ref29	10	0	3.72	4.68	0.1352	0.0636	28
Ref30	10	0	3.75	4.91	0.1350	0.0632	207
Ex30	3	7	3.75	4.81	0.1370	0.0582	373

As shown in Table 7, in comparison to the OLED in Comparative Examples 25, 26, 28 and 29, the emitting efficiency of the OLED in Comparative Examples 27 and 30 is improved, and the lifespan of the OLED in Comparative Examples 27 and 30 is significantly increased.

In addition, when the EML of the OLED includes the first host of Formula 1, which is an anthracene derivative, and the second host of Formula 3, which is an anthracene derivative substituted with deuterium, as Examples 29 and 30, the emitting efficiency and the lifespan are further improved.

FIG. 4 is a schematic cross-sectional view illustrating an OLED having a tandem structure of two emitting units according to the first embodiment of the present disclosure.

As shown in FIG. 4, the OLED D includes the first and second electrodes 160 and 164 facing each other and the organic emitting layer 162 between the first and second electrodes 160 and 164. The organic emitting layer 162 includes a first emitting part 310 including a first EML 320, a second emitting part 330 including a second EML 340 and a charge generation layer (CGL) 350 between the first and second emitting parts 310 and 330.

The first electrode 160 may be formed of a conductive material having a relatively high work function to serve as an anode for injecting a hole into the organic emitting layer 162. The second electrode 164 may be formed of a conductive material having a relatively low work function to serve as a cathode for injecting an electron into the organic emitting layer 162.

The CGL 350 is positioned between the first and second emitting parts 310 and 330, and the first emitting part 310, the CGL 350 and the second emitting part 330 are sequentially stacked on the first electrode 160. Namely, the first emitting part 310 is positioned between the first electrode 160 and the CGL 350, and the second emitting part 320 is positioned between the second electrode 160 and the CGL 350.

The first emitting part 310 includes a first EML 320, a first EBL 316 between the first electrode 160 and the first EML 320 and a first HBL between the first EML 320 and the CGL 350.

In addition, the first emitting part 310 may further include a first HTL 314 between the first electrode 160 and the first EBL 316 and an HIL 312 between the first electrode 160 and the first HTL 314.

The first EML 320 includes a first host 322, which is an anthracene derivative, a second host 324, which is a deuterated anthracene derivative, and a blue dopant (not shown) such that blue light is provided from the first EML 320.

Namely, the first EML 320 may include the compound of Formula 1 as the first host 322, the compound of Formula 3 as the second host 324 and the compound of Formula 5-1 or Formula 5-2 as the blue dopant.

In the first EML 320, a weight % ratio of the first host 322 to the second host 324 may be about 1:9 to about 9:1, preferably about 1:9 to about 7:3. To provide sufficient emitting efficiency and lifespan of the OLED D and the organic light emitting display device, the weight % ratio of the first host 322 to the second host 324 may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host 322 to the second host 324 may be about 7:3.

The first EBL 316 may include an electron blocking material of Formula 7. The first HBL 318 may include at least one of a hole blocking material of Formula 9 and a hole blocking material of Formula 11.

The second emitting part 330 includes the second EML 340, a second EBL 334 between the CGL 350 and the second EML 340 and a second HBL 336 between the second EML 340 and the second electrode 164.

In addition, the second emitting part 330 may further include a second HTL 332 between the CGL 350 and the second EBL 334 and an EIL 338 between the second HBL 336 and the second electrode 164.

The second EML 340 includes a first host 342, which is an anthracene derivative, a second host 344, which is a deuterated anthracene derivative, and a blue dopant (not shown) such that blue light is provided from the second EML 340.

Namely, the second EML 340 may include the compound of Formula 1 as the first host 342, the compound of Formula 3 as the second host 344 and the compound of Formula 5-1 or Formula 5-2 as the blue dopant.

In the second EML 340, a weight % ratio of the first host 342 to the second host 344 may be about 1:9 to about 9:1, preferably about 1:9 to about 7:3. To provide sufficient emitting efficiency and lifespan of the OLED D and the organic light emitting display device, the weight % ratio of the first host 342 to the second host 344 may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host 342 to the second host 344 may be about 7:3.

The first host 342 of the second EML 340 may be same as or different from the first host 322 of the first EML 320, and the second host 344 of the second EML 340 may be same as or different from the second host 324 of the first EML 320. In addition, the blue dopant of the second EML 340 may be same as or different from the blue dopant of the first EML 320.

The second EBL 334 may include an electron blocking material of Formula 7. The second HBL 336 may include at least one of a hole blocking material of Formula 9 and a hole blocking material of Formula 11.

The CGL 350 is positioned between the first and second emitting parts 310 and 330. Namely, the first and second emitting parts 310 and 330 are connected through the CGL 350. The CGL 350 may be a P-N junction CGL of an N-type CGL 352 and a P-type CGL 354.

The N-type CGL 352 is positioned between the first HBL 318 and the second HTL 332, and the P-type CGL 354 is positioned between the N-type CGL 352 and the second HTL 332.

In the OLED D, since each of the first and second EMLs 320 and 340 includes the first host 322 and 342, each of which is an anthracene derivative, and the second host 324 and 344, each of which is a deuterated anthracene derivative, the OLED D and the organic light emitting display device 100 have advantages in the emitting efficiency and the lifespan.

In addition, at least one of the first and second EBLs **316** and **334** includes an amine derivative of Formula 7, and at least one of the first and second HBLs **318** and **336** includes at least one of a hole blocking material of Formula 9 and a hole blocking material of Formula 11. As a result, the lifespan of the OLED D and the organic light emitting display device **100** is further improved.

Moreover, since the first and second emitting parts **310** and **330** for emitting blue light are stacked, the organic light emitting display device **100** provides an image having high color temperature.

FIG. 5 is a schematic cross-sectional view illustrating an organic light emitting display device according to a second embodiment of the present disclosure, and FIG. 6 is a schematic cross-sectional view illustrating an OLED for the organic light emitting display device according to the second embodiment of the present disclosure.

As shown in FIG. 5, the organic light emitting display device **400** includes a first substrate **410**, where a red pixel BP, a green pixel GP and a blue pixel BP are defined, a second substrate **470** facing the first substrate **410**, an OLED D, which is positioned between the first and second substrates **410** and **470** and providing white emission, and a color filter layer **480** between the OLED D and the second substrate **470**.

Each of the first and second substrates **410** and **470** may be a glass substrate or a plastic substrate. For example, each of the first and second substrates **410** and **470** may be a polyimide substrate.

A buffer layer **420** is formed on the substrate, and the TFT Tr corresponding to each of the red, green and blue pixels RP, GP and BP is formed on the buffer layer **420**. The buffer layer **420** may be omitted.

A semiconductor layer **422** is formed on the buffer layer **420**. The semiconductor layer **422** may include an oxide semiconductor material or polycrystalline silicon.

A gate insulating layer **424** is formed on the semiconductor layer **422**. The gate insulating layer **424** may be formed of an inorganic insulating material such as silicon oxide or silicon nitride.

A gate electrode **430**, which is formed of a conductive material, e.g., metal, is formed on the gate insulating layer **424** to correspond to a center of the semiconductor layer **422**.

An interlayer insulating layer **432**, which is formed of an insulating material, is formed on the gate electrode **430**. The interlayer insulating layer **432** may be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride, or an organic insulating material, e.g., benzocyclobutene or photo-acryl.

The interlayer insulating layer **432** includes first and second contact holes **434** and **436** exposing both sides of the semiconductor layer **422**. The first and second contact holes **434** and **436** are positioned at both sides of the gate electrode **430** to be spaced apart from the gate electrode **430**.

A source electrode **440** and a drain electrode **442**, which are formed of a conductive material, e.g., metal, are formed on the interlayer insulating layer **432**.

The source electrode **440** and the drain electrode **442** are spaced apart from each other with respect to the gate electrode **430** and respectively contact both sides of the semiconductor layer **422** through the first and second contact holes **434** and **436**.

The semiconductor layer **422**, the gate electrode **430**, the source electrode **440** and the drain electrode **442** constitute

the TFT Tr. The TFT Tr serves as a driving element. Namely, the TFT Tr may correspond to the driving TFT Td (of FIG. 1).

Although not shown, the gate line and the data line cross each other to define the pixel, and the switching TFT is formed to be connected to the gate and data lines. The switching TFT is connected to the TFT Tr as the driving element.

In addition, the power line, which may be formed to be parallel to and spaced apart from one of the gate and data lines, and the storage capacitor for maintaining the voltage of the gate electrode of the TFT Tr in one frame may be further formed.

A passivation layer **450**, which includes a drain contact hole **452** exposing the drain electrode **442** of the TFT Tr, is formed to cover the TFT Tr.

A first electrode **460**, which is connected to the drain electrode **442** of the TFT Tr through the drain contact hole **452**, is separately formed in each pixel. The first electrode **460** may be an anode and may be formed of a conductive material having a relatively high work function. For example, the first electrode **460** may be formed of a transparent conductive material such as indium-tin-oxide (ITO) or indium-zinc-oxide (IZO).

A reflection electrode or a reflection layer may be formed under the first electrode **460**. For example, the reflection electrode or the reflection layer may be formed of aluminum-palladium-copper (APC) alloy.

A bank layer **466** is formed on the passivation layer **450** to cover an edge of the first electrode **460**. Namely, the bank layer **466** is positioned at a boundary of the pixel and exposes a center of the first electrode **460** in the red, green and blue pixels RP, GP and BP. The bank layer **466** may be omitted.

An organic emitting layer **462** is formed on the first electrode **460**.

Referring to FIG. 6, the organic emitting layer **462** includes a first emitting part **530** including a first EML **520**, a second emitting part **550** including a second EML **540**, a third emitting part **570** including a third EML **560**, a first CGL **580** between the first and second emitting parts **530** and **550** and a second CGL **590** between the second and third emitting parts **550** and **570**.

The first electrode **460** may be formed of a conductive material having a relatively high work function to serve as an anode for injecting a hole into the organic emitting layer **462**. The second electrode **464** may be formed of a conductive material having a relatively low work function to serve as a cathode for injecting an electron into the organic emitting layer **462**.

The first CGL **580** is positioned between the first and second emitting parts **530** and **550**, and the second CGL **590** is positioned between the second and third emitting parts **550** and **570**. Namely, the first emitting part **530**, the first CGL **580**, the second emitting part **550**, the second CGL **590** and the third emitting part **570** are sequentially stacked on the first electrode **460**. In other words, the first emitting part **530** is positioned between the first electrode **460** and the first CGL **580**, the second emitting part **550** is positioned between the first and second CGLs **580** and **590**, and the third emitting part **570** is positioned between the second electrode **460** and the second CGL **590**.

The first emitting part **530** may include an HIL **532**, a first HTL **534**, a first EBL **536**, the first EML **520** and a first HBL **538** sequentially stacked on the first electrode **460**. Namely, the HIL **532**, the first HTL **534** and the first EBL **536** are positioned between the first electrode **460** and the first EML

520, and the first HBL **538** is positioned between the first EML **520** and the first CGL **580**.

The first EML **520** includes a first host **522**, which is an anthracene derivative, a second host **524**, which is a deuterated anthracene derivative, and a blue dopant (not shown) such that blue light is provided from the first EML **520**.

Namely, the first EML **520** may include the compound of Formula 1 as the first host **522**, the compound of Formula 3 as the second host **524** and the compound of Formula 5-1 or Formula 5-2 as the blue dopant.

In the first EML **520**, a weight % ratio of the first host **522** to the second host **524** may be about 1:9 to about 9:1, preferably about 1:9 to about 7:3. To provide sufficient emitting efficiency and lifespan of the OLED D and the organic light emitting display device, the weight % ratio of the first host **522** to the second host **524** may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host **522** to the second host **524** may be about 7:3.

The first EBL **536** may include an electron blocking material of Formula 7. The first HBL **538** may include at least one of a hole blocking material of Formula 9 and a hole blocking material of Formula 11.

The second EML **550** may include a second HTL **552**, the second EML **540** and an electron transporting layer (ETL) **554**. The second HTL **552** is positioned between the first CGL **580** and the second EML **540**, and the ETL **554** is positioned between the second EML **540** and the second CGL **590**.

The second EML **540** may be a yellow-green EML. For example, the second EML **540** may include a host and a yellow-green dopant. Alternatively, the second EML **540** may include a host, a red dopant and a green dopant. In this instance, the second EML **540** may include a lower layer including the host and the red dopant (or the green dopant) and an upper layer including the host and the green dopant (or the red dopant).

The third emitting part **570** may include a third HTL **572**, a second EBL **574**, the third EML **560**, a second HBL **576** and an EIL **578**. The third EML **560** (or the third emitting part **570**) includes a first host **562**, which is an anthracene derivative, a second host **564**, which is a deuterated anthracene derivative, and a blue dopant (not shown) such that blue light is provided from the third EML **560**. Namely, the third EML **560** may include the compound of Formula 1 as the first host **562**, the compound of Formula 3 as the second host **564** and the compound of Formula 5-1 or Formula 5-2 as the blue dopant.

In the third EML **560**, a weight % ratio of the first host **562** to the second host **564** may be about 1:9 to about 9:1, preferably about 1:9 to about 7:3. To provide sufficient emitting efficiency and lifespan of the OLED D and the organic light emitting display device, the weight % ratio of the first host **562** to the second host **564** may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host **562** to the second host **564** may be about 7:3.

The first host **562** of the third EML **560** may be same as or different from the first host **522** of the first EML **520**, and the second host **564** of the third EML **560** may be same as or different from the second host **524** of the first EML **520**. In addition, the blue dopant of the third EML **560** may be same as or different from the blue dopant of the first EML **520**.

The second EBL **574** may include an electron blocking material of Formula 7. The second HBL **576** may include at least one of a hole blocking material of Formula 9 and a hole

blocking material of Formula 11. The electron blocking material of the second EBL **574** may be same as or different from the electron blocking material of the first EBL **536**, and the hole blocking material of the second HBL **576** may be same as or different from the hole blocking material of the first HBL **538**.

The first CGL **580** is positioned between the first emitting part **530** and the second emitting part **550**, and the second CGL **590** is positioned between the second emitting part **550** and the third emitting part **570**. Namely, the first and second emitting stacks **530** and **550** are connected through the first CGL **580**, and the second and third emitting stacks **550** and **570** are connected through the second CGL **590**. The first CGL **580** may be a P-N junction CGL of a first N-type CGL **582** and a first P-type CGL **584**, and the second CGL **590** may be a P-N junction CGL of a second N-type CGL **592** and a second P-type CGL **594**.

In the first CGL **580**, the first N-type CGL **582** is positioned between the first HBL **538** and the second HTL **552**, and the first P-type CGL **584** is positioned between the first N-type CGL **582** and the second HTL **552**.

In the second CGL **590**, the second N-type CGL **592** is positioned between the ETL **554** and the third HTL **572**, and the second P-type CGL **594** is positioned between the second N-type CGL **592** and the third HTL **572**.

In the OLED D, since each of the first and third EMLs **520** and **560** includes the first host **522** and **562**, each of which is an anthracene derivative, the second host **524** and **564**, each of which is a deuterated anthracene derivative, and the blue dopant.

Accordingly, the OLED D including the first and third emitting parts **530** and **570** with the second emitting part **550**, which emits yellow-green light or red/green light, can emit white light.

In FIG. 6, the OLED D has a triple-stack structure of the first, second and third emitting parts **530**, **550** and **570**. Alternatively, the OLED D may have a double-stack structure without the first emitting part **530** and the third emitting part **570**.

Referring to FIG. 5 again, a second electrode **464** is formed over the substrate **110** where the organic emitting layer **162** is formed.

In the organic light emitting display device **400**, since the light emitted from the organic emitting layer **462** is incident to the color filter layer **480** through the second electrode **464**, the second electrode **464** has a thin profile for transmitting the light.

The first electrode **460**, the organic emitting layer **462** and the second electrode **464** constitute the OLED D.

The color filter layer **480** is positioned over the OLED D and includes a red color filter **482**, a green color filter **484** and a blue color filter **486** respectively corresponding to the red, green and blue pixels RP, GP and BP.

Although not shown, the color filter layer **480** may be attached to the OLED D by using an adhesive layer. Alternatively, the color filter layer **480** may be formed directly on the OLED D.

An encapsulation film (not shown) may be formed to prevent penetration of moisture into the OLED D. For example, the encapsulation film may include a first inorganic insulating layer, an organic insulating layer and a second inorganic insulating layer sequentially stacked, but it is not limited thereto. The encapsulation film may be omitted.

A polarization plate (not shown) for reducing an ambient light reflection may be disposed over the top-emission type OLED D. For example, the polarization plate may be a circular polarization plate.

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In FIG. 5, the light from the OLED D passes through the second electrode **464**, and the color filter layer **480** is disposed on or over the OLED D. Alternatively, when the light from the OLED D passes through the first electrode **460**, the color filter layer **480** may be disposed between the OLED D and the first substrate **410**.

A color conversion layer (not shown) may be formed between the OLED D and the color filter layer **480**. The color conversion layer may include a red color conversion layer, a green color conversion layer and a blue color conversion layer respectively corresponding to the red, green and blue pixels RP, GP and BP. The white light from the OLED D is converted into the red light, the green light and the blue light by the red, green and blue color conversion layer, respectively.

As described above, the white light from the organic light emitting diode D passes through the red color filter **482**, the green color filter **484** and the blue color filter **486** in the red pixel RP, the green pixel GP and the blue pixel BP such that the red light, the green light and the blue light are provided from the red pixel RP, the green pixel GP and the blue pixel BP, respectively.

In FIGS. 5 and 6, the OLED D emitting the white light is used for a display device. Alternatively, the OLED D may be formed on an entire surface of a substrate without at least one of the driving element and the color filter layer to be used for a lightening device. The display device and the lightening device each including the OLED D of the present disclosure may be referred to as an organic light emitting device.

FIG. 7 is a schematic cross-sectional view illustrating an organic light emitting display device according to a third embodiment of the present disclosure.

As shown in FIG. 7, the organic light emitting display device **600** includes a first substrate **610**, where a red pixel BP, a green pixel GP and a blue pixel BP are defined, a second substrate **670** facing the first substrate **610**, an OLED D, which is positioned between the first and second substrates **610** and **670** and providing white emission, and a color conversion layer **680** between the OLED D and the second substrate **470**.

Although not shown, a color filter may be formed between the second substrate **670** and each color conversion layer **680**.

A TFT Tr, which corresponding to each of the red, green and blue pixels RP, GP and BP, is formed on the first substrate **610**, and a passivation layer **650**, which has a drain contact hole **652** exposing an electrode, e.g., a drain electrode, of the TFT Tr is formed to cover the TFT Tr.

The OLED D including a first electrode **660**, an organic emitting layer **662** and a second electrode **664** is formed on the passivation layer **650**. In this instance, the first electrode **660** may be connected to the drain electrode of the TFT Tr through the drain contact hole **652**.

An bank layer **666** covering an edge of the first electrode **660** is formed at a boundary of the red, green and blue pixel regions RP, GP and BP.

The OLED D emits a blue light and may have a structure shown in FIG. 3 or FIG. 4. Namely, the OLED D is formed in each of the red, green and blue pixels RP, GP and BP and provides the blue light.

The color conversion layer **680** includes a first color conversion layer **682** corresponding to the red pixel RP and a second color conversion layer **684** corresponding to the green pixel GP. For example, the color conversion layer **680** may include an inorganic color conversion material such as a quantum dot.

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The blue light from the OLED D is converted into the red light by the first color conversion layer **682** in the red pixel RP, and the blue light from the OLED D is converted into the green light by the second color conversion layer **684** in the green pixel GP.

Accordingly, the organic light emitting display device **600** can display a full-color image.

On the other hand, when the light from the OLED D passes through the first substrate **610**, the color conversion layer **680** is disposed between the OLED D and the first substrate **610**.

While the present disclosure has been described with reference to exemplary embodiments and examples, these embodiments and examples are not intended to limit the scope of the present disclosure. Rather, it will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure without departing from the spirit or scope of the invention. Thus, it is intended that the present disclosure cover the modifications and variations of the present disclosure provided they come within the scope of the appended claims and their equivalents.

The various embodiments described above can be combined to provide further embodiments. All of patents, patent application publications, patent applications, foreign patents, foreign patent applications and non-patent publications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

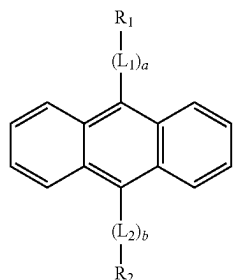
These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

The invention claimed is:

1. An organic light emitting diode (OLED), comprising:
 - a first electrode;
 - a second electrode facing the first electrode;
 - a first emitting material layer including a first host, a second host and a blue dopant and positioned between the first and second electrodes;
 - a first electron blocking layer including an electron blocking material of an amine derivative and positioned between the first electrode and the first emitting material layer; and
 - a first hole blocking layer including at least one of a first hole blocking material and a second hole blocking material and positioned between the second electrode and the first emitting material layer,
 wherein the first host is a non-deuterated anthracene derivative, and the second host is a deuterated anthracene derivative,
 wherein a weight % ratio of the first host to the second host is 3:7 to 9:1,
 wherein the first hole blocking material is an azine derivative, and the second hole blocking material is a benzimidazole derivative, and
 wherein the second host has only difference of substitution of deuterium atom from the first host.
2. The OLED of claim 1, wherein the first host is represented by Formula 1:

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Formula 1

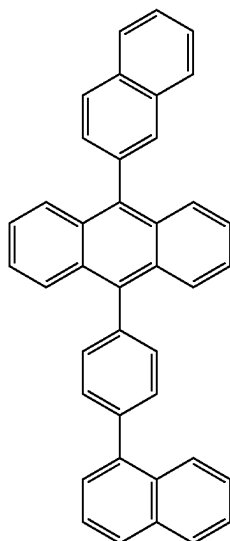
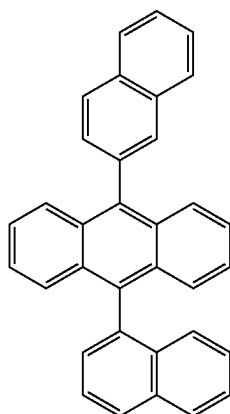


wherein each of R_1 and R_2 is independently $C_6\sim C_{30}$ aryl group or $C_5\sim C_{30}$ heteroaryl group, and each of L_1 and L_2 is independently $C_6\sim C_{30}$ arylene group, and

wherein each of a and b is an integer of 0 or 1, and at least one of a and b is 0.

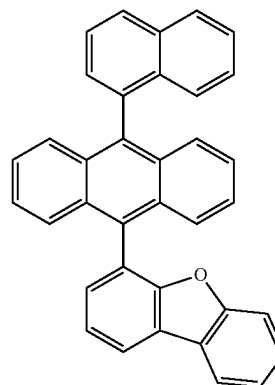
3. The OLED of claim 2, wherein the first host is a compound being one of the followings of Formula 2:

Formula 2

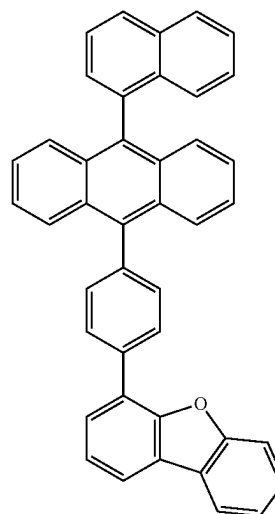
**156**

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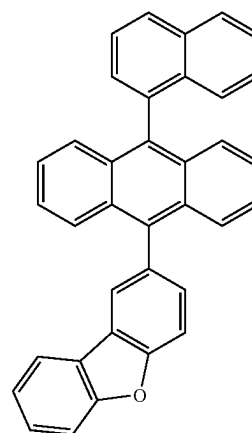
Host 3



Host 4

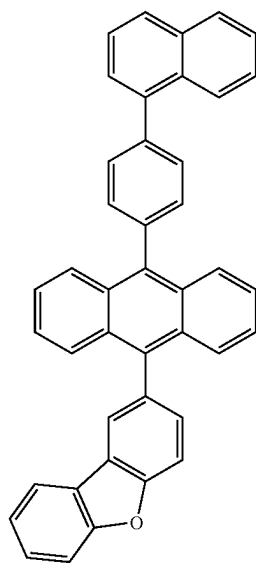
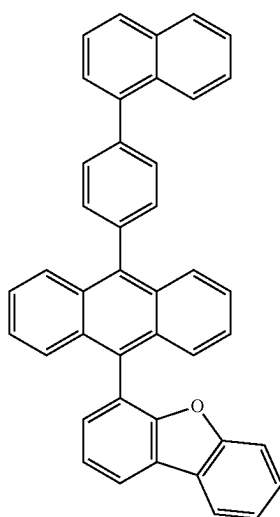
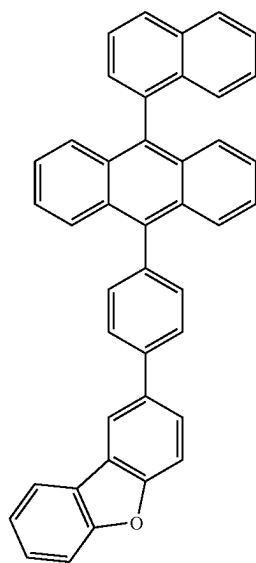


Host 5



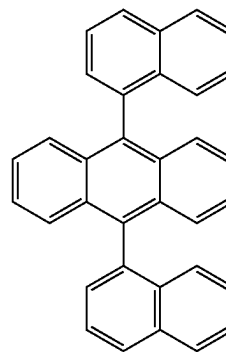
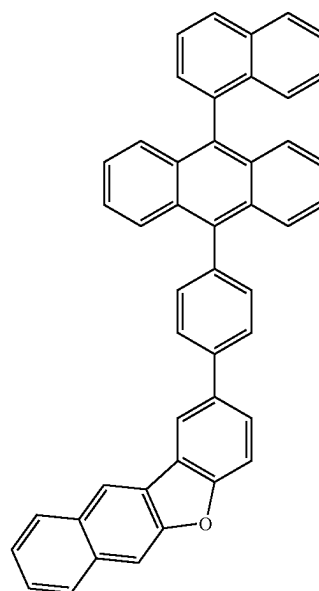
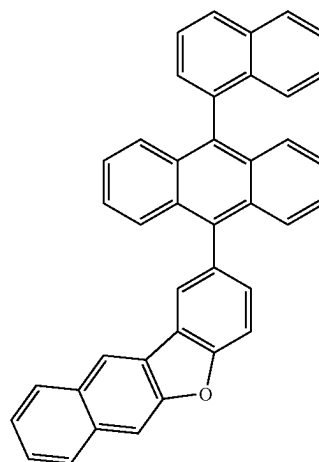
157

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Host 6

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Host 7

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Host 8

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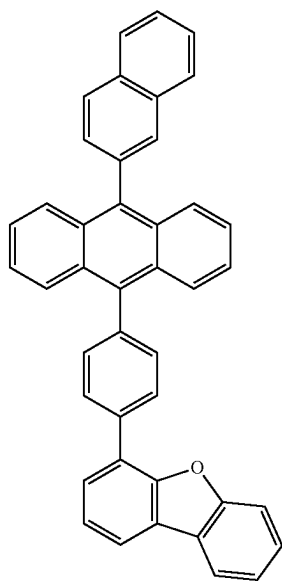
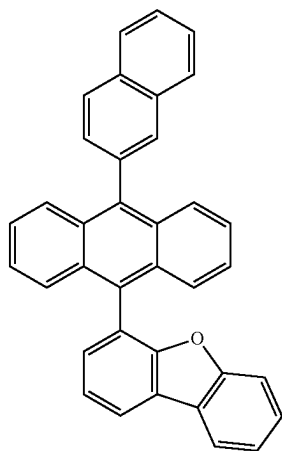
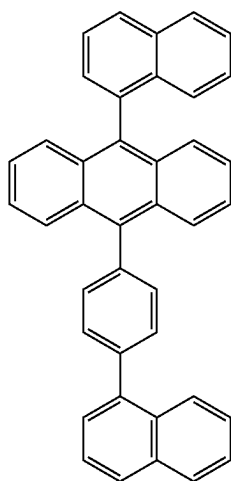
Host 9

Host 10

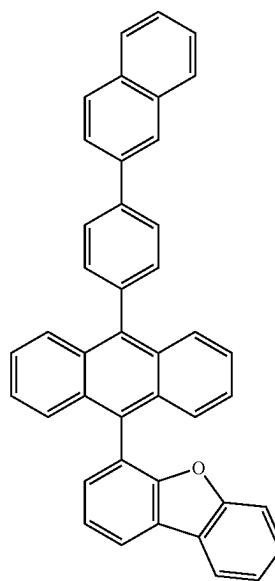
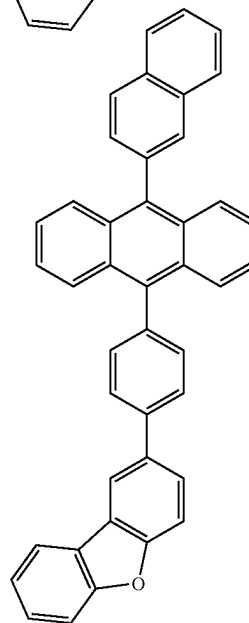
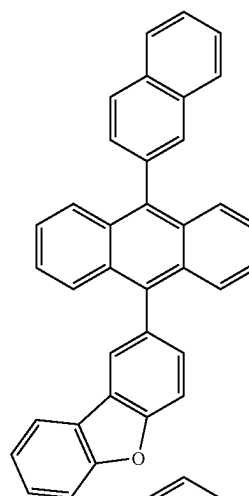
Host 11

159

-continued

**160**

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Host 12

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Host 13

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Host 14

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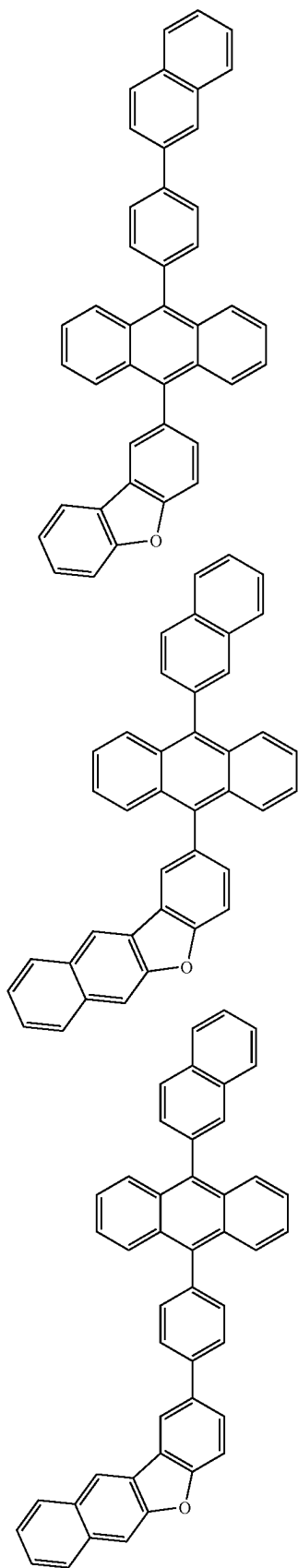
Host 15

Host 16

Host 17

161

-continued



162

-continued

Host 18

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Host 19

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Host 20

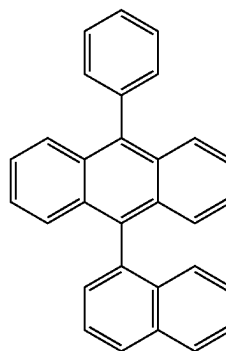
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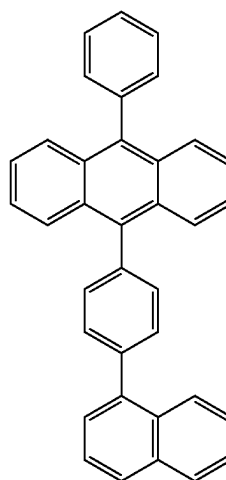
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60

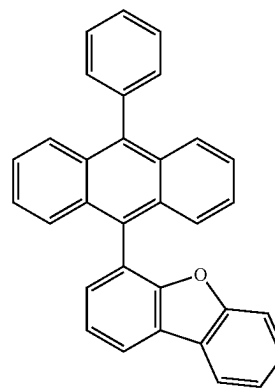
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Host 21



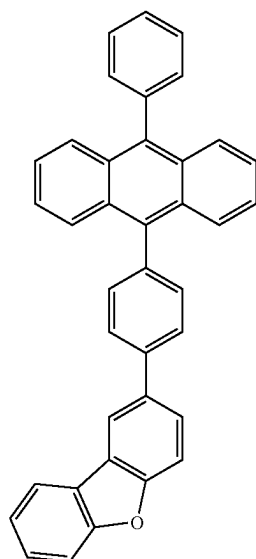
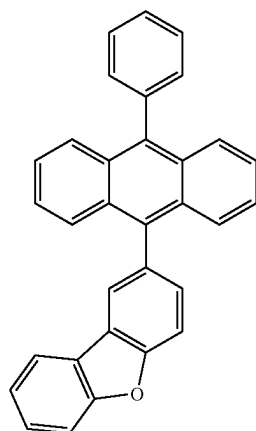
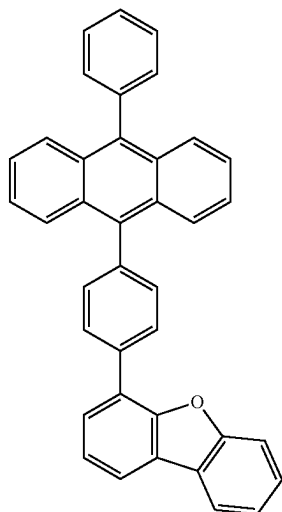
Host 22



Host 23

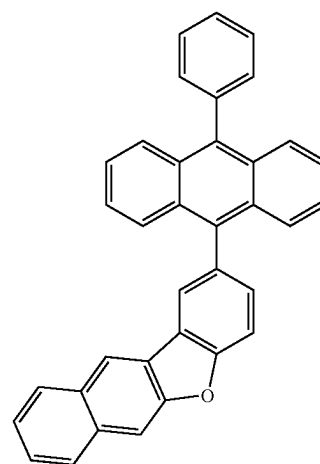
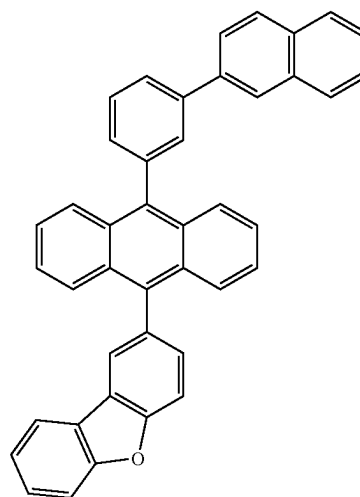
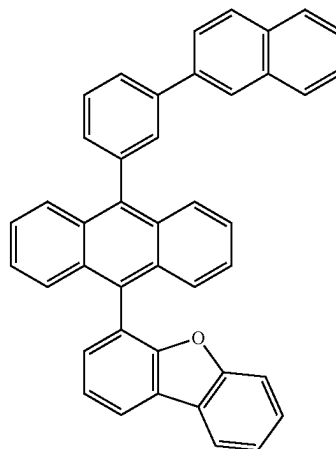
163

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164

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Host 24

Host 27

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Host 25

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Host 26

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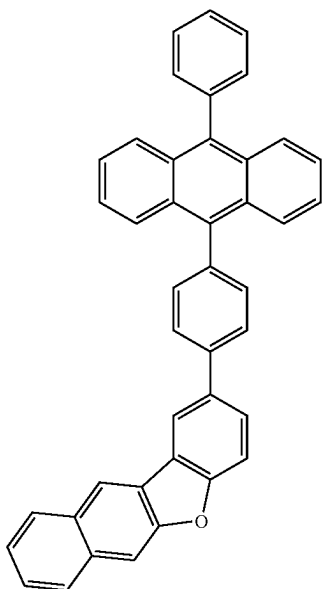
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Host 28

Host 29

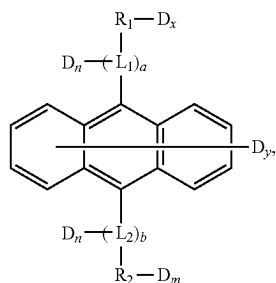
165

-continued



4. The OLED of claim 2, wherein the second host is represented by Formula 3:

Formula 3



wherein the definition of R_1 , R_2 , L_1 , L_2 , a and b is same as Formula 1, and

wherein each of x , y , m and n is independently a positive integer, and a summation of x , y , m and n may be 15 to 29.

5. The OLED of claim 4, wherein the second host is a compound being one of the followings of Formula 12:

166

Host 30 Formula 12

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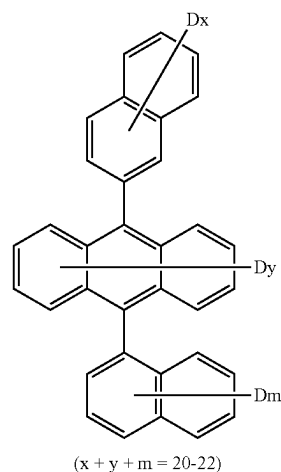
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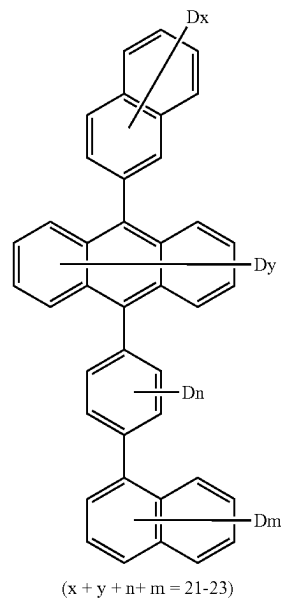
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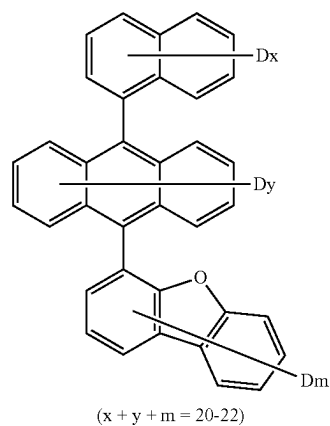


Host 31

Host 32

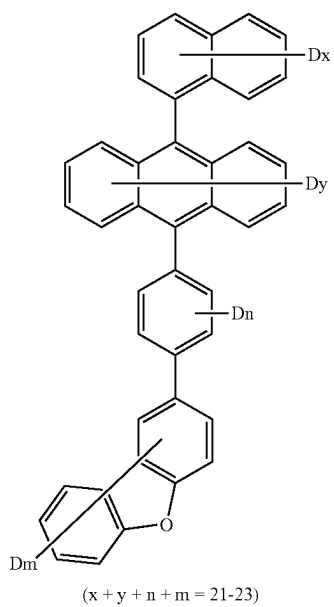
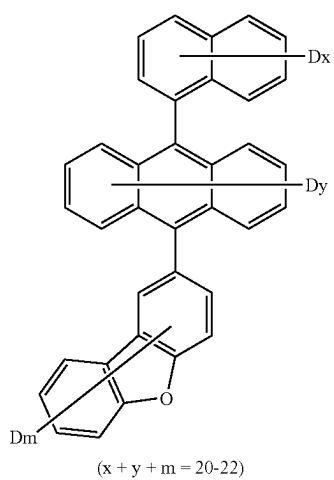
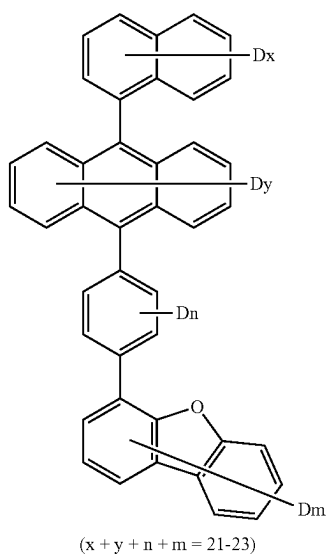


Host 33



167

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**168**

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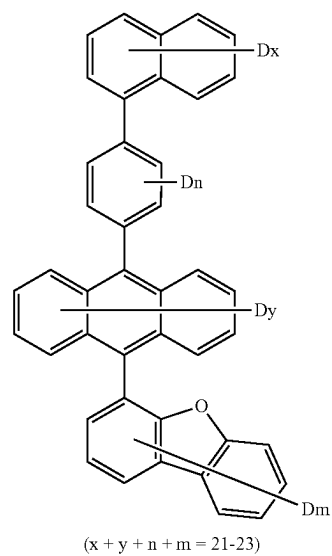
Host 34

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Host 37

Host 35

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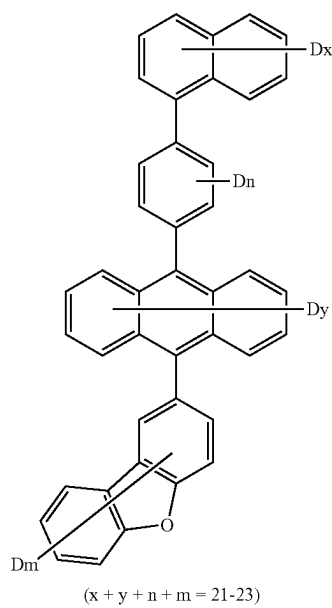
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Host 36

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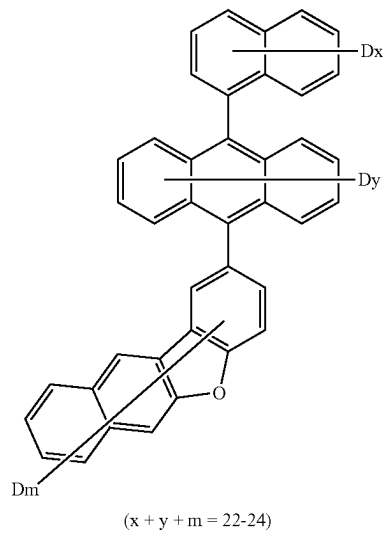
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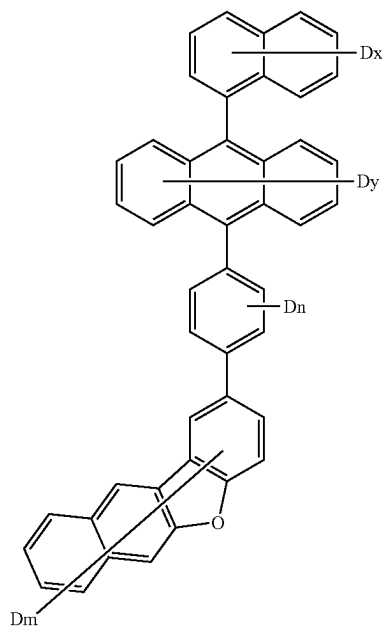
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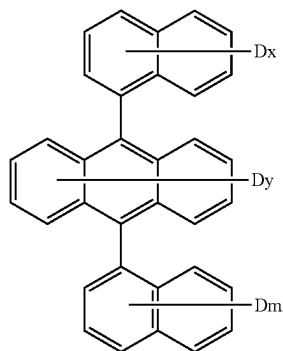
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169

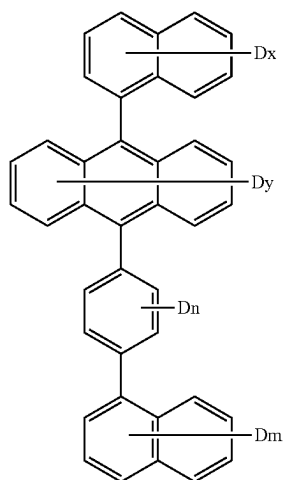
-continued



$$(x + y + n + m = 23-25)$$



$$(x + y + m = 20-22)$$



$$(x + y + n + m = 21-23)$$

170

-continued

Host 40

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Host 41

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Host 42

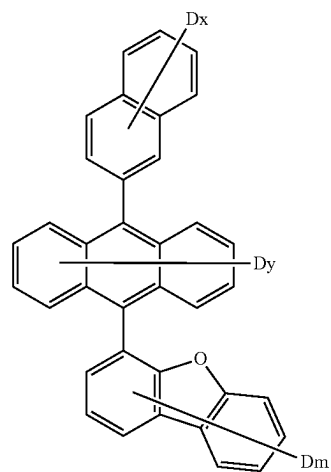
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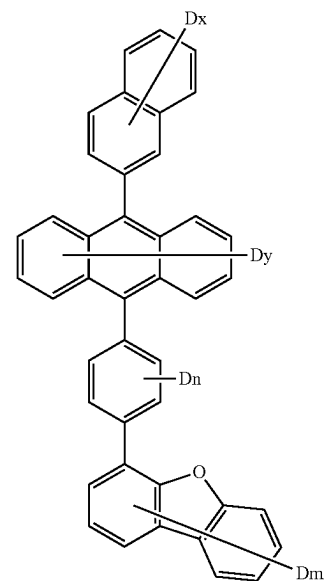
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Host 43



$$(x + y + m = 20-22)$$

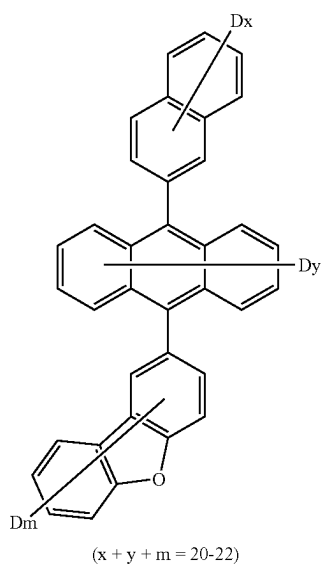
Host 44



$$(x + y + n + m = 21-23)$$

171

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172

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Host 45

Host 47

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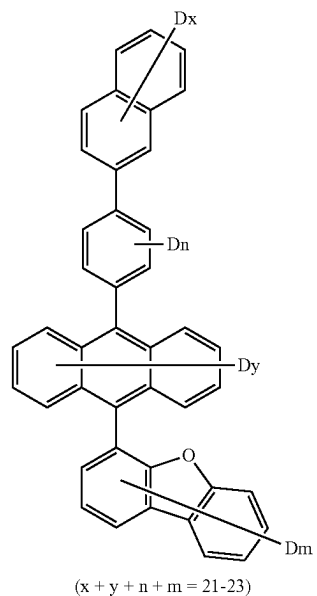
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Host 46

Host 48

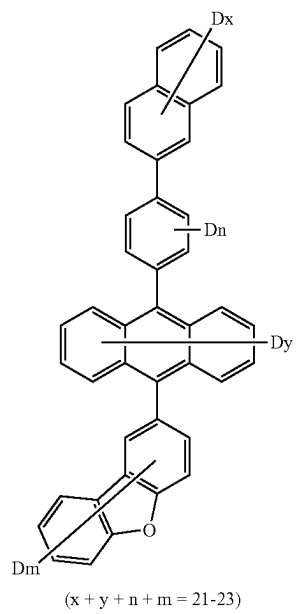
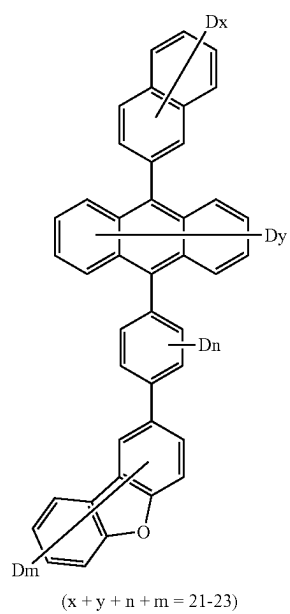
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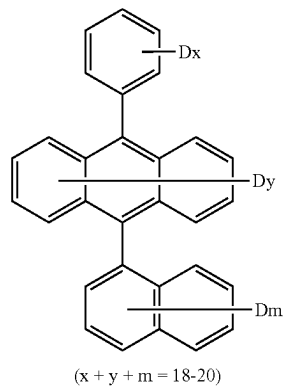
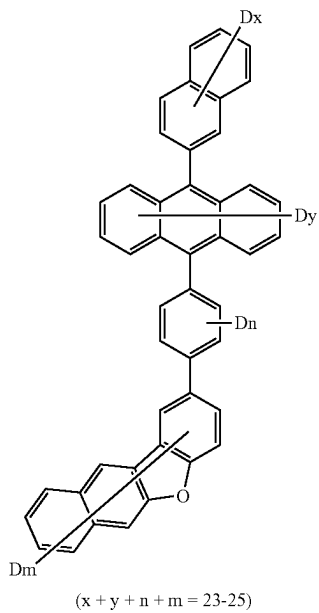
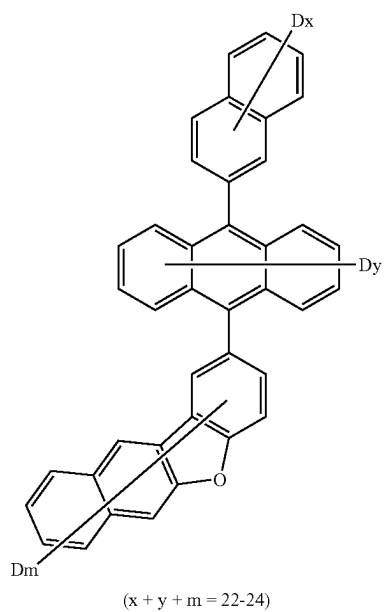
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173

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**174**

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Host 49

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Host 50

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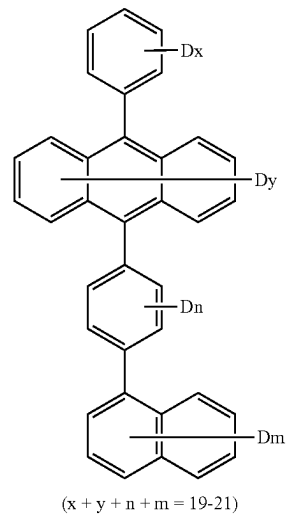
Host 51

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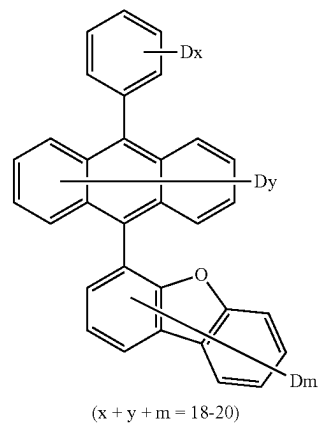
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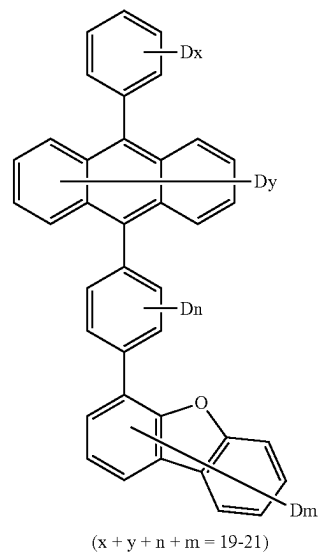
Host 52



Host 53

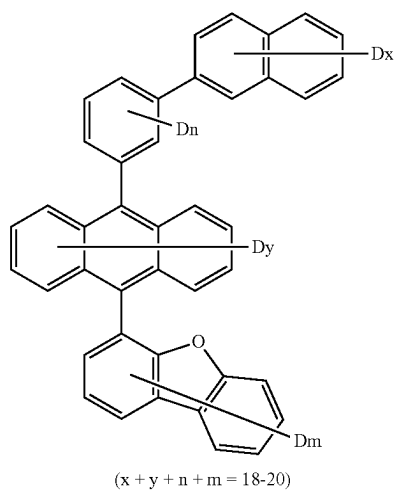
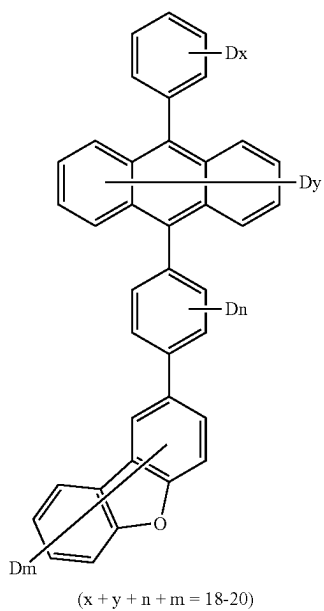
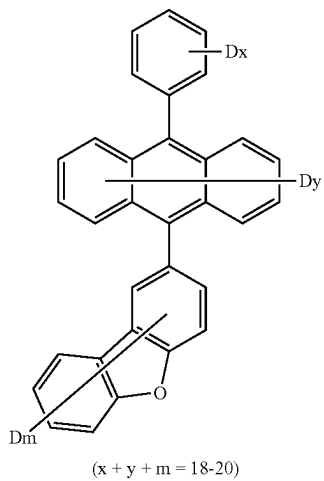


Host 54



175

-continued



176

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Host 55

Host 58

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Host 56

($x + y + n + m = 21-23$)

Host 59

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($x + y + m = 20-22$)

Host 60

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Host 57

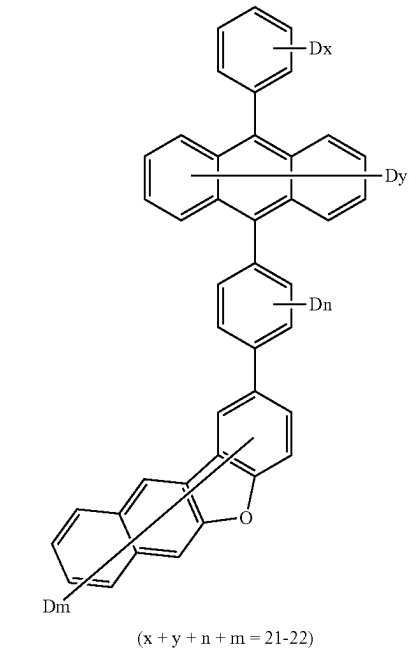
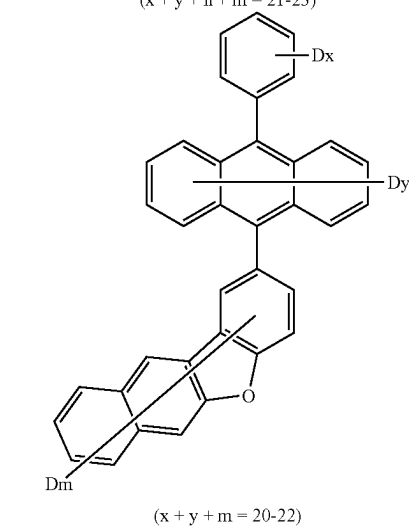
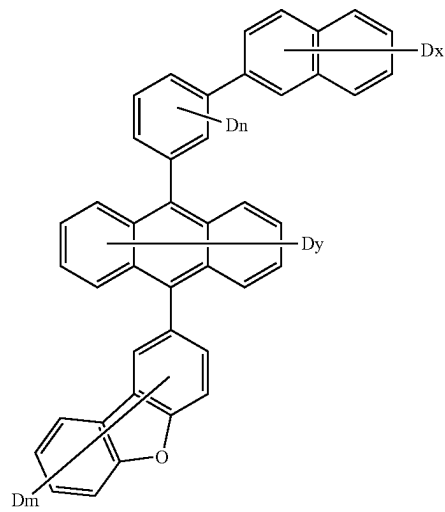
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($x + y + n + m = 21-22$)



177

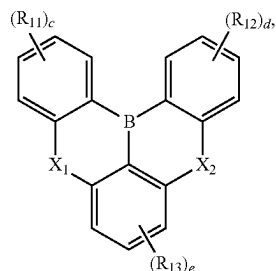
6. The OLED of one of claim 1, wherein the second host is a compound in which all or part of hydrogen of the first host is substituted with deuterium.

7. The OLED of claim 1, wherein the weight % ratio of the first host to the second host is 3:7.

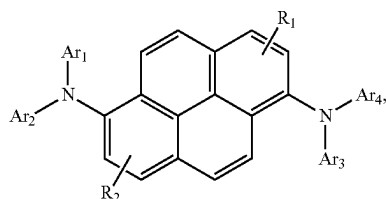
8. The OLED of claim 1, wherein the weight % ratio of the first host to the second host is 7:3.

9. The OLED of claim 1, wherein the blue dopant is represented by Formula 4 or Formula 5:

Formula 4



Formula 5



wherein in Formula 4, each of c and d is independently an integer of 0 to 4, and e is an integer of 0 to 3,

wherein each of R₁₁ and R₁₂ is independently selected from the group consisting of C₁~C₂₀ alkyl group, C₆~C₃₀ aryl group, C₅~C₃₀ hetero aryl group and C₆~C₃₀ aryl amino group, or adjacent two among R₁₁ or adjacent two among R₁₂ form a fused aromatic ring or a hetero-aromatic ring,

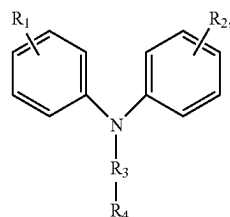
wherein R₁₃ is selected from the group consisting of C₁~C₁₀ alkyl group, C₆~C₃₀ aryl group, C₅~C₃₀ hetero aryl group and C₅~C₃₀ aromatic amino group,

wherein each of X₁ and X₂ is independently oxygen (O) or NR₁₄, and R₁₄ is C₆~C₃₀ aryl group, and

wherein in Formula 5, each of Ar₁, Ar₂, Ar₃ and Ar₄ is independently selected from the group consisting of C₆~C₃₀ aryl group and C₅~C₃₀ hetero aryl group, and each of R₁ and R₂ is independently selected from the group consisting of hydrogen, C₁~C₂₀ alkyl group and C₆~C₃₀ aryl group.

10. The OLED of claim 1, wherein the electron blocking material is represented by Formula 6:

Formula 6

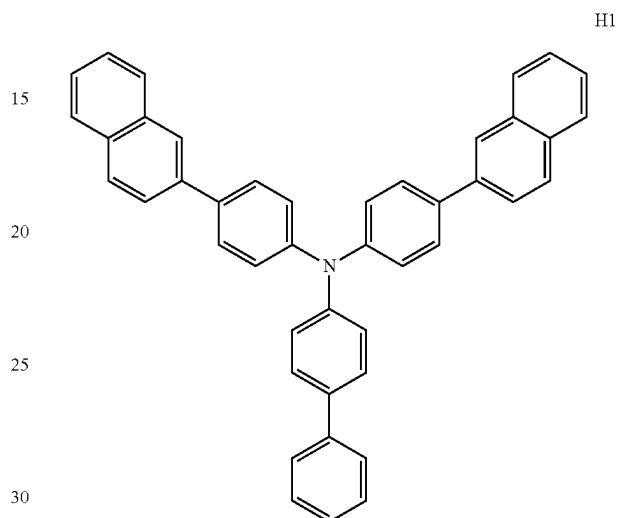


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wherein each of R₁, R₂, R₃ and R₄ is independently selected from the group consisting of monocyclic aryl group or polycyclic aryl group, and at least one of R₁, R₂, R₃ and R₄ is polycyclic aryl group.

11. The OLED of claim 10, wherein the electron blocking material is a compound being one of the followings of Formula 7:

Formula 7



H1

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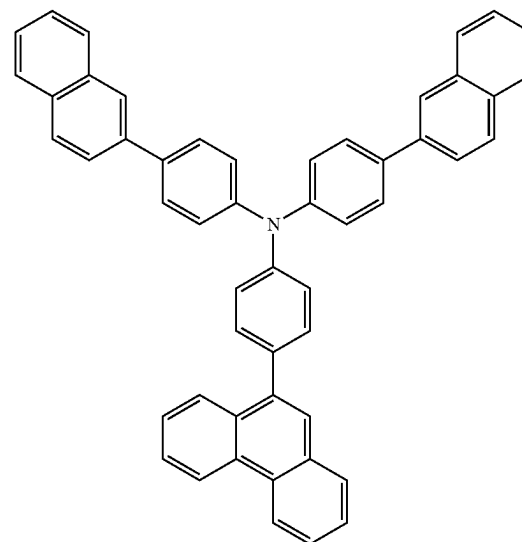
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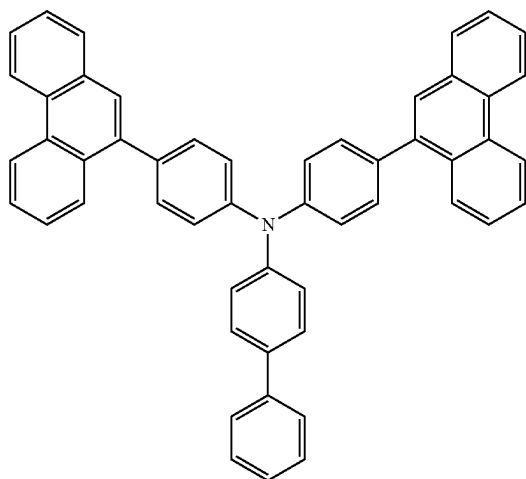
H2



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H3



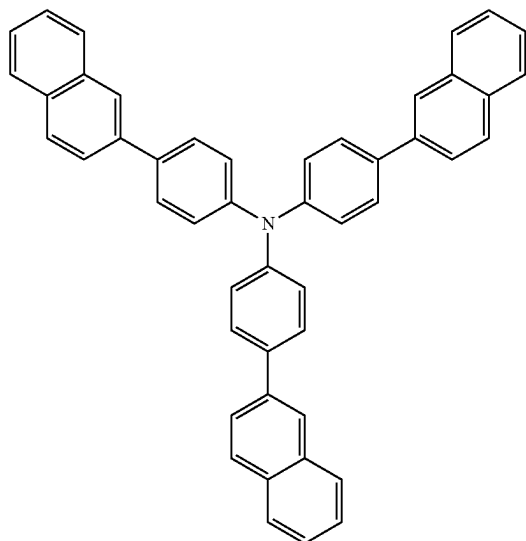
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H4



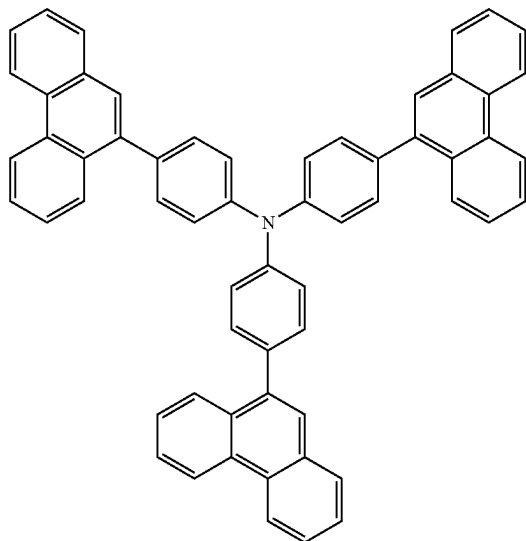
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H5



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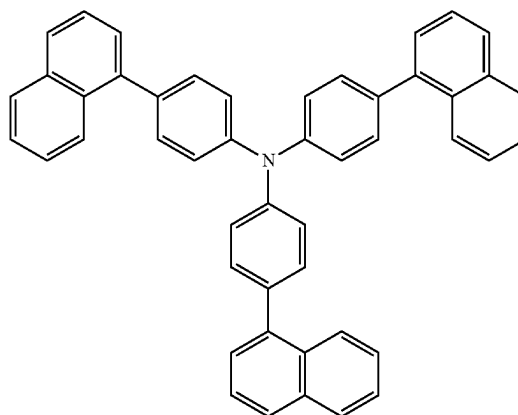
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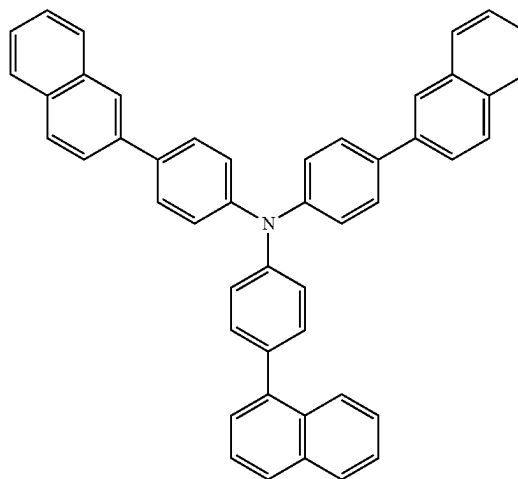
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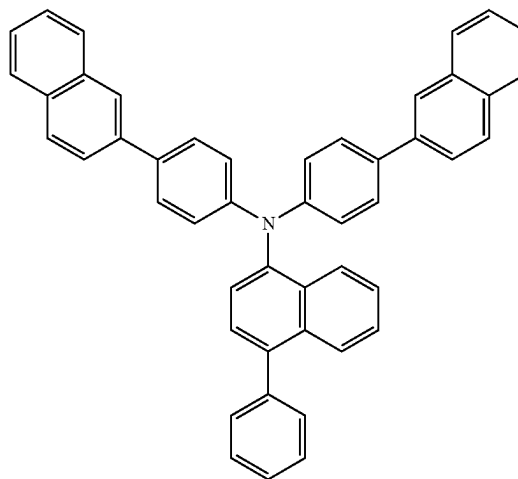
H6



H7

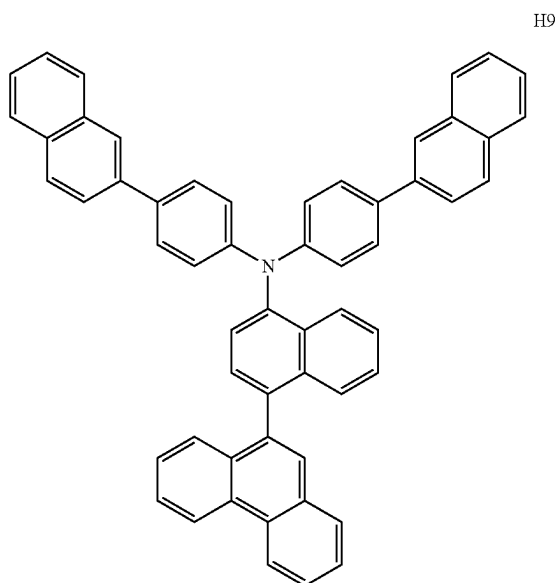


H8

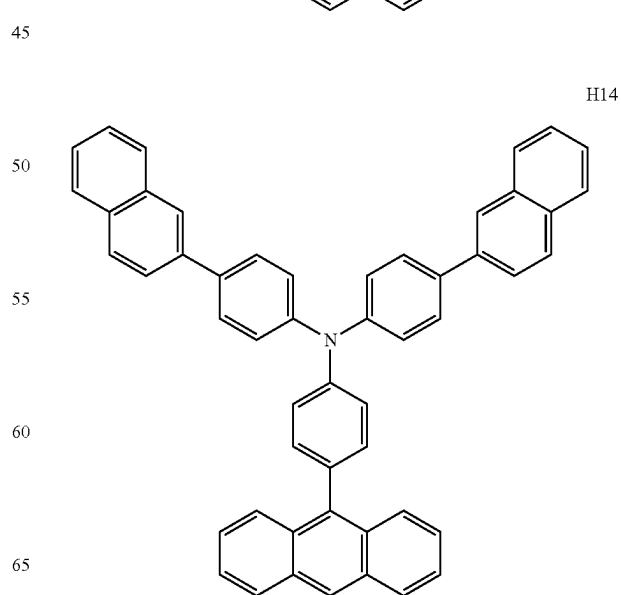
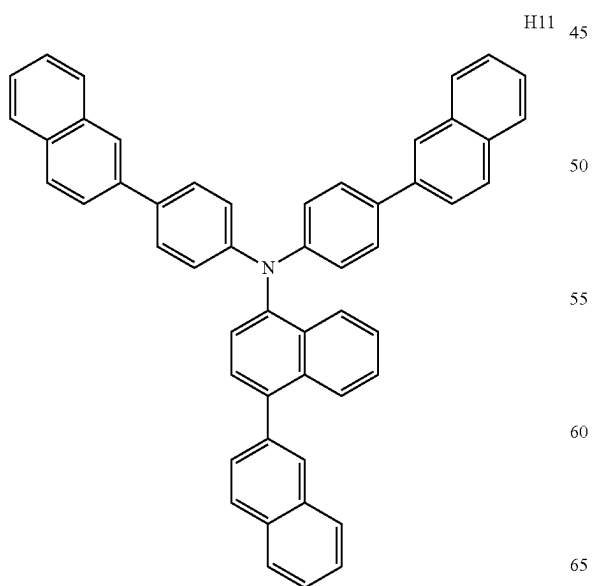
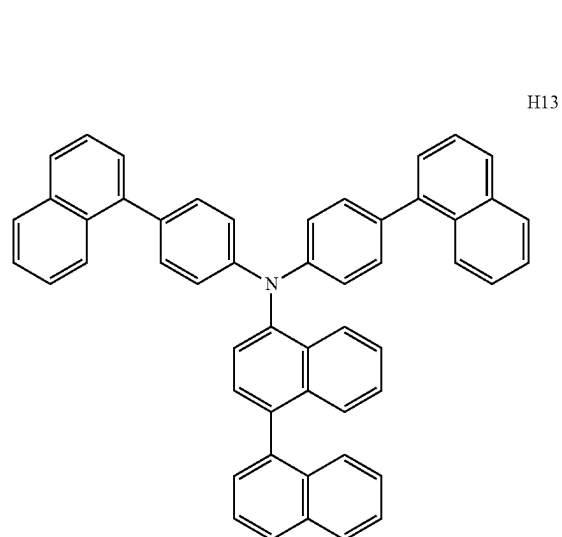
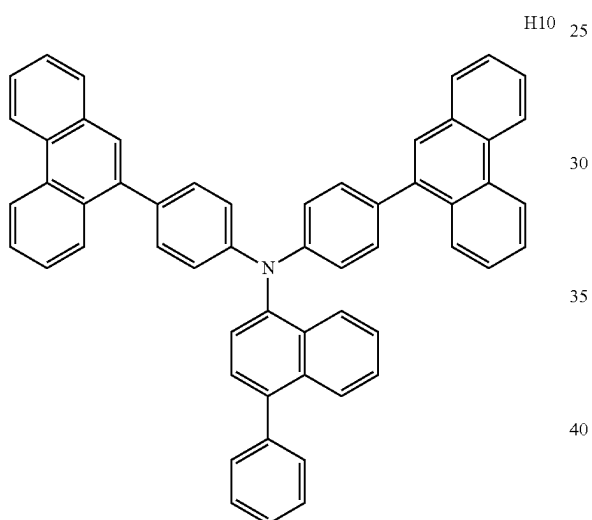
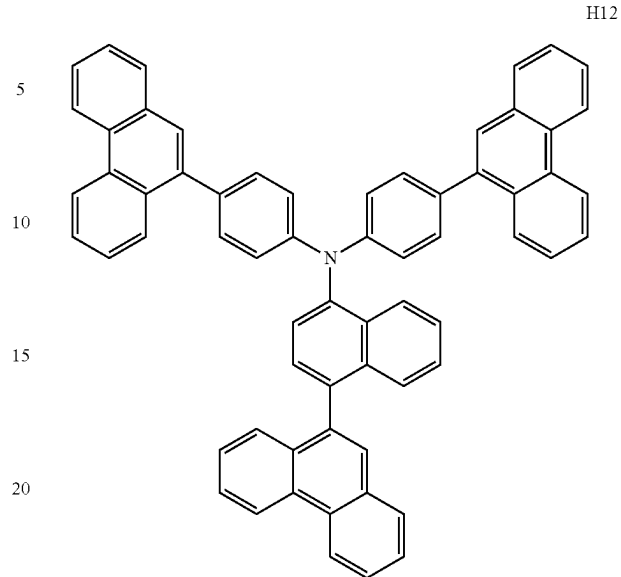


181

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**182**

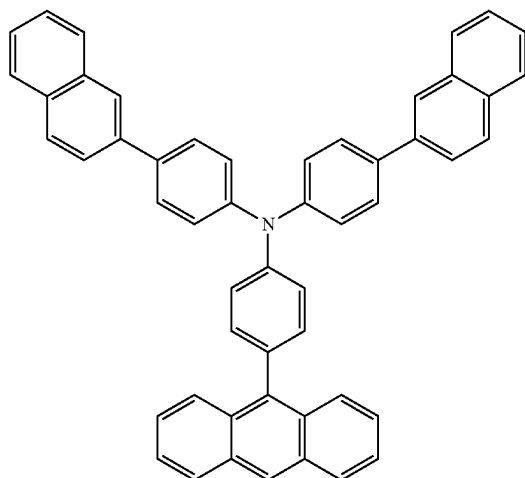
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183

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H15



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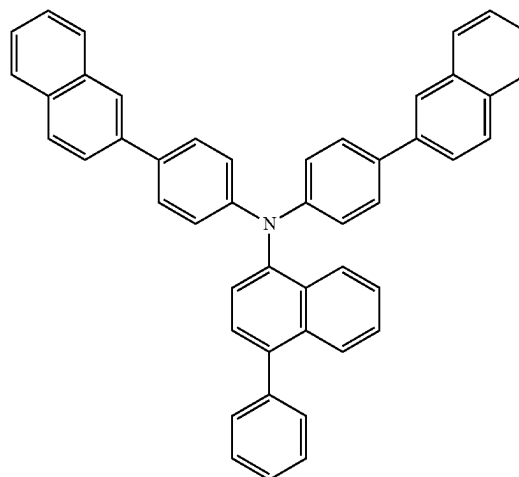
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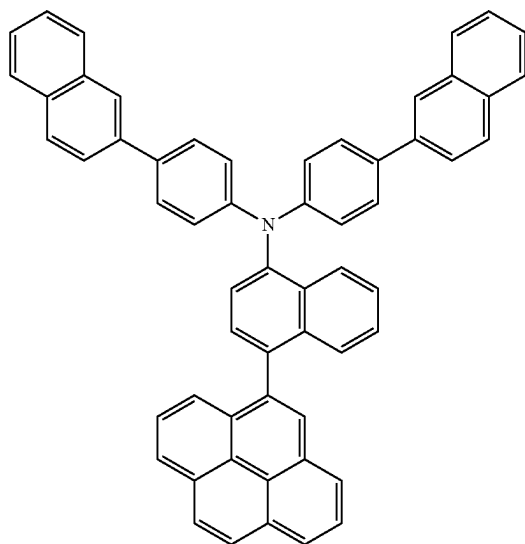
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H18



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H16



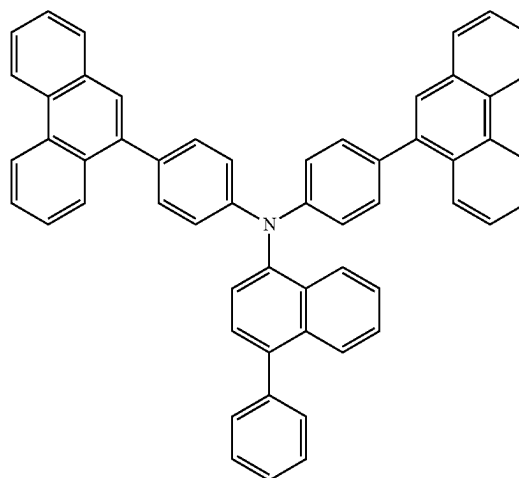
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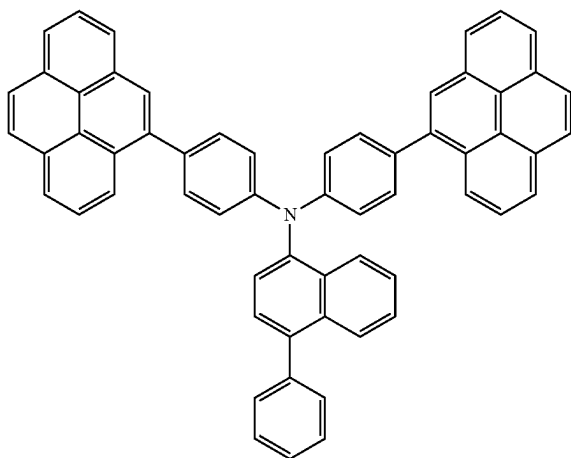
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H19



H17



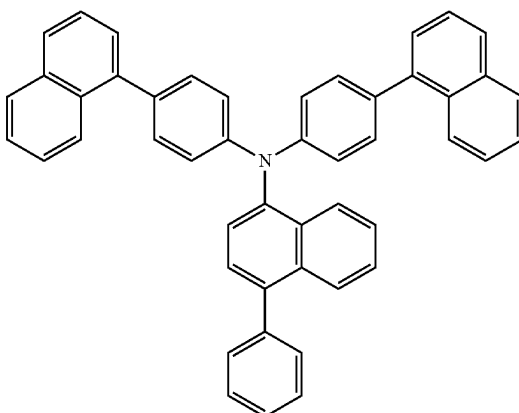
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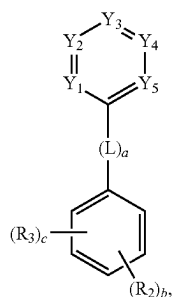
H20



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12. The OLED of claim 1, wherein the first hole blocking material is represented by Formula 8:

Formula 8



wherein each of Y_1 to Y_5 are independently CR_1 or N, and one to three of Y_1 to Y_5 is N,

wherein R_1 is independently hydrogen or $C_6\sim C_{30}$ aryl group,

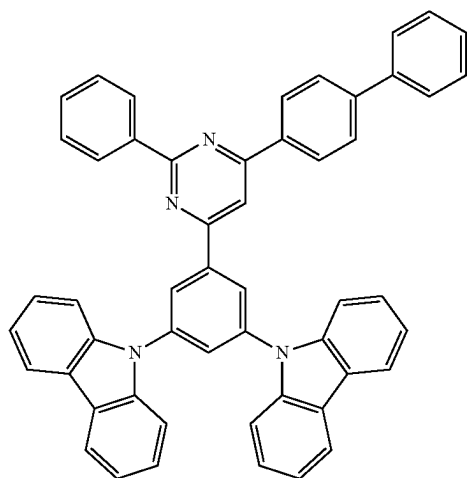
wherein L is $C_6\sim C_{30}$ arylene group, and R_2 is $C_6\sim C_{30}$ aryl group or $C_5\sim C_{30}$ hetero aryl group,

wherein R_3 is hydrogen, or adjacent two of R_3 form a fused ring, and

wherein "a" is 0 or 1, "b" is 1 or 2, and "c" is an integer of 0 to 4.

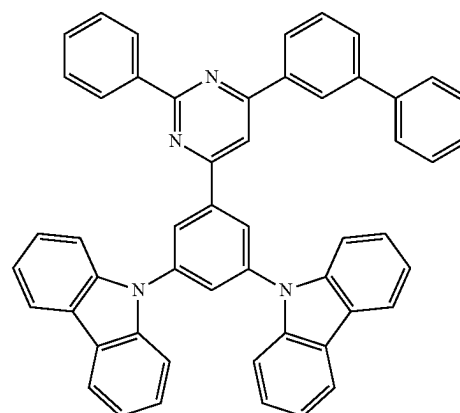
13. The OLED of claim 12, wherein the first hole blocking material is a compound being one of the followings of Formula 9:

Formula 9

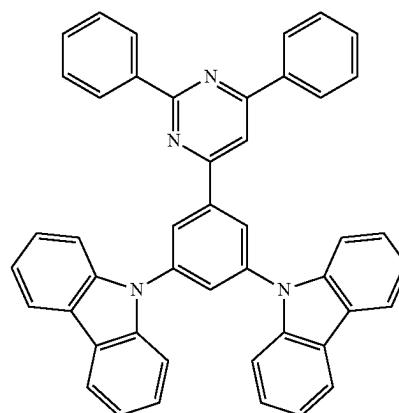
**186**

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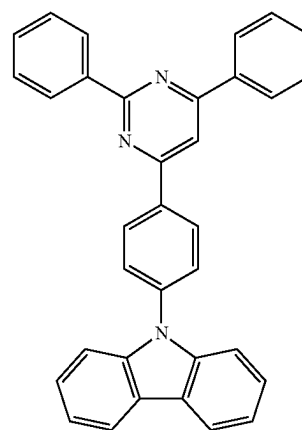
E2



E3

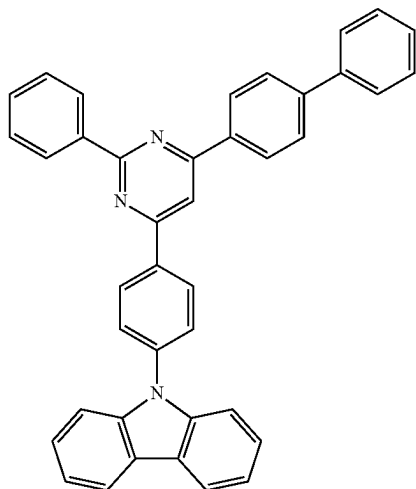


E4



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E5

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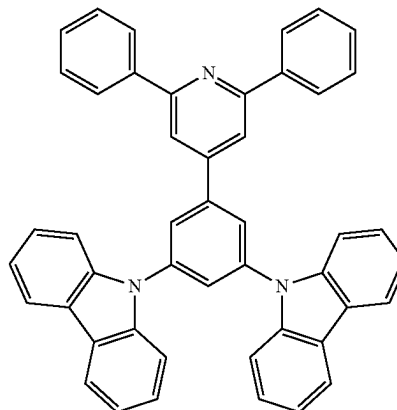
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188

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E8

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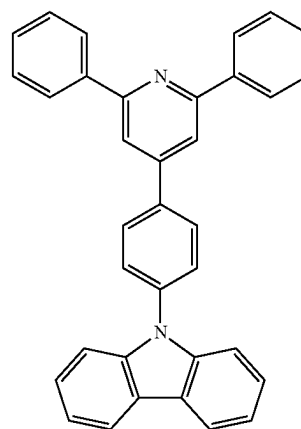
E6

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E9

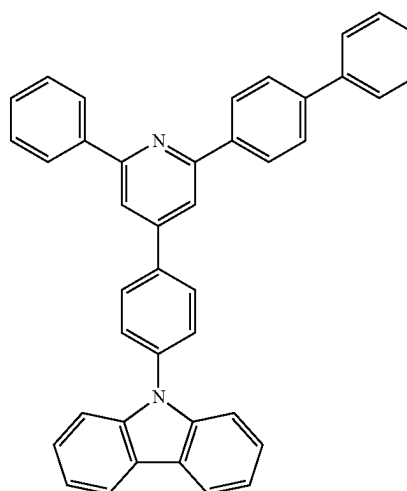
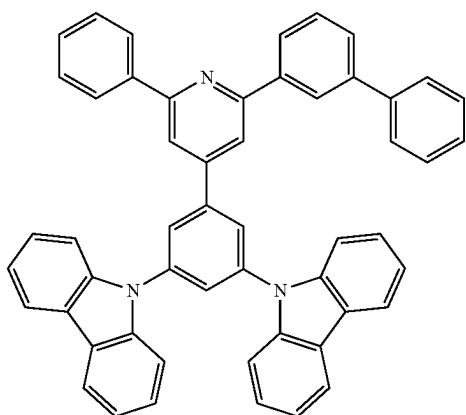
E7

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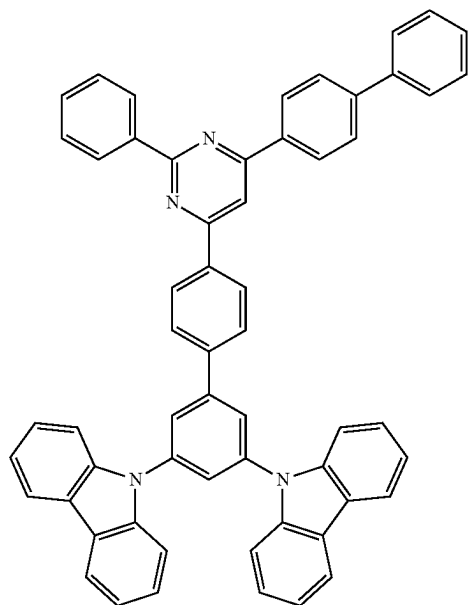
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E10

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-continued



E11

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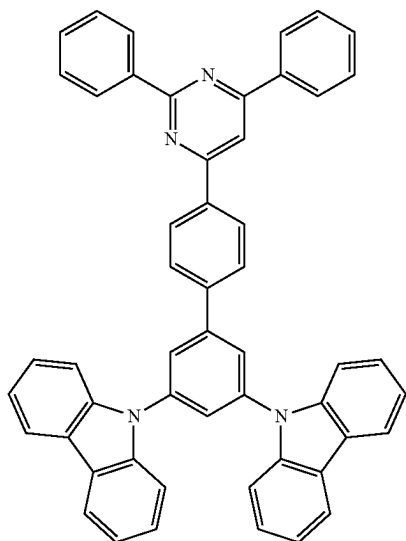
E12

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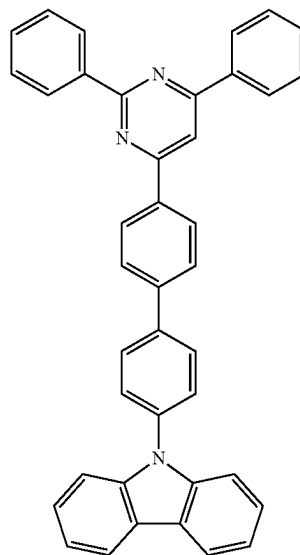
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**190**

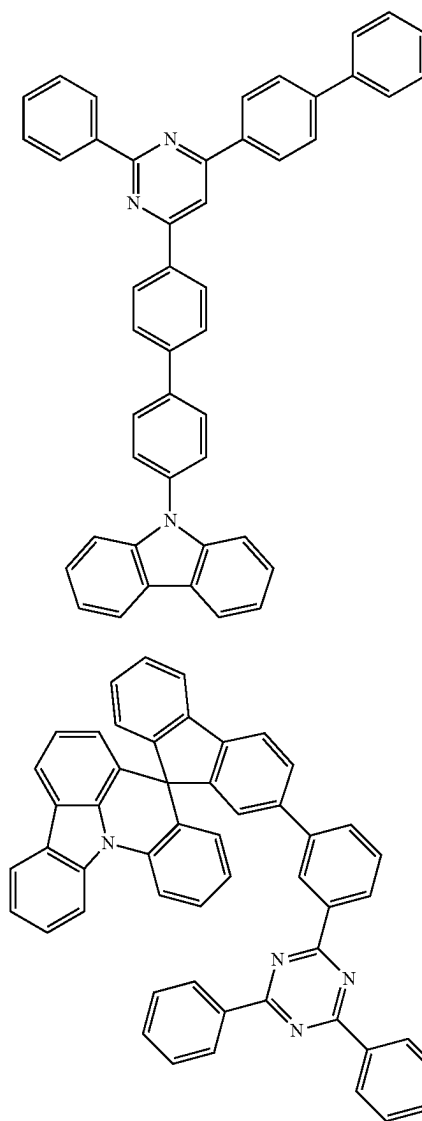
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E13

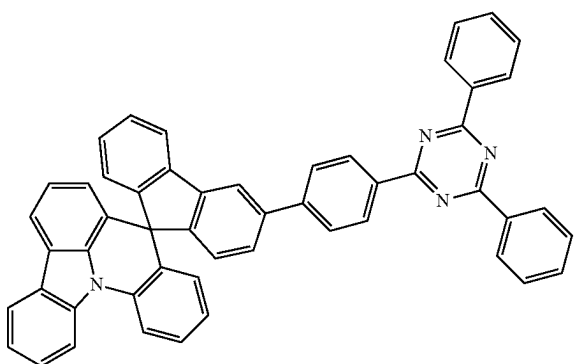
E14

E15



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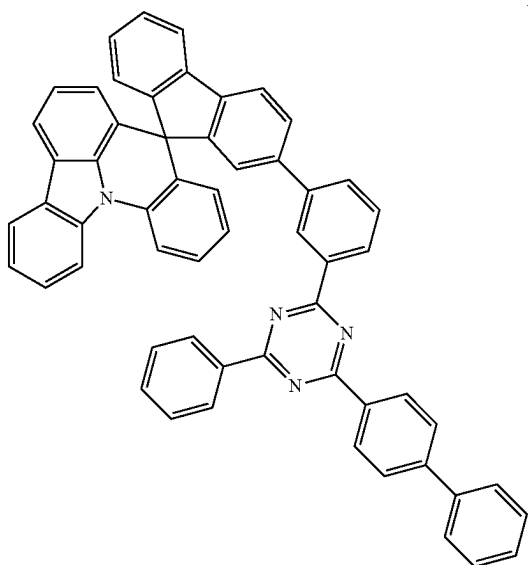
E16

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E17

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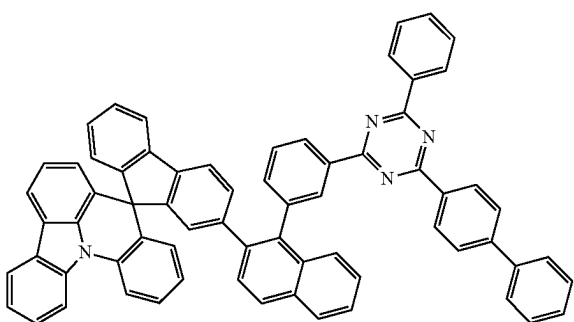
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E18

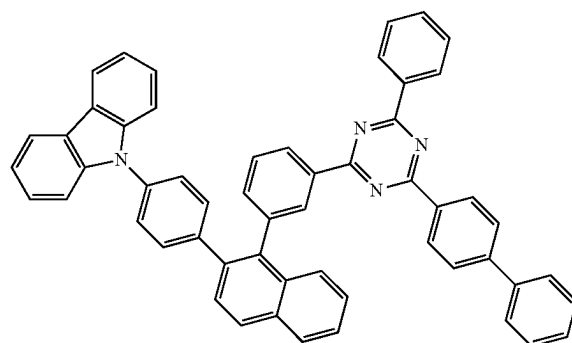
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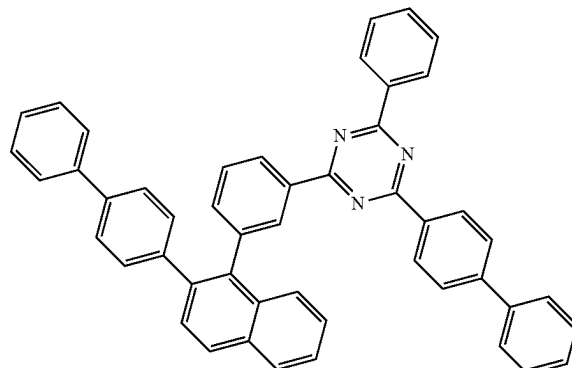
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192

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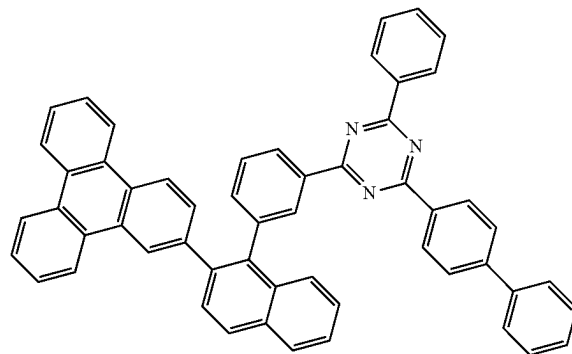


E19

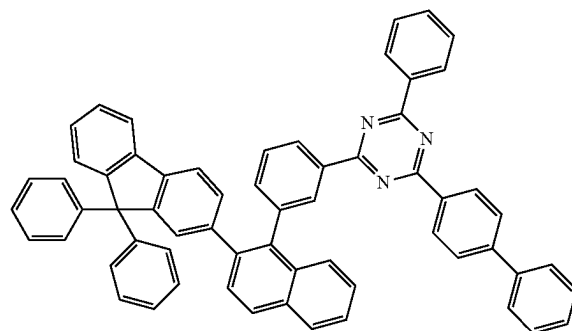


E20

E21



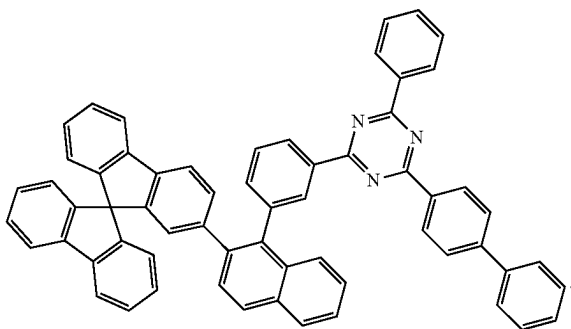
E22



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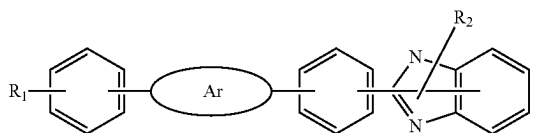
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E23



14. The OLED of claim 1, wherein the second hole blocking material is represented by Formula 10:

Formula 10

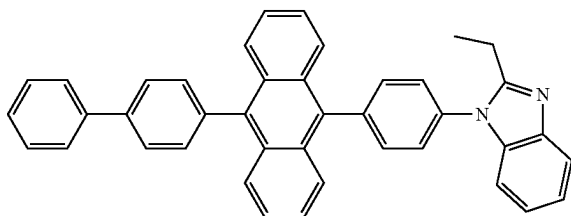


wherein Ar is C₁₀~C₃₀ arylene group, R₁ is C₆~C₃₀ aryl group or C₅~C₃₀ hetero aryl group, and

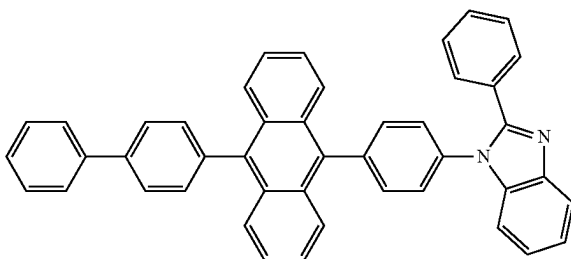
wherein R₂ is C₁~C₁₀ alkyl group or C₆~C₃₀ aryl group.

15. The OLED of claim 14, wherein the second hole blocking material is a compound being one of the followings of Formula 11:

Formula 11



F1

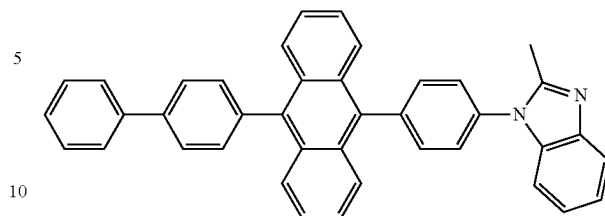


F2

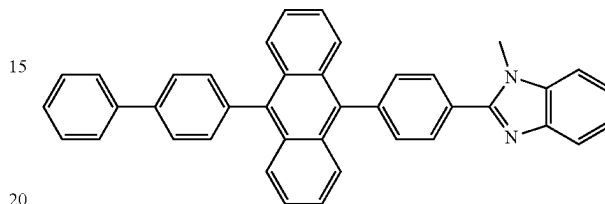
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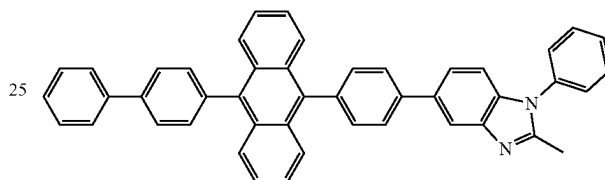
F3



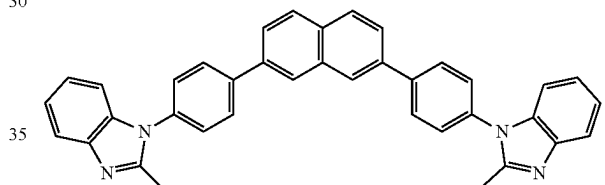
F1



F2



F6



F35

16. The OLED of claim 1, further comprising:

a second emitting material layer including the first host, the second host and the blue dopant and positioned between the first emitting material layer and the second electrode; and

a first charge generation layer between the first and second emitting material layers.

17. The OLED of claim 16, further comprising:

a third emitting material layer emitting a yellow-green light and positioned between the first charge generation layer and the second emitting material layer; and

a second charge generation layer between the second and third emitting material layers.

18. The OLED of claim 16, further comprising:

a third emitting material layer emitting a red light and a green light and positioned between the first charge generation layer and the second emitting material layer; and

a second charge generation layer between the second and third emitting material layers.

19. An organic light emitting device, comprising:

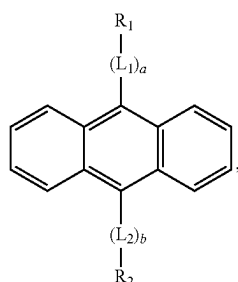
a substrate; and

an organic light emitting diode positioned on the substrate and including a first electrode; a second electrode facing the first electrode; a first emitting material layer including a first host, a second host and a blue dopant and positioned between the first and second electrodes; a first electron blocking layer including an electron

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blocking material of an amine derivative and positioned between the first electrode and the first emitting material layer; and a first hole blocking layer including at least one of a first hole blocking material and a second hole blocking material and positioned between the second electrode and the first emitting material layer, wherein the first host is an anthracene derivative, and the second host is a deuterated anthracene derivative, wherein the first hole blocking material is an azine derivative, and the second hole blocking material is a benzimidazole derivative, wherein the first host is represented by Formula 1:

Formula 1

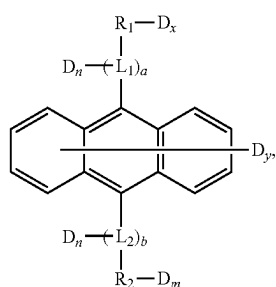


wherein in Formula 1, R_1 is $C_6\sim C_{30}$ aryl group or $C_5\sim C_{30}$ heteroaryl group, and R_2 is $C_6\sim C_{30}$ aryl group, and each of L_1 and L_2 is independently $C_6\sim C_{30}$ arylene group, and

wherein each of a and b is an integer of 0 or 1, and at least one of a and b is 0,

wherein the second host is represented by Formula 3:

Formula 3



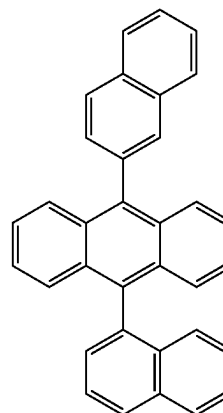
wherein in Formula 3, R_1 is $C_6\sim C_{30}$ aryl group or $C_5\sim C_{30}$ heteroaryl group, and R_2 is dibenzofuranyl or benzo[b]naphthofuranyl,

wherein each of L_1 and L_2 is independently $C_6\sim C_{30}$ arylene group, and each of a and b is an integer of 0 or 1, and at least one of a and b is 0, and

wherein each of x, y, m and n is independently a positive integer, and a summation of x, y, m and n may be 15 to 29.

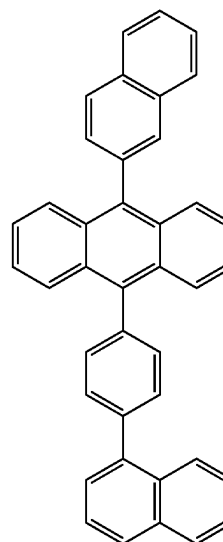
20. The organic light emitting device of claim 19, wherein the first host is a compound being one of the followings of Formula 2:

Formula 2

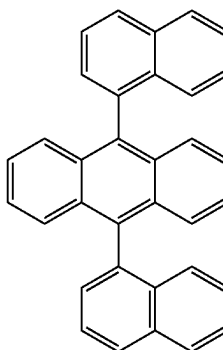


Host 1

Host 2

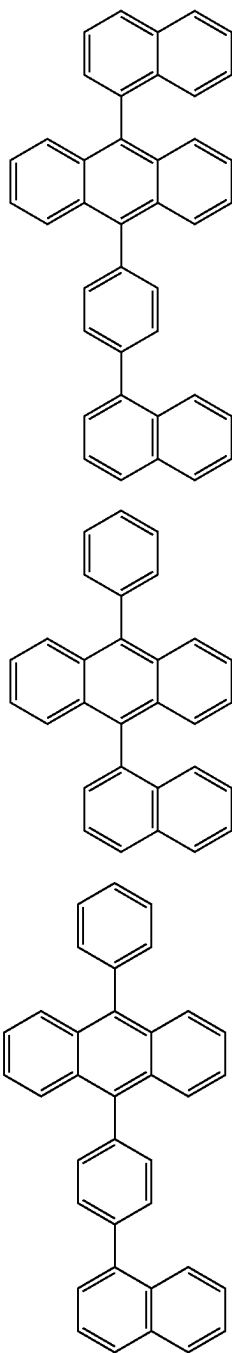


Host 11



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-continued



21. The organic light emitting device of claim 19, wherein a weight % ratio of the first host to the second host is 1:9 to 9:1.

22. The organic light emitting device of claim 21, wherein the weight % ratio of the first host to the second host is 1:9 to 7:3.

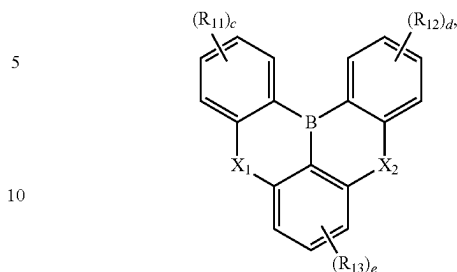
23. The organic light emitting device of claim 21, wherein the weight % ratio of the first host to the second host is 3:7.

24. The organic light emitting device of claim 21, wherein the weight % ratio of the first host to the second host is 7:3.

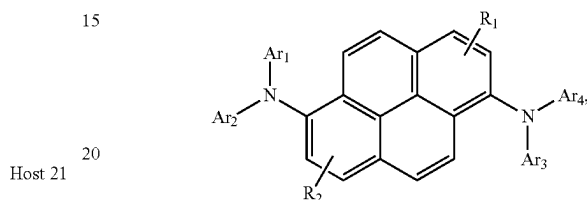
25. The organic light emitting device of claim 19, wherein the first blue dopant is represented by Formula 4 or Formula 5:

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Host 12 Formula 4



Formula 5



wherein in Formula 4, each of c and d is independently an integer of 0 to 4, and e is an integer of 0 to 3,

wherein each of R₁₁ and R₁₂ is independently selected from the group consisting of C₁~C₂₀ alkyl group, C₆~C₃₀ aryl group, C₅~C₃₀ hetero aryl group and C₆~C₃₀ aryl amino group, or adjacent two among R₁₁ or adjacent two among R₁₂ form a fused aromatic ring or a hetero-aromatic ring,

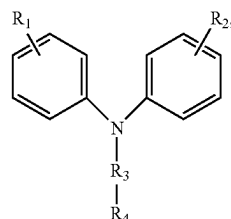
wherein R₁₃ is selected from the group consisting of C₁~C₁₀ alkyl group, C₆~C₃₀ aryl group, C₅~C₃₀ hetero aryl group and C₅~C₃₀ aromatic amino group,

wherein each of X₁ and X₂ is independently oxygen (O) or NR₁₄, and R₁₄ is C₆~C₃₀ aryl group, and

wherein in Formula 5, each of Ar₁, Ar₂, Ar₃ and Ar₄ is independently selected from the group consisting of C₆~C₃₀ aryl group and C₅~C₃₀ hetero aryl group, and each of R₁ and R₂ is independently selected from the group consisting of hydrogen, C₁~C₂₀ alkyl group and C₆~C₃₀ aryl group.

26. The organic light emitting device of claim 19, wherein the electron blocking material is represented by Formula 6:

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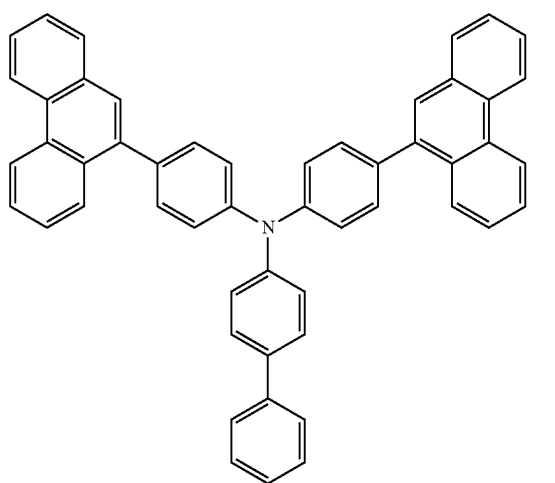
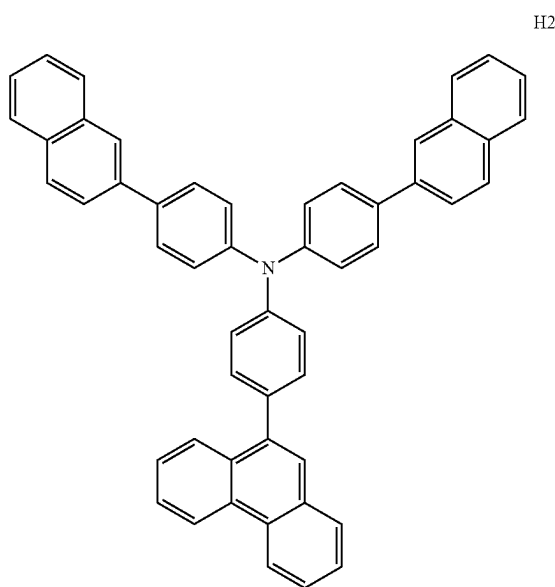
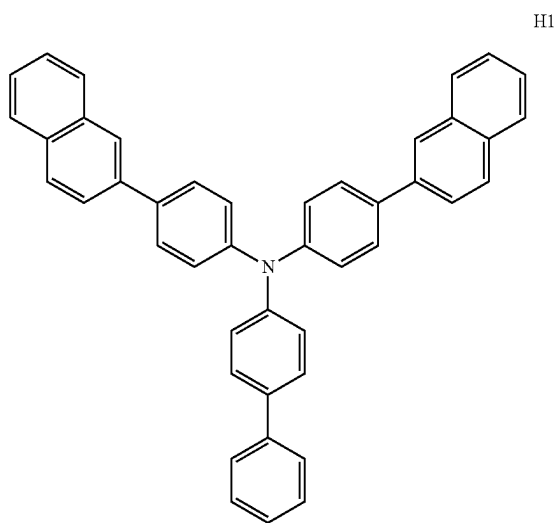
Formula 6

wherein each of R₁, R₂, R₃ and R₄ is independently selected from the group consisting of monocyclic aryl group or polycyclic aryl group, and at least one of R₁, R₂, R₃ and R₄ is polycyclic aryl group.

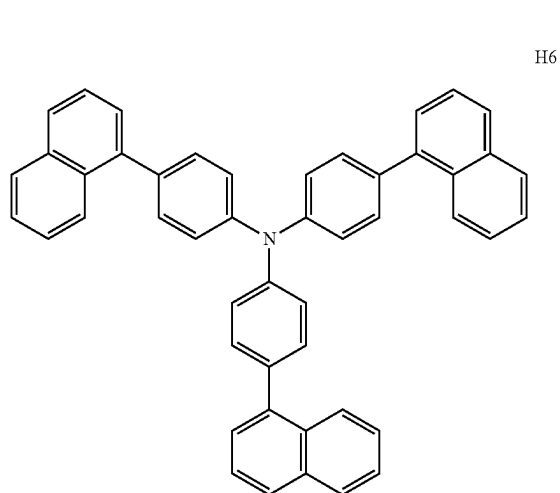
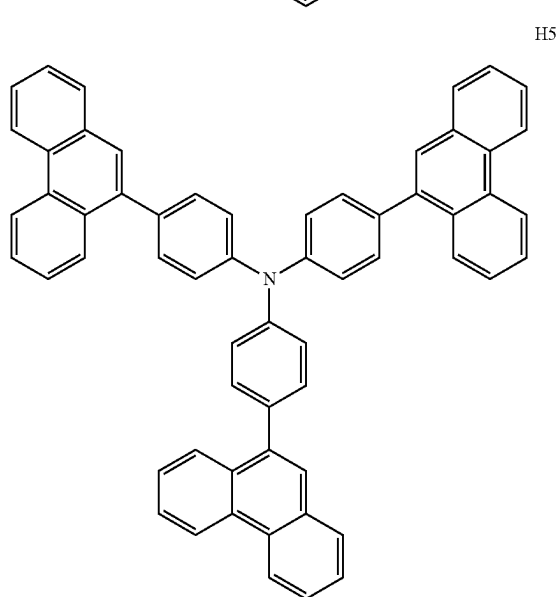
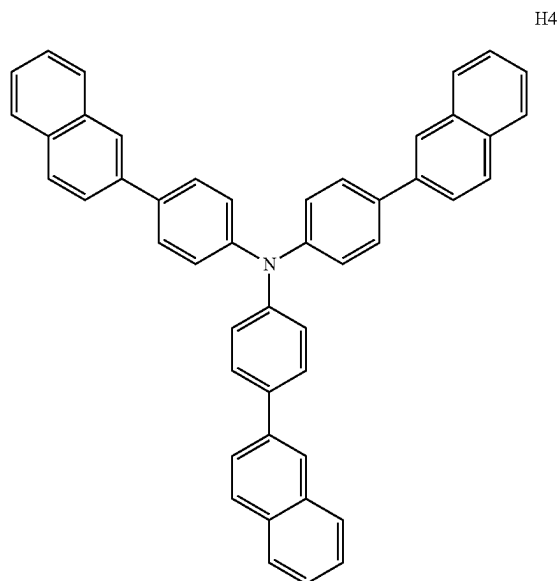
27. The organic light emitting device of claim 26, wherein the electron blocking material is a compound being one of the followings of Formula 7:

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Formula 7

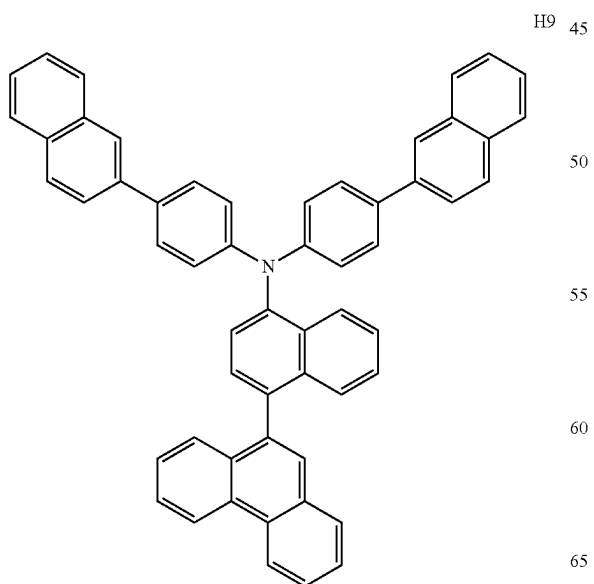
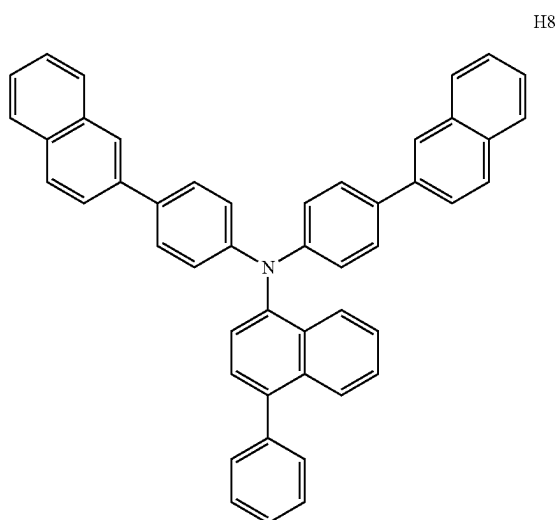
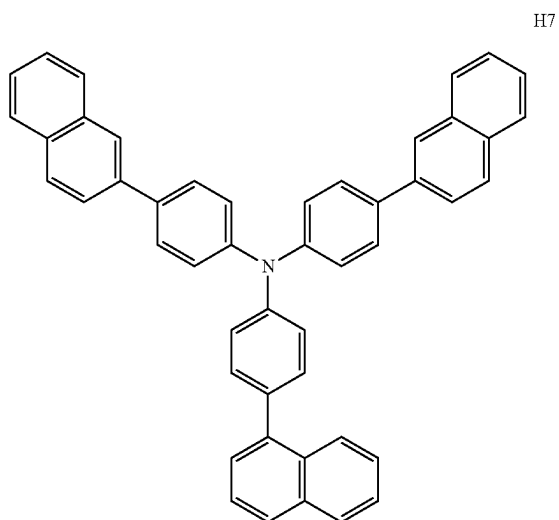
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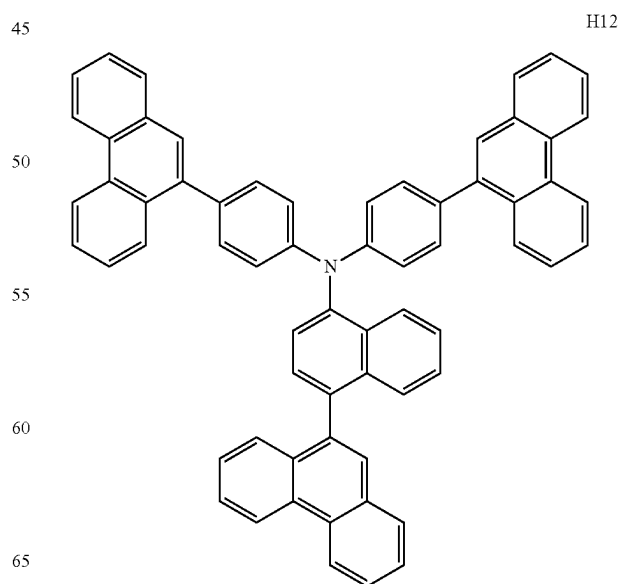
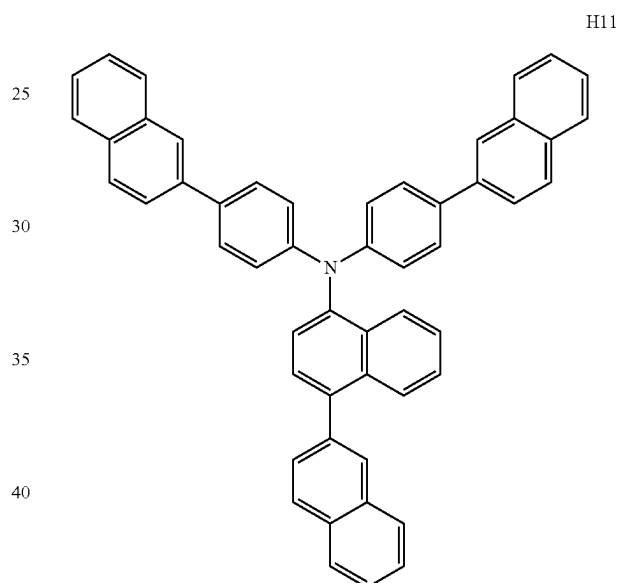
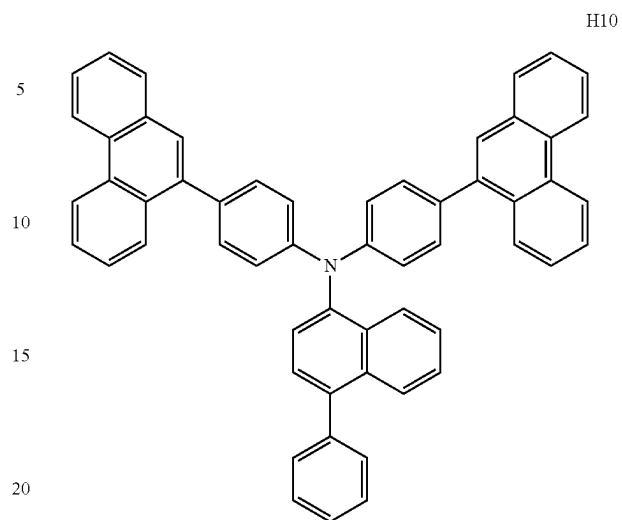


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**202**

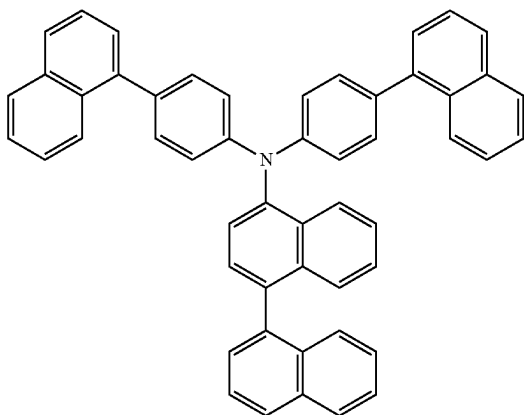
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H13



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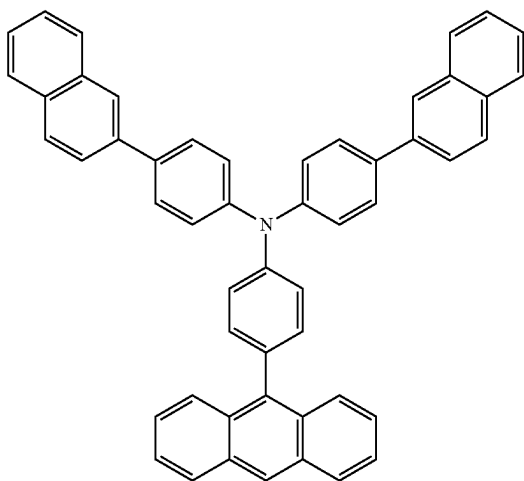
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H14

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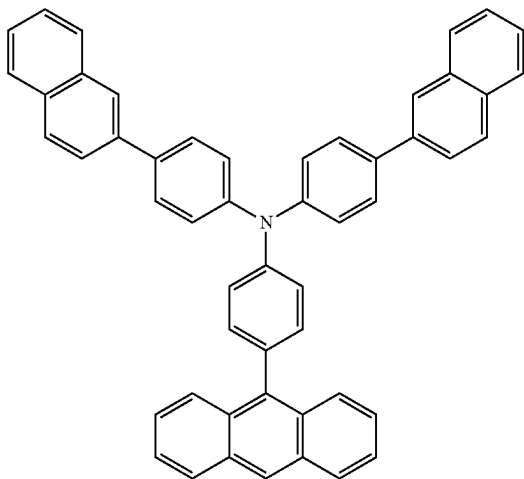
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H15

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204

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H16

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H17

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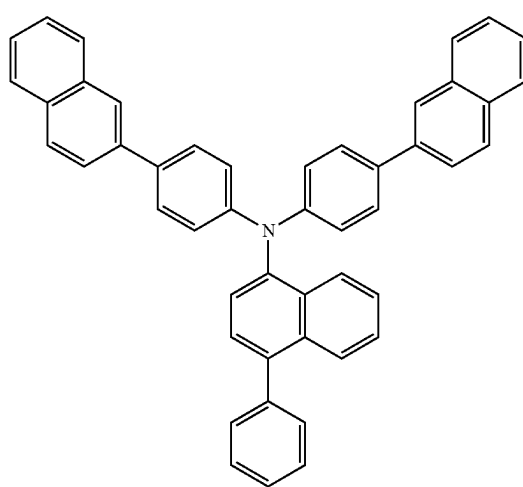
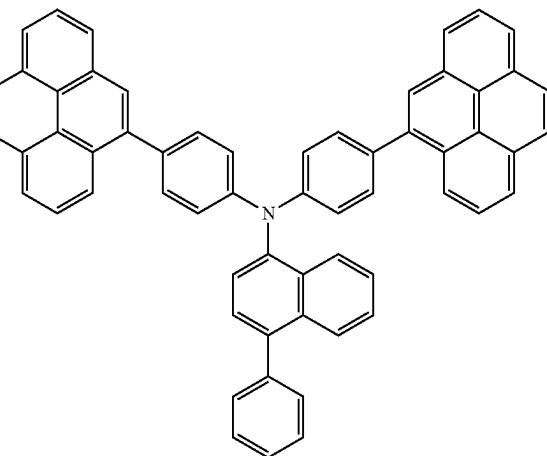
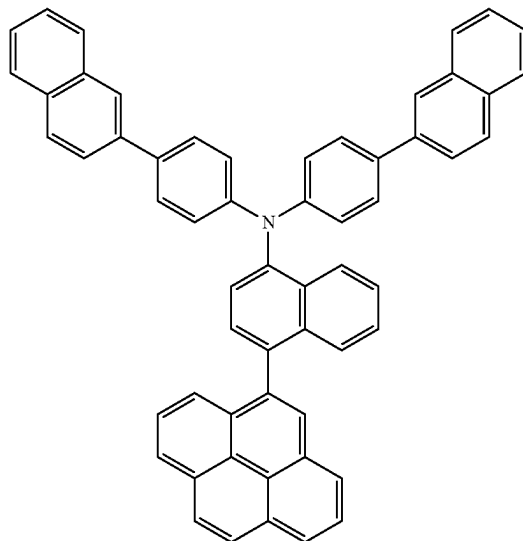
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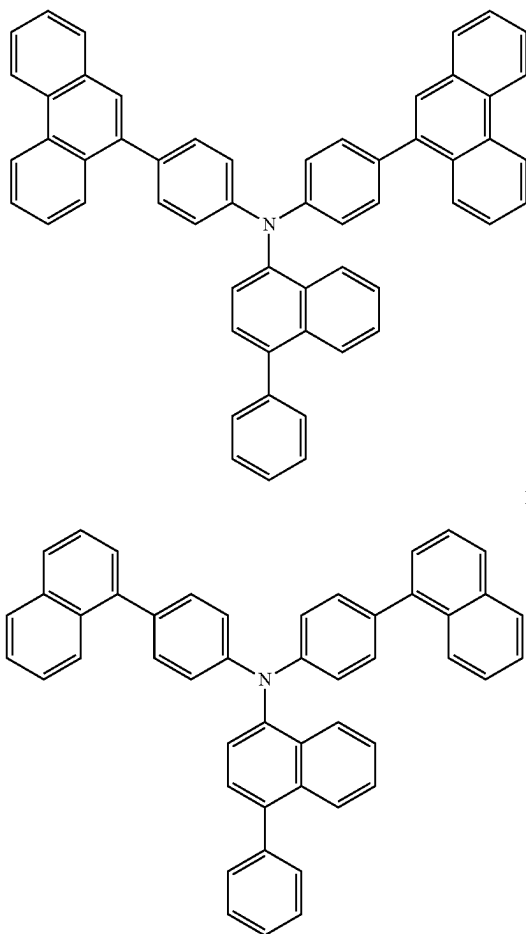
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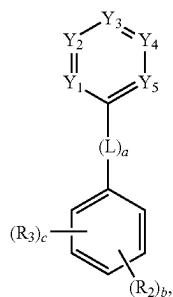
205

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28. The organic light emitting device of claim 19, wherein the first hole blocking material is represented by Formula 8:

Formula 8

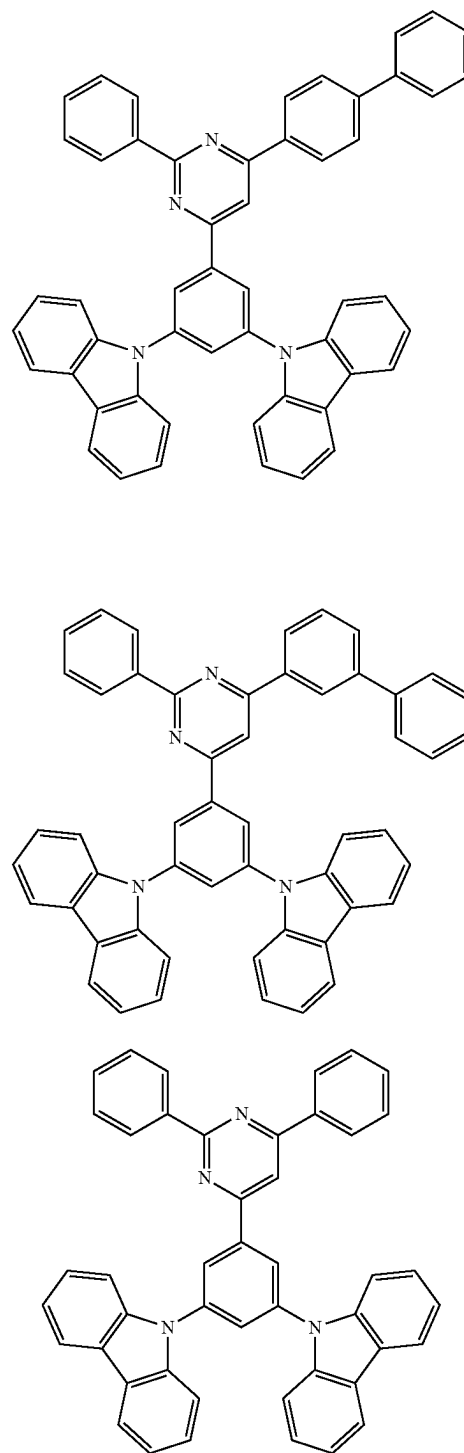


wherein each of Y_1 to Y_5 are independently CR_1 or N, and one to three of Y_1 to Y_5 is N, wherein R_1 is independently hydrogen or $C_6\text{--}C_{30}$ aryl group, wherein L is $C_6\text{--}C_{30}$ arylene group, and R_2 is $C_6\text{--}C_{30}$ aryl group or $C_5\text{--}C_{30}$ hetero aryl group, wherein R_3 is hydrogen, or adjacent two of R_3 form a fused ring, and wherein “a” is 0 or 1, “b” is 1 or 2, and “c” is an integer of 0 to 4.

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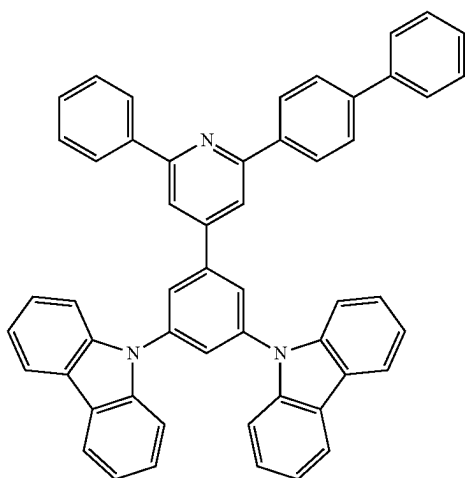
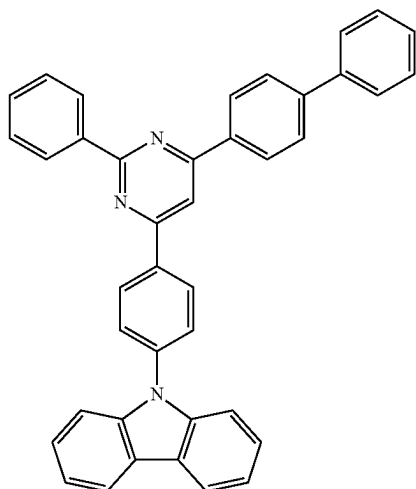
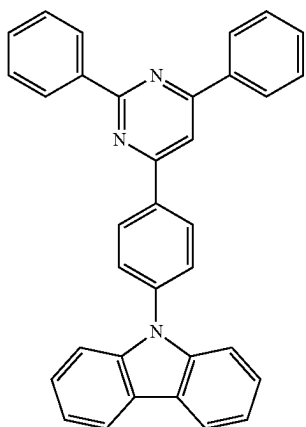
29. The organic light emitting device of claim 28, wherein the first hole blocking material is a compound being one of the followings of Formula 9:

Formula 9

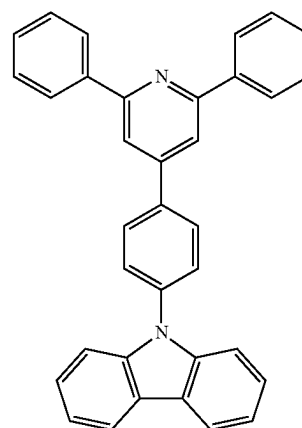
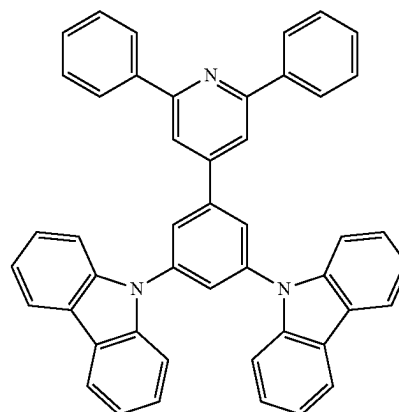
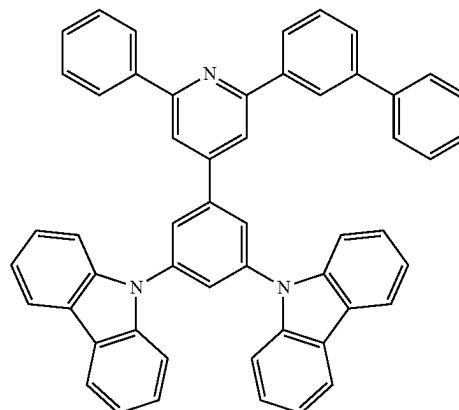


207

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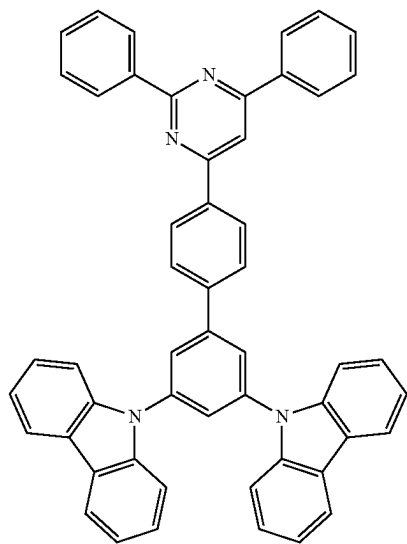
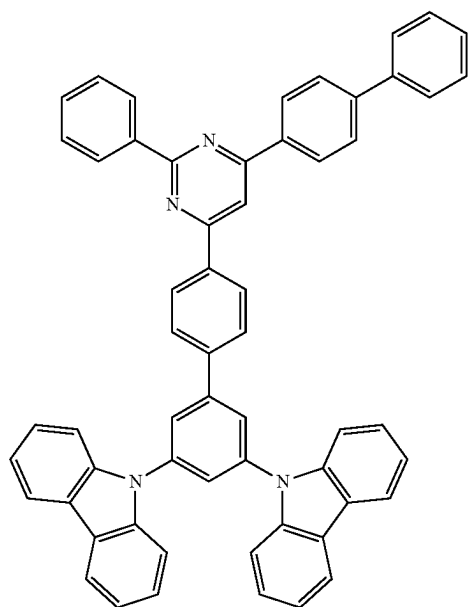
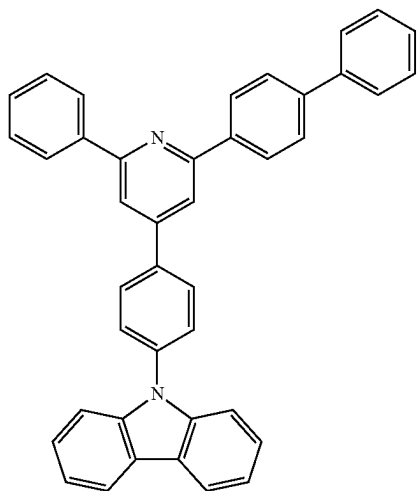
**208**

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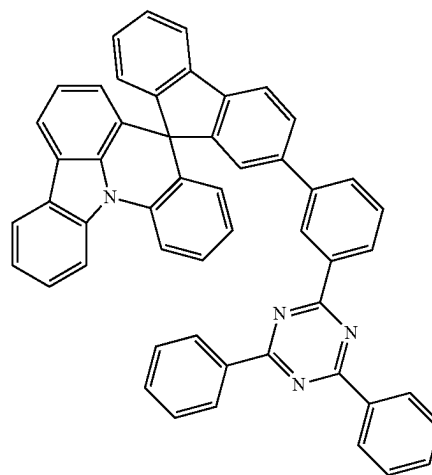
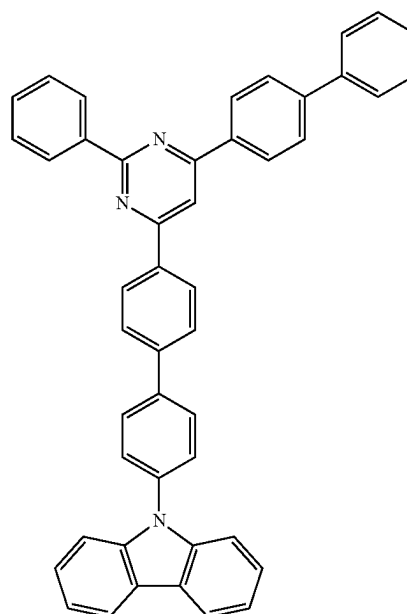
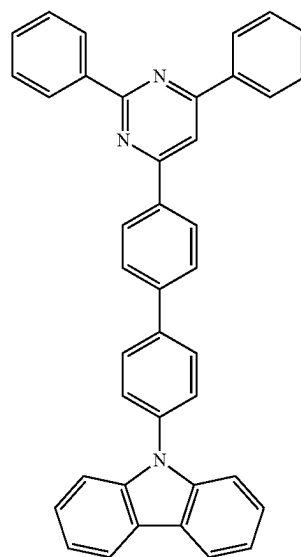


209

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**210**

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E10

E13

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E11

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E12

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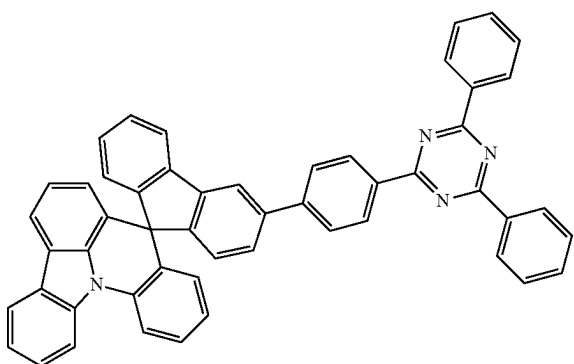
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E14

E15

211

-continued



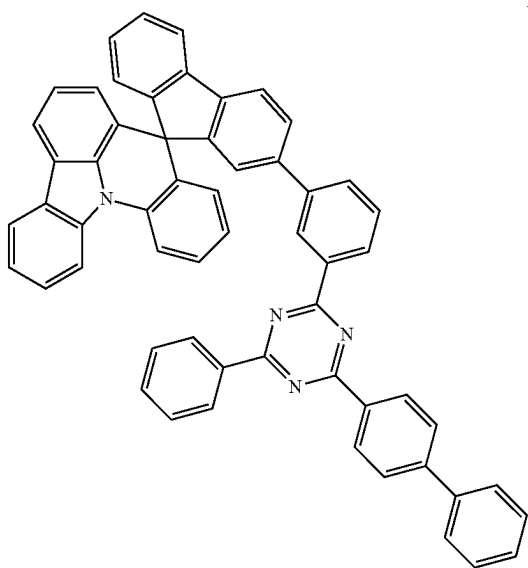
E16

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E17

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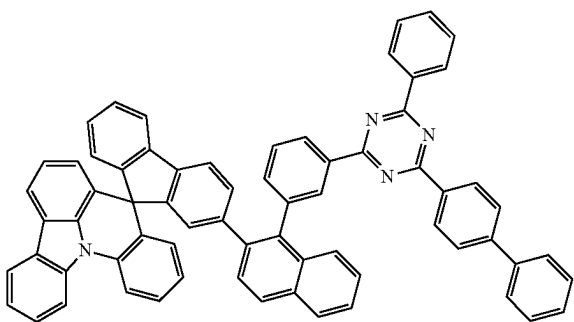
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E18

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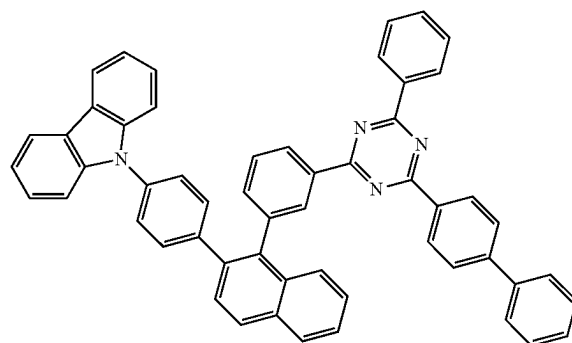


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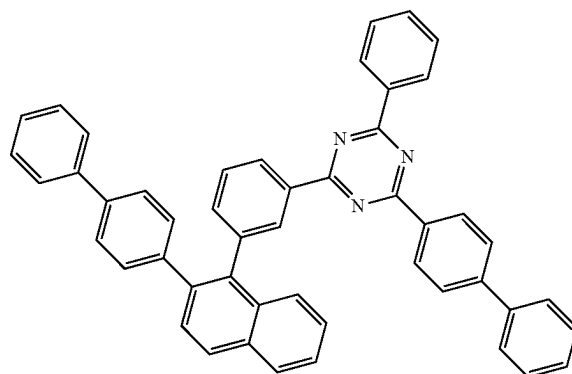
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212

-continued



E19



E20

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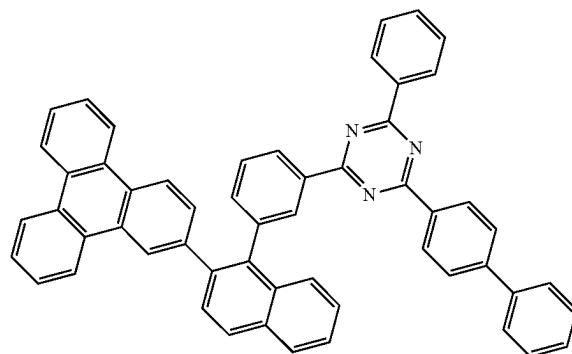
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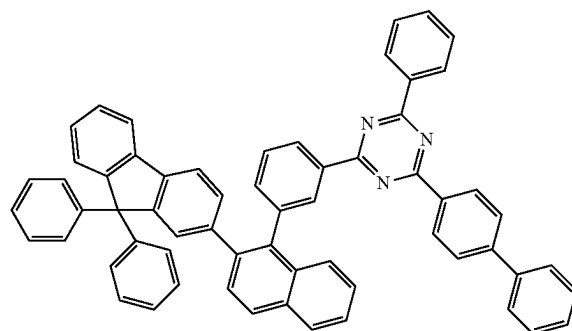
E21



E22

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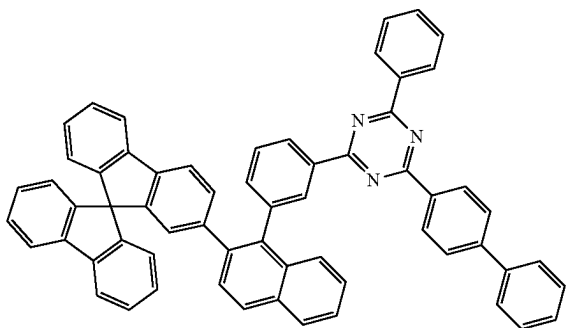
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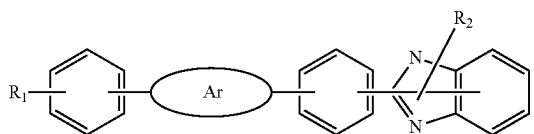
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E23



30. The organic light emitting device of claim 19, wherein the second hole blocking material is represented by Formula 10:

Formula 10

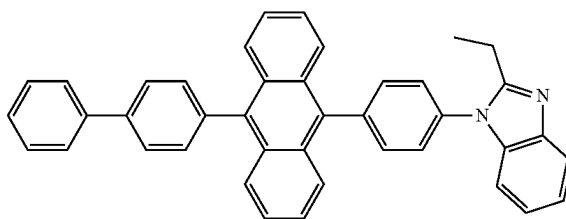


wherein Ar is C₁₀~C₃₀ arylene group, R₁ is C₆~C₃₀ aryl group or C₅~C₃₀ hetero aryl group, and

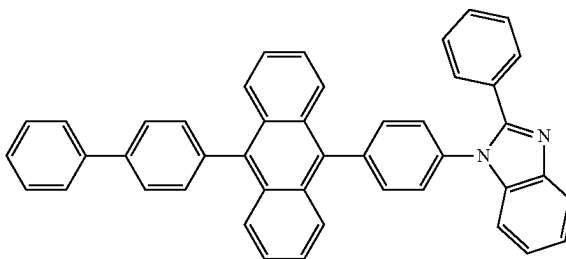
wherein R₂ is C₁~C₁₀ alkyl group or C₆~C₃₀ aryl group.

31. The organic light emitting device of claim 30, wherein the second hole blocking material is a compound being one of the followings of Formula 11:

Formula 11



F1

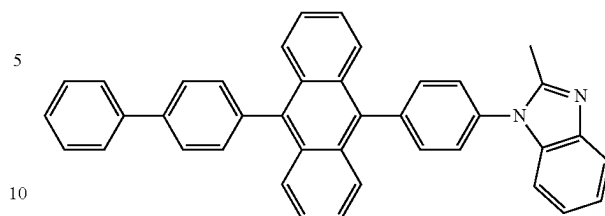


F2

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-continued

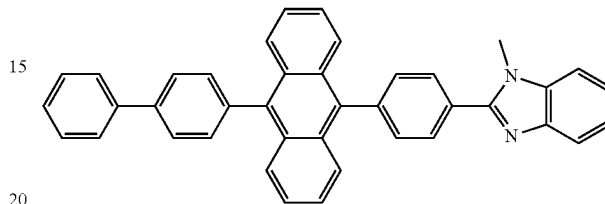
F3



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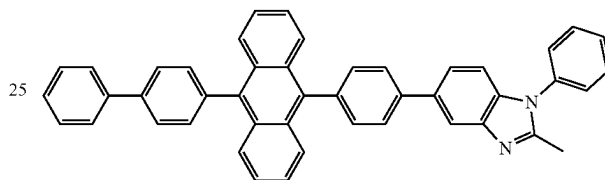
F1



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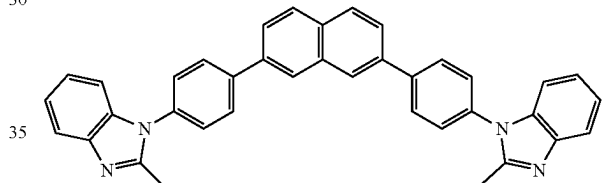
F2



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F6



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32. The organic light emitting device of claim 19, wherein the organic light emitting diode further includes:

a second emitting material layer including a third host, a fourth host and a second blue dopant and positioned between the first emitting material layer and the second electrode; and

a first charge generation layer between the first and second emitting material layers,

wherein the third host is an anthracene derivative, and the fourth host is a deuterated anthracene derivative.

33. The organic light emitting device of claim 32, wherein the organic light emitting diode further includes:

a third emitting material layer emitting a yellow-green light and positioned between the first charge generation layer and the second emitting material layer; and

a second charge generation layer between the second and third emitting material layers.

34. The organic light emitting device of claim 33, wherein a red pixel, a green pixel and a blue pixel are defined on the substrate, and the organic light emitting diode corresponds to each of the red, green and blue pixels, and

wherein the organic light emitting device further includes: a color filter layer disposed between the substrate and the organic light emitting diode or on the organic light emitting diode and corresponding to the red, green and blue pixels.

35. The organic light emitting device of claim 32, wherein the organic light emitting diode further includes:

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a third emitting material layer emitting a red light and a green light and positioned between the first charge generation layer and the second emitting material layer; and

a second charge generation layer between the second and third emitting material layers.

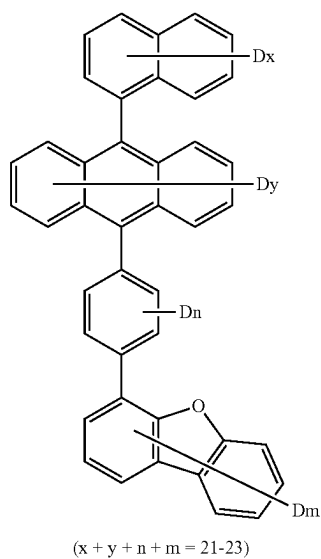
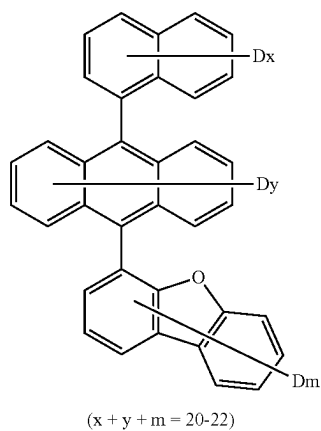
36. The organic light emitting device of claim 19, wherein a red pixel, a green pixel and a blue pixel are defined on the substrate, and the organic light emitting diode corresponds to each of the red, green and blue pixels, and

wherein the organic light emitting device further includes:

a color conversion layer disposed between the substrate and the organic light emitting diode or on the organic light emitting diode and corresponding to the red and green pixels.

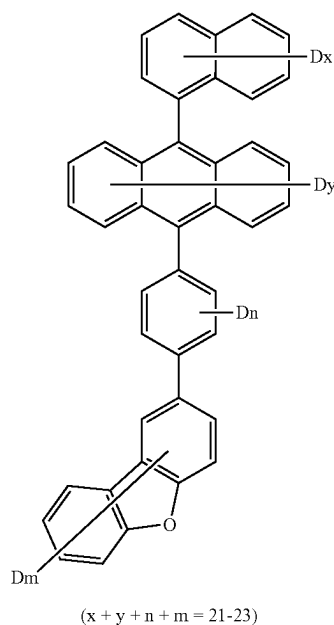
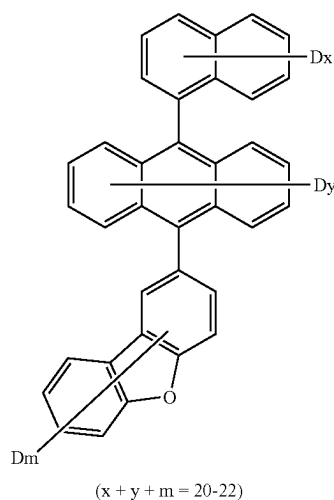
37. The organic light emitting device of claim 19, wherein the second host is a compound being one of the followings of Formula 12:

Formula 12

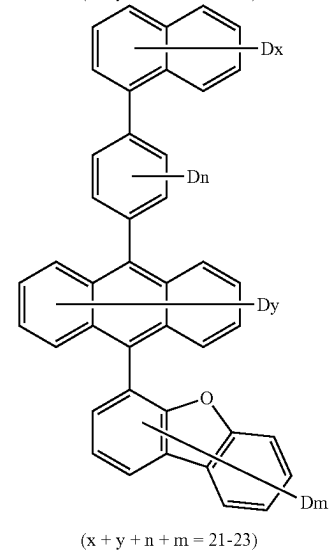


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-continued



Host 34



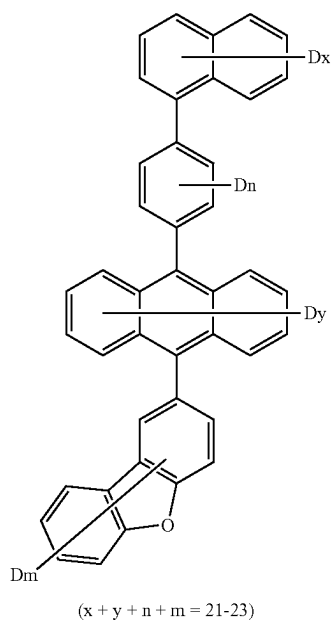
Host 35

Host 36

Host 37

217

-continued



218

-continued

Host 38

Host 40

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Host 39

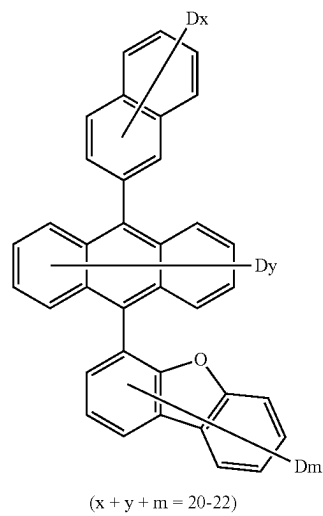
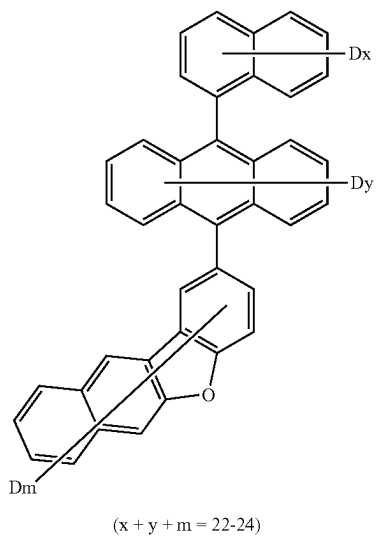
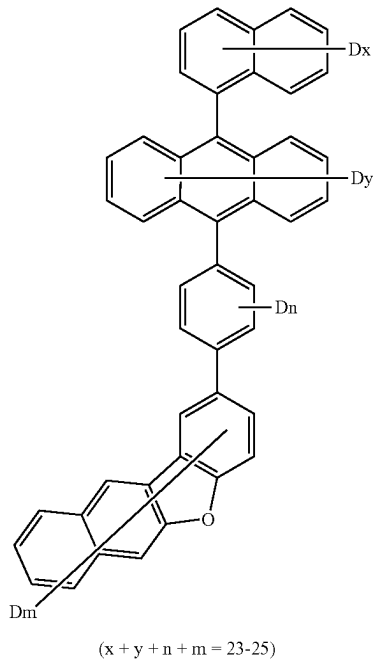
Host 43

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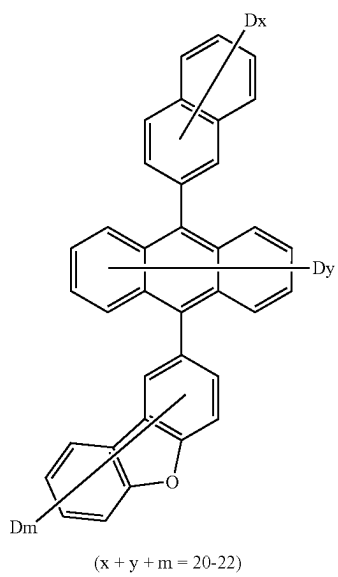
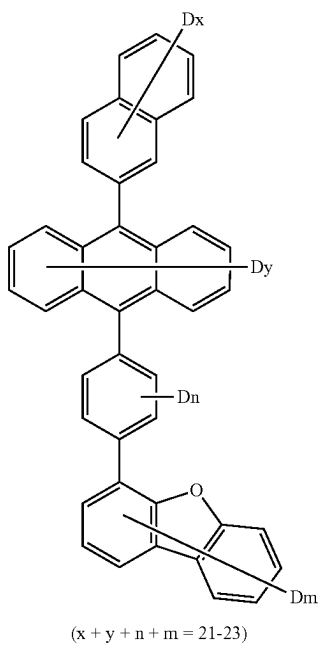
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219

-continued



220

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Host 44

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Host 45

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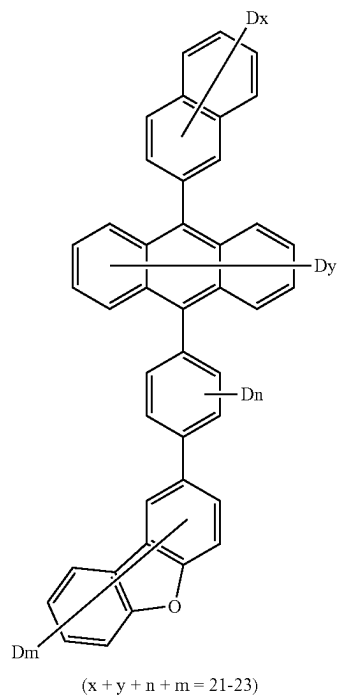
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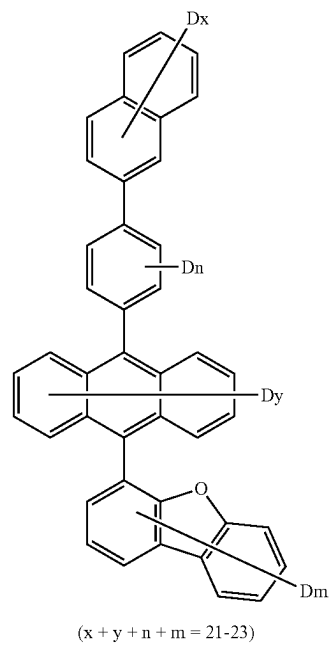
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Host 46

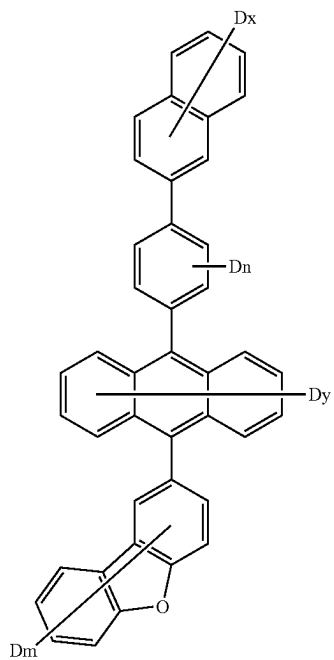


Host 47



221

-continued



$$(x + y + n + m = 21-23)$$

222

-continued

Host 48

Host 50

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Host 49

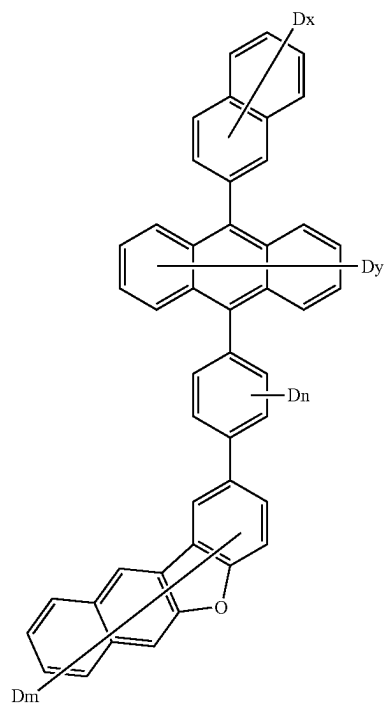
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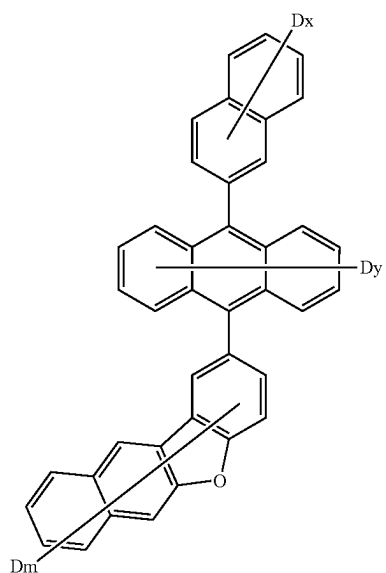
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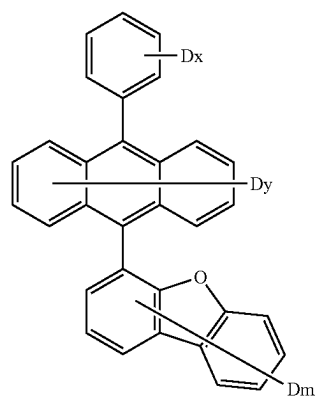
65



$$(x + y + n + m = 23-25)$$



$$(x + y + m = 22-24)$$

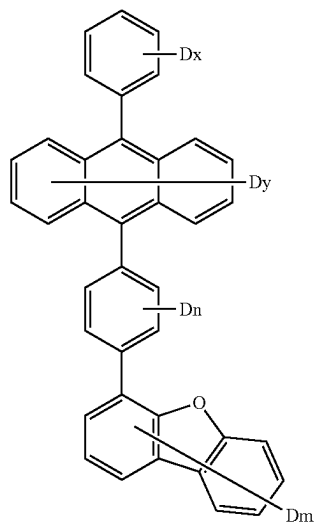


$$(x + y + m = 18-20)$$

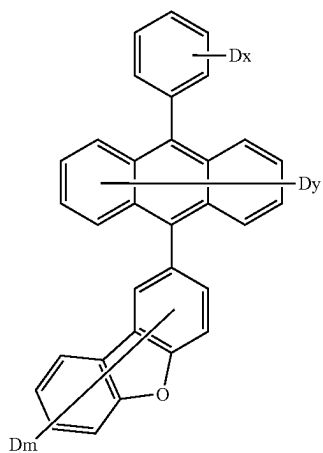
Host 53

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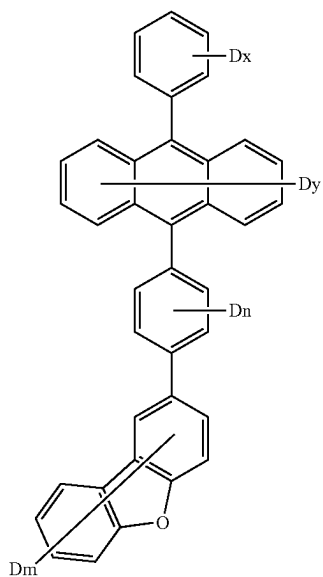
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$$(x + y + n + m = 19-21)$$



$$(x + y + m = 18-20)$$



$$(x + y + n + m = 18-20)$$

224

-continued

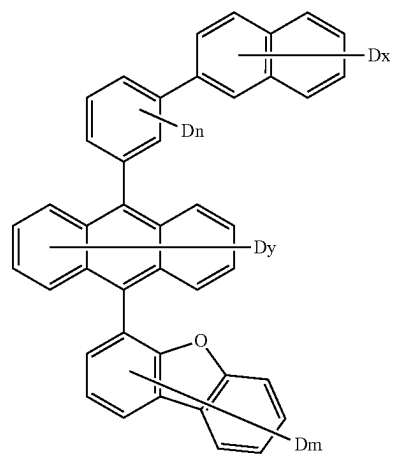
Host 54

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$$(x + y + n + m = 18-20)$$

Host 57

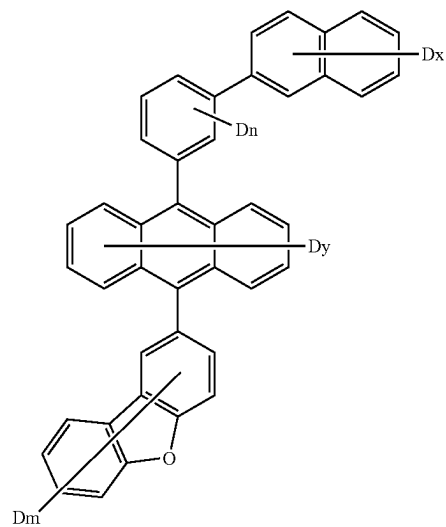
Host 55

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$$(x + y + n + m = 21-23)$$

Host 58

Host 56

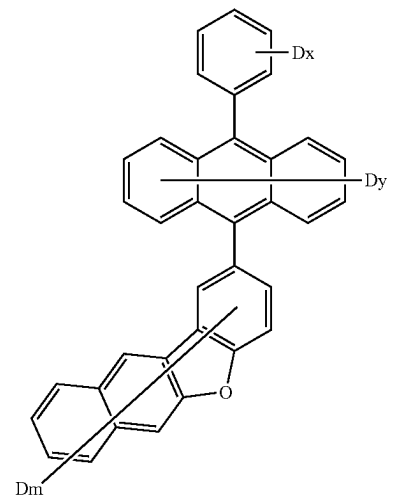
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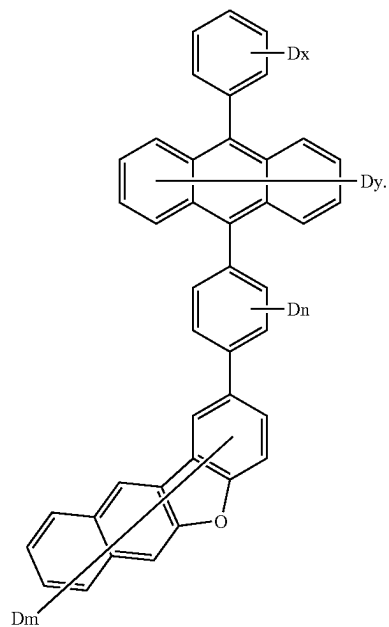


$$(x + y + m = 20-22)$$

Host 59

225

-continued



$$(x + y + n + m = 21-22)$$

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Host 60

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